

Provenance 15 — Fermion fable and the primordial gravitational field: the coupled field equations, the canonical spin connection of the interacting system, the Kohn–Sham source, the density-functional ground and first excited states, the solution from the beginning of the present universe to today, and the answers to questions [1] and [2]

Effort B of 2026-09-24

2026-09-24

Effort B of 2026-09-24. The quantized fermion fable (a complex 16-component Grassmann spinor, canonically quantized with the observer’s time x_4 as time in the $4 + 4$ -dimensional pre-universe) is put into the Einstein equations of the author’s primordial gravitational field as their source. The Einstein tensor of the author’s canonical frame is computed in full, with the free function a_4 never given a value, and what it demands of any source is proved. The frame is generalized to a family on which fable can act back; its asymptotic region is an 8-dimensional Bianchi-I universe. The canonical spin connection of that interacting system is computed and discussed completely. The fable’s energy–momentum tensor operator is evaluated in its Kohn–Sham ground state, and the coupled equations are solved from before nucleosynthesis to today, three times independently (Rust/CVODE, Python/scipy, Mathematica/NDSolve). Ideas of density-functional theory give the fable’s ground and first excited states: in the homogeneous gas and in the discrete states bound along the hidden coordinate x_0 . The author’s two questions — [1] a time-varying dark-energy equation of state, [2] a time-varying dark-matter equation of state — are answered directly and quantitatively, with every caveat.

This page is complete on its own. Everything a reader needs is restated here, including the fermion field itself (section 3) and the canonical spin connection (section 6). Code, logs, data files, notebooks, and the TeX and PDF files are named only as the places where things are computed, recorded or produced.

How to read the evidence. Every mathematical statement on this page carries one of these marks:

mark	meaning
[A], [proved P §n]	an assertion <code>cfAssert[label, test]</code> of the Mathematica notebook, Part P, Section n. The run log prints <code>PASS label</code> ; the label is quoted verbatim , in typewriter type, exactly as the log prints it. The Part VIII labels on this page were spliced into it by a script from the final run log <code>claude-fable/run_fermion_fable_final.log</code> (commit 3a5020d), not retyped
[D], [displayed P §n]	a displayed or printed output of the notebook (a table, a count, a closed form, a <code>Print</code> line of the run log), not an assertion
[P], [prose P §n]	stated only in a <code>Text</code> cell (or a code comment) of the notebook; not checked by an assertion

mark	meaning
[nb05 §n], [nb06 §n], [nb07 §n]	a number printed by an executed cell of the fable-cosmology notebook 05_fermion_fable_quantum_eos.ipynb, 06_fable_primordial_gravity_8d.ipynb or 07_fable_dft_states.ipynb, section n, in the executed versions committed in aca5102; every such cell asserts the claims it prints (a failed assertion stops the notebook). A number quoted from the wall-state report and followed by [nb07 §n] is printed again by notebook 07 (or by a result file it writes in that section), to the digits it prints
[WG n]	a result of the wall-state solver, quoted from its report fable-cosmology/fermion/waveguide_REPORT.txt, section n, which names the log of the run that produced it
[commit h]	stated in the message of commit h of the repository (git log --format=%B -1 h prints it)
[derived here]	a short derivation carried out on this page from marked results, every step shown
[file]	read from a named file of the repository (code, CSV, log), which is named

Numbers are quoted from the files and logs named next to them. Nothing is invented. a_4 is never given a value anywhere: the notebook’s last assertion of Part VIII checks that it is still undefined, PART VIII [control]: a_4 is STILL UNDEFINED – nothing in Part VIII gave it a value; the scale factors scA , scB , scC were only ever substituted inside assertions, and none of them entered the canonical frame.

Notebook 07. The fable-cosmology notebook 07_fable_dft_states.ipynb reproduces the wall-state and density-functional numbers of section 8 from the solver fable-cosmology/fermion/waveguide.py. It was executed and committed in aca5102. The numbers of section 8 are quoted from waveguide_REPORT.txt and its logs; each one that notebook 07 prints again carries the mark [nb07 §n], and the few it does not print again are named at the end of section 8.

Two kinds of “section”. “Section n” with a capital S is a section of the Mathematica notebook (Sections 1–33; Part VII is Sections 26–29, Part VIII is Sections 30–33). “section n” in lower case is a section of this page. The mathematics is written in standard notation; where the notebook’s Wolfram names matter they are given in typewriter type. $T_{16}[a]$ (T16[a]) are the 16×16 Dirac matrices, σ_{16} (sigma16) the spinor metric, $\Psi^\dagger$ the classical conjugate, $\bar{\Psi}$ the Dirac conjugate, ∂_μ a partial derivative, D_μ the spin-covariant derivative, $\sqrt{g}$ (Sqrt[g]) the volume factor $\sqrt{|\det g|}$, and $G^\mu{}_\nu$ the Einstein tensor with one index up and one down.

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Where things live

what	where (relative to the repository root)
the Mathematica notebook (generated, never edited by hand)	claude-fable/claude-fable_Einstein-Rosen-2-Planes.nb
its cell manifests; Part VIII (Sections 30–33) is this effort’s symbolic side, Part VII (Sections 26–29) the fermion field	claude-fable/cells_part1.wl ... cells_part8.wl; claude-fable/cells_part8.wl, claude-fable/cells_part7.wl
the builder, the run harness, the runner, the checker, the section map	claude-fable/build_tools.py, claude-fable/runner_header.wl, claude-fable/run_from_nb.wls, claude-fable/verify_nb.wls, claude-fable/nb_section_map.wls
the final run log of the whole notebook, Parts I–VIII (217 Input cells, 644/644 assertions, 0 cells with messages, 198.352 s, 46 non-vanishing witnesses; commit 3a5020d)	claude-fable/run_fermion_fable_final.log
Part VIII’s acceptance run (217 Input cells, 642/642: the two comparisons with the Rust solver’s CSVs skipped, because those CSVs did not yet exist, and three spin-connection labels as they read before 3a5020d; commit bc04ed1)	claude-fable/run_fermion_fable_part8.log
the run log through Part VII, Part VII’s own acceptance run (198 Input cells, 528/528)	claude-fable/run_fermion_fable_part7.log
the checker’s log and the section map	claude-fable/verify_nb_part8.log; claude-fable/nb_section_map.log

what	where (relative to the repository root)
the TeX/text export of the fermion field's equations	provenance-latex/generated/*.tex, *.txt (written by claude-fable/export_fermion_fable_tex.wls)
the solver of the coupled system (Rust, pure-Rust SUNDIALS 7.8.0 CVODE)	fable-cosmology/rust/fable_fermion/ (src/constants.rs, numerics.rs, potentials.rs, kohn_sham.rs, models.rs, run.rs, cvode_driver.rs, main.rs; tests/cosmology.rs)
the independent Python implementation of the coupled system	fable-cosmology/fermion/crosscheck.py
the Mathematica reference runs of the coupled system	fable-cosmology/reference/make_reference_fermion.wls → mathematica_fable4d_mass30eV.csv, mathematica_fable4d_power.csv
the wall-state (x_0) solver, its derivation, its Mathematica check, its report	fable-cosmology/fermion/waveguide.py; waveguide_derivation.wls → waveguide_derivation.log; waveguide_check.wls; waveguide_REPORT.txt
the notebooks (generated by notebooks/_build/nbgen.py, checked by notebooks/_build/nbcheck.py)	fable-cosmology/notebooks/05_fermion_fable_quantum_eos.ipynb, 06_fable_primordial_gravity_8d.ipynb, 07_fable_dft_states.ipynb
every CSV table and PNG figure the notebooks write	fable-cosmology/results/nb05_*, nb06_*, nb07_*
setup and the complete reproduction	fable-cosmology/setup.sh, setup.ps1; fable-cosmology/run_all.sh, run_all.ps1
the LaTeX twin of this page, its preamble and build scripts	provenance-latex/PROVENANCE-15-FERMION-FABLE-AND-THE-PRIMORDIAL-GRAVITATIONAL-FIELD.tex → .pdf; provenance-latex/preamble.tex; provenance-latex/build_all.sh, build_all.ps1

<repo> below is wherever the repository is cloned. Every shell block finds it with `git rev-parse --show-toplevel`.

0 The task, in the author's words

On 2026-09-24 the author gave three instructions. This page answers the second, and it follows the third. Their operative parts, verbatim (the ellipses [...] are cuts in the quotation, not in the instruction's meaning):

B. “Include fable as a source in the Einstein Gravitational Field Equations for this primordial Gravitational Field, and solve the coupled field equations over a time interval from the beginning of the present universe until the present time. Try to use some of the relevant ideas from Density Functional Theory to obtain the fable ground and first excited states. Create another new provenance markdown file, latex, and pdf files [...] the interacting (fermion fable field, primordial Gravitational Field) system [...] coupled equations, canonical spin connection for the interacting system, energy-momentum tensor operator, pressure, density and equations of state [...] complete discussion of the canonical spin connections. Answer: [1] Does this system provide a physical mechanism for a time-varying dark energy equation of state (w)? [2] [...] time-varying dark matter equation of state (w)?”

C. “Thoroughly discuss the commands that you use to solve the coupled system, and to test, verify, execute, and display your calculations of the coupled field equations. Create another new provenance markdown file, latex, and pdf files that contain your complete solution. Each of your major efforts should have its own section in the

markdown file, latex, and pdf files. Do not ever refer the USER to another markdown file for any reason. The full set of 100% complete ideas must be given its own section in each markdown file.” And: push, check and verify the repository. “DO NOT TAKE SHORTCUTS.”

The first instruction (Effort A) asked for the fermion fable itself: a complex 16-spinor, quantized canonically in $4 + 4$ dimensions, with its field equations in the primordial gravitational field, its energy–momentum tensor operator, pressure, density and equations of state, and a complete discussion of the canonical spin connection. Effort B stands on it, so this page restates all of it that the coupled system uses: section 3 (the field) and section 6 (the canonical spin connection, followed by the connection of the interacting system). The complete set of ideas of all three efforts is section 1.

The work of Effort B, in the order it was done.

1. A design (revision 1) was written for the whole of 2026-09-24’s work. Four adversarial reviewers examined it — quantization; field equations, energy–momentum tensor and spin connection; Einstein equations and cosmology (findings E-*); density-functional theory and numerics (findings DFT-*) — each followed by an independent skeptic. No finding was refuted; several were confirmed only in part, and the corrected fix was adopted. Every adopted correction that concerns Effort B is stated where it applies on this page.
2. The solver `fable-cosmology/rust/fable_fermion` was written on the vendored pure-Rust SUNDIALS 7.8.0 CVODE of the rustSolveIt repositories, with the independent Python check `fable-cosmology/fermion/crosscheck.py` (commit c5892bd).
3. Part VIII of the Mathematica notebook (Sections 30–33, manifest `claudio-fable/cells_part8.wl`) proves the symbolic side: the Einstein tensor, the three theorems, the generalized frame, the Bianchi-I system, the spin connection of the interacting system, and the coupled equations for the Kohn–Sham fluid, and it solves two reference cases with NDSolve (commit bc04ed1).
4. The fable-cosmology notebooks `05_fermion_fable_quantum_eos.ipynb` and `06_fable_primordial_gravity_8d.ipynb` run the solver on every case, re-compute every run independently, and state the answers.
5. The wall-state solver `fable-cosmology/fermion/waveguide.py`, its symbolic derivation and its Mathematica check compute the discrete states along x_0 and their density-functional treatment (commit 18a2501). Notebook `07_fable_dft_states.ipynb` reproduces them.
6. Six defects of the solver that the notebooks exposed were fixed (commit 2bc936d); notebooks 05–07 were then executed on the final solver and committed with their results (commit aca5102).
7. The whole Mathematica notebook, Parts I–VIII, was run once more after notebook 06 had written the Rust solver’s CSVs, so that Part VIII’s two comparisons with them ran (commit 1671155); the fact-check of this page then found that three Part VIII labels wrote spin-connection components with upper flat indices where the compared array has both indices down, the labels and their Text cell were corrected, and the final run was recorded again (commit 3a5020d, 644/644).

1 The complete set of ideas

{{COMPLETE-SET-OF-IDEAS}}

2 Foundations: the pre-universe, its algebra, the canonical frame, the author's L_a and eLa, and the classical fable of Part VI

This section sets out everything that the new work on fermion fable stands on. It restates the author's pre-universe (a curved $4 + 4$ -dimensional spacetime), the real Clifford algebra and split-octonion spinors built on it, the author's own wave function, Lagrangian and field equations, the canonical frame field, and the classical field fable of 2026-09-16. It ends with the reason why that classical field has to be refined into a complex 16-component spinor. The reader needs nothing outside this document to follow it.

2.1 How the evidence is cited

All the mathematics lives in one Mathematica notebook, `claude-fable/claude-fable_Einstein-Rosen-2-Planes.nb`. That notebook is never edited by hand. Its Parts I–VI, which this section restates, are generated from the cell manifests `claude-fable/cells_part1.wl ... cells_part6.wl`. In a manifest, `Text` cells explain and `Input` cells compute. Every claim the notebook checks is an assertion `cfAssert[label, test]`, which prints `PASS label` or `FAIL label`. The complete run of Parts I–VI is recorded in `claude-fable/run_fable51_part6.log`. It evaluated 172 Input cells in 162.5s with no cell raising a message, and its final tally reads (the witness line is shortened here; in the log it continues with a one-sentence explanation of what a witness is):

```
identities accepted on numerical evidence alone : 0    (stage 3 returned True)
non-vanishing witnesses                        : 27
PASS no identity in this notebook was accepted on numerical evidence alone
Assertions run: 358   passed: 358   failed: 0
```

Each statement below is tagged with one of five kinds of evidence:

- **[A]** an assertion. Its label is quoted exactly as the log prints it after `PASS`.
- **[D]** a displayed result: an Output the notebook prints but does not assert. Where a displayed result was re-read for the provenance pages of 2026-09-24 from the saved kernel state, that is said.
- **[P]** prose only: a statement that appears in a `Text` cell and is not checked by any assertion.
- **[VII]** proved in Part VII, the new part of the notebook written for fermion fable (Sections 26–29). Its labels are quoted verbatim from `claude-fable/run_fermion_fable_part7.log`, which reports `Assertions run: 528 passed: 528 failed: 0` for Parts I–VII together.
- **[VIII]** proved in Part VIII, the notebook's coupled system (Sections 30–33). Its labels are quoted verbatim from the final run of the whole notebook, `claude-fable/run_fermion_fable_final.log` (commit 3a5020d), which reports `Assertions run: 644 passed: 644 failed: 0` for Parts I–VIII together (sections 4–7 of this page). Part VIII's acceptance run, `claude-fable/run_fermion_fable_part8.log` (commit bc04ed1), reports 642/642: in it the two comparisons with the Rust solver's CSVs were skipped, because those CSVs did not yet exist, and three spin-connection labels still wrote the components with upper flat indices (section 6.15).

How an identity is certified (restated from Section 15 of the notebook). `cfZeroQ` and `cfZeroArrayQ` try three stages in turn:

1. exact structural zero;
2. Simplify under the geometric assumptions, with a 5-second budget;
3. a 30-digit numerical evaluation at three rational probe points, with test functions substituted for the undetermined a_4 (and for the undetermined rapidity of the notebook's boosted-frame bridge); the result is accepted as zero only if its magnitude is below 10^{-22} .

An identity accepted at stage 3 counts as a *numerical certificate*, and the run reports zero of those: every identity in Parts I–VI is closed symbolically. A claim that something is *not* zero is made with `cfNonZeroWitnessQ`. It passes only if, at some probe point, every entry evaluates to a number and at least one of them has magnitude above 10^{-10} . That is a complete proof of non-vanishing; the run has 27 such witnesses. The probe functions are test inputs only, never definitions.

The session policy is the original notebook’s, unchanged: `Simplify` has a 1-second budget and `FullSimplify` a 3-second budget. `DIM8 = 8`, and the model’s three constants M (a mass), K and H (a Hubble-like constant) are `Protected`, so no cell can assign them (Section 1: a `Text` cell and the `Input` cells that set this up; no assertion) [P].

2.2 Coordinates, the flat 4+4 metric, and which directions are timelike

The coordinates are $X = \{x_0, x_1, \dots, x_7\}$. The flat tangent-space metric is

$$\text{eta4488} = \eta = \text{diag}(+1, +1, +1, +1, -1, -1, -1, -1), \quad \text{signature } (4, 4). \quad (1)$$

coordinate	η	name in the original notebook (Section 2)	reading used in this work
x_0	+1	hidden space	the hidden spacelike coordinate
x_1, x_2, x_3	+1	ordinary 3-space	the observed sheet
x_4	−1	time	the observer’s time
x_5, x_6, x_7	−1	superluminal deflating time directions	the hidden timelike sheet

[A] `eta4488 . eta4488 == ID8; eta4488 is symmetric; signature of eta4488 is (4,4).`

Index conventions (restated from Section 2). Latin indices a, b, A, B are flat and run 0..7. Greek indices μ, ν are curved and run 0..7. Wolfram arrays are 1-based, so index A sits at array position $A + 1$. For example, x_4 is entry 5 of X , so Part VI’s energy density $\rho = T_{44}$ is the array element `T[[5,5]]`.

The notebook uses two sets of assumptions:

- `ssX = H > 0 && sX0 && 6 H x0 > 0 && 2 la[x4] > 0 && Cot[6 H x0] > 0 && Sin[6 H x0] > 0`, with `sX0 = x0 > 0 && x1 > 0 && ... && x7 > 0` (all eight coordinates positive), for the (x_0, x_4) chart;
- `constraintVars = x0 > 0 && x4 > 0 && z > 0 && t > 0`, for the chart $z = 6Hx_0, t = Hx_4$.

`la` is a free function inherited from the original notebook. It occurs only inside `ssX`. It is never given a value, and nothing here invents one [P, Section 2].

2.3 $\text{SO}(4)$: the self-dual and anti-self-dual blocks

The six antisymmetric real 4×4 matrices split under the Hodge star into a self-dual 3-space and an anti-self-dual 3-space, the two commuting $\mathfrak{su}(2)$ factors of $\mathfrak{so}(4)$. The original notebook builds both from two elementary tensors (Section 3):

$$Q_a[h, p, q] = \text{Signature}[\{h, p, q, 4\}], \quad Q_b[h, p, q] = \delta_{p4}\delta_{qh} - \delta_{ph}\delta_{q4}, \quad (2)$$

$$\text{s4by4}[h] = Q_a - Q_b \text{ (self-dual)}, \quad \text{t4by4}[h] = Q_a + Q_b \text{ (anti-self-dual)}, \quad (3)$$

$$h = 1, 2, 3. \quad (4)$$

Explicitly [D]:

$$s_1 = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & -1 & 0 & 0 \\ -1 & 0 & 0 & 0 \end{pmatrix}, \quad s_2 = \begin{pmatrix} 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \end{pmatrix}, \quad s_3 = \begin{pmatrix} 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & -1 & 0 \end{pmatrix},$$

$$t_1 = \begin{pmatrix} 0 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 \\ 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix}, \quad t_2 = \begin{pmatrix} 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix}, \quad t_3 = \begin{pmatrix} 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 \end{pmatrix},$$

with $s_h = \text{s4by4}[h]$ and $t_h = \text{t4by4}[h]$.

[A] `s4by4` is self-dual; `t4by4` is anti-self-dual; `s4by4` and `t4by4` are antisymmetric; the two `su(2)` factors commute; and, for the nine mixed products `st[J,K] = s_J t_K`, `st[J,K]` is symmetric.

2.4 The real 8×8 Clifford generators τ , $\bar{\tau}$, and the spinor metric σ

The $O(4,4)$ spinor metric is

$$\sigma = \begin{pmatrix} 0 & I_4 \\ I_4 & 0 \end{pmatrix}. \quad (5)$$

[A] `sigma . sigma == ID8`; `sigma` is symmetric; `Tr[sigma] == 0`. The type-1 and type-2 split-octonion spinors both carry this same σ , because $\sigma = \sigma^{-1}$ [P].

The generators (Section 4) are

$$\tau[0] = I_8, \quad (6)$$

$$\tau[h] = \begin{pmatrix} 0 & s_h \\ s_h & 0 \end{pmatrix}, \quad h = 1, 2, 3, \quad (7)$$

$$\tau[3+k] = \begin{pmatrix} 0 & t_{4-k} \\ -t_{4-k} & 0 \end{pmatrix}, \quad k = 1, 2, 3 \quad (t_3, t_2, t_1), \quad (8)$$

$$\tau[7] = \tau[1]\tau[2]\tau[3]\tau[4]\tau[5]\tau[6] = \text{diag}(-I_4, +I_4) \quad [\text{D}], \quad (9)$$

$$\bar{\tau}[A] = \sigma (\sigma \tau[A])^T, \quad A = 0, \dots, 7 \quad (\text{the } \sigma\text{-adjoint}). \quad (10)$$

The order 3,2,1 is the original's. It is what makes $\tau[7]$ the product of the first six and $\sigma = \tau[1]\tau[2]\tau[3]$ [P]. The original calls the list of the six blocks `sixAntiSymmetric8by8`, but that name fits only half of it: [A] blocks 1-3 (built from `s4by4`) are antisymmetric; blocks 4-6 (built from `t4by4`) are symmetric.

The defining relation is the split Clifford algebra of the $4 + 4$ quadratic form in its real 8×8 “half-spinor” form:

$$\frac{1}{2}(\tau[A] \bar{\tau}[B] + \tau[B] \bar{\tau}[A]) = \eta_{AB} I_8. \quad (11)$$

[A] `(1/2)(tau[A].taubar[B] + tau[B].taubar[A]) == eta4488[[A+1,B+1]] ID8`.

The ordering identities [A]:

- `sigma == tau[1].tau[2].tau[3]`
- `sigma == tau[4].tau[5].tau[6].tau[7]`
- `tau[1]...tau[7] == tau[0] == ID8`
- `sigma . taubar[A] == Transpose[sigma . tau[A]]`
- `-eta4488[[A+1,A+1]] tau[A] == Transpose[tau[A]]` for $A = 1..7$

The last one says that $\tau[1..3]$ are antisymmetric and $\tau[4..7]$ are symmetric. The second invariant of the original is $\Omega = \sigma \tau[7]$: [A] `Omega == tau[4].tau[5].tau[6]`.

A complete basis of the 8×8 algebra (Section 7) is the 63 ordered products of 1, 2 or 3 of $\tau[1..7]$, together with I_8 . [A] the 8×8 basis has 64 elements; the 64 elements are distinct; `(1/8) Tr[bas64[A].bas64[B]] == eta64[[A,B]]`; symmetric 8×8 basis elements number 36; antisymmetric 8×8 basis elements number 28.

2.5 The 16×16 Dirac matrices, chirality, σ_{16} , and $\mathfrak{so}(4, 4)$

Following E. A. Lord's reduced Brauer–Weyl construction (Math. Proc. Camb. Phil. Soc. 64 (1968) 765–778), the 8×8 generators are doubled off-diagonally (Section 5):

$$T_{16}[A] = \begin{pmatrix} 0 & \bar{\tau}[A] \\ \tau[A] & 0 \end{pmatrix} \quad (A = 0, \dots, 7), \quad (12)$$

$$T_{16}[8] = T_{16}[0]T_{16}[1] \cdots T_{16}[7], \quad \sigma_{16} = T_{16}[0]T_{16}[1]T_{16}[2]T_{16}[3]. \quad (13)$$

The $T_{16}[A]$ are real 16×16 matrices, and “ $\gamma[a]$ ” means $T_{16}[a]$. $T_{16}[8]$ is the chirality operator and σ_{16} is the 16×16 spinor metric.

Assertions [A]:

- `{T16[A], T16[B]}/2 == eta4488[[A+1,B+1]] ID16`
- `sigma16 == ArrayFlatten[{{-sigma,0},{0,sigma}}]`
- `sigma16 . T16[A] is antisymmetric for A = 0..7`
- `T16[8] == sigma16 . T16[4].T16[5].T16[6].T16[7]`
- `sigma16 is symmetric and sigma16 . sigma16 == ID16`
- `sigma16 . covariantDiffMatrix is symmetric, with covariantDiffMatrix = T16[5]T16[6]T16[7]`

The two chiralities are the two split-octonion spinors. The notebook defines $P_L = \frac{1}{2}(I_{16} - T_{16}[8])$ and $P_R = \frac{1}{2}(I_{16} + T_{16}[8])$. [A] `PL + PR == ID16`; `PL is idempotent`; `PR is idempotent`; `PL . PR == PR . PL == 0`; `PL projects onto the upper (type-1) components, PR onto the lower (type-2) components.`

The last assertion is equivalent to

$$T_{16}[8] = \text{diag}(-I_8, +I_8). \quad (14)$$

A 16-component spinor is therefore the direct sum $\Psi = (\psi_1, \psi_2)$:

- the upper eight components are the **type-1** split-octonion spinor, of chirality -1 ;
- the lower eight components are the **type-2** spinor, of chirality $+1$.

Every $T_{16}[A]$ exchanges the two halves, since it is block off-diagonal [P].

The $\mathfrak{so}(4,4)$ generators are

$$S^{AB} = \text{SAB}[[A+1,B+1]] = \frac{1}{4} [T_{16}[A], T_{16}[B]]. \quad (15)$$

Assertions [A]:

- `SAB is antisymmetric in its two flat indices`
- `sigma16 . SAB is antisymmetric`
- `T16[8] commutes with every SAB: the chiral halves are each Spin(4,4)-invariant`
- `[SAB,SAB] closes on the so(4,4) algebra, which is the relation  $[S_{AB}, S_{CD}] = -(\eta_{AC}S_{BD} - \eta_{AD}S_{BC} - \eta_{BC}S_{AD} + \eta_{BD}S_{AC})$`
- `[SAB, T16] reproduces the vector representation, which is  $[S_{AB}, \gamma_C] = -\eta_{CA}\gamma_B + \eta_{CB}\gamma_A$`

The spin connection of the frame field enters as $\frac{1}{8} \omega_{\mu ab}[\gamma^a, \gamma^b] = \frac{1}{2} \omega_{\mu ab} S^{ab}$.

The 256-element basis (Section 6) is the ordered products of k distinct generators, $k = 1..8$, followed by I_{16} , taken in the original's order. [A] `base16[[93]]` is `sigma16` (the word $\{0,1,2,3\}$); `base16[[255]]` is the chirality operator `T16[8]`; `base16[[256]]` is the identity; `Tr[M.M]/16` is $+1$ or -1 for every basis element; 136 basis elements have `Tr[M.M] > 0`; 120 basis elements have `Tr[M.M] < 0`; symmetric basis elements number $16 \cdot 17/2 = 136$; antisymmetric basis elements number $16 \cdot 15/2 = 120$; `Tr[M.M] > 0` exactly when M is symmetric; `Tr[M.M] < 0` exactly when M is antisymmetric.

Section 8 also builds the ordinary 4×4 Dirac algebra of the (x_1, x_2, x_3, x_4) block, whose metric is $g_{44} = \text{diag}(1, 1, 1, -1)$. For example [A]: `{gamma4[h], gamma4[k]} == 2 g44[[h,k]]` ID4 for $h, k = 1..4$; [S44,S44] closes on the $\text{so}(3,1)$ algebra. It plays no role below.

2.6 Cartan triality and the split octonions (Section 9): what is proved

The vector, the type-1 spinor and the type-2 spinor are three 8-dimensional representations of $\text{Spin}(4,4)$ that triality relates. The original makes the vector-to-spinor bridge concrete with one distinguished spinor, `unit`. It is row 8 of the eigenvectors of σ , rescaled by $1/\sqrt{2}$. Assertions [A]:

- `unit` is pinned to the author's spinor $\{1/\sqrt{2}, 0, 0, 0, 1/\sqrt{2}, 0, 0, 0\}$ - the row order of `Eigensystem` is not relied on silently
- each `uEig` row is sigma-normalized to ± 1
- `unit` is sigma-normalized to $+1$

The bridge and its inverse are

$$\text{triVecToSpin}[[A + 1, a + 1]] = (\text{unit} \cdot \sigma \cdot \bar{\tau}[A])[[a + 1]], \quad (16)$$

$$\text{triSpinToVec} = \text{triVecToSpin}^{-1}, \quad (17)$$

where the row index A of `triVecToSpin` is a flat vector index and the column index a is a type-1 spinor index. Assertions [A]:

- `triVecToSpin . triSpinToVec == ID8`
- `triSpinToVec . triVecToSpin == ID8`
- `triSpinToVec == Transpose[ eta_AA tau[A].unit ]` (the original's closed form)
- `etaTri` is symmetric; `etaTri` has signature $(4,4)$
- `triVecToSpin` transports `etaTri` back to `eta4488`
- THE TRIALITY METRIC IS EXACTLY THE SPINOR METRIC: `etaTri == sigma`

The split-octonion structure constants are read off from the generators through the bridge. In the type-1 spinor basis they are `mabc`; in the vector basis they are `mABC`, the usual multiplication table. Assertions [A]:

- `eA[1]` is a two-sided identity for the vector-basis product
- `<eA, eB>` composition is compatible with `eta` (basis-unit norm form)
- `m_{ABC}` restricted to imaginary directions is totally antisymmetric

The scope of what is proved: the norm is multiplicative *on the basis units*, and the imaginary structure constants are totally antisymmetric. The general composition law $N(xy) = N(x)N(y)$ for arbitrary elements, and alternativity, are **not** asserted.

2.7 The wave function of the un-universe and the author's Lagrangian L_a (Section 11)

The original notebook calls `Psi16` the wave function of the “un-universe”. It is sixteen scalar functions of the hidden coordinate x_0 and the time x_4 :

$$\Psi_{16} = (f_0, \dots, f_{15}), \quad f_k \equiv \text{f16}[k][x_0, x_4], \quad (18)$$

where the upper eight components are the type-1 spinor and the lower eight the type-2 spinor. [A] `Psi16` is the direct sum of the type-1 and type-2 spinors. Two re-parametrizations are recorded:

- `f16[k] -> Z[k][6 H #1, H #2] &`, which is the one used downstream;
- `f16[k] -> nZ[k][6 H #1, H #2]/Sqrt[Sin[6 H #1]] &`, the author's alternative normalization.

The author's Lagrangian, exactly as defined in Section 11:

```
La[] := ((1/H) Transpose[Psi16] . sigma16 .
  Sum[T16[alpha1 - 1] . D[Psi16, X[[alpha1]]], {alpha1, 1, Length[X]}]
  + 2 (M/H) Transpose[Psi16] . sigma16 . Psi16
) // Simplify[#, constraintVars] &
```

that is,

$$L_a = \frac{1}{H} \Psi_{16}^T \sigma_{16} T_{16}[\alpha] \partial_\alpha \Psi_{16} + \frac{2M}{H} \Psi_{16}^T \sigma_{16} \Psi_{16} \quad (19)$$

$$= \frac{1}{H} \Psi_{16}^T \sigma_{16} (T_{16}[0] \partial_{x_0} + T_{16}[4] \partial_{x_4}) \Psi_{16} + \frac{2M}{H} s, \quad s = \Psi_{16}^T \sigma_{16} \Psi_{16}. \quad (20)$$

The second line holds because Ψ_{16} depends only on x_0 and x_4 . It was checked for this document against the evaluated `La[]` (a `Simplify` difference of exactly 0); it is not a notebook assertion. The mass term is $(2M/H) s$: it is $+2M/H$ times the scalar bilinear s .

The evaluated `La[] [D]` is shown below. Write $f_k = \text{f16}[k][x_0, x_4]$, $\partial_0 f_k = \text{Derivative}[1, 0][\text{f16}[k]][x_0, x_4]$ and $\partial_4 f_k = \text{Derivative}[0, 1][\text{f16}[k]][x_0, x_4]$. Then $H L_a$ is exactly

$$\begin{aligned} H L_a = & -4M(f_0 f_4 + f_1 f_5 + f_2 f_6 + f_3 f_7 - f_8 f_{12} - f_9 f_{13} - f_{10} f_{14} - f_{11} f_{15}) \\ & + f_{12}(\partial_4 f_5 + \partial_0 f_0) + f_{13}(-\partial_4 f_4 + \partial_0 f_1) + f_{14}(-\partial_4 f_7 + \partial_0 f_2) \\ & + f_{15}(\partial_4 f_6 + \partial_0 f_3) + f_8(-\partial_4 f_1 + \partial_0 f_4) + f_9(\partial_4 f_0 + \partial_0 f_5) \\ & + f_{10}(\partial_4 f_3 + \partial_0 f_6) + f_{11}(-\partial_4 f_2 + \partial_0 f_7) + f_4(\partial_4 f_{13} - \partial_0 f_8) \\ & - f_5(\partial_4 f_{12} + \partial_0 f_9) - f_6(\partial_4 f_{15} + \partial_0 f_{10}) + f_7(\partial_4 f_{14} - \partial_0 f_{11}) \\ & - f_0(\partial_4 f_9 + \partial_0 f_{12}) + f_1(\partial_4 f_8 - \partial_0 f_{13}) + f_2(\partial_4 f_{11} - \partial_0 f_{14}) \\ & - f_3(\partial_4 f_{10} + \partial_0 f_{15}). \end{aligned}$$

The structural identity behind it. $\sigma_{16} T_{16}[A]$ is antisymmetric (Section 5). An antisymmetric form vanishes on the diagonal, so for commuting components $\Psi_{16}^T \sigma_{16} T_{16}[\alpha] \Psi_{16} = 0$ identically. The kinetic term therefore equals minus its transpose, with no boundary term. [A] `Psi16 . sigma16 . T16[alpha] . Psi16 == 0` identically (antisymmetric form); the kinetic term equals minus its transpose, exactly; `La[] ==` its conjugate form when the coefficients match. The last one is $L_a = -\frac{1}{H} (\partial_\alpha \Psi_{16})^T \sigma_{16} T_{16}[\alpha] \Psi_{16} + \frac{2M}{H} \Psi_{16}^T \sigma_{16} \Psi_{16}$. The original's scratch cell compared L_a with a form carrying mismatched factors of H and so found a non-zero difference [P].

2.8 The Euler–Lagrange equations eLa (Section 12)

The original’s Euler–Lagrange operator, transcribed exactly:

```
eL[Lagrangian_Symbol, detsqrt_] := Module[{L, t},
  L = Lagrangian[];
  t = Table[
    FullSimplify[
      (1/detsqrt) ( D[L, f16[k][x0, x4]]
        - D[D[L, Derivative[1, 0][f16[k]]][x0, x4]], x0]
        - D[D[L, Derivative[0, 1][f16[k]]][x0, x4]], x4 ) ,
    constraintVars],
  {k, 0, 15}];
  Return[t /. subsDefects]; (* subsDefects = {} *)
eLa = eL[La, 1]
```

[A] there are sixteen field equations. $\text{eLa}[[k]]$ is the Euler–Lagrange expression for the component f_{k-1} , and the field equations are $\text{eLa}[[k]] = 0$. **The sixteen expressions, exactly as the notebook computes them** [D], in the notation above:

$$\begin{aligned}
\text{eLa}[[1]] &= -\frac{2}{H}(2Mf_4 + \partial_4 f_9 + \partial_0 f_{12}) & (\text{varied: } f_0) \\
\text{eLa}[[2]] &= -\frac{2}{H}(2Mf_5 - \partial_4 f_8 + \partial_0 f_{13}) & (\text{varied: } f_1) \\
\text{eLa}[[3]] &= -\frac{2}{H}(2Mf_6 - \partial_4 f_{11} + \partial_0 f_{14}) & (\text{varied: } f_2) \\
\text{eLa}[[4]] &= -\frac{2}{H}(2Mf_7 + \partial_4 f_{10} + \partial_0 f_{15}) & (\text{varied: } f_3) \\
\text{eLa}[[5]] &= -\frac{2}{H}(2Mf_0 - \partial_4 f_{13} + \partial_0 f_8) & (\text{varied: } f_4) \\
\text{eLa}[[6]] &= -\frac{2}{H}(2Mf_1 + \partial_4 f_{12} + \partial_0 f_9) & (\text{varied: } f_5) \\
\text{eLa}[[7]] &= -\frac{2}{H}(2Mf_2 + \partial_4 f_{15} + \partial_0 f_{10}) & (\text{varied: } f_6) \\
\text{eLa}[[8]] &= -\frac{2}{H}(2Mf_3 - \partial_4 f_{14} + \partial_0 f_{11}) & (\text{varied: } f_7) \\
\text{eLa}[[9]] &= \frac{2}{H}(2Mf_{12} - \partial_4 f_1 + \partial_0 f_4) & (\text{varied: } f_8) \\
\text{eLa}[[10]] &= \frac{2}{H}(2Mf_{13} + \partial_4 f_0 + \partial_0 f_5) & (\text{varied: } f_9) \\
\text{eLa}[[11]] &= \frac{2}{H}(2Mf_{14} + \partial_4 f_3 + \partial_0 f_6) & (\text{varied: } f_{10}) \\
\text{eLa}[[12]] &= \frac{2}{H}(2Mf_{15} - \partial_4 f_2 + \partial_0 f_7) & (\text{varied: } f_{11}) \\
\text{eLa}[[13]] &= \frac{2}{H}(2Mf_8 + \partial_4 f_5 + \partial_0 f_0) & (\text{varied: } f_{12}) \\
\text{eLa}[[14]] &= \frac{2}{H}(2Mf_9 - \partial_4 f_4 + \partial_0 f_1) & (\text{varied: } f_{13}) \\
\text{eLa}[[15]] &= \frac{2}{H}(2Mf_{10} - \partial_4 f_7 + \partial_0 f_2) & (\text{varied: } f_{14}) \\
\text{eLa}[[16]] &= \frac{2}{H}(2Mf_{11} + \partial_4 f_6 + \partial_0 f_3) & (\text{varied: } f_{15})
\end{aligned}$$

In Wolfram InputForm the first reads, character for character (broken over two lines here),

```
eLa[[1]] = (-2*(2*M*f16[4][x0, x4] + Derivative[0, 1][f16[9]][x0, x4]
+ Derivative[1, 0][f16[12]][x0, x4]))/H
```

and the other fifteen follow the same pattern with the indices and signs above.

These were printed for the provenance pages of 2026-09-24 by loading the notebook’s own DumpSave file `claude-fable/claude-fable_Einstein-Rosen-2-Planes-eLa.mx`. That file is SameQ to the author’s original `Pre-gravityPre-Big_Bang_M6=3-Generations_of_Einstein-Rosen-2-Planes-eLa.mx` in the repository root; this was re-checked for the provenance pages of 2026-09-24 with SameQ, which returned True. So they are the author’s equations, unchanged. In matrix form they read

$$\text{eLa} = \frac{2}{H} \sigma_{16} (T_{16}[0] \partial_{x_0} + T_{16}[4] \partial_{x_4} + 2M) \Psi_{16}, \quad \text{i.e.} \quad (21)$$

$$(T_{16}[0] \partial_{x_0} + T_{16}[4] \partial_{x_4}) \Psi_{16} = -2M \Psi_{16}. \quad (22)$$

This matrix form was checked for the provenance pages of 2026-09-24 (a `Simplify` difference of exactly 0); it is not a notebook assertion. In Part VI's language (below) it is the flat-frame Dirac equation $\gamma^\mu \partial_\mu \Psi = H V'(s) \Psi$ with $\gamma^\mu = T_{16}[\mu]$ and $V'(s) = -2M/H$. What Part VI asserts is the fidelity anchor below: the Euler–Lagrange expressions of its fable Lagrangian with $V = -(2M/H)s$, on the flat frame and with no volume factor, are exactly `eLa`.

The coupling pattern. Iterating `MergeSetsStep` to a fixed point merges any two sets of component indices that share an index. The result partitions the sixteen components into [D]

$$\text{eLaCouplings} = \{\{0, 5, 8, 13\}, \{1, 4, 9, 12\}, \{2, 7, 10, 15\}, \{3, 6, 11, 14\}\}. \quad (23)$$

[A] the coupled blocks partition all sixteen components; there are four blocks of four. The notebook writes `eLa` and `eLazt` to `.mx` files beside itself, as the original does.

2.9 The (z, t) chart, four blocks of four, and the closed-form solutions (Section 13)

The chart. The chart is $z = 6Hx_0$, $t = Hx_4$. Substituting `f16[k] → Z[k] [6 H x0, H x4]` and then $x_0 \rightarrow z/(6H)$, $x_4 \rightarrow t/H$, with the original's normalization $1/(2H)$, gives `eLazt =  $\frac{1}{2H}$  eLa` in the new chart. The first equation, for example, reads [D]

$$\text{eLazt}[[1]] = -\frac{2M Z_4 + H \partial_t Z_9 + 6H \partial_z Z_{12}}{H^2}, \quad Z_k \equiv Z[k] [z, t]. \quad (24)$$

Decoupling. Solving $0 = \text{eLazt}$ for the sixteen t -derivatives gives [A] all sixteen t -derivatives are determined. Relabelling each coupling set as a consecutive block, `sZt0yZ` maps Z_k to yZ_j as follows [D]:

- $Z_0, Z_5, Z_8, Z_{13} \rightarrow yZ_{0..yZ_3}$;
- $Z_1, Z_4, Z_9, Z_{12} \rightarrow yZ_{4..yZ_7}$;
- $Z_2, Z_7, Z_{10}, Z_{15} \rightarrow yZ_{8..yZ_{11}}$;
- $Z_3, Z_6, Z_{11}, Z_{14} \rightarrow yZ_{12..yZ_{15}}$.

Assertions [A]:

- the relabelling is a bijection of the sixteen components
- the system splits into four blocks of four equations
- each 4x4 block is closed in its own four components

The relabelling is a linear map. [A] the `yZ` coefficient matrix is the identity; `ORTHOGONAL: Transpose[S] . S == ID16`; `ORTHOGONAL: S . Transpose[S] == ID16`; but NOT a `O(8,8)` transformation: it does not preserve `sigma16`; and NOT a direct sum: it mixes `type-1` with `type-2`.

Closed forms. `DSolve`, wrapped in a 60-second `TimeConstrained`, returns every block unevaluated. The log prints `block 1: returned unevaluated through block 4: returned unevaluated`. The author obtained closed forms with Maple and pasted them into the original as strings. The notebook reproduces them verbatim, parses them with the author's `ConvertMapleToMathematicaV2`, and then substitutes them back into the four blocks. That substitution, not the parser, is what certifies them. [A] all sixteen components have a closed form; there are sixteen pure-function rules; `THE MAPLE CLOSED FORMS SOLVE ALL SIXTEEN EQUATIONS`.

Block j has one constant C_j and four constants $c_{j1}..c_{j4}$. Every component of block j carries the same exponential

$$\begin{aligned} E_j &= \exp\left(\frac{6(z/6 + t)H^2C_j^2 - 6(-z/6 + t)M^2}{6C_jH^2}\right) \\ &= \exp\left(C_j\left(t + \frac{z}{6}\right) - \frac{M^2}{C_jH^2}\left(t - \frac{z}{6}\right)\right). \end{aligned} \quad (25)$$

Block 1, verbatim from the Maple string:

$$\begin{aligned} yZ_0 &= E_1 \frac{H^2(c_{11}c_{12} - c_{13}c_{14})C_1^2 - M^2(c_{11}c_{12} + c_{13}c_{14})}{2HC_1M}, & yZ_2 &= c_{13}c_{14} E_1, \\ yZ_1 &= -E_1 \frac{H^2(c_{11}c_{12} - c_{13}c_{14})C_1^2 + M^2(c_{11}c_{12} + c_{13}c_{14})}{2HC_1M}, & yZ_3 &= c_{11}c_{12} E_1. \end{aligned}$$

Blocks 2, 3 and 4 have the same shape with their own constants and signs (Section 13 prints all four strings).

2.10 Bilinears, the solution branches, and $M_6 = 3$ generations of Einstein–Rosen 2-planes (Section 14)

Five bilinears are formed from the solution written in the relabelled basis. `psi1sol` and `psi2sol` are its first and second halves of eight components **in the relabelled yZ basis**, that is, coupling blocks 1–2 and coupling blocks 3–4. They are not the type-1 and type-2 spinors. [A] the `yZ` split at 8 is NOT the type-1/type-2 split; the first relabelled block holds four type-1 and four type-2 components.

For the full sixteen-component norm, [A]:

- `psi . sigma16 . psi == psi2.sigma.psi2 - psi1.sigma.psi1`
- `sigma16` alone is SYMMETRIC, so `psi . sigma16 . psi` is not forced to vanish
- and for this solution it does NOT vanish: it is not the zero expression

With the light-cone coordinates $ztadv = 6t + z$ and $ztret = 6t - z$, the notebook requires the two block norms to be light-cone constants. [A] **the light-cone-constant condition has solutions**. `Solve` returns four branches [D], among them $\{C_2 \rightarrow -C_1, C_4 \rightarrow -C_3\}$, the original's branch. [A] **on the branch $C_2 = -C_1, C_4 = -C_3$ the two norms are independent of $ztadv$ and $ztret$** .

The displayed values on that branch [D, re-read for the provenance pages of 2026-09-24 from the saved kernel state] are sharper than the assertion. There, $\psi \cdot \sigma_{16} \cdot \psi$, $\psi_1 \cdot \sigma \cdot \psi_1$ and $\psi_2 \cdot \sigma \cdot \psi_2$ are all identically 0, while $\psi_2 \cdot \sigma \cdot \psi_1$ is not.

The two mass branches [D] are, both taken on the branch $C_2 \rightarrow -C_1, C_4 \rightarrow -C_3$:

- **Psi16ab**: $M \rightarrow H M_{ab} \sqrt{C_1 C_3}$, which keeps a free dimensionless ratio M_{ab} ;
- **Psi16aa**: $M \rightarrow H \sqrt{C_1 C_3}$, the case $M_{ab} = 1$.

In the (x_0, x_4) chart the advanced and retarded coordinates are $\xi_{adv} = x_0 - x_4$ and $\xi_{ret} = x_0 + x_4$. Every component of **Psi16aa** is a constant times one of $\exp(\pm H(C_1 \xi_{ret} + C_3 \xi_{adv}))$ or $\exp(\pm H(C_3 \xi_{ret} + C_1 \xi_{adv}))$ [D, re-read for the provenance pages of 2026-09-24 from the saved kernel state; the notebook prints **Short forms**]. The special case $C_3 \rightarrow C_1$, **Psi16x0ONLY**, depends on x_0 alone. It is exactly [D, re-read in full from the saved kernel state], with $e_{\pm} \equiv e^{\pm 2C_1 H x_0}$:

$$\begin{aligned} \text{Psi16x0ONLY} &= (-c_{13}c_{14}e_+, c_{23}c_{24}e_-, -c_{33}c_{34}e_+, c_{43}c_{44}e_-, \\ &\quad c_{21}c_{22}e_-, -c_{11}c_{12}e_+, c_{41}c_{42}e_-, -c_{31}c_{32}e_+, \end{aligned}$$

$$c_{13}c_{14}e_+, c_{23}c_{24}e_-, c_{33}c_{34}e_+, c_{43}c_{44}e_-, \\ c_{21}c_{22}e_-, c_{11}c_{12}e_+, c_{41}c_{42}e_-, c_{31}c_{32}e_+).$$

Triality turns each spinor of the solution into a vector. The genuine type-1 and type-2 spinors are $\psi_1 = \text{Psi16ab}[[1;;8]]$ and $\psi_2 = \text{Psi16ab}[[9;;16]]$. From them:

$$\text{vectorFormOfType2} = \text{triVecToSpin} \cdot \psi_2, \quad (26)$$

$$\text{vectorFormOfType1} = \eta \cdot (\psi_1^T \cdot \sigma \cdot \text{triSpinToVec}). \quad (27)$$

Assertions [A]:

- both triality vectors have eight components
- `vectorFormOfType2 == psi2` contracted on the SPINOR index of `triVecToSpin`
- `triVecToSpin` and `triSpinToVec` really are mutually inverse

Each vector is normal to a 7-plane through the origin of the split-octonion algebra, the set of position vectors `Zoct` with $\langle \text{vector}, \text{Zoct} \rangle = 0$. [A] `sevenPlane2` is annihilated by `vectorFormOfType2`; `sevenPlane1` is annihilated by `vectorFormOfType1`.

Two non-parallel 7-planes in an 8-space meet in a 6-plane, **M6**. [A] **M6** is a non-empty solution set; **M6** has six free parameters. The intersection determines `zoct[7]` and `zoct[8]` as linear combinations of the free `zoct[1]..zoct[6]`, with coefficients that depend on x_0, x_4 and the constants C_1, C_3, M_{ab}, c_{jk} [D].

What is proved is that **M6** exists and is a 6-parameter set. The reading of **M6** as the home of “the three generations of Einstein–Rosen 2-plane bridges”, which gives the original notebook its title, is the author’s interpretation and is stated in prose only [P].

Finally [A]: `cfResolution[unit]`: the closed-form bridge equals the matrix inverse; `cfResolution[unit]`: the two bridges are mutually inverse; `cfResolution[unit]` reproduces `triVecToSpin`.

2.11 The canonical frame field (Section 15), restated

At every point of the curved $4 + 4$ spacetime there is a local flat $4 + 4$ Minkowski coordinate system. The frame field installs it; in eight dimensions a vielbein is simply an 8-dimensional vierbein. The frame is stored with the curved index μ as the row and the flat index a as the column. The canonical frame is the one the original notebook records as the value of its symbol `gtrye`:

$$e = \text{frameCanonical} = \text{diag}(\tan 6Hx_0, q, q, q, 1, p, p, p), \quad (28)$$

$$q = \frac{e^{-a_4(Hx_4)}}{(\sin 6Hx_0)^{1/6}}, \quad p = \frac{e^{+a_4(Hx_4)}}{(\sin 6Hx_0)^{1/6}}, \quad g = e \eta e^T, \quad (29)$$

$$ds^2 = \tan^2(6Hx_0) dx_0^2 + q^2(dx_1^2 + dx_2^2 + dx_3^2) - dx_4^2 - p^2(dx_5^2 + dx_6^2 + dx_7^2). \quad (30)$$

Assertions [A]:

- `CANONICAL [structural]: frame . coframe == ID8` (certifies only that the frame is invertible)
- `CANONICAL [definition]: g == frame . eta4488 . Transpose[frame]` (bridge [d]; this IS how `g` was defined, so it cannot fail)
- `CANONICAL [has content]: g == diag(Tan[6 H x0]^2, q^2, q^2, q^2, -1, -p^2, -p^2, -p^2)`, the stated line element
- `CANONICAL [has content]: Diagonal[g] == Diagonal[eta4488] * Diagonal[frame]^2` exactly

- CANONICAL [has content]: g has signature (4,4), GIVEN a4 real and $0 < 6 H x_0 < \pi/2$
- Inverse[g] == Transpose[coframe] . eta4488 . coframe
- g is symmetric
- the metric is non-degenerate
- FABLE-5.1: det g == Sec[6 H x0]^2 (positive, as signature 4+4 requires)

So $\sqrt{|\det g|} = \sec 6Hx_0$ on $0 < 6Hx_0 < \pi/2$. This is the notebook's volume factor `sqrtDetgTele` = Sec[6 H x0], defined in Section 21 and used throughout Part VI (Section 15 itself displays Abs[Sec[6 H x0]]).

Two facts follow from the definitions of q and p . First, $qp = (\sin 6Hx_0)^{-1/3}$: the two exponentials cancel [P, Section 15]. Hence $q^3 p^3 = 1/\sin 6Hx_0$, and the 8-volume element factorizes. [A]:

- FABLESCALAR [has content]: Sqrt[det g_8] == q^3 * (Tan[6 H x0] p^3): observed 3-volume density times hidden 4-volume density
- FABLESCALAR [has content]: the 8-volume element is independent of x4: d(q^3 p^3)/dx4 == 0
- FABLESCALAR [has content]: the hidden 4-volume density scales as Exp[+3 a4[H x4]] and the observed 3-volume as Exp[-3 a4[H x4]]: Vhid Exp[-3 a4] and q^3 Exp[+3 a4] are x4-independent

One Text cell of Section 23 writes $q^3 p^3 = 1/\sqrt{\sin 6Hx_0}$. That is a typo in the prose: from $qp = (\sin 6Hx_0)^{-1/3}$ the cube is $1/\sin 6Hx_0$, which is what makes $\sqrt{\det g} = q^3 \tan(6Hx_0) p^3 = \sec 6Hx_0$. The assertions above are unaffected.

The curved Dirac matrices are $\gamma^\mu = \sum_a \text{coframe}[[a, \mu]] T_{16}[a]$, with `coframe` = e^{-1} . On the canonical frame they are as follows (read for the provenance pages of 2026-09-24 from the saved kernel state; they follow directly from the diagonal coframe):

$$\begin{aligned} \gamma^0 &= \cot(6Hx_0) T_{16}[0], & \gamma^i &= \frac{1}{q} T_{16}[i] = e^{a_4} (\sin 6Hx_0)^{1/6} T_{16}[i], \\ \gamma^4 &= T_{16}[4], & \gamma^h &= \frac{1}{p} T_{16}[h] = e^{-a_4} (\sin 6Hx_0)^{1/6} T_{16}[h], \end{aligned} \quad (31)$$

for $i = 1, 2, 3$ and $h = 5, 6, 7$. [A] {gammaCurved[mu], gammaCurved[nu]} == 2 gInv[mu, nu] ID16; and, from Part VI, FABLE [has content]: (gamma^4)^2 == -ID16, so gamma^4 d_4 Psi' == H V'(s) Psi' is solved by d_4 Psi' == -H V'(s) gamma^4 Psi'.

The free function a_4 . a_4 is a free, differentiable scalar function of Hx_4 . The original notebook never defines it; it occurs only inside the recorded frame. It is carried symbolically everywhere and is never given a value, and every result holds for arbitrary a_4 . The notebook says so on screen: the log prints NOTE a4 is an undefined scalar function inherited from the original notebook together with the explanation, which also names la. Near its end Part VI checks [A] PART VI [control]: a4 is STILL UNDEFINED - nothing in Part VI gave it a value; and the FLRW scale factor aF never entered the canonical frame.

The physical reading. The observer is the geodesic observer $u = \partial/\partial x_4$: $g_{44} = -1$, and x_0 is constant along the worldline [P, Section 23]. The cosmological time is $t = x_4$, the proper time of that observer [P, Section 25]. The observed sheet is x_1, x_2, x_3 ; the hidden spacelike coordinate is x_0 ; the hidden timelike sheet is x_5, x_6, x_7 . Part VI makes the kinematic identification $a(t) \sim q \sim e^{-a_4(Ht)}$, $H_{\text{obs}} = -H a'_4(Ht)$. Assertions [A]:

- KINEMATIC [definition]: $\ln q == -a_4[H x_4] + \text{const}(x_0): D[\text{Log}[q], x_4] == -H a_4' [H x_4] == H_{\text{obs}}$
- KINEMATIC [has content]: the second sheet contracts at exactly the opposite rate: $D[\text{Log}[p], x_4] == +H a_4' [H x_4] == -H_{\text{obs}}$
- KINEMATIC [has content]: $H_{\text{obs}} > 0$ exactly when $a_4' < 0$ (and $H_{\text{obs}} < 0$ when $a_4' > 0$) - the sign HYPOTHESIS of this Part, stated, not derived

So which sheet grows is set by the sign of a_4' , which is never fixed. Section 15's prose describes the 3-space contracting while the deflating directions expand. Part VI reads the same frame the other way round (the observed sheet expands when $a_4' < 0$) and states that sign as a hypothesis [P].

What the canonical spin connection contributes. The spin connection of this frame follows from the zero-torsion vielbein postulate. Only the facts that the classical fable uses are restated here:

$$\Gamma_{\mu}^{\text{spin}} = \frac{1}{8} \omega_{\mu ab} [T_{16}[a], T_{16}[b]], \quad D_{\mu} \Psi = \partial_{\mu} \Psi + \Gamma_{\mu}^{\text{spin}} \Psi, \quad (32)$$

where each $\Gamma_{\mu}^{\text{spin}}$ is block diagonal: it never mixes type-1 with type-2. Its 24 non-zero components [D, Section 16] are listed below, with $c = \cos 6Hx_0$, $s = \sin 6Hx_0$, $A = a_4(Hx_4)$, $A' = a_4'(Hx_4)$, $j = 1, 2, 3$ and $k = 5, 6, 7$:

component	value
$\omega_{j;0j} = -\omega_{j;j0}$	$Hc^2/(e^A s^{13/6})$
$\omega_{j;4j} = -\omega_{j;j4}$	$HA'/(e^A s^{1/6})$
$\omega_{k;0k} = -\omega_{k;k0}$	$-e^A Hc^2/s^{13/6}$
$\omega_{k;4k} = -\omega_{k;k4}$	$e^A HA'/s^{1/6}$

Assertions [A]:

- CANONICAL [has content]: $\omega_{\mu, a, b} == -\omega_{\mu, b, a}$ (metric compatibility)
- INDEPENDENT CHECK: the frame-only derivation reproduces `omegaCanonical` exactly
- [has content] the spinor connection is compatible with ω : $[\text{Gamma}^{\text{spin}}_{\mu}, \text{gamma}^a] + \omega_{\mu}^a{}_b \text{gamma}^b == 0$
- $\text{Gamma}^{\text{spin}}_{\mu}$ is block diagonal: it never mixes type-1 with type-2
- [has content] and therefore the curved Dirac matrices are covariantly constant: $D_{\mu} \text{gammaCurved}^{\mu} == 0$

From Part V (the fable-5.1 bridge), three results are used below [A]:

- FABLE-5.1 [THE RESULT]: $\text{gamma}^{\mu} \text{Gamma}^{\text{spin}}_{\mu} == -(1/2) T_{\mu} \text{gamma}^{\mu}$, one vector term, with the Weitzenböck torsion vector $T_{\mu} = \partial_{\mu} \ln \sin 6Hx_0$, i.e. FABLE-5.1 torsion vector has exactly one non-zero component, $6 H \cot[6 H x_0]$ along x_0 . Hence $\gamma^{\mu} \Gamma_{\mu}^{\text{spin}} = -3H \cot^2(6Hx_0) T_{16}[0]$.
- FABLE-5.1 [THE RESULT]: $\text{DiracCanonical}[\text{Sqrt}[\text{Sin}[6Hx_0]] \text{Psi}'] == \text{Sqrt}[\text{Sin}[6Hx_0]] \text{DiracFable51}[\text{Psi}']$, for a generic Psi'
- FABLE-5.1 [THE RESULT]: $\text{Psi}^T \text{sigma16} \text{gamma}^{\mu} \text{Gamma}^{\text{spin}}_{\mu} \text{Psi} == 0$ for EVERY Psi - the spin connection drops out of the Lagrangian, because FABLE-5.1: the reason is Section 11's identity, $\text{Psi}^T \text{sigma16} T_{16}[a] \text{Psi} == 0$ ($\text{sigma16} T_{16}[a]$ antisymmetric). Note that this reason is a statement about **commuting** components.

The Ricci scalar of the canonical metric is [D, Section 16]

$$R = -3H^2 \left((55 + 7 \cos 12Hx_0) \cot^2(6Hx_0) \csc^2(6Hx_0) - 2a'_4(Hx_4)^2 \right). \quad (33)$$

2.12 The classical field fable of 2026-09-16 (Part VI, Section 24)

Part VI (Sections 23–25) places two matter fields on the pre-universe. It asks of each the cosmologist's question: what are the energy density ρ , the pressure P , and $w = P/\rho$? The first field, **fableScalar**, is a real scalar with potential $V(\phi)$. The second, **fable**, is the subject of this document.

fableScalar, in brief. It has $\hat{L}_\phi = -\frac{1}{2}g^{\mu\nu}\partial_\mu\phi\partial_\nu\phi - V(\phi)$ and the Hilbert tensor $T_{\mu\nu} = \partial_\mu\phi\partial_\nu\phi + g_{\mu\nu}\hat{L}_\phi$. Its central theorem is [A] **FABLESCALAR [THE RESULT]: the null energy condition $\rho + P_{-1} == 2$ KE holds IDENTICALLY, for every $\phi(x_0, x_4, x_5)$ and every V , so it cannot cross $w = -1$ with positive energy density.**

2.12.1 The field and its Lagrangian

fable is a **real, commuting** 16-component spinor $\Psi(x)$ that transforms under $\text{Spin}(4,4)$ exactly as Ψ_{16} . It uses $T_{16}[a]$, σ_{16} , the curved γ^μ , the scalar bilinear $s = \Psi^T \sigma_{16} \Psi$, and a general potential $V(s)$:

$$L_\Psi = \sqrt{\det g} \hat{L}_\Psi, \quad \hat{L}_\Psi = \frac{1}{H} \Psi^T \sigma_{16} \gamma^\mu D_\mu \Psi - V(s), \quad (34)$$

$$D_\mu = \partial_\mu + \Gamma_\mu^{\text{spin}}, \quad s = \Psi^T \sigma_{16} \Psi. \quad (35)$$

In Part VI's generic computations $\Psi = \Psi(x_0, x_4)$ has sixteen arbitrary component functions. Assertions [A]:

- **FABLE [definition]:** `s = Psi^T sigma16 Psi` is a genuine scalar with content: on the constant spinor `e1 + e5`, `s == -2`
- **FABLE [has content]:** `Lhat` with the covariant derivative `== Lhat` with the partial derivative, for the generic `Psi(x0, x4)` (Part V's theorem, re-asserted here)

The fidelity anchor. The author's Lagrangian is the special case $V(s) = -(2M/H)s$, taken on the flat frame ($e \rightarrow I_8$) and with no volume factor ($\sqrt{g} \rightarrow 1$). With that potential $-V = +(2M/H)s$, which is exactly L_a 's mass term. [A]:

- **FABLE [fidelity]:** with `V = -(2M/H) s`, `frame -> ID8` and `Sqrt[g] -> 1`, on the model's `Psi16`, the Lagrangian helper `IS` the author's `La[]` of Section 11
- **FABLE [fidelity]:** and its Euler-Lagrange equations, formed with the author's own `eL` operator, `ARE` the sixteen equations `eLa` of Section 12
- **FABLE [solver regression]:** the new helper `cfEulerLagrange` reproduces `eLa` too
- **FABLE-5.1:** the covariant Lagrangian with `frame -> ID8` and volume factor `-> 1` `IS` the author's `La` of Section 11, term for term

2.12.2 The field equation

The equation is obtained by explicit variation of $\sqrt{g} \hat{L}_\Psi$. The kinetic matrix is $A^\mu = \sigma_{16} \gamma^\mu$, which is antisymmetric; the mass matrix σ_{16} is symmetric. [A]:

- FABLE [THE RESULT]: the Euler-Lagrange equations by explicit variation == $\text{Sqrt}[g] \left[(2/H) (A^\mu d_\mu \Psi + (1/2) D \Psi) - 2 V'(s) \sigma_{16} \Psi \right]$, where $D = \frac{1}{\sqrt{g}} \partial_\mu (\sqrt{g} A^\mu)$
- FABLE [has content]: $A^\mu = \sigma_{16} \gamma^\mu$ is antisymmetric and σ_{16} is symmetric - the structure that makes the variation close
- FABLE [has content]: in general $(1/\text{Sqrt}[g]) d_\mu (\text{Sqrt}[g] \gamma^\mu) == [\gamma^\mu, \Gamma^\mu_{\text{spin}}]$ (covariant constancy of γ^μ , Section 17)
- FABLE [has content]: the HYPOTHESIS that makes it a product - the anticommutator $\{\gamma^\mu, \Gamma^\mu_{\text{spin}}\} == 0$ on the canonical frame (no totally antisymmetric part of the connection)
- FABLE [has content]: hence $(1/2)(1/\text{Sqrt}[g]) d_\mu (\text{Sqrt}[g] \gamma^\mu) == \gamma^\mu \Gamma^\mu_{\text{spin}}$, and on this geometry == $-3 H \cot[6 H x_0]^2 T_{16}[0]$
- FABLE [THE RESULT]: $EL == \text{Sqrt}[g] (2/H) \sigma_{16} \cdot (\gamma^\mu D_\mu \Psi - H V'(s) \Psi)$: the field equation of fable is EXACTLY the Levi-Civita covariant Dirac equation $\gamma^\mu D_\mu \Psi == H V'(s) \Psi$
- FABLE [has content]: the connection term does not vanish on an x_0 -independent spinor - witnessed on the constant spinor e_1 : a spinor homogeneous in everything but x_4 is NOT a solution

So the field equation of the classical fable is

$$\gamma^\mu D_\mu \Psi = H V'(s) \Psi, \quad \text{i.e. on the canonical frame} \quad (36)$$

$$\begin{aligned} \cot(6Hx_0) T_{16}[0] \partial_0 \Psi + \frac{1}{q} T_{16}[i] \partial_i \Psi + T_{16}[4] \partial_4 \Psi + \frac{1}{p} T_{16}[h] \partial_h \Psi \\ - 3H \cot^2(6Hx_0) T_{16}[0] \Psi = H V'(s) \Psi \end{aligned} \quad (37)$$

(summed over $i = 1, 2, 3$ and $h = 5, 6, 7$; for $\Psi(x_0, x_4)$ only the ∂_0 and ∂_4 terms survive).

The rescaling. [A]:

- FABLE [definition]: under $\Psi = \text{Sqrt}[\text{Sin}[6 H x_0]] \Psi'$, $s == \text{Sin}[6 H x_0]$ $\Psi'^T \sigma_{16} \Psi'$
- FABLE [THE RESULT]: $(\gamma^\mu D_\mu - H V'(s)) [\text{Sqrt}[\text{Sin}[6 H x_0]] \Psi'] == \text{Sqrt}[\text{Sin}[6 H x_0]] (\gamma^\mu d_\mu \Psi' - H V'(s) \Psi')$ for a GENERIC $\Psi'(x_0, x_4)$: the rescaling removes the connection term exactly
- FABLE [has content]: for $\Psi' = \Psi'(x_4)$ the field equation reduces to $\text{Sqrt}[\text{Sin}[6 H x_0]] (\gamma^4 d_4 \Psi' - H V'(s) \Psi')$, $s = \text{Sin}[6 H x_0]$ $\Psi'^T \sigma_{16} \Psi'$
- FABLE [has content]: the reduced equation still depends on x_0 through $V'(\text{Sin}[6 H x_0] s')$: for a NONLINEAR V a $\Psi'(x_4)$ alone is not a solution - witnessed with $V = s^2$ on $\Psi' = e_1 + e_5$
- FABLE [has content]: for the mass term $V = -(2M/H) s$, V' is constant and $\Psi'(x_4)$ IS an exact solution

2.12.3 The energy–momentum tensor

The tensor is built with the covariant derivative and symmetrized:

$$T_{\mu\nu}^{\text{cov}} = -\frac{1}{2H}(\Psi^T \sigma_{16} \gamma_\mu D_\nu \Psi + \Psi^T \sigma_{16} \gamma_\nu D_\mu \Psi) + g_{\mu\nu} \hat{L}_\Psi, \quad (38)$$

with $\gamma_\mu = g_{\mu\nu} \gamma^\nu$. Assertions [A]:

- FABLE [definition]: T_cov is symmetric by construction
- FABLE [has content]: the diagonal of T_cov equals the diagonal of the partial-derivative tensor, for the generic Psi(x0, x4): rho and every pressure are the same in both
- FABLE [has content]: but the two tensors DIFFER off the diagonal - witnessed on the (x3, x4) component T[[4,5]], a momentum density along the observed sheet, for the constant spinor e1 + e13 and V = s^2
- FABLE [THE RESULT]: nabla^mu T_{mu nu} == 0 ON SHELL in all eight components, for a generic Psi(x0, x4) - the covariant tensor is the conserved one

“On shell” means the following. The field equation is solved for $\partial_4 \Psi = F = -\gamma^4 (H V'(s) \Psi - \gamma^0 \partial_0 \Psi - \frac{1}{2} \text{div} G \Psi)$, where $\text{div} G = \frac{1}{\sqrt{g}} \partial_\mu (\sqrt{g} \gamma^\mu)$ (on this geometry $\frac{1}{2} \text{div} G = -3H \cot^2(6Hx_0) T_{16}[0]$, by the assertion quoted above), and cfOnShell replaces every x_4 -derivative of the field by the corresponding derivative of F , repeatedly, until none is left. [A] FABLE [definition]: F solves the field equation for d_4 Psi: $\gamma^4 F + \gamma^0 d_0 \Psi + (1/2) \text{div} G \Psi - H V'(s) \Psi == 0$; FABLE [definition]: cfOnShell leaves no x_4 -derivative of the field in rho. The partial-derivative tensor was checked symbolically while Part VI was designed: its x_0 and x_4 components are conserved and the other six are not. That check takes minutes and is not repeated in the notebook [P].

2.12.4 ρ , P , and w on shell

The observer is $u = \partial/\partial x_4$, so $\rho = T_{44}$ (T[[5,5]]) and $P_i = T^i_i$ (no sum, $i = 1, 2, 3$). Two kinetic bilinears organize the answer:

$$K_\Psi = \frac{1}{H} \Psi^T \sigma_{16} \gamma^4 \partial_4 \Psi \quad (\text{along the observer's time}), \quad (39)$$

$$K_h = -\frac{1}{H} \Psi^T \sigma_{16} \gamma^0 \partial_0 \Psi \quad (\text{along the hidden coordinate } x_0). \quad (40)$$

Assertions [A]:

- FABLE [THE RESULT]: on shell, Lhat == s V'(s) - V(s)
- FABLE [THE RESULT]: on shell, rho == V(s) + K_h, K_h = -(1/H) Psi^T sigma16 gamma^0 d_0 Psi
- FABLE [THE RESULT]: on shell, P_1 == s V'(s) - V(s), and P_2 == P_3 == P_1
- FABLE [has content]: on shell, K_Psi == s V'(s) + K_h - the x4 kinetic bilinear is fixed by the potential and the hidden one
- FABLE [has content]: K_h == 0 for Psi = Sqrt[Sin[6 H x0]] Psi'(x4) (d_0 Psi is proportional to Psi, and Psi^T sigma16 gamma^0 Psi == 0)
- FABLE [THE RESULT]: so for those solutions rho == V(s), hence w == s V'(s)/V(s) - 1, and rho > 0 requires V(s) > 0

In summary:

$$\rho = V(s) + K_h, \quad P = sV'(s) - V(s), \quad (41)$$

$$\text{and for } \Psi = \sqrt{\sin 6Hx_0} \Psi'(x_4) : \quad \rho = V(s), \quad w = \frac{sV'(s)}{V(s)} - 1. \quad (42)$$

2.12.5 No dilution on the pre-universe

Assertions [A]:

- FABLE [THE RESULT]: on shell, $ds/dx_4 == 2 \Psi^T \sigma_1 \gamma^4 \gamma^0 d_0 \Psi$ for a generic $\Psi(x_0, x_4)$
- FABLE [has content]: the two other terms of $d_4 \Psi$ drop out because $\sigma_1 \gamma^4$ and $\sigma_1 \gamma^4 \gamma^0$ are ANTISYMMETRIC
- FABLE [control]: that bilinear is NOT identically zero for an x_0 -dependent Ψ - witnessed on $\Psi = e_1 + x_0 e_2$: ' $ds/dx_4 == 0$ for generic $\Psi(x_0, x_4)$ ' would be FALSE
- FABLE [THE RESULT]: for $\Psi = \text{Sqrt}[\text{Sin}[6 H x_0]] \Psi'(x_4)$ it vanishes: s is CONSTANT IN x_4 - no dilution on the pre-universe
- FABLE [THE RESULT]: the same from the reduced equation: $ds'/dx_4 == 0$ on shell for $\Psi'(x_4)$, because $\Psi'^T \sigma_1 \gamma^4 \Psi' == 0$

So on those solutions s , V , ρ and w are all constant along x_4 . This is the spinor analogue of the scalar's "no Hubble friction" (for the scalar, [A] FABLESCALAR [THE RESULT]: for $\phi = \phi(x_4)$ the field equation is $\phi'' == -V'(\phi)$: NO HUBBLE FRICTION on the pre-universe). For the scalar, Section 23's prose gives the reason as geometric: the factor $\sqrt{g} g^{44} = -\sec 6Hx_0$ does not depend on x_4 , because the 8-volume element is x_4 -independent [P].

2.12.6 Dust for the author's mass term

[A]:

- FABLE [THE RESULT]: the author's mass term $V = -(2M/H) s$ is DUST: on shell $P_1 == 0$ identically and $\rho == -(2M/H) s$ ($w == 0$, exactly)
- FABLE [has content]: that density is not zero - witnessed on $\Psi' = e_1 + e_5$ at $M = 1$ ($\rho = (2M/H) * 2 \text{Sin}[6 H x_0]$ there)

The density $\rho = -(2M/H)s$ is positive only on the branch $Ms < 0$ [P, Section 24].

2.12.7 $w = sV'/V - 1$ for the six potentials, and the classical phantom crossing

With $u \equiv (s/s_1)^2$:

mass:	$V = ms,$	$w = 0,$
lambda-mass:	$V = V_0 + ms,$	$w = -\frac{V_0}{V_0 + ms},$
power:	$V = ms + \lambda s^n,$	$w = \frac{ms + n\lambda s^n}{ms + \lambda s^n} - 1,$
hilltop:	$V = V_0 - \mu(s - s_*)^2,$	$w = \frac{-2\mu s(s - s_*)}{V_0 - \mu(s - s_*)^2} - 1,$
lorentz:	$V = V_0 + \frac{ms}{1 + u},$	$w = \frac{ms(1 - u)}{(1 + u)^2(V_0 + ms/(1 + u))} - 1,$

$$\text{expdamp:} \quad V = V_0 + m s e^{-s/s_1}, \quad w = \frac{m s e^{-s/s_1} (1 - s/s_1)}{V_0 + m s e^{-s/s_1}} - 1.$$

Assertions [A]:

- FABLE [has content]: the closed forms of $w_{\text{Psi}}(s) = s V'(s)/V(s) - 1$ for the six potentials of the numerical work (mass, lambda-mass, power, hilltop, lorentz, expdamp)
- FABLE [THE RESULT]: mass $\rightarrow w == 0$ (dust); $V = \text{lam } s^n$ alone $\rightarrow w == n - 1$ (accelerating for $n < 2/3$, phantom for $n < 0$); $n = 0.236 \rightarrow w == -0.764$ exactly
- FABLE [THE RESULT]: $w + 1 == s V'/V$ vanishes where V' does, with $V > 0$ there: lorentz and expdamp cross $w = -1$ at $s = s_1$, hilltop at $s = s^*$ - the phantom divide is crossed with no wrong-sign kinetic term
- FABLE [has content]: lorentz and expdamp are bounded below by $V_0 > 0$ for $s \geq 0$ ($\rho > 0$ on the whole history), and $w < -1$ for $s > s_1$ (the past), $w > -1$ for $0 < s < s_1$ (the present)
- FABLE [has content]: the limits - lorentz and expdamp: $w \rightarrow -1$ both as $s \rightarrow \text{infinity}$ and as $s \rightarrow 0$; lambda-mass: $w \rightarrow 0$ (dust) as $s \rightarrow \text{infinity}$ and $\rightarrow -1$ (Lambda) as $s \rightarrow 0$
- FABLE [has content]: expdamp with $(V_0, m, s_1) = (1.566, 0.839, 2.21)$ at $s = 1$ (today) gives $w == -0.8608$, Unite's $w_0 = -0.861$ to three decimals

The assumptions for the sign statements are $V_0 > 0$, $m > 0$, $s_1 > 0$. This is the **classical phantom crossing**: $w + 1 = sV'/V$ changes sign where V' does, with $V > 0$ there, and no kinetic term has the wrong sign. It is the spinor-quintom mechanism of Cai and Wang, Class. Quantum Grav. 25 (2008) 165014. The stability of perturbations is not examined in Part VI [P]. The V_0 terms of lambda-mass, lorentz and expdamp are a bare cosmological constant and are reported as such [D, the summary table printed in Section 25].

2.12.8 The FLRW reference model (Model B)

Because $ds/dx_4 = 0$ on the pre-universe, a history $w(a)$ needs a different background. Part VI builds it as a **separate** frame, $\text{frameFLRW} = \text{diag}(1, a_F(x_4), a_F(x_4), a_F(x_4), 1, 1, 1, 1)$, with its own scale factor a_F , and feeds it to the same spin-connection solver. The canonical frame and a_4 are not touched. Assertions [A]:

- FLRW [definition]: the reference frame does not mention a_4 , and the canonical frame is untouched
- FLRW [solver regression]: vielbein postulate residual is zero
- FLRW [has content]: omega is antisymmetric (metric compatible)
- FLRW [THE RESULT]: $\gamma^\mu \Gamma_{\text{spin}_\mu} == (3/2) (a'/a) \gamma^4$ - the Hubble term of the FLRW Dirac equation
- FLRW [definition]: the rule solves $\gamma^\mu D_\mu \text{Psi} == H V'(s) \text{Psi}$ for $d_4 \text{Psi}$
- FLRW [THE RESULT]: $d(a^3 s)/dx_4 == 0$ on shell, for EVERY V : $s = s_0 a^{-3}$, the dust dilution law, and $w_{\text{Psi}}(a) = s V'(s)/V(s) - 1$ on it
- FLRW [control]: without the Hubble term, $d(a^3 s)/dx_4 == 3 (a'/a) a^3 s$ instead (i.e. $ds/dx_4 == 0$, the pre-universe law): the dilution IS the Hubble term

Model B, solved independently with NDSolve. The units are $8\pi G = 1$, $H_0 = 1$ and $\rho_{\text{crit}} = 3$. The independent variable is $N = \ln a$, and $s = e^{-3N}$ is exact, so only the cosmic time needs integrating:

$$\frac{dt}{dN} = \frac{1}{H}, \quad H^2 = \Omega_m e^{-3N} + \Omega_r e^{-4N} + \frac{V(s)}{3}, \quad \Omega_m = \frac{3}{10}, \quad \Omega_r = \frac{84}{10^6}, \quad (43)$$

$$V = \text{expdamp with } (V_0, m, s_1) = \left(\frac{1566}{1000}, \frac{839}{1000}, \frac{221}{100}\right), \quad (44)$$

$$N \in [-7, 0], \quad 701 \text{ rows,} \quad t(N_0) = \frac{1}{2H(N_0)} \text{ added.} \quad (45)$$

All inputs are exact rationals, and `WorkingPrecision -> 24`. Assertions [A]:

- MODEL B [definition]: 701 rows from $N = -7$ to 0, $s(0) == 1$ and $\rho_{\Psi}(0) == V(1) == 1.566 + 0.839 \text{ Exp}[-1/2.21]$
- MODEL B [THE RESULT]: $w \leq -1$ at every grid point with $s > s_1$ and $w > -1$ at every grid point with $s < s_1$: the crossing is at $s = s_1$, $a = s_1^{(-1/3)} = 0.7677$
- MODEL B [has content]: a genuine phantom phase: min $w = -1.302$ (to 0.001) at $a = 0.543$, and $w(a = 1) = -0.8608$
- MODEL B [has content]: $V(s) > 0$ at every row - $\rho_{\Psi} > 0$ on the whole history
- MODEL B [fidelity]: the reference CSV has the expected header and the same 701 values of N
- MODEL B [fidelity]: w agrees with `fable-cosmology/reference/mathematica_spinor_expdamp.csv` at EVERY row to $1e-9$
- MODEL B [fidelity]: so do t , H and ρ_{Ψ} , to $1e-9$ (s differs only by the file's 16-digit rounding of numbers of order $1e9$)

The run prints [D] (one line in the log, broken here):

```
Model B (expdamp):
w(a=1) = -0.86084564037764806560795392619028548152`20.
min w   = -1.30209881490918676547015655743122703056`20.   at N = -0.61
      (a = 0.54335086907449978711266408781657974806`20.)
t(a=1) = 0.96433642136311524576147498023448633382`20.
H(a=1) = 0.99998205003431888956547950340514648567`20.
```

Model B applies no normalization: $V(1) = 1.566 + 0.839 e^{-1/2.21} \approx 2.0996 = 3 \Omega_{\Psi_0}$ with $\Omega_{\Psi_0} = 0.6999$ [P, Section 24]. That is why the printed $H(a=1)$ is 0.99998 and not exactly 1: $H(a=1)^2 = 3/10 + 84/10^6 + V(1)/3 \approx 0.99996$ (this arithmetic was added for this document; it is not a notebook statement).

2.12.9 What is proved on the pre-universe, and what belongs to the reference model

Proved on the 8-manifold, for arbitrary a_4 , with every identity closed symbolically:

- the Lagrangian and its fidelity to L_a and $\mathbf{eLa}$;
- the covariant Dirac equation and its rescaling;
- the covariant energy-momentum tensor and its conservation;
- $\rho = V + K_h$ and $P = sV' - V$;
- dust for the author's mass term;

- the constancy of s in x_4 ;
- the volume factorization.

The history $s = s_0 a^{-3}$, the phantom crossing at $a = 0.768$, and all contact with the Unite fits $(w_0, w_a) = (-0.861, -0.60)$ belong to 4-dimensional physics on the separate FLRW frame [P, Section 25].

The volume bookkeeping is proved [A]:

- VOLUME [has content]: the reduced density $\rho_4 = V^{-1} \rho_8$ is NOT separately conserved: $d \rho_4 / dx_4 + 3 H_{\text{obs}} (\rho_4 + P_4) = 3 H_{\text{obs}} P_4$, for every w
- VOLUME [has content]: equivalently $d \rho_4 / dx_4 = -3 H_{\text{obs}} \rho_4$: ρ_4 falls as a^{-3} for EVERY w - a volume effect, the same for a cosmological constant as for dust

Which of ρ_4 and ρ_8 gravitates is left undetermined in Part VI, because Part VI has no gravitational sector: no Einstein equations, and a_4 free [P, Section 25].

2.13 Why the classical fable must be refined

The fable of Part VI is a field of **real, commuting** numbers. A fermion needs **anticommuting** (Grassmann) components, and the change is not cosmetic. It rests on two algebraic facts about the author's Clifford algebra.

F1, the symmetry of CT . Write $\Gamma_{(r)}$ for an antisymmetrized product of r of the $T_{16}[a]$. With $C_- = \sigma_{16}$:

- σ_{16} is symmetric;
- $\sigma_{16} T_{16}[a]$ is **antisymmetric** (this much is already Part I's `sigma16 . T16[A]` is antisymmetric for `A = 0..7`);
- $\sigma_{16} \Gamma_{ab}$ is antisymmetric;
- $\sigma_{16} \Gamma_{abc}$ and $\sigma_{16} \Gamma_{abcd}$ are symmetric;
- $\sigma_{16} T_{16}[8]$ is symmetric;
- $\sigma_{16} T_{16}[a] \sigma_{16}^{-1} = -T_{16}[a]^T$.

The second conjugation is $C_+ = \sigma_{16} T_{16}[8]$:

- it is symmetric, and $C_+ T_{16}[a]$ is **symmetric**;
- $C_+ \Gamma_{ab}$ and $C_+ \Gamma_{abc}$ are antisymmetric;
- $C_+ T_{16}[a] C_+^{-1} = +T_{16}[a]^T$.

The $\text{Spin}(4,4)$ -invariant bilinear forms on the 16-spinor form a two-dimensional space, spanned by σ_{16} and $\sigma_{16} T_{16}[8]$ ($C_+ T_{16}[8] = \sigma_{16}$). Both are symmetric, and both are chirality-diagonal. [VII] REFINEMENT [definition]: $T_{16}[8]^2 = \text{sigma16}^2 = C_+^2 = \text{ID16}$, C_+ is symmetric, and $C_+ . T_{16}[8] = \text{sigma16}$ - the four candidate mass matrices are only two, REFINEMENT [has content]: $C_- T_{16}[a] C_-^{-1} = -T_{16}[a]^T$ and $C_+ T_{16}[a] C_+^{-1} = +T_{16}[a]^T$ for $a = 0..7$, REFINEMENT [THE RESULT]: the symmetry of C Gamma depends only on the rank r of Gamma: C_- : $S A A S S A A S S$, C_+ : $S S A A S S A A S$ ($r = 0..8$), REFINEMENT [THE RESULT]: the $\text{Spin}(4,4)$ -invariant bilinear forms on the 16-spinor form a 2-dimensional space, spanned by $C_- = \text{sigma16}$ and $C_+ = \text{sigma16} T_{16}[8]$, REFINEMENT [has content]: every invariant form is SYMMETRIC and chirality-diagonal, and the two chiral pieces $PL \text{ sigma16 } PL$ and $PR \text{ sigma16 } PR$ are invariant separately.

F2, what anticommuting components do to a bilinear. For Grassmann θ :

- $\theta^T N \theta$ sees only the **antisymmetric** part of N ;
- $\theta^T N d\theta = \frac{1}{2} d(\theta^T N \theta) + \theta^T N_{\text{sym}} d\theta$.

For commuting components the roles are exactly reversed: $\theta^T N \theta$ sees only the symmetric part, and the Euler–Lagrange expression of $\theta^T N \phi$ is $2N_{\text{antisym}}\phi$. That reversal is why Part VI’s real commuting fable has both a mass term (σ_{16} symmetric) and a kinetic term ($\sigma_{16}T_{16}[a]$ antisymmetric). [VII] GRASSMANN [THE RESULT]: $\theta^T N \theta == \theta^T N_A \theta$ and $\theta^T N_S \theta == 0$ for a general 16×16 N – an anticommuting bilinear sees only the ANTISYMMETRIC part, GRASSMANN [THE RESULT]: $\theta^T N \phi - (1/2) d(\theta^T N \theta) == \theta^T N_S \phi$ for a general N , GRASSMANN [THE RESULT]: the Euler–Lagrange expression of $L = \theta^T N \phi$ (left derivatives) is $(N + N^T) \phi$: EMPTY field equations exactly when N is antisymmetric, GRASSMANN [control]: for COMMUTING components the roles are reversed – $\theta^T N \theta$ sees only N_S , and the Euler–Lagrange expression of $\theta^T N \phi$ is $2 N_A \phi$.

The consequences:

- A real Grassmann 16-spinor with the author’s σ_{16} has no dynamics.** Its mass term $\theta^T \sigma_{16} \theta$ vanishes identically, because σ_{16} is symmetric. Its kinetic term $\theta^T \sigma_{16} T_{16}[a] d\theta$ is a total derivative, because $\sigma_{16} T_{16}[a]$ is antisymmetric. So its Euler–Lagrange expression is identically empty, even though the kinetic term itself is not zero. The statement holds on **every** frame. $\sigma_{16} \Gamma_{abc}$ is symmetric, so the connection term $\theta^T \sigma_{16} \{\gamma^\mu, \Gamma_\mu^{\text{spin}}\} \theta$ vanishes too. [VII] REFINEMENT (a) [THE RESULT]: real Grassmann + sigma16: both invariant mass terms vanish, and $\theta^T \text{sigma16 } T_{16}[a] \phi == (1/2) d(\theta^T \text{sigma16 } T_{16}[a] \theta)$ for every a (a total derivative), REFINEMENT (a) [has content]: its Euler–Lagrange expression is identically empty in every direction – while the kinetic term itself is NOT zero, REFINEMENT (a) [THE RESULT]: FRAME-INDEPENDENT – $\text{sigma16 } \{T_{16}[a], S^{bc}\}$ is SYMMETRIC for all a, b, c (so $\theta^T \text{sigma16 } \{\gamma^\mu, \Gamma_\mu\} \theta == 0$ on every frame), and $d_\mu(\text{Sqrt}[g] \text{sigma16 } \gamma^\mu) == \text{Sqrt}[g] \text{sigma16 } [\gamma^\mu, \Gamma_\mu]$ on the canonical frame.
- A Majorana 16-spinor built with C_+ propagates but is massless.** Its kinetic term is genuine, but every invariant bilinear vanishes on it, so there is no s and $V(s)$ is undefined. It has exactly one quartic Lorentz scalar: the quartic scalars $v \cdot v$, $B \cdot B$ and $C \cdot C$ built from it are all proportional to one quartic. Its commuting shadow has no kinetic term at all. [VII] REFINEMENT (b) [THE RESULT]: Majorana with C_+ : the kinetic term is genuine – Euler–Lagrange expression $2 C_+ T_{16}[a] \phi$ with $C_+ T_{16}[a]$ invertible – but every invariant bilinear vanishes (massless, no $V(s)$), REFINEMENT (b) [has content]: the scope – the quartic Lorentz scalars $v.v$, $B.B$, $C.C$ of the Majorana spinor are NOT zero and are all proportional to ONE quartic (same monomials, one constant ratio each), REFINEMENT (b) [has content]: and the commuting shadow of the C_+ Majorana spinor has NO kinetic term: for commuting components $\theta^T C_+ T_{16}[a] \phi$ is the total derivative $(1/2) d(\theta^T C_+ T_{16}[a] \theta)$.
- Weyl and Majorana–Weyl 8-spinors (one chirality) cannot propagate.** Both C s are chirality-diagonal and γ^μ is chirality-odd, so $P_L C T_{16}[a] P_L = P_R C T_{16}[a] P_R = 0$. [VII] REFINEMENT (c) [THE RESULT]: Weyl and Majorana–Weyl: $P_L C T_{16}[a] P_L == P_R C T_{16}[a] P_R == 0$ for $C = C-, C_+$ and every a (C chirality-diagonal, $T_{16}[a]$ chirality-odd) – no kinetic term.
- Fermion fable is the complex 16-component spinor.** It is $\Psi = (\theta_1 + i\theta_2)/\sqrt{2}$, built from two real Grassmann 16-spinors, with the conjugate $\bar{\Psi} = \Psi^\dagger \sigma_{16}$. Here † is the

conjugation of the complex Grassmann algebra; the Hilbert-space adjoint is fixed in the quantization, section 3.4.

- It has two scalar bilinears. $s = \bar{\Psi}\Psi = i\theta_1^T\sigma_{16}\theta_2$ is non-zero and Hermitian under $\dagger$, and so is $p = \bar{\Psi}T_{16}[8]\Psi$. (With the Hilbert-space adjoint that the quantization fixes, p is odd under the fundamental symmetry J and becomes an anti-Hermitian operator, while s stays Hermitian.)
- The symmetrized kinetic term is Hermitian and genuine.
- It reduces **exactly** to Part VI’s real commuting fable: its Lagrangian, its field equation, and its covariant energy–momentum tensor. Through that reduction it reduces to the author’s L_a and eLa.

[VII] REFINEMENT (d) [THE RESULT]: $s = \text{Psibar Psi} == i \text{ theta1}^T \text{ sigma16 theta2}$: NON-ZERO (16 monomials) and HERMITIAN; $p = \text{Psibar T16}[8] \text{ Psi}$ is non-zero and Hermitian too, REFINEMENT (d) [THE RESULT]: its symmetric part is $A_S == (i/2) \Omega$, $\Omega = [[0, K], [-K, 0]]$ real SYMMETRIC and invertible: a genuine kinetic term, REFINEMENT (d) [has content]: the symmetrized kinetic term is HERMITIAN and equals $\text{Psi}^{\text{ddag}} K \text{ phi} - (1/2) d(\text{Psi}^{\text{ddag}} K \text{ Psi})$; the unsymmetrized one is not Hermitian, FIDELITY [1] [fidelity]: $\text{Psi} \rightarrow \text{psi}$, $\text{Psibar} \rightarrow \text{psi}^T \text{ sigma16}$ turns the symmetrized covariant Lhat into Part VI’s cfLhatFable EXACTLY (canonical frame, generic V), FIDELITY [2] [fidelity]: on the flat frame with $V = -(2M/H) s$ and no volume factor, the same limit on Psi16 IS the author’s La[] of Section 11, FIDELITY [3] [fidelity]: $\text{sigma16} \cdot \text{EL_Psibar} + \text{EL_Psi}$ in the real limit == Part VI’s explicit variation cfELFable, component for component, FIDELITY [3] [fidelity]: and on the flat frame with $V = -(2M/H) s$ the same combination IS the author’s eLa of Section 12, FIDELITY [4] [fidelity]: the energy-momentum tensor That in the real limit IS Part VI’s covariant tensor cfTCov, all 64 components.

The claim must be stated with its scope. The complex 16-spinor is the **minimal** field that:

1. uses the author’s σ_{16} conjugation;
2. propagates;
3. admits a potential $V(s)$;
4. reduces to Part VI’s classical fable as its real commuting shadow.

It is **not** the unique field with dynamics: a Majorana spinor with C_+ also propagates. The author chose the complex 16-spinor on 2026-09-24.

One more consequence matters for everything that follows. Part VI’s mechanism for the spin connection dropping out of the Lagrangian was $\Psi^T \sigma_{16} T_{16}[a] \Psi = 0$, which is a property of commuting components. It does not carry over to complex Grassmann fields. There, the drop of the connection is a property of the matrix $\{\gamma^\mu, \Gamma_\mu^{\text{spin}}\}$ itself, and it has to be proved again. Whether the classical results of this section, among them w and the phantom crossing, survive quantization is decided in sections 3.6 and 7. The results of the refinement are restated in section 3.1.

3 The fermion fable in brief

This section restates, complete enough to be read alone, the quantized fermion fable that is the source of the Einstein equations in the rest of this page: the field, its Lagrangian, its field equations in the primordial gravitational field, its canonical anticommutator, its energy–momentum tensor

operator, and its density, pressures and equation of state. Everything here is proved in Part VII of the notebook (Sections 26–29, manifest `claude-fable/cells_part7.wl`, Input cells 173–198, 170 assertions); the labels are quoted verbatim from `claude-fable/run_fermion_fable_part7.log`, whose tally reads **Assertions run: 528 passed: 528 failed: 0** for Parts I–VII together. Section 2.13 gave the algebraic reasons (F1, F2) for the refinement; this section states the results.

3.1 The field

Fermion fable is a **complex 16-component Grassmann spinor** Ψ , two real Grassmann 16-spinors θ_1, θ_2 combined as $\Psi = (\theta_1 + i\theta_2)/\sqrt{2}$, with the classical conjugate $\Psi^\dagger = (\theta_1 - i\theta_2)^T/\sqrt{2}$ and the Dirac conjugate

$$\bar{\Psi} = \Psi^\dagger \sigma_{16}, \quad \sigma_{16} = T_{16}[0] T_{16}[1] T_{16}[2] T_{16}[3] = \text{diag}(-\sigma, \sigma).$$

It has two Lorentz-scalar bilinears, $s = \bar{\Psi}\Psi$ and $p = \bar{\Psi} T_{16}[8] \Psi$; the self-interaction depends on s only (section 3.4). Written as $\Psi = (\psi_1, \psi_2)$ (upper eight components the type-1 split-octonion spinor of chirality -1 , lower eight the type-2 spinor of chirality $+1$), $s = -\psi_1^\dagger \sigma \psi_1 + \psi_2^\dagger \sigma \psi_2$.

Why complex, and with what scope. The $\text{Spin}(4,4)$ -invariant bilinear forms on the 16-spinor are exactly $\text{span}\{\sigma_{16}, \sigma_{16} T_{16}[8]\}$, both symmetric and chirality-diagonal. Therefore a real Grassmann 16-spinor with the author's σ_{16} has no dynamics on any frame (its mass term vanishes and its kinetic term is a total derivative), a Majorana 16-spinor built with $C_+ = \sigma_{16} T_{16}[8]$ propagates but is massless and has no $V(s)$, and a one-chirality (Weyl or Majorana–Weyl) spinor cannot propagate. The complex 16-spinor is the **minimal** field that uses the author's σ_{16} conjugation, propagates, admits $V(s)$, and reduces to Part VI's classical fable as its real commuting shadow; it is **not** the unique field with dynamics. The author chose it on 2026-09-24. **[proved VII §26]**

- REFINEMENT [THE RESULT]: the $\text{Spin}(4,4)$ -invariant bilinear forms on the 16-spinor form a 2-dimensional space, spanned by $C_- = \text{sigma16}$ and $C_+ = \text{sigma16 } T_{16}[8]$
- REFINEMENT (a) [THE RESULT]: real Grassmann + sigma16 : both invariant mass terms vanish, and $\text{theta}^T \text{sigma16 } T_{16}[a] \text{ phi} == (1/2) \text{ d}(\text{theta}^T \text{sigma16 } T_{16}[a] \text{ theta})$ for every a (a total derivative)
- REFINEMENT (b) [THE RESULT]: Majorana with C_+ : the kinetic term is genuine – Euler-Lagrange expression $2 C_+ T_{16}[a] \text{ phi}$ with $C_+ T_{16}[a]$ invertible – but every invariant bilinear vanishes (massless, no $V(s)$)
- REFINEMENT (c) [THE RESULT]: Weyl and Majorana–Weyl: $\text{PL } C T_{16}[a] \text{ PL} == \text{PR } C T_{16}[a] \text{ PR} == 0$ for $C = C_-, C_+$ and every a (C chirality-diagonal, $T_{16}[a]$ chirality-odd) – no kinetic term
- REFINEMENT (d) [THE RESULT]: $s = \text{Psibar Psi} == i \text{ theta1}^T \text{sigma16 theta2}$: NON-ZERO (16 monomials) and HERMITIAN; $p = \text{Psibar } T_{16}[8] \text{ Psi}$ is non-zero and Hermitian too

Four four-dimensional Dirac fermions. The observer's Clifford algebra $T_{16}[1..4]$ is $M_4(\mathbb{C})$; its commutant in $M_{16}(\mathbb{C})$ is generated by the hidden gammas $h_A = \Omega_4 T_{16}[A]$ ($A = 0, 5, 6, 7$, $\Omega_4 = T_{16}[1]T_{16}[2]T_{16}[3]T_{16}[4]$). So $\mathbb{C}^{16} = \mathbb{C}^4$ (observer Dirac index) $\otimes \mathbb{C}^4$ (hidden flavour index): under the observer's $\text{Spin}(3,1)$ of x_1, x_2, x_3, x_4 fermion fable is four Dirac fermions, $g = 4 \times 2 = 8$ particle states (and 8 antiparticle states) per momentum. **[proved VII §26]** 4D [THE RESULT]: the 16 products of the hidden gammas span the commutant, and the two algebras meet only in the multiples of ID16 (rank of the union $16 + 16 - 1 = 31$): $\mathbb{C}^{16} = \mathbb{C}^4(x) \mathbb{C}^4$, four 4D Dirac fermions.

Fidelity. In the real commuting limit ($\Psi \rightarrow \psi$, $\bar{\Psi} \rightarrow \psi^T \sigma_{16}$) the Lagrangian, the field equations and the energy-momentum tensor below become Part VI's classical fable exactly (section 2.12), and through it the author's L_a and eLa (sections 2.7, 2.8). [proved VII §26]

- FIDELITY [1] [fidelity]: $\Psi \rightarrow \psi$, $\bar{\Psi} \rightarrow \psi^T \sigma_{16}$ turns the symmetrized covariant Lhat into Part VI's cfLhatFable EXACTLY (canonical frame, generic V)
- FIDELITY [2] [fidelity]: on the flat frame with $V = -(2M/H)$ s and no volume factor, the same limit on Psi16 IS the author's La[] of Section 11
- FIDELITY [4] [fidelity]: the energy-momentum tensor That in the real limit IS Part VI's covariant tensor cfTCov, all 64 components

3.2 The Lagrangian

The only admissible Lagrangian is the **symmetrized covariant** one:

$$L = \sqrt{g} \hat{L}, \quad \hat{L} = \frac{1}{2H} \left[\bar{\Psi} \gamma^\mu D_\mu \Psi - (D_\mu \bar{\Psi}) \gamma^\mu \Psi \right] - V(s), \quad s = \bar{\Psi} \Psi,$$

$$D_\mu \Psi = \partial_\mu \Psi + \Gamma_\mu \Psi, \quad D_\mu \bar{\Psi} = \partial_\mu \bar{\Psi} - \bar{\Psi} \Gamma_\mu,$$

with Γ_μ the canonical spin connection of section 6 (on the dynamical frames of this page, the connection of those frames, section 6.14 onwards). There is no factor i , as in the author's L_a : the Lagrangian is Hermitian because $\sigma_{16} \gamma^a$ and $\sigma_{16} \Gamma_\mu$ are real antisymmetric matrices. It contains the connection only through the matrix $\sum_\mu \{\gamma^\mu, \Gamma_\mu\}$, which vanishes for **each** μ on the canonical frame (and on the warped and Bianchi-I frames of this page, section 6.17), so there it equals the partial-derivative symmetrized Lagrangian. The unsymmetrized covariant Lagrangian differs from it by a total divergence; the unsymmetrized *partial* Lagrangian is inadmissible (its two field equations are not Dirac conjugates). [proved VII §27]

- LAGRANGIAN (b) [THE RESULT]: Lsym is HERMITIAN - real for $\Psi = u + i v$, $\bar{\Psi} = (u - i v)^T \sigma_{16}$, generic u, v of all eight coordinates
- LAGRANGIAN (c) [THE RESULT]: Lsym - Lsym_d == (1/(2H)) Sum_mu Psi_bar {gamma^mu, Gamma_mu} Psi identically, and {gamma^mu, Gamma_mu} == 0 for EACH mu on the canonical frame: Lsym == Lsym_d
- LAGRANGIAN (f) [control]: so Lun_d is INADMISSIBLE - its two field equations are not Dirac conjugates: they differ by 2 Psi_bar gamma^mu Gamma_mu = -6 H Cot^2 Psi_bar T16[0], non-zero (witness u = e1, v = 0, V = s^2)

3.3 The field equations in the primordial gravitational field

Varying $\bar{\Psi}$ and Ψ independently:

$$\gamma^\mu D_\mu \Psi = H V'(s) \Psi, \quad (D_\mu \bar{\Psi}) \gamma^\mu = -H V'(s) \bar{\Psi}.$$

[proved VII §27]

- FIELD EQUATION [THE RESULT]: EL_Psi_bar == (Sqrt[g]/H) (gamma^mu D_mu Psi - H V'(s) Psi) for generic Psi, Psi_bar of all eight coordinates: the field equation is gamma^mu D_mu Psi == H V'(s) Psi
- FIELD EQUATION [THE RESULT]: EL_Psi == -(Sqrt[g]/H) ((D_mu Psi_bar) gamma^mu + H V'(s) Psi_bar): the adjoint equation is (D_mu Psi_bar) gamma^mu == -H V'(s) Psi_bar

- **FIELD EQUATION** [has content]: the adjoint equation IS the Dirac conjugate of the field equation: with $\text{Psi} = u + i v$, $\text{Psibar} = (u - i v)^T \text{sigma16}$, adjoint residual == $-(\text{field residual})^* \cdot \text{sigma16}$

On the author's canonical frame (section 2.11; $q = e^{-a_4(Hx_4)}/\sin^{1/6}(6Hx_0)$, $p = e^{+a_4(Hx_4)}/\sin^{1/6}(6Hx_0)$, a_4 free; this p is the frame factor, not the bilinear of section 3.1), with $\gamma^\mu \Gamma_\mu = -3H \cot^2(6Hx_0) T_{16}[0]$ (section 6.8), summed over $i = 1, 2, 3$ and $h = 5, 6, 7$:

$$\cot(6Hx_0) T_{16}[0] \partial_0 \Psi + \frac{1}{q} \sum_i T_{16}[i] \partial_i \Psi + T_{16}[4] \partial_4 \Psi + \frac{1}{p} \sum_h T_{16}[h] \partial_h \Psi - 3H \cot^2(6Hx_0) T_{16}[0] \Psi = H V'(s) \Psi.$$

[**proved VII §27**] WRITTEN OUT (i) [THE RESULT]: the field equation on the canonical frame is $\text{Cot } T_{16}[0] \text{ d_0 Psi} + (1/q) T_{16}[i] \text{ d_i Psi} + T_{16}[4] \text{ d_4 Psi} + (1/p) T_{16}[h] \text{ d_h Psi} - 3 H \text{ Cot}^2 T_{16}[0] \text{ Psi} == H V'(s) \text{ Psi}$.

In the split-octonion $8 + 8$ form, covariantly on any frame ($\Gamma_\mu = \text{diag}(\Gamma_\mu^{(1)}, \Gamma_\mu^{(2)})$), the connection never mixes the two types; $\tau_a = \text{tau}[a]$, $\bar{\tau}_a = \text{taubar}[a]$:

$$\begin{aligned} e_a^\mu \bar{\tau}_a (\partial_\mu + \Gamma_\mu^{(2)}) \psi_2 &= H V'(s) \psi_1, & e_a^\mu \tau_a (\partial_\mu + \Gamma_\mu^{(1)}) \psi_1 &= H V'(s) \psi_2, \\ (\partial_\mu \bar{\psi}_2 - \bar{\psi}_2 \Gamma_\mu^{(2)}) \tau_a e_a^\mu &= -H V'(s) \bar{\psi}_1, & (\partial_\mu \bar{\psi}_1 - \bar{\psi}_1 \Gamma_\mu^{(1)}) \bar{\tau}_a e_a^\mu &= -H V'(s) \bar{\psi}_2, \\ \bar{\psi}_1 &= -\psi_1^\dagger \sigma, & \bar{\psi}_2 &= \psi_2^\dagger \sigma, \end{aligned}$$

and on the canonical frame the connection term is $-3H \cot^2 \psi_2$ in the upper and $-3H \cot^2 \psi_1$ in the lower rows. [**proved VII §27**] WRITTEN OUT (ii) [THE RESULT]: the covariant $8+8$ form - rows 1..8 are $e_a^\mu \text{taubar}[a] (\text{d_mu} + \text{Gamma}^{(2)}_\mu) \text{psi2} - H V' \text{psi1}$, rows 9..16 are $e_a^\mu \text{tau}[a] (\text{d_mu} + \text{Gamma}^{(1)}_\mu) \text{psi1} - H V' \text{psi2}$; and the conjugate rows.

All sixteen component equations on the canonical frame. They are the notebook's export `provenance-latex/generated/fermion_fable_field_equations.tex`, included here unchanged (the same equations in plain text are the file `provenance-latex/generated/fermion_fable_field_equations.txt`; both are written by `claude-fable/export_fermion_fable_tex.wls` from the asserted term lists). ψ_k is component k of Ψ (the notebook's 0-based component index), $\partial_\mu \psi_k = \partial \psi_k / \partial x_\mu$, $e^{a_4(Hx_4)}$ is the exponential of the free function a_4 at Hx_4 , and $V'(s)$ is evaluated at $s = \bar{\Psi} \Psi$:

$$\begin{aligned} (E_0) \quad & \cot(6Hx_0) \partial_0 \psi_8 + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_7 \psi_8 - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_3 \psi_{13} \\ & - \partial_4 \psi_{13} + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_2 \psi_{14} + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_5 \psi_{14} \\ & - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_1 \psi_{15} + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_6 \psi_{15} - 3H \cot^2(6Hx_0) \psi_8 \\ & = H V'(s) \psi_0 \end{aligned}$$

$$\begin{aligned} (E_1) \quad & \cot(6Hx_0) \partial_0 \psi_9 + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_7 \psi_9 + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_3 \psi_{12} \\ & + \partial_4 \psi_{12} - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_1 \psi_{14} - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_6 \psi_{14} \\ & - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_2 \psi_{15} + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_5 \psi_{15} - 3H \cot^2(6Hx_0) \psi_9 \\ & = H V'(s) \psi_1 \end{aligned}$$

$$\begin{aligned} (E_2) \quad & \cot(6Hx_0) \partial_0 \psi_{10} + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_7 \psi_{10} - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_2 \psi_{12} \\ & - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_5 \psi_{12} + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_1 \psi_{13} + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_6 \psi_{13} \\ & - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_3 \psi_{15} + \partial_4 \psi_{15} - 3H \cot^2(6Hx_0) \psi_{10} \end{aligned}$$

$$= HV'(s) \psi_2$$

$$(E_3) \quad \cot(6Hx_0) \partial_0 \psi_{11} + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_7 \psi_{11} + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_1 \psi_{12} \\ - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_6 \psi_{12} + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_2 \psi_{13} - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_5 \psi_{13} \\ + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_3 \psi_{14} - \partial_4 \psi_{14} - 3H \cot^2(6Hx_0) \psi_{11} \\ = HV'(s) \psi_3$$

$$(E_4) \quad - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_3 \psi_9 + \partial_4 \psi_9 + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_2 \psi_{10} \\ - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_5 \psi_{10} - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_1 \psi_{11} - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_6 \psi_{11} \\ + \cot(6Hx_0) \partial_0 \psi_{12} - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_7 \psi_{12} - 3H \cot^2(6Hx_0) \psi_{12} \\ = HV'(s) \psi_4$$

$$(E_5) \quad e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_3 \psi_8 - \partial_4 \psi_8 - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_1 \psi_{10} \\ + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_6 \psi_{10} - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_2 \psi_{11} - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_5 \psi_{11} \\ + \cot(6Hx_0) \partial_0 \psi_{13} - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_7 \psi_{13} - 3H \cot^2(6Hx_0) \psi_{13} \\ = HV'(s) \psi_5$$

$$(E_6) \quad - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_2 \psi_8 + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_5 \psi_8 + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_1 \psi_9 \\ - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_6 \psi_9 - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_3 \psi_{11} - \partial_4 \psi_{11} \\ + \cot(6Hx_0) \partial_0 \psi_{14} - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_7 \psi_{14} - 3H \cot^2(6Hx_0) \psi_{14} \\ = HV'(s) \psi_6$$

$$(E_7) \quad e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_1 \psi_8 + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_6 \psi_8 + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_2 \psi_9 \\ + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_5 \psi_9 + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_3 \psi_{10} + \partial_4 \psi_{10} \\ + \cot(6Hx_0) \partial_0 \psi_{15} - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_7 \psi_{15} - 3H \cot^2(6Hx_0) \psi_{15} \\ = HV'(s) \psi_7$$

$$(E_8) \quad \cot(6Hx_0) \partial_0 \psi_0 - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_7 \psi_0 + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_3 \psi_5 \\ + \partial_4 \psi_5 - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_2 \psi_6 - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_5 \psi_6 \\ + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_1 \psi_7 - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_6 \psi_7 - 3H \cot^2(6Hx_0) \psi_0 \\ = HV'(s) \psi_8$$

$$(E_9) \quad \cot(6Hx_0) \partial_0 \psi_1 - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_7 \psi_1 - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_3 \psi_4 \\ - \partial_4 \psi_4 + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_1 \psi_6 + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_6 \psi_6 \\ + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_2 \psi_7 - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_5 \psi_7 - 3H \cot^2(6Hx_0) \psi_1 \\ = HV'(s) \psi_9$$

$$(E_{10}) \quad \cot(6Hx_0) \partial_0 \psi_2 - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_7 \psi_2 + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_2 \psi_4 \\ + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_5 \psi_4 - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_1 \psi_5 - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_6 \psi_5 \\ + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_3 \psi_7 - \partial_4 \psi_7 - 3H \cot^2(6Hx_0) \psi_2 \\ = HV'(s) \psi_{10}$$

$$(E_{11}) \quad \cot(6Hx_0) \partial_0 \psi_3 - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_7 \psi_3 - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_1 \psi_4$$

$$\begin{aligned}
& + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_6 \psi_4 - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_2 \psi_5 + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_5 \psi_5 \\
& - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_3 \psi_6 + \partial_4 \psi_6 - 3H \cot^2(6Hx_0) \psi_3 \\
& = HV'(s) \psi_{11} \\
(E_{12}) \quad & e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_3 \psi_1 - \partial_4 \psi_1 - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_2 \psi_2 \\
& + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_5 \psi_2 + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_1 \psi_3 + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_6 \psi_3 \\
& + \cot(6Hx_0) \partial_0 \psi_4 + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_7 \psi_4 - 3H \cot^2(6Hx_0) \psi_4 \\
& = HV'(s) \psi_{12} \\
(E_{13}) \quad & - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_3 \psi_0 + \partial_4 \psi_0 + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_1 \psi_2 \\
& - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_6 \psi_2 + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_2 \psi_3 + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_5 \psi_3 \\
& + \cot(6Hx_0) \partial_0 \psi_5 + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_7 \psi_5 - 3H \cot^2(6Hx_0) \psi_5 \\
& = HV'(s) \psi_{13} \\
(E_{14}) \quad & e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_2 \psi_0 - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_5 \psi_0 - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_1 \psi_1 \\
& + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_6 \psi_1 + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_3 \psi_3 + \partial_4 \psi_3 \\
& + \cot(6Hx_0) \partial_0 \psi_6 + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_7 \psi_6 - 3H \cot^2(6Hx_0) \psi_6 \\
& = HV'(s) \psi_{14} \\
(E_{15}) \quad & - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_1 \psi_0 - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_6 \psi_0 - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_2 \psi_1 \\
& - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_5 \psi_1 - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_3 \psi_2 - \partial_4 \psi_2 \\
& + \cot(6Hx_0) \partial_0 \psi_7 + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_7 \psi_7 - 3H \cot^2(6Hx_0) \psi_7 \\
& = HV'(s) \psi_{15}
\end{aligned}$$

The sixteen adjoint equations $\bar{E}_r$ (provenance-latex/generated/fermion_fable_adjoint_equations.txt) contain, for each r , the same eight (direction, component) pairs with ψ_k in place of ψ_k : the terms along the spacelike directions $x_0..x_3$ have the same sign, those along the timelike directions $x_4..x_7$ the opposite sign, the connection term $-3H \cot^2$ is the same, and the V' term changes sign ($T_{16}[a]^T = \eta_{aa} T_{16}[a]$). [proved VII §27] WRITTEN OUT (iii) [definition]: the term lists reproduce all sixteen field equations and all sixteen adjoint equations exactly.

The rescaled form. $\Psi = \sqrt{\sin 6Hx_0} \Psi'$ removes the connection term exactly, because $\gamma^\mu \Gamma_\mu = -\frac{1}{2} T_\mu \gamma^\mu$ with the gradient Weitzenböck torsion vector $T_\mu = \partial_\mu \ln \sin(6Hx_0)$:

$$\gamma^\mu D_\mu \Psi - H V'(s) \Psi = \sqrt{\sin 6Hx_0} (\gamma^\mu \partial_\mu \Psi' - H V'(\sin(6Hx_0) s') \Psi'), \quad s = \sin(6Hx_0) s'.$$

[proved VII §27] WRITTEN OUT (iv) [THE RESULT]: $\Psi = \text{Sqrt}[\text{Sin}[6Hx_0]] \Psi'$ turns the field equation into $\text{Sqrt}[\text{Sin}] (\text{Cot } T_{16}[0] \text{ d}_0 \Psi' + (1/q) T_{16}[i] \text{ d}_i \Psi' + T_{16}[4] \text{ d}_4 \Psi' + (1/p) T_{16}[h] \text{ d}_h \Psi' - H V'(\text{Sin } s') \Psi')$: NO connection term. The x_4 -only family $\Psi = \sqrt{\sin 6Hx_0} \Psi'(x_4)$ solves the equation if and only if $V''s' = 0$: for linear V (the author's mass term) or for null data. [proved VII §29] PRE-UNIVERSE (2) [THE RESULT]: $\Psi = \text{Sqrt}[\text{Sin}] \Psi'(x_4)$ satisfies the field equation iff $\text{d}_x \Psi' V'(\text{Sin}[6Hx_0] s') == 0$, and $\text{d}_x \Psi' V'(\text{Sin } s') == 6 H \text{Cos}[6 H x_0] s' V''(\text{Sin } s')$: V linear OR $s' == 0$.

The operator form. As an operator equation the right-hand side is $H:V'(s)\Psi$: with the Weyl-symmetrized product: exact for linear V ; for nonlinear V it differs by Hartree and Fock contractions with the coincident J -vacuum propagator (a scalar part that renormalizes V' , and a γ^4 part, the sea charge, removed by charge-symmetric ordering). In the Kohn–Sham mean field $V'(s) \rightarrow V'(\langle s \rangle)$ and the equation is linear [prose VII §27].

The currents. The only internal $U(1)$ is the vector phase; its current $j^\mu = i\bar{\Psi}\gamma^\mu\Psi$ is real and conserved on shell. The particle-number current is $n^\mu = -\frac{i}{H}\bar{\Psi}\gamma^\mu\Psi$, with $n^4 = \Psi^\dagger\Psi/H \geq 0$ and $N = \int \sqrt{g} n^4 d^7x$ the normal-ordered number of particles minus antiparticles. [proved VII §27, §28]

- CURRENT [THE RESULT]: OFF SHELL (1/Sqrt[g]) d_mu(Sqrt[g] j^mu) == i (Ebar Psi + Psibar E) for generic fields of all eight coordinates - conserved on shell
- CHARGE [THE RESULT]: n^4 = -(i/H) Psibar gamma^4 Psi == (1/H) Psi^ddag G Psi == (1/H) Psi^dagger Psi, whose mode form Sum_u eta_u b_u^ddag b_u == Sum_u b_u^dagger b_u is the particle number (normal-ordered: particles - antiparticles)

3.4 The canonical anticommutator in 4+4 dimensions

Time is x_4 (lapse 1, shift 0 on every frame of this page). The slice $x_4 = \text{const}$ is seven-dimensional with signature (4, 3) — $x_0..x_3$ spacelike, x_5, x_6, x_7 timelike — and is not a Cauchy surface. The Dirac bracket of the symmetrized Lagrangian gives

$$\{\Psi_a(x), \Psi_b^\dagger(y)\}_{x_4=y_4} = \frac{H}{\sqrt{g}} G_{ab} \delta^7(x-y), \quad G = -i\sigma_{16}\gamma^4, \quad \{\Psi_a, \Psi_b\} = 0.$$

On the canonical frame $\gamma^4 = T_{16}[4]$ and $\sqrt{g} = \sec 6Hx_0$, so the matrix is

$$G = J := -iT_{16}[0]T_{16}[1]T_{16}[2]T_{16}[3]T_{16}[4], \quad J \text{ Hermitian}, \quad J^2 = \text{ID}_{16},$$

eigenvalues +1 (8), -1 (8): signature (8, 8),

$$\{\Psi_a(x), \Psi_b^\dagger(y)\} = H \cos(6Hx_0) J_{ab} \delta^7(x-y).$$

[proved VII §28]

- QUANTIZATION [THE RESULT]: $G = -i \text{sigma}_{16} \gamma^4 == J = -i T_{16}[0].T_{16}[1].T_{16}[2].T_{16}[3].T_{16}[4]$ on the canonical frame; J is Hermitian, $J^2 == \text{ID}_{16}$, eigenvalues +1 (8) and -1 (8): signature (8, 8)
- ANTICOMMUTATOR [THE RESULT]: on the canonical frame ($c = \text{Sqrt}[g]/H$) $\{\Psi, \Psi^{\text{ddag}}\} == (H/\text{Sqrt}[g]) J == H \text{Cos}[6 H x_0] J$ ($J^{-1} == J$), which IS the design's $(H/\text{Sqrt}[g])(-i \text{sigma}_{16} \gamma^4)$

The spin connection does not enter it ($\Gamma_4 = 0$; the anticommutator comes from the time-derivative term alone).

The fundamental symmetry and the Hilbert adjoint. $J = -i\Omega_4 T_{16}[0]$ is a hidden flavour gamma, of signature (2, 2) on the flavour factor; $\sigma_{16} = J\beta$ with $\beta = -iT_{16}[4]$ the observer's Dirac-adjoint matrix. The Hilbert adjoint is defined by the J -involution, $\Psi^\dagger := \Psi^\ddagger J$; then $\bar{\Psi} = \Psi^\dagger\beta$, $s = \Psi^\dagger\beta\Psi$, and $\{\Psi_a, \Psi_b^\dagger\} = (H/\sqrt{g})\delta_{ab}\delta^7$: **positive**. The adjoint is mode-independent, and a mode basis must consist of J -eigenvectors. For Grassmann fields the $*$ -structure is a free choice and the filled Dirac sea makes the Hamiltonian bounded below; for a commuting field the same construction fails. Under this adjoint a bilinear $\Psi^\dagger M \Psi$ (with M classically Hermitian) is a

Hermitian operator iff $[J, M] = 0$ and anti-Hermitian iff $\{J, M\} = 0$: $s = \bar{\Psi}\Psi$ is Hermitian and $p = \bar{\Psi} T_{16}[8] \Psi$ anti-Hermitian, which is why the self-interaction depends on s only. [proved VII §28]

- J [THE RESULT]: with $\Psi^\dagger := \Psi^{\text{ddag}} J$: $\text{Psibar} = \Psi^\dagger \text{beta}$ ($J \text{sigma16} == \text{beta}$), $\Psi^{\text{ddag}} G = \Psi^\dagger$ ($J G == \text{ID16}$), $\{\Psi, \Psi^\dagger\} == (H/\text{Sqrt}[g]) \text{ID16}$ ($G^{-1} J == \text{ID16}$): POSITIVE
- J [has content]: the Hermiticity rule - for classically Hermitian M , $(J M)^\dagger == J M$ iff $[J, M] == 0$ and $== -J M$ iff $\{J, M\} == 0$: so $s = \Psi^{\text{ddag}} \text{sigma16} \Psi$ is Hermitian and $p = \Psi^{\text{ddag}} C+ \Psi$ is ANTI-Hermitian
- KREIN MODES [control]: a Krein-boosted pair $u_1 = (5/4)u_+ + (3/4)u_-$, $u_2 = (3/4)u_+ + (5/4)u_-$ in the $\omega = +5$ eigenspace is still G -orthonormal and still consists of modes, but not of J -eigenvectors; the induced $J_U = G U U^\dagger$ differs from J , and under it s is NOT Hermitian

The admissible sector (a theorem). A positive Fock quantization in which the local $s(x)$, the local energy-momentum tensor and the free Hamiltonian are Hermitian exists if and only if the span of the momenta carried by the modes is spacelike-definite. On the canonical frame: no momentum along x_5, x_6, x_7 , and J is then unique. With a hidden momentum above threshold the frequencies are imaginary ($\omega^2 = k_s^2 - k_h^2 + m^2$; at $k = (0, \frac{3}{4}, 1, 0 \mid k_5 = 2)$, $m = 1$: $\pm 1.19896 i$), at threshold the evolution is nilpotent. The admissible theory is a truncation (dimensional reduction to $\Psi(x_0, \dots, x_4)$) with a finite hidden coordinate volume $V_{\text{hid}} = \int dx_5 dx_6 dx_7$ — an assumption, and compact timelike directions contain closed timelike curves — and its anticommutator is

$$\{\Psi_a(x), \Psi_b^\dagger(y)\} = \frac{H}{V_{\text{hid}} \sqrt{|g|}} G_{ab} \delta^4(x_{0..3} - y_{0..3}).$$

[proved VII §28]

- ADMISSIBILITY (1) [THE RESULT]: $E_k^2 == (k_s^2 - k_h^2 + m^2) \text{ID16}$ for every k : $\omega^2 = k_s^2 - k_h^2 + m^2$
- ADMISSIBILITY (5) [THE RESULT]: ABOVE threshold ($k = (0, 3/4, 1, 0 \mid k_5 = 2)$, $m = 1$) $\omega^2 == -23/16$: every eigenvalue of E_k is $\pm i \text{Sqrt}[23]/4 = \pm 1.19896 i$ and every eigenvector is G -NEUTRAL ($u^\dagger G u == 0$)
- ADMISSIBILITY (7) [THE RESULT]: J is UNIQUE - the joint commutant of beta and the admissible E_k is 8-dimensional, contains J , and every element of it commutes with J (so $X^2 == 1$ with $J X > 0$ forces $X == J$)

The Fock space. Per spatial momentum the sixteen J -eigenmodes are 8 of frequency $+\omega$ and 8 of frequency $-\omega$, $\omega = \sqrt{k \cdot k + m^2}$; every mode has energy equal to its frequency, the negative-frequency modes are filled (the Dirac sea, the unique ground state), and the first excitations are 8 particles and 8 antiparticles. A finite Jordan-Wigner Fock space at one momentum ($k = 3e_1$, $m = 4$, dimension $2^{16} = 65536$) checks all of it exactly, including the sign of the anticommutator through the Heisenberg equation. [proved VII §28]

- FOCK (mode level) [THE RESULT]: the spectrum - a UNIQUE ground state at -8 $\omega = -40$, the Dirac sea (modes 9..16 filled, 1..8 empty); $:\text{Ham}: = \text{Ham} + 40 >= 0$
- FOCK (mode level) [THE RESULT]: the first excited level, $\omega = 5$ above the sea, is 16-fold: 8 particle states ($N = +1$) and 8 antiparticle states ($N = -1$), with $N = \text{Sum } b^\dagger b - 8$

- FOCK (field level) [THE RESULT]: Heisenberg - $[H_{\text{free}}, \Psi_a] == -(G^{-1} \hbar \Psi)_a$ for all 16 components, so $i \hbar d_4 \Psi = i [H, \Psi]$ IS the field equation $i \hbar d_4 \Psi = H \Psi$

What survives. 13 of the 28 Lorentz generators commute with J (Spin(4,1) of $x_0..x_4$ and Spin(3) of $x_5..x_7$); the 12 boosts between $x_0..x_3$ and $x_5..x_7$ and the 3 rotations between x_4 and $x_5..x_7$ are lost. The wall $x_0 = 0$ of the canonical frame is at finite proper distance $z = -\ln \cos(6Hx_0)/(6H)$; the one-particle space of $\Psi' = \Psi/\sqrt{\sin}$ is $L^2(dz)$, the x_0 -operator is regular at the wall and limit-point at infinity, and the Lorentz-covariant, J -compatible wall conditions are $T_{16}[0] \Psi' = \pm \Psi'$. [proved VII §28]

- J [THE RESULT]: 13 of the 28 Lorentz generators commute with J (a, b both in {0..4}: Spin(4,1), or both in {5,6,7}: Spin(3)); the other 15 anticommute - 12 boosts $x_0..x_3 \leftrightarrow x_5..x_7$ and 3 rotations $x_4 \leftrightarrow x_5..x_7$
- BOUNDARY [THE RESULT]: the bag condition $T_{16}[0] \Psi' = \pm \Psi'$ at the wall - $T_{16}[0]^2 == 1$, $\{\beta, T_{16}[0]\} == 0$, the normal-flux form vanishes on both eigenspaces ($P \beta T_{16}[0] P == 0$), and $[J, T_{16}[0]] == 0$

The anticommutator on the time-dependent frames of the coupled system is section 6.19.

3.5 The energy-momentum tensor operator

$$\hat{T}_{\mu\nu} = -\frac{1}{4H} [\bar{\Psi} \gamma_\mu D_\nu \Psi - (D_\nu \bar{\Psi}) \gamma_\mu \Psi + (\mu \leftrightarrow \nu)] + g_{\mu\nu} \hat{L}, \quad \gamma_\mu = g_{\mu\nu} \gamma^\nu,$$

normal-ordered with respect to the J -vacuum when it is an operator. It is the **Hilbert** (symmetric-tetrad) tensor of the symmetrized action, equal to this covariant expression off shell; the variation of the spin connection contributes exactly nothing to it (the Lagrangian sees the connection only through its totally antisymmetric part, which a symmetric tetrad variation does not change); it is **not** the partial-derivative tensor. It is symmetric, Hermitian under the classical conjugation, and conserved on shell. [proved VII §29]

- EMT [THE RESULT]: the Hilbert (symmetric-tetrad) tensor of the symmetrized action IS That, OFF shell - components (x1,x1), (x4,x4), (x1,x4), (x0,x5), (x5,x4)
- EMT [has content]: the spin-connection variation contributes NOTHING - dropping it leaves the Hilbert tensor unchanged at (x1,x4), (x0,x5), (x5,x4)
- EMT CONSERVATION [THE RESULT]: $\nabla^\mu \text{That}_{\{\mu\nu\}} == 0$ ON SHELL in all eight components, for generic $\Psi(x_0, x_4)$, $\bar{\Psi}(x_0, x_4)$ and generic V

On shell, for fields of (x_0, x_4) and generic V :

$$\begin{aligned} \hat{L} &= s V'(s) - V(s) & (\text{the kinetic bilinear is } s V'(s)), \\ \rho = \hat{T}_{44} &= V(s) + K_h, & K_h = -\frac{1}{2H} (\bar{\Psi} \gamma^0 \partial_0 \Psi - \partial_0 \bar{\Psi} \gamma^0 \Psi), \\ T^i_i = T^h_h &= s V'(s) - V(s) & (\text{observed } i = 1, 2, 3 \text{ and hidden } h = 5, 6, 7), \\ T^0_0 &= s V'(s) - V(s) + K_h, & \text{trace} = 7 s V'(s) - 8 V(s). \end{aligned}$$

[proved VII §29]

- EMT ON SHELL [THE RESULT]: $\rho = \text{That}_{44} == V(s) + K_h$
- EMT ON SHELL [THE RESULT]: $T^i_i == T^h_h == s V'(s) - V(s)$ for the three observed and the three hidden directions, and $T^0_0 == s V'(s) - V(s) + K_h$

$\hat{T}_{\mu\nu}$ is a Hermitian operator for μ, ν in $\{0..4\}$; its components that mix 0..4 with the hidden sheet are anti-Hermitian, with zero expectation value in every state invariant under the hidden rotations. [proved VII §29] EMT HERMITICITY [THE RESULT]: J-parity table - (mu, nu) in $\{0..4\}$: EVEN (Hermitian); (mu in 0..4, h in 5..7): ODD (anti-Hermitian); (h, h'), h != h': ZERO identically; (h, h): even.

3.6 Density, pressures and the equation of state

The observer is $u = \partial/\partial x_4$: $\rho = \langle \hat{T}_{44} \rangle$, $P_j = \langle \hat{T}^j_j \rangle$ (no sum), $w = P_{\text{obs}}/\rho$ with P_{obs} the pressure along the observed sheet. With the canonical normalization $\chi = \Psi/\sqrt{H}$ and $W(\sigma) := V(H\sigma)$, $\sigma = \langle \bar{\chi}\chi \rangle$, each occupied positive-frequency mode contributes ω to the energy density, k_j^2/ω to the pressure along j , and m/ω to σ . [proved VII §29] MODE SUMS [has content]: with $\text{ubar} = \text{u}^\dagger \text{dag}$ sigma16 these are omega u^\dagger \text{dag} G u and k_j u^\dagger \text{dag} (dh/dk_j) u, so by Section 28 each occupied mode contributes omega, k_j^2/omega and (to sigma) m/omega.

The Kohn–Sham Fermi sea. Fill the positive-frequency modes with $|k| < k_F$ along the observed sheet (zero modes along x_0 and x_5, x_6, x_7), $g = 8$, mass m , $\omega_F = \sqrt{k_F^2 + m^2}$, $\mathcal{L} = \ln((k_F + \omega_F)/m)$:

$$\begin{aligned} n &= \frac{g k_F^3}{6\pi^2}, \\ \sigma_{\text{KS}} &= \frac{g m}{4\pi^2} [k_F \omega_F - m^2 \mathcal{L}], \\ \varepsilon_{\text{KS}} &= \frac{g}{16\pi^2} [k_F \omega_F (2k_F^2 + m^2) - m^4 \mathcal{L}], \\ P_{\text{KS}} &= \frac{g}{48\pi^2} [k_F \omega_F (2k_F^2 - 3m^2) + 3m^4 \mathcal{L}], \end{aligned}$$

$$\varepsilon_{\text{KS}} + P_{\text{KS}} = \omega_F n, \quad \varepsilon_{\text{KS}} - 3P_{\text{KS}} = m \sigma_{\text{KS}}, \quad \partial \varepsilon_{\text{KS}} / \partial m = \sigma_{\text{KS}}, \quad \partial \varepsilon_{\text{KS}} / \partial k_F = \omega_F dn / dk_F.$$

[proved VII §29]

- KOHN-SHAM [THE RESULT]: each closed form has k_F -derivative $(g/(2\pi^2)) k_F^2 f(k_F)$ with $f = 1, m/\omega_F, \omega_F, k_F^2/(3\omega_F)$, and vanishes at $k_F = 0$ - so it IS the integral $g \int d^3k / (2\pi)^3 f$
- KOHN-SHAM [THE RESULT]: $\varepsilon + P == w_F n$ (Gibbs), $\varepsilon - 3P == m \sigma$ (trace), $d\varepsilon/dm == \sigma$, $d\varepsilon/dk_F == w_F dn/dk_F$

In the mean field $m = W'(\sigma)$:

$$\rho = \varepsilon_{\text{KS}} + W - \sigma W', \quad P_{\text{obs}} = P_{\text{KS}} + \sigma W' - W, \quad P_{\text{hid}} = P_{x_0} = \sigma W' - W, \quad \rho + P_{\text{obs}} = \omega_F n.$$

[proved VII §29] MEAN FIELD [THE RESULT]: $\rho = \varepsilon + W - \sigma W'$, $P_{\text{obs}} = P_{\text{KS}} + \sigma W' - W$ give $\rho + P_{\text{obs}} == w_F n$ exactly; and $d\rho/dn == w_F + (\sigma_{\text{KS}} - \sigma) W'' \sigma'(n)$, i.e. $== w_F$ at the self-consistent point $\sigma = \sigma_{\text{KS}}$ ($m = W'(\sigma)$ throughout). Hence $w_f + 1 = \omega_F n / \rho \geq 0$ wherever $\rho > 0$: **the quantized fable cannot cross the phantom divide**. For the author's mass term $W = m\sigma$ the condensate $W - \sigma W'$ vanishes, $\rho = \varepsilon_{\text{KS}} > 0$, $P_{\text{obs}} = P_{\text{KS}} \geq 0$, and $w = P_{\text{KS}}/\varepsilon_{\text{KS}}$ rises monotonically with $x = k_F/|m|$ from 0 (dust) to 1/3 (radiation); the classical limit is the non-relativistic limit $k_F/|m| \rightarrow 0$, which reproduces Part VI's $\rho = V(s)$, $P = sV' - V$, with the sign rule $\text{sign}(\sigma) = \text{sign}(m) = \text{sign}(W')$. [proved VII §29]

- AUTHOR'S MASS TERM [THE RESULT]: w rises MONOTONICALLY - $dw/dx == x^2 \Delta / (3 w_x I_1^2)$ with $\Delta = x^2 I_1 - w_x^2 I_2$, $\Delta(0) = 0$ and $\Delta' = 2 x I_3 > 0$ - from $w \rightarrow 0$ ($x \rightarrow 0$, $w = x^2/5 + 0(x^4)$: dust) to $w \rightarrow 1/3$ ($x \rightarrow \text{infinity}$: radiation)

- CLASSICAL LIMIT [THE RESULT]: at the self-consistent point $\rho == W + 3 P_{KS}$ and $P_{obs} == (\sigma W' - W) + P_{KS}$ exactly ($\sigma = \sigma_{KS}$, $m = W'$); Part VI's $\rho = V(s)$, $P = s V' - V$ is the limit $P_{KS}/\epsilon_{KS} \rightarrow 0$, i.e. $k_F/m \rightarrow 0$

On the pre-universe (the canonical frame, a_4 free) the charge per coordinate volume is conserved, the observed-sheet momenta redshift as e^{a_4} , and the quantum gas **cools** ($w : 1/3 \rightarrow 0$) when $a_4' < 0$, while the classical Part VI fable's w is frozen. [proved VII §29] PRE-UNIVERSE (4) [THE RESULT]: $x = k_{F,phys}/m = x_c/(q m)$ obeys $d \log[x]/dx_4 == H a_4' [H x_4]$; since $dw/dx > 0$, $dw/dx_4 = w'(x) x H a_4' < 0$ exactly when $a_4' < 0$ (the observed sheet expanding): the quantum gas COOLS.

Section 7 turns all of this into the per-7-volume source of the coupled system and proves the properties the Einstein equations need.

4 The Einstein equations of the primordial gravitational field, with fable as their source

This is Section 30 of the notebook (manifest `claude-fable/cells_part8.wl`, Input cells 199–203, 28 assertions, 3.534s in `claude-fable/run_fermion_fable_final.log`). Part VIII opens with a statement of purpose: Parts III–V built the $4 + 4$ geometry of the pre-universe but never asked what **sources** it; Part VI put fields on it without letting them act back; Part VII made fable a quantized fermion with an energy–momentum tensor operator; Part VIII closes the loop [prose VIII, introduction]. It never redefines anything of Parts I–VII, and it never gives a_4 a value.

4.1 The semiclassical system and its conventions

The metric stays classical; fable is quantized. The coupled equations are

$$\begin{aligned} G^\mu{}_\nu &= \kappa \langle \hat{T}^\mu{}_\nu \rangle && \text{(semiclassical gravity),} \\ \kappa &= 8\pi G_8 > 0 && \text{(the eight-dimensional gravitational coupling, a plain symbol),} \end{aligned}$$

with the expectation value taken in a state of the fable Fock space of section 3.4 — in section 7 and after, the Kohn–Sham ground state. The Einstein tensor is used with one index up and one down, $G^\mu{}_\nu$, because then the answer does not depend on how the metric signs are distributed. For a source at rest in the observer's frame the energy density and the pressures are the diagonal of the mixed tensor [prose VIII, introduction]:

$$\rho = -T^4{}_4, \quad P_{obs} = T^i{}_i \quad (i = 1, 2, 3), \quad P_{hid} = T^h{}_h \quad (h = 5, 6, 7), \quad P_C = T^0{}_0,$$

so $G^4{}_4 = -\kappa\rho$, $G^i{}_i = \kappa P_{obs}$, $G^h{}_h = \kappa P_{hid}$, $G^0{}_0 = \kappa P_C$. The dynamical frames of Sections 31–33 carry their own scale factors **scA**, **scB**, **scC** (functions of x_4 , separate symbols); the canonical frame is recovered from them only by an explicit, labelled substitution inside an assertion.

4.2 The tool, and the three tests it must pass

One helper, `cfEinsteinData`, computes every Einstein tensor of Part VIII from the Christoffel symbols of Section 16's solver:

$$\begin{aligned} R_{\mu\nu} &= \partial_\lambda \Gamma^\lambda{}_{\mu\nu} - \partial_\nu \Gamma^\lambda{}_{\mu\lambda} + \Gamma^\lambda{}_{\lambda\sigma} \Gamma^\sigma{}_{\mu\nu} - \Gamma^\lambda{}_{\nu\sigma} \Gamma^\sigma{}_{\mu\lambda}, \\ R &= g^{\mu\nu} R_{\mu\nu}, \quad G^\mu{}_\nu = g^{\mu\kappa} R_{\kappa\nu} - \frac{1}{2} \delta^\mu{}_\nu R. \end{aligned}$$

It is not trusted until it passes three tests, each of which a wrong index or sign would fail [proved VIII §30]:

1. on the separate FLRW frame of Part VI it returns the Friedmann equations with the textbook signs: EINSTEIN [solver regression]: on the FLRW frame of Section 24, $G^4_4 == -3 (a'/a)^2$ and $G^i_i == -(2 a''/a + (a'/a)^2)$: the Friedmann equations with the textbook signs; EINSTEIN [solver regression]: ... and $G == -3 (a''/a + (a'/a)^2)$ along the four static directions x_0, x_5, x_6, x_7 , with no off-diagonal component;
2. on the canonical frame its Ricci scalar equals the one Section 16 obtained from the curvature 2-form of the spin connection: EINSTEIN [fidelity]: the Ricci scalar from the Christoffel symbols == RicciScalarCanonical of Section 16 (from the curvature 2-form of the spin connection): two routes, one answer;
3. its output obeys the contracted Bianchi identity: EINSTEIN [has content]: the contracted Bianchi identity $\nabla_\mu G^\mu_\nu == 0$ holds for the canonical Einstein tensor, all eight components, a_4 arbitrary.

And the canonical metric is not a vacuum solution: EINSTEIN [control]: the canonical metric is NOT a vacuum solution - G^0_0 is not zero (witnessed).

4.3 The Einstein tensor of the canonical metric: all sixty-four components

With $\cot = \cot(6Hx_0)$, $\csc = \csc(6Hx_0)$, $a'_4 = a'_4(Hx_4)$, $a''_4 = a''_4(Hx_4)$ (derivatives with respect to the argument of a_4), the mixed Einstein tensor of the canonical metric

$$ds^2 = \tan^2(6Hx_0) dx_0^2 + q^2 dx_{\text{obs}}^2 - dx_4^2 - p^2 dx_{\text{hid}}^2$$

is **diagonal** — all 56 off-diagonal components vanish identically — and

$$\begin{aligned} G^0_0 &= 3H^2(5\cot^4 - a_4'^2), \\ G^i_i &= \frac{H^2}{2} \left(15(9 + \cos(12Hx_0)) \cot^2 \csc^2 - 6a_4'^2 - 2a_4''^2 \right) & i = 1, 2, 3, \\ G^4_4 &= \frac{3H^2}{2} \left((55 + 7\cos(12Hx_0)) \cot^2 \csc^2 + 2a_4'^2 \right), \\ G^h_h &= \frac{H^2}{2} \left(15(9 + \cos(12Hx_0)) \cot^2 \csc^2 - 6a_4'^2 + 2a_4''^2 \right) & h = 5, 6, 7, \\ R &= -3H^2 \left((55 + 7\cos(12Hx_0)) \cot^2 \csc^2 - 2a_4'^2 \right) \end{aligned}$$

for every a_4 [prose VIII §30], and [proved VIII §30]

- EINSTEIN [THE RESULT]: all 64 components of G^μ_ν of the canonical metric, a_4 arbitrary: 8 diagonal ones as stated, 56 off-diagonal ones identically zero
- EINSTEIN [THE RESULT]: the Ricci scalar $R == -3 H^2 ((55 + 7 \cos[12 H x_0]) \cot^2 \csc^2 - 2 a_4'^2)$

The split, with no cross terms. Every component is a function of x_0 alone plus a function of x_4 alone. The statement is made without giving a_4 a value: the x_0 -part is the Einstein tensor of the **separate** frame $\text{diag}(\tan 6Hx_0, S^{-1/6}, S^{-1/6}, S^{-1/6}, 1, S^{-1/6}, S^{-1/6}, S^{-1/6})$ ($S = \sin 6Hx_0$), which does not contain a_4 , and the a_4 -part is the Einstein tensor of the **separate** frame $\text{diag}(1, e^{-a_4}, e^{-a_4}, e^{-a_4}, 1, e^{+a_4}, e^{+a_4}, e^{+a_4})$, which does not contain x_0 ; the two frames multiply to the canonical frame. [proved VIII §30]

- EINSTEIN [definition]: the two factor frames multiply to the canonical frame, and neither is it: the x_0 -only frame contains no a_4 , the a_4 -only frame contains no x_0

- EINSTEIN [THE RESULT]: NO CROSS TERMS - $G^{\mu}_{\nu}(\text{canonical}) == G^{\mu}_{\nu}(x_0\text{-only frame}) + G^{\mu}_{\nu}(a_4\text{-only frame})$, all 64 components
- EINSTEIN [has content]: the x_0 -part is the same on the observed and the hidden sheet ($G^i_i == G^h_h$), and the a_4 -part is x_0 -independent

The two parts, component by component [derived here: the x_0 -part is the canonical formula at $a'_4 = a''_4 = 0$, since the a_4 -part depends on a_4 only through the rates a'_4 and a''_4 (it is a Bianchi-I tensor, section 5.1) and vanishes for constant a_4 , where the a_4 -only frame is constant; the a_4 -part is the difference]:

component	x_0 -part G_{x_0} (the x_0 -only frame)	a_4 -part G_{a_4} (the a_4 -only frame)
G^0_0	$15H^2 \cot^4$	$-3H^2 a_4'^2$
G^i_i ($i = 1, 2, 3$)	$\frac{15H^2}{2} (9 + \cos(12Hx_0)) \cot^2 \csc^2$	$-3H^2 a_4'^2 - H^2 a_4''$
G^4_4	$\frac{3H^2}{2} (55 + 7 \cos(12Hx_0)) \cot^2 \csc^2$	$+3H^2 a_4'^2$
G^h_h ($h = 5, 6, 7$)	$\frac{15H^2}{2} (9 + \cos(12Hx_0)) \cot^2 \csc^2$	$-3H^2 a_4'^2 + H^2 a_4''$

The x_0 -part does not distinguish the observed from the hidden sheet; only a_4'' does.

4.4 Theorem (a): energy — the canonical metric demands negative total energy

$G^4_4 = -\kappa \rho_{\text{total}}$, and the minimum of $55 + 7 \cos(12Hx_0)$ is 48. Therefore, for every a_4 and everywhere in $0 < 6Hx_0 < \pi/2$,

$$\rho_{\text{total}} = -G^4_4/\kappa \leq -(72H^2 \cot^2 \csc^2 + 3H^2 a_4'^2)/\kappa < 0,$$

$$G^4_4 - (72H^2 \cot^2 \csc^2 + 3H^2 a_4'^2) = \frac{21}{2} H^2 (1 + \cos(12Hx_0)) \cot^2 \csc^2 \geq 0 \quad \text{exactly.}$$

The total energy density that sources the canonical metric is **negative everywhere**. No source with $\rho \geq 0$ can be the sole source of this geometry — in particular not the Kohn–Sham fable with the author’s mass term, whose $\rho = \varepsilon_{\text{KS}}$ is positive. [proved VIII §30]

- ENERGY [has content]: min over x_0 of $55 + 7 \cos[12 H x_0]$ is 48 (attained where $\cos[12 H x_0] = -1$)
- ENERGY [THE RESULT]: $G^4_4 - (72 H^2 \cot^2 \csc^2 + 3 H^2 a_4'^2) == (21/2) H^2 (1 + \cos[12 H x_0]) \cot^2 \csc^2$, which is ≥ 0
- ENERGY [THE RESULT]: $72 H^2 \cot^2 \csc^2 + 3 H^2 a_4'^2 > 0$ in $0 < 6 H x_0 < \pi/2$ for real a_4 , hence $\rho_{\text{total}} = -G^4_4/\kappa < 0$: the canonical metric demands **NEGATIVE** total energy density
- ENERGY [has content]: Section 29’s ε_{KS} vanishes at $kF = 0$ and $d\varepsilon_{\text{KS}}/d kF == g kF^2 wF/(2 \pi^2) > 0$, so the KS fable with the author’s mass term ($\rho = \varepsilon_{\text{KS}}$) cannot be the sole source

4.5 The background-stress reading (an illustration; the splitting is a choice)

One may split the source as $T = T_{\text{bg}} + T_{\text{rest}}$ with $T_{\text{bg}} := G_{x_0}/\kappa$, the x_0 -part held as a fixed background. Its energy density is negative everywhere:

$$\rho_{\text{bg}} = -\frac{3H^2}{2\kappa} (55 + 7 \cos(12Hx_0)) \cot^2 \csc^2 < 0.$$

The remaining equations $G_{a_4} = \kappa T_{\text{rest}}$ then demand a stress that does not depend on x_0 , because G_{a_4} does not; the fable solutions of Part VI do — the rescaled solution $\Psi = \sqrt{\sin 6Hx_0} \Psi'(x_4)$

has $\rho = V(\sin(6Hx_0)s')$, which depends on x_0 for the author's mass term. So fable would have to deform the x_0 -profile. This is an illustration of the splitting, not a theorem about the full coupled problem, and the notebook states it as such [prose VIII §30]. [proved VIII §30]

- BACKGROUND [has content]: with the splitting $T_{bg} := G_{x0}/\kappa$, $\rho_{bg} == -(3 H^2/(2 \kappa)) (55 + 7 \cos[12 H x_0]) \cot^2 \csc^2 < 0$ everywhere
- BACKGROUND [has content]: the illustration - G_{a4} is x_0 -independent, while the fable density of Section 24, $\rho = V(\sin[6 H x_0] s')$ on $\Psi = \text{Sqrt}[\sin] \Psi'(x_4)$, depends on x_0 for the mass term (witnessed on $\Psi' = e_1 + e_5$, $M = 1$)

4.6 Theorem (b): anisotropy

Subtracting the hidden-sheet equation from the observed-sheet one, the x_0 -parts cancel:

$$G^i_i - G^h_h = -2H^2 a''_4 \quad \text{so} \quad a''_4 = -\frac{\kappa(P_{\text{obs}} - P_{\text{hid}})}{2H^2} \quad \text{for ANY source.}$$

Take fable as the source, as a c-number configuration $\Psi(x_0, x_4), \bar{\Psi}(x_0, x_4)$ (independent functions) of the symmetrized Lagrangian. Along a direction on which the fields do not depend, $D_i \Psi = \Gamma_i \Psi$ and $D_i \bar{\Psi} = -\bar{\Psi} \Gamma_i$, so the kinetic part of $\hat{T}_{ii}$ is $-\frac{1}{2H} \bar{\Psi} \{\gamma_i, \Gamma_i\} \Psi$, which vanishes because $\{\gamma_\mu, \Gamma_\mu\} = 0$ for each μ separately. Hence, **off shell** and for every V ,

$$P_{\text{obs}} = T^i_i = \hat{L} = T^h_h = P_{\text{hid}},$$

and such configurations force $a''_4 = 0$: the anisotropy of the canonical frame cannot be sourced by them. The Kohn–Sham Fermi gas is different: its quasiparticles move along the observed sheet only, so $P_{\text{obs}} - P_{\text{hid}} = P_{\text{KS}}/v > 0$ (with v the hidden 4-volume factor of section 7) and it **can** source $a''_4 < 0$. [proved VIII §30]

- ANISOTROPY [THE RESULT]: $G^i_i - G^h_h == -2 H^2 a''_4[H x_4]$ for every observed i and hidden h , hence $a''_4 == -\kappa (P_{\text{obs}} - P_{\text{hid}})/(2 H^2)$ for any source
- ANISOTROPY [has content]: the premise, re-checked with the index lowered - $\{\gamma_\mu, \Gamma_\mu\} == 0$ for EACH μ separately on the canonical frame ($\gamma_\mu = g_{\mu\nu} \gamma^\nu$; Section 27 (c))
- ANISOTROPY [THE RESULT]: OFF SHELL, for generic independent $\Psi(x_0, x_4), \bar{\Psi}(x_0, x_4)$ and every V : $T^i_i == \hat{L} == T^h_h$ ($i = 1, 2, 3$; $h = 5, 6, 7$), so $P_{\text{obs}} == P_{\text{hid}}$ and such fable configurations force $a''_4 == 0$
- ANISOTROPY [control]: T^0_0 is NOT $\hat{L}$ off shell - the x_0 -derivatives enter it (witnessed on $\Psi = x_0 e_1$, $\bar{\Psi}$ = the first column of $T_{16}[0]$, $V = s^2$): the statement is special to the directions on which the fields do not depend
- ANISOTROPY [has content]: Section 29's P_{KS} vanishes at $k_F = 0$ and $dP_{\text{KS}}/dk_F == g k_F^4/(6 \pi^2 w_F) > 0$, so $P_{\text{KS}} > 0$ for $k_F > 0$
- ANISOTROPY [THE RESULT]: for the KS fluid $P_{\text{obs}} - P_{\text{hid}} == P_{\text{KS}}/v$ identically, for every W : the Fermi gas CAN source $a''_4 < 0$

4.7 Theorem (c): momentum

$G^0_4 = G^4_0 = 0$ on the canonical frame, so the source must carry no momentum along the hidden coordinate, $T^0_4 = 0$. That is a real condition on fable: its T_{04} is not identically zero. [proved VIII §30]

- MOMENTUM [THE RESULT]: $G^0_4 = G^4_0 = 0$ on the canonical frame, so $T^0_4 = 0$ is required of the source
- MOMENTUM [has content]: $T^0_4 = 0$ is a genuine condition - fable's T_{04} is not identically zero (witnessed on $\Psi = x_4 e_1$, $\Psi_{\text{sib}} = T_{16}[0] e_1$, $V = s^2$)

4.8 The asymptotic canonical frame is sourced by a ghost stiff fluid

The a_4 -part of the Einstein tensor is the Einstein tensor of $\text{diag}(1, e^{-a_4}, e^{-a_4}, e^{-a_4}, 1, e^{a_4}, e^{a_4}, e^{a_4})$, which section 5.3 shows to be the asymptotic ($6Hx_0 \rightarrow \pi/2$) form of the canonical frame. With $h = H a'_4(Hx_4)$ (so that the observer's Hubble rate is $H_{\text{obs}} = -h$) and $h' = dh/dx_4 = H^2 a''_4$, the stress it demands is

$$\rho = P_C = -\frac{3h^2}{\kappa}, \quad P_{\text{obs}} = \rho - \frac{h'}{\kappa}, \quad P_{\text{hid}} = \rho + \frac{h'}{\kappa}.$$

Its x_0 -pressure equals its energy density (stiff, $w_C = +1$), and both are negative. When $h' = 0$ all seven pressures equal $\rho = -3h^2/\kappa$: exactly the energy-momentum tensor of a **ghost** scalar $\phi(x_4)$ (a scalar with the wrong sign of its kinetic term, $L = +\frac{1}{2} g^{mn} \partial_m \phi \partial_n \phi$, $T_{mn} = -\partial_m \phi \partial_n \phi + g_{mn} L$) with $\phi'^2 = 6h^2/\kappa$. In general the demanded stress is the ghost tensor plus the traceless anisotropic stress (h'/κ) $\text{diag}(0, -1, -1, -1, 0, +1, +1, +1)$. The volume-preserving expansion of the observed sheet ($h < 0$, $H_{\text{obs}} > 0$) is sourced by negative energy. **[proved VIII §30]**

- GHOST [THE RESULT]: the a_4 -part demands $\rho = P_C = -3 h^2/\kappa$, $P_{\text{obs}} = \rho - h'/\kappa$, $P_{\text{hid}} = \rho + h'/\kappa$, with $h = H a'_4[H x_4]$, $h' = dh/dx_4$
- GHOST [THE RESULT]: demanded stress == ghost-scalar tensor with $\phi'^2 = 6 h^2/\kappa + (h'/\kappa) \text{diag}(0, -1, -1, -1, 0, 1, 1, 1)$: negative-energy, stiff ($P_C = \rho$), plus a traceless anisotropic part
- GHOST [has content]: the ghost scalar's energy density is $-\phi'^2/2 < 0$ and all seven of its pressures equal it

and PART VIII [control]: a_4 is STILL UNDEFINED after Section 30.

4.9 What Section 30 proves

Proved symbolically, with a_4 arbitrary: the Einstein tensor of the canonical metric (all 64 components, the $x_0 + a_4$ split with no cross terms, the Ricci scalar two ways, the Bianchi identity); negative total energy for every source; the anisotropy law, with $P_{\text{obs}} = P_{\text{hid}}$ for every c-number fable configuration of (x_0, x_4) and $P_{\text{obs}} - P_{\text{hid}} = P_{\text{KS}}/v$ for the Kohn-Sham gas; zero x_0 -momentum; and the ghost-scalar reading of the asymptotic canonical frame. The conclusion that drives the rest of this page: **fable, whose energy is positive, cannot source the canonical frame alone; the frame must be generalized** (section 5).

5 The generalized frame, the (0,4) equation, the asymptotic region, and the 8-dimensional Bianchi-I system

This is Section 31 of the notebook (Input cells 204–207, 21 assertions, 11.310s). Section 30 showed that the canonical frame, with its single free function a_4 , cannot absorb a positive energy density. Section 31 builds the family of frames on which fable can act back.

5.1 The generalized warped frame and its Einstein tensor

The generalization keeps the x_0 -profile of the canonical frame and gives each block its own scale factor, a function of the observer's time x_4 :

$$\begin{aligned} e &= \text{diag}(\tan(6Hx_0)C, AS^{-1/6}, AS^{-1/6}, AS^{-1/6}, 1, BS^{-1/6}, BS^{-1/6}, BS^{-1/6}), \\ S &= \sin 6Hx_0, \\ A &= \text{scA}[x_4] \text{ (observed sheet)}, \quad B = \text{scB}[x_4] \text{ (hidden timelike sheet)}, \\ C &= \text{scC}[x_4] \text{ (hidden } x_0 \text{ direction)}, \\ H_A &= A'/A, \quad H_B = B'/B, \quad H_C = C'/C, \quad \Theta = 3H_A + 3H_B + H_C \quad (' = d/dx_4). \end{aligned}$$

The canonical frame is the member $A = e^{-a_4}$, $B = e^{+a_4}$, $C = 1$ — a labelled substitution, used only inside assertions. The theorem:

$$\begin{aligned} \text{diag } G^\mu{}_\nu &= \text{diag } G_{x_0}/C^2 + \text{diag } G_{\text{BI}}(A, B, C) \quad \text{EXACTLY,} \\ G^4{}_0 &= 3H \cot(6Hx_0) (2H_C - H_A - H_B), \quad G^0{}_4 = -3H \cot^3(6Hx_0) (2H_C - H_A - H_B)/C^2 \\ &\quad \text{(the only off-diagonal components),} \end{aligned}$$

where G_{x_0} is the x_0 -part of section 4.3 and G_{BI} is the Einstein tensor of the unwarped 8-dimensional Bianchi-I frame $\text{diag}(C, A, A, A, 1, B, B, B)$. The contracted Bianchi identity holds for the whole family, and on the canonical member everything reduces to Section 30. [**proved VIII §31**]

- **WARPED [definition]:** the canonical frame IS the member $A = \text{Exp}[-a_4]$, $B = \text{Exp}[+a_4]$, $C = 1$ of the family (labelled substitution)
- **WARPED [has content]:** the contracted Bianchi identity $\text{nabla}_\mu \hat{G}^\mu{}_\nu == 0$ holds for the warped family, A, B, C arbitrary
- **WARPED [THE RESULT]:** $\text{diag } \hat{G}^\mu{}_\nu == \text{diag } G_{x_0}/C^2 + \text{diag } G_{\text{BI}}(A, B, C)$ EXACTLY (the x_0 -part of Section 30 and the unwarped Bianchi-I tensor)
- **WARPED [THE RESULT]:** the only off-diagonal components are $\hat{G}^4{}_0 == 3H \cot(2H_C - H_A - H_B)$ and $\hat{G}^0{}_4 == -3H \cot^3(2H_C - H_A - H_B)/C^2$
- **WARPED [fidelity]:** on the canonical member the warped Einstein tensor IS Section 30's, all 64 components, and the Bianchi-I part IS Section 30's a_4 -part

(the two geometries took 8.727s and 0.569s, `claude-fable/run_fermion_fable_final.log`).

The Bianchi-I part, component by component [derived here from the Ricci components asserted in section 5.4, $R^i{}_i = H'_i + H_i\Theta$ and $R^4{}_4 = \sum_7(H'_i + H_i^2)$, with $R = R^4{}_4 + 3R^A{}_A + 3R^B{}_B + R^C{}_C$ and $G^\mu{}_\mu = R^\mu{}_\mu - R/2$; $\Sigma_2 = 3H_A^2 + 3H_B^2 + H_C^2$, $\dot{S} = 3H'_A + 3H'_B + H'_C$]:

$$\begin{aligned} G_{\text{BI}}^4{}_4 &= -\frac{1}{2}(\Theta^2 - \Sigma_2) = -(3H_A^2 + 3H_B^2 + 9H_AH_B + 3H_AH_C + 3H_BH_C), \\ G_{\text{BI}}^i{}_i &= H'_A + H_A\Theta - \dot{S} - \frac{1}{2}(\Sigma_2 + \Theta^2) \quad (i = 1, 2, 3), \\ G_{\text{BI}}^h{}_h &= H'_B + H_B\Theta - \dot{S} - \frac{1}{2}(\Sigma_2 + \Theta^2) \quad (h = 5, 6, 7), \\ G_{\text{BI}}^0{}_0 &= H'_C + H_C\Theta - \dot{S} - \frac{1}{2}(\Sigma_2 + \Theta^2). \end{aligned}$$

Checks [**derived here**]: on the canonical member ($H_A = -h$, $H_B = h$, $H_C = 0$, $h = Ha'_4$, $\Theta = 0$) these give $G^4{}_4 = 3h^2$, $G^i{}_i = -h' - 3h^2$, $G^h{}_h = h' - 3h^2$, $G^0{}_0 = -3h^2$ — the a_4 -part of section 4.3 ($h' = H^2a''_4$); on the FLRW frame ($A = a$, $B = C = 1$, $H_A = a'/a$) they give $G^4{}_4 = -3H_A^2$, $G^i{}_i = -(2a''/a + H_A^2)$, $G^0{}_0 = G^h{}_h = -3(a''/a + H_A^2)$ — the solver regression of section 4.2.

5.2 The (0,4) equation singles out the author's volume-preserving structure; it does not derive it

The Einstein equation $G^4_0 = \kappa T^4_0$ with a source that carries no momentum along x_0 ($T^4_0 = 0$: required on the canonical frame by theorem (c), and true of the Kohn–Sham fluid, whose quasiparticles move along the observed sheet only) gives, since $\cot(6Hx_0)$ is not zero,

$$2H_C = H_A + H_B.$$

With the x_0 direction **static** ($C = 1$, a labelled substitution) this is $H_A + H_B = 0$, i.e. $AB = \text{const}$: exactly the author's volume-preserving pairing $A = e^{-a_4}$, $B = e^{+a_4}$. So the author's ansatz is singled out by the momentum constraint once $C = 1$ is imposed; it is **not** derived, because $C = 1$ is itself an assumption. With C dynamical the constraint is $H_C = (H_A + H_B)/2$, and the energy bill changes sign. The Bianchi-I part of $\kappa\rho = -G^4_4$ is the sum over the 21 pairs of the seven directions:

$$\kappa\rho = -G_{x_0}^4/C^2 + 3H_A^2 + 3H_B^2 + 9H_AH_B + 3H_AH_C + 3H_BH_C,$$

pair sum = $-3H_A^2 < 0$	on the volume-preserving branch ($H_B = -H_A$, $H_C = 0$): section 4.4's negative energy,
pair sum = $\frac{9}{2}(H_A^2 + H_B^2) + 12H_AH_B$	on the branch $H_C = (H_A + H_B)/2$
$= +\frac{9}{2}H_A^2 > 0$	when the hidden sheet is static ($H_B = 0$).

With C dynamical a **positive-energy** expansion of the observed sheet is allowed (the x_0 -term $-G_{x_0}^4/C^2$ is still negative; section 5.3 shows how fast it decays). [**proved VIII §31**]

- (0,4) [THE RESULT]: with $T^4_0 = 0$ the (0,4) Einstein equation is $2H_C == H_A + H_B$ ($\cot[6Hx_0]$ is not zero for $0 < 6Hx_0 < \pi/2$)
- (0,4) [THE RESULT]: with $C = 1$ (labelled substitution) it reads $H_A + H_B == 0$, i.e. $d(A B)/dx_4 == 0$: the volume-preserving pairing is SINGLED OUT, and the canonical member satisfies it
- (0,4) [has content]: $\kappa\rho == -G^4_4 == -G_{x_0}^4/C^2 + (\text{sum over the 21 pairs of the seven directions of } H_i H_j)$
- (0,4) [THE RESULT]: the pair sum is $-3H_A^2$ on the volume-preserving branch (negative energy) and $(9/2)(H_A^2 + H_B^2) + 12H_AH_B$ on the branch $H_C = (H_A + H_B)/2$, i.e. $+(9/2)H_A^2 > 0$ for a static hidden sheet

5.3 The asymptotic region, and the decay rates

The proper distance along x_0 at fixed x_4 ($C = 1$) is

$$z = \int_0^{x_0} \tan(6Hy) dy = -\frac{\ln \cos(6Hx_0)}{6H}, \quad x_0(z) = \frac{\arccos(e^{-6Hz})}{6H},$$

and $6Hx_0 \rightarrow \pi/2$ is $z \rightarrow \infty$. In the coordinate z the warped metric is **exactly**

$$ds^2 = C^2 dz^2 + A^2(1 - e^{-12Hz})^{-1/6} dx_{\text{obs}}^2 - dx_4^2 - B^2(1 - e^{-12Hz})^{-1/6} dx_{\text{hid}}^2,$$

so it tends to the unwarped Bianchi-I metric, and every x_0 -term of the Einstein tensor decays with $\cot^2(6Hx_0) = e^{-12Hz}/(1 - e^{-12Hz})$. As $6Hx_0 \rightarrow \pi/2$:

$$G_{x_0}^0 \sim 15H^2 \cot^4, \quad G_{x_0}^i{}_i = G_{x_0}^h{}_h \sim 60H^2 \cot^2, \quad G_{x_0}^4{}_4 \sim 72H^2 \cot^2,$$

$$G^4_0 \sim \cot, \quad G^0_4 \sim \cot^3.$$

Read as a background stress $T_{\text{bg}} = G_{x_0}/(\kappa C^2)$, in units of $H^2 \cot^2/(\kappa C^2)$: $\rho_{\text{bg}} \rightarrow -72$, $P_{\text{obs}} = P_{\text{hid}} \rightarrow 60$, $P_C \rightarrow 15 \cot^2 \rightarrow 0$. If T_{bg} is **not** counted as a source, the x_0 -terms act on the Bianchi-I evolution equations (section 5.4) as residual drivers $\kappa(P_i - T/6)|_{\text{bg}} \rightarrow -12H^2 \cot^2/C^2$ for A and for B , and $-72H^2 \cot^2/C^2$ for C . The present universe may be treated as the 8-dimensional Bianchi-I region when the background is negligible against the observer's critical density:

$$\frac{\kappa |\rho_{\text{bg}}|}{3H_A^2} \rightarrow 24 \left(\frac{H}{H_A} \right)^2 \frac{\cot^2}{C^2} \ll 1 \quad (\text{the validity inequality of the models of section 9}).$$

[proved VIII §31]

- ASYMPTOTIC [definition]: $z = \text{Int}_0^x \text{Tan}[6 H y] dy == -\text{Log}[\text{Cos}[6 H x_0]]/(6 H)$, and $x_0(z) = \text{ArcCos}[\text{Exp}[-6 H z]]/(6 H)$ inverts it
- ASYMPTOTIC [THE RESULT]: in the coordinate z , $g_{zz} == C^2$ exactly and $g_{ii} = A^2 (1 - \text{Exp}[-12 H z])^{(-1/6)}$, $g_{hh} = -B^2 (1 - \text{Exp}[-12 H z])^{(-1/6)}$: the metric tends to the unwarped Bianchi-I metric
- ASYMPTOTIC [has content]: $\text{Cot}[6 H x_0]^2 == \text{Exp}[-12 H z]/(1 - \text{Exp}[-12 H z])$: every x_0 -term decays like $\text{Exp}[-12 H z]$
- ASYMPTOTIC [THE RESULT]: decay rates as $6 H x_0 \rightarrow \text{Pi}/2$: $G_{x_0^0_0}/\text{Cot}^4 \rightarrow 15 H^2$, $G_{x_0^i_i}/\text{Cot}^2 \rightarrow 60 H^2$, $G_{x_0^h_h}/\text{Cot}^2 \rightarrow 60 H^2$, $G_{x_0^4_4}/\text{Cot}^2 \rightarrow 72 H^2$; G^4_0/Cot and G^0_4/Cot^3 are free of x_0
- ASYMPTOTIC [THE RESULT]: in units of $H^2 \text{Cot}^2/(\kappa C^2)$: $\rho_{\text{bg}} \rightarrow -72$, $P_{\text{obs}} = P_{\text{hid}} \rightarrow 60$, $P_C/\text{Cot}^2 \rightarrow 15$
- ASYMPTOTIC [THE RESULT]: residual drivers $\kappa(P_i - T/6)|_{\text{bg}} \rightarrow -12 H^2 \text{Cot}^2/C^2$ for A and B , and $-72 H^2 \text{Cot}^2/C^2$ for C (if T_{bg} is not counted as a source)
- ASYMPTOTIC [THE RESULT]: validity of 'present universe = Bianchi-I': $\kappa |\rho_{\text{bg}}|/(3 H_A^2)$ divided by $24 (H/H_A)^2 \text{Cot}^2/C^2 \rightarrow 1$

What the asymptotic reading neglects [derived here]. The Bianchi-I system below treats the rates H_A, H_B, H_C as free; the (0,4) equation of section 5.2 is then satisfied only to the extent that its coefficient $3H \cot(6Hx_0) = 3H e^{-6Hz}/\sqrt{1 - e^{-12Hz}}$ is negligible. In the stabilized model of section 9 ($H_B = H_C = 0, H_A > 0$) the bracket is $2H_C - H_A - H_B = -H_A$, so at finite z the equation would demand an x_0 -momentum $\kappa T^4_0 = -3H \cot H_A$, of order $\cot$ — one power of $\cot$ less suppressed than the diagonal terms of order $\cot^2$ in the validity inequality. The author's constant H and the observer's position z along x_0 are not fixed by anything in the notebook, so the size of these terms is a parameter of the reading, not a computed number.

5.4 The 8-dimensional Bianchi-I system

In the asymptotic region the primordial field is

$$ds^2 = C^2 dz^2 + A^2(dx_1^2 + dx_2^2 + dx_3^2) - dx_4^2 - B^2(dx_5^2 + dx_6^2 + dx_7^2),$$

and its mixed Einstein tensor does **not** depend on whether the hidden directions are timelike or spacelike: flipping the three hidden signs of η leaves every G^μ_ν unchanged. With the source $T^\mu_\nu = \text{diag}(P_C, P_{\text{obs}}, P_{\text{obs}}, P_{\text{obs}}, -\rho, P_{\text{hid}}, P_{\text{hid}}, P_{\text{hid}})$ and $T = -\rho + 3P_{\text{obs}} + 3P_{\text{hid}} + P_C$, the Einstein equations are equivalent to

$$\text{constraint: } 3H_A^2 + 3H_B^2 + 9H_A H_B + 3H_A H_C + 3H_B H_C = \kappa \rho$$

$$(-G^4_4 = \frac{1}{2}(\Theta^2 - \sum_i H_i^2), \text{ the 21 pairs),}$$

$$\text{evolution: } H'_i + H_i \Theta = \kappa(P_i - T/6), \quad i = A, B, C, \quad \Theta = 3H_A + 3H_B + H_C,$$

the evolution form being the Ricci form $R^i_i = \kappa(T^i_i - T/(D-2))$ with $D-2=6$, and $R^4_4 =$ the sum over the seven directions of $(H'_i + H_i^2)$. [proved VIII §31]

- BIANCHI-I [THE RESULT]: the mixed Einstein tensor does NOT depend on the signs of the hidden directions (timelike x5..x7 or spacelike: identical G^{μ}_{μ})
- BIANCHI-I [THE RESULT]: $-G^4_4 == 3 H_A^2 + 3 H_B^2 + 9 H_A H_B + 3 H_A H_C + 3 H_B H_C == (\Theta^2 - \text{Sum}_i H_i^2)/2$, the sum over the 21 pairs of the seven directions
- BIANCHI-I [has content]: $R^i_i == H_i' + H_i \Theta$ for $i = A$ (x1), B (x5), C (x0), and $R^4_4 == \text{Sum over the seven directions of } (H_i' + H_i^2)$
- BIANCHI-I [THE RESULT]: $G^{\mu}_{\mu} == \kappa T^{\mu}_{\mu}$ is EQUIVALENT to the constraint $\text{sum-over-pairs} == \kappa \rho$ plus the evolution equations $H_i' + H_i \Theta == \kappa (P_i - T/6)$, $i = A, B, C$

(the sign-flipped geometry took 0.562s).

The same Bianchi-I Einstein tensor was re-computed numerically, from the metric, in notebook 06 (section 4, for test functions A, B, C), and every statement agreed to machine precision [nb06 §4]:

off-diagonal $G^{\mu}_{\mu} = 0$	0.00e+00
$-G^4_4 - (3H_A^2 + 3H_B^2 + 9H_A H_B + 3H_A H_C + 3H_B H_C)$	1.63e-16
$R^0_0 - (dH_C/dt + H_C \Theta)$	2.78e-17
$R^1_1 - (dH_A/dt + H_A \Theta)$	5.55e-17
$R^5_5 - (dH_B/dt + H_B \Theta)$	2.43e-17
$G^{\mu}_{\mu}(\text{hidden timelike}) - G^{\mu}_{\mu}(\text{hidden spacelike})$	0.00e+00
$\text{div } T (\mu = 4) + (d\rho/dt + 3H_A(\rho + P_{\text{obs}}) + 3H_B(\rho + P_{\text{hid}}) + H_C(\rho + P_C))$	2.22e-16
the Bianchi-I Einstein tensor, computed from the metric, is the one the solver integrates	

Constraint propagation. With S the pair sum, the evolution equations and the conservation law give $\frac{d}{dt}(S - \kappa\rho) = -2\Theta(S - \kappa\rho)$: the constraint decays forward in time (for $\Theta > 0$) and grows backward [file: fable-cosmology/rust/fable_fermion/src/models.rs, module documentation; prose nb06 §4]. The solver therefore monitors the constraint and does not impose it, and it integrates the unstabilized model forward only (section 9.11).

5.5 The 8-dimensional conservation law for anisotropic pressures

The source must obey the $\nu = 4$ component of $\nabla_{\mu} T^{\mu}_{\nu} = 0$:

$$\rho' + 3H_A(\rho + P_{\text{obs}}) + 3H_B(\rho + P_{\text{hid}}) + H_C(\rho + P_C) = 0,$$

and the other seven components vanish identically for a diagonal, x_4 -dependent source. [proved VIII §31] BIANCHI-I [THE RESULT]: 8D conservation for anisotropic pressures - $\text{nabla}_{\mu} T^{\mu}_4 == -(\rho' + 3 H_A (\rho + P_{\text{obs}}) + 3 H_B (\rho + P_{\text{hid}}) + H_C (\rho + P_C))$, and the other seven components vanish identically. Radiation (on the observed sheet: $P_{\text{obs}} = \rho_r/3$, $P_{\text{hid}} = P_C = 0$) obeys it with $\rho_r \propto A^{-4}(B^3 C)^{-1}$, and baryons (dust in every direction) with $\rho_b \propto A^{-3}(B^3 C)^{-1}$ [file: fable-cosmology/rust/fable_fermion/src/models.rs, module documentation]; section 7.5 proves it for the Kohn–Sham fable.

6 The canonical spin connection of the interacting system: a complete discussion

This section covers the spin connection in full, in two parts. Subsections 6.1–6.13 are the **canonical** spin connection of the $4+4$ pre-universe: the frame it comes from, how it is computed, every non-zero component, its 16×16 spinor form, the gauge-covariant derivative of the 16-component spinor, how it relates to the three bridges of Part IV and to the fable-5.1 (Weitzenböck) bridge

6.14–6.22 are the spin connection of the **interacting system**: the connection of the generalized warped frame and of the 8-dimensional Bianchi-I frame on which fable acts back on the primordial field (section 5), with every non-zero component, its action on the Dirac operator, its Weitzenböck form, the rescaling that removes it, the anticommutator of the rescaled field, its relation to the fundamental symmetry J , and the recovery of the canonical connection. It stands on its own. Every object is defined here and every result is restated here, with the place where it is proved.

Throughout, “Section n” (capital S) means a section of the notebook. The numbered parts of this discussion are called “subsections” 6.1–6.22.

6.1 Sources, and how the status of each statement is marked

Where the proofs live. The proofs are in the Mathematica notebook `claude-fable/claude-fable_Einstein-Rosen-2-Planes.nb`. That notebook is generated from cell manifests and is never edited by hand:

manifest	Part	Sections	subject
<code>claude-fable/cells_part1.wl</code>	I	0–4	session, coordinates, eta4488 , tau , sigma
<code>claude-fable/cells_part2.wl</code>	I	5–9	T16 , sigma16 , SAB , matrix bases, Cartan triality
<code>claude-fable/cells_part3.wl</code>	II	10–14	the author’s physics reproduced (Psi16 , La , eLa)
<code>claude-fable/cells_part4.wl</code>	III, IV	15–17, 18–20	frame, Christoffel symbols, canonical spin connection, spinor covariant derivative; the three bridges
<code>claude-fable/cells_part5.wl</code>	V	21–22	the fable-5.1 (Weitzenböck) bridge; the master comparison
<code>claude-fable/cells_part6.wl</code>	VI	23–25	fableScalar and the real (classical) fable spinor
<code>claude-fable/cells_part7.wl</code>	VII	26–29	the complex fermion fable (new): refinement, Lagrangian and field equations, canonical quantization, energy–momentum tensor
<code>claude-fable/cells_part8.wl</code>	VIII	30–33	fable as a source of the Einstein equations of the primordial field (the coupled system; its spin connection is Section 32)

The run of Parts I–VI, `claude-fable/run_fable51_part6.log`, evaluates 172 Input cells straight out of the `.nb`. It reports Assertions run: 358 passed: 358 failed: 0 and cells w/ msgs : 0, with identities accepted on numerical evidence alone : 0 and non-vanishing witnesses : 27. The Part VII run is recorded in `claude-fable/run_fermion_fable_part7.log`. It evaluates 198 Input cells and reports Assertions run: 528 passed: 528 failed: 0 and cells w/ msgs : 0, with

identities accepted on numerical evidence alone : 0 and non-vanishing witnesses : 38 (Part VII adds 170 assertions and 11 witnesses to Parts I–VI). The acceptance run of Part VIII, `claude-fable/run_fermion_fable_part8.log`, reported Assertions run: 642 passed: 642 failed: 0; its two comparisons of the Rust solver with the Mathematica reference runs were skipped, because the solver’s CSVs did not exist yet. The final run of the whole notebook, `claude-fable/run_fermion_fable_final.log`, evaluates 217 Input cells and reports Assertions run: 644 passed: 644 failed: 0 and cells w/ msgs : 0, with identities accepted on numerical evidence alone : 0 and non-vanishing witnesses : 46.

How an identity is certified. Each identity is certified in up to three stages:

1. exact structural equality (the entry is literally 0);
2. Simplify under the geometric assumptions, with a 5-second budget;
3. a numerical check at three rational probe points to 30 significant digits.

Stage 3 has two uses, and the notebook counts them separately. The first is to accept an identity. The notebook reports *zero* of those, and its last assertion fails if that count is ever non-zero, so every identity quoted below is closed symbolically. The second is to witness that something is *not* zero, through `cfNonZeroWitnessQ`. A witness succeeds only if every entry evaluates to an actual number at some probe point and at least one of them exceeds 10^{-10} in magnitude. That is a complete proof of non-vanishing.

The probe functions used in stage 3 are test inputs only, never definitions: `a4 -> Function[u, 3/7 + u/5 + Sin[u]/11]`, a probe for the boost rapidity of Bridge 1, and probe values of the Bridge 3 parameter `lambdaOct`.

Status marks used below.

mark	meaning
[proved P §n]	“label” or label
[displayed P §n]	
[prose P §n]	
[derived here]	

The notebook grades its own assertion labels. `[definition]` means true by construction. `[structural]` means true for any input of the right shape. `[solver regression]` is a check of the code, not of the geometry. `[has content]` could have failed. `[THE RESULT]` marks a headline result. `[control]` shows that the wrong alternative fails.

6.2 Conventions

- Coordinates $X = \{x_0, x_1, \dots, x_7\}$. The Wolfram array position is the coordinate index plus one, so `x4` is entry 5. Greek indices are curved and Latin indices are flat; both run 0..7. x_4 is the observer’s time ($g_{44} = -1$), x_1, x_2, x_3 are the observed 3-space, x_0 is the hidden spacelike direction, and x_5, x_6, x_7 are the hidden timelike sheet. The notebook’s own words for $x_5..x_7$ are “the superluminal deflating directions”.

- Flat metric: `eta4488 = DiagonalMatrix[{1,1,1,1,-1,-1,-1,-1}]`. [proved I §2] “signature of eta4488 is (4,4)”, “eta4488 . eta4488 == ID8”.
- The 8×8 generators have `tau[0] = ID8`. The conjugates are `taubar[A] = sigma . Transpose[sigma . tau[A]]`, with `sigma = ArrayFlatten[{0,ID4},{ID4,0}]` ([proved I §4] “sigma . sigma == ID8”, “sigma is symmetric”). It follows that `taubar[0] = ID8`.
- The 16×16 Dirac matrices are E. A. Lord’s reduced Brauer–Weyl doubling, `T16[A] = ArrayFlatten[{0, taubar[A]}, {tau[A], 0}]` for $A = 0..7$. They are real, and `T16[0] = ArrayFlatten[{0,ID8},{ID8,0}]`. [proved I §5] “{T16[A], T16[B]}/2 == eta4488[[A+1,B+1]] ID16”. Throughout, $T_{16}[a]$ is γ^a , with the flat index *up*. Since eta4488 is its own inverse, $\gamma_a = \eta_{ab} T_{16}[b]$ differs from $T_{16}[a]$ by the sign η_{aa} .
- Spinor metric: `sigma16 = T16[0].T16[1].T16[2].T16[3]`. [proved I §5] “sigma16 == ArrayFlatten[{-sigma,0},{0,sigma}]”, “sigma16 . T16[A] is antisymmetric for A = 0..7”, “sigma16 is symmetric and sigma16 . sigma16 == ID16”.
- Chirality: `T16[8] = T16[0].T16[1]...T16[7] = diag(-ID8, +ID8)`. [proved I §5] “PL projects onto the upper (type-1) components, PR onto the lower (type-2) components”. A 16-spinor is $\Psi = (\psi_1, \psi_2)$. The upper 8 components ψ_1 are a split-octonion type-1 spinor, and the lower 8 components ψ_2 are a type-2 spinor.
- $so(4,4)$ generators: `SAB[[A+1,B+1]] = (1/4) (T16[A].T16[B] - T16[B].T16[A])`. [proved I §5] “SAB is antisymmetric in its two flat indices”, “sigma16 . SAB is antisymmetric”, “T16[8] commutes with every SAB: the chiral halves are each Spin(4,4)-invariant”, “[SAB, T16] reproduces the vector representation”.
- H , K and M are Protected constants of the original notebook.

6.3 The canonical frame (vielbein), the metric, and the volume factor

A vielbein here is simply an 8-dimensional vierbein, that is, an 8-dimensional frame field. The canonical frame is the diagonal matrix that the original notebook records as the value of the symbol it writes “gtrye with indices alpha and (A)” [prose III §15]:

$$e = \text{diag}(\tan 6Hx_0, q, q, q, 1, p, p, p), \quad (46)$$

$$q = \frac{e^{-a_4(Hx_4)}}{\sin^{1/6} 6Hx_0}, \quad p = \frac{e^{+a_4(Hx_4)}}{\sin^{1/6} 6Hx_0}. \quad (47)$$

In the notebook: `frameCanonical = DiagonalMatrix[{Tan[6 H x0], q, q, q, 1, p, p, p}], q = cfQminus (observed directions  $x_1, x_2, x_3$ ), p = cfQplus (hidden directions  $x_5, x_6, x_7$ ).`

Rows curved, columns flat. `frameCanonical[[mu+1, a+1]] =  $e_\mu^a$` . The coframe is the matrix inverse, stored with the flat index on the row: `coframeCanonical = Inverse[frameCanonical]`, `coframeCanonical[[a+1, mu+1]] =  $e_a^\mu$` . [proved III §15] “CANONICAL [structural]: frame . coframe == ID8 (certifies only that the frame is invertible)”.

The free function a_4 . The original notebook never defines a_4 . It appears only inside the recorded frame. It is carried symbolically everywhere and is *never given a value*. The notebook prints a notice to this effect when Section 15 runs [prose III §15], and Part VI ends with [proved VI §25] “PART VI [control]: a_4 is STILL UNDEFINED - nothing in Part VI gave it a value; and the FLRW scale factor a_F never entered the canonical frame”. Every result in this section holds for an arbitrary differentiable a_4 . Below, a'_4 means `Derivative[1][a4][H x4]`.

Reciprocity. $qp = \sin^{-1/3} 6Hx_0$ does not depend on x_4 : the observed directions contract exactly as fast as the hidden directions expand [prose III §15].

The metric. The bridge between the curved and flat metrics is

$$[d] \quad g_{\mu\nu} = e_\mu^a \eta_{ab} e_\nu^b, \quad (48)$$

that is, `gCanonical = frameCanonical . eta4488 . Transpose[frameCanonical]`, and the line element is

$$ds^2 = \tan^2(6Hx_0) dx_0^2 + q^2 (dx_1^2 + dx_2^2 + dx_3^2) - dx_4^2 - p^2 (dx_5^2 + dx_6^2 + dx_7^2). \quad (49)$$

- [proved III §15] “CANONICAL [definition]: `g == frame . eta4488 . Transpose[frame]` (bridge [d]; this IS how `g` was defined, so it cannot fail)”.
- [proved III §15] “CANONICAL [has content]: `g == diag(Tan[6 H x0]^2, q^2, q^2, q^2, -1, -p^2, -p^2, -p^2)`, the stated line element”.
- [proved III §15] “CANONICAL [has content]: `Diagonal[g] == Diagonal[eta4488] * Diagonal[frame]^2` exactly” (Sylvester’s law made explicit).
- [proved III §15] “CANONICAL [has content]: `g` has signature (4,4), GIVEN `a4` real and $0 < 6 H x_0 < \pi/2$ ”.
- [proved III §15] “Inverse[g] == Transpose[coframe] . eta4488 . coframe”, “`g` is symmetric”, “the metric is non-degenerate”.

The volume factor. Section 15 displays `Det[g] = Sec[6 H x0]^2` and `Sqrt[Abs[Det[g]]] = Abs[Sec[6 H x0]]` [displayed III §15]. [proved V §21] “FABLE-5.1: `det g == Sec[6 H x0]^2` (positive, as signature 4+4 requires)”. On the domain $0 < 6Hx_0 < \pi/2$ the root is $\sec 6Hx_0$. From Part V on, the notebook uses `sqrtDetgTele = Sec[6 H x0]` rather than `Abs[Sec[...]]`, so that `Abs` is never differentiated [prose V §21]. The same result follows from the frame directly [derived here]:

$$\det e = \tan(6Hx_0) q^3 p^3 = \frac{\tan 6Hx_0}{\sin 6Hx_0} = \sec 6Hx_0 = \sqrt{|\det g|}. \quad (50)$$

The volume factor does not depend on x_4 .

The curved Dirac matrices. `gammaCurvedCanonical[[mu+1]] = Sum_a coframe[[a,mu]] T16[a]`. The frame is diagonal, so [derived here]

$$\begin{aligned} \gamma^0 &= \cot(6Hx_0) T_{16}[0], \\ \gamma^j &= \frac{1}{q} T_{16}[j] = e^{a4} \sin^{1/6}(6Hx_0) T_{16}[j] \quad (j = 1, 2, 3), \\ \gamma^4 &= T_{16}[4], \\ \gamma^k &= \frac{1}{p} T_{16}[k] = e^{-a4} \sin^{1/6}(6Hx_0) T_{16}[k] \quad (k = 5, 6, 7). \end{aligned} \quad (51)$$

- [proved III §17] “{`gammaCurved[mu]`, `gammaCurved[nu]`} == 2 `gInv[mu,nu]` ID16”.
- [proved VI §24] “FABLE [has content]: `(gamma^4)^2 == -ID16`, so `gamma^4 d_4 Psi' == H V'(s) Psi'` is solved by `d_4 Psi' == -H V'(s) gamma^4 Psi'`”.

6.4 The generalized Christoffel symbols

These are the Levi-Civita symbols of g :

$$\Gamma^\rho_{\mu\nu} = \frac{1}{2} g^{\rho s} (\partial_\mu g_{s\nu} + \partial_\nu g_{s\mu} - \partial_s g_{\mu\nu}). \quad (52)$$

`cfChristoffel[g, ginv, coords]` computes the first derivatives $\text{dg}[[k,m,p]] = D[g[[m,p]]]$, `coords[[k]]` once, reuses them in the triple loop, and simplifies each entry under the notebook's assumptions. The result is `GammaCanonical[[rho+1, mu+1, nu+1]]`. Section 16 lists its 37 non-zero entries (out of 512) [**displayed III §16**]; here `Gamma[r;m,n]` stands for Γ^r_{mn} :

```
Gamma[0;0,0] = 12*H*Csc[12*H*x0]
Gamma[0;j,j] = (H*Cos[6*H*x0]^3)/(E^(2*a4[H*x4])*Sin[6*H*x0]^(10/3))    j = 1,2,3
Gamma[0;k,k] = -(E^(2*a4[H*x4])*H*Cos[6*H*x0]^3)/Sin[6*H*x0]^(10/3)    k = 5,6,7
Gamma[j;0,j] = Gamma[j;j,0] = -H*Cot[6*H*x0]                            j = 1,2,3
Gamma[j;j,4] = Gamma[j;4,j] = -H*a4'[H*x4]                             j = 1,2,3
Gamma[4;j,j] = -(H*a4'[H*x4])/(E^(2*a4[H*x4])*Sin[6*H*x0]^(1/3))        j = 1,2,3
Gamma[4;k,k] = -(E^(2*a4[H*x4])*H*a4'[H*x4])/Sin[6*H*x0]^(1/3)          k = 5,6,7
Gamma[k;0,k] = Gamma[k;k,0] = -H*Cot[6*H*x0]                            k = 5,6,7
Gamma[k;4,k] = Gamma[k;k,4] = H*a4'[H*x4]                               k = 5,6,7
```

The count is $1 + 3 + 3 + 6 + 6 + 3 + 3 + 6 + 6 = 37$.

- [**proved III §16**] “Gamma is symmetric in its two lower indices (torsion-free affine connection)”. This check is structural: swapping the two lower indices in the formula reproduces the formula whenever g is symmetric.
- [**proved VI §23**] “HELPER [solver regression]: `cfCovariantDivergence` of the metric itself vanishes (metric compatibility of Section 16’s Gamma; a wrong index slot in the helper fails this)”.

6.5 The vielbein postulate, and how `cfSpinConnection` solves it

The spin connection is fixed by the zero-torsion (Levi-Civita) condition, written as the *vielbein postulate*. The total covariant derivative of the frame vanishes when Γ acts on the curved index and ω acts on the flat index [**prose III §16**]:

$$\partial_\mu e_\nu^a - \Gamma^\rho_{\mu\nu} e_\rho^a + \omega_\mu^a{}_b e_\nu^b = 0. \quad (53)$$

In the notebook: `D[e[nu,a], x[mu]] - Gamma[rho,mu,nu] e[rho,a] + omegaMixed[mu,a,b] e[nu,b] == 0`, with `omegaMixed[mu,a,b] =  $\omega_\mu^a{}_b$` . Contracting with the coframe frees ν :

$$\omega_\mu^a{}_c = \sum_\nu \left(\sum_\rho \Gamma^\rho_{\mu\nu} e_\rho^a - \partial_\mu e_\nu^a \right) e_c^\nu. \quad (54)$$

The notebook’s solver is exactly this formula (Section 16):

```
cfSpinConnection[frame_, coords_, etaFlat_, gamma_] :=
Module[{n = Length[coords], cofr, dE},
  cofr = cfSimp[Inverse[frame]]; (* cofr[[a,nu]] = e_a^nu *)
  dE = Table[D[frame[[nu, a]], coords[[mu]]], {mu, n}, {a, n}];
  Table[
    cfSimp[Sum[(Sum[gamma[[r, mu, nu]] frame[[r, a]], {r, n}]
      - dE[[mu, nu, a]] cofr[[c, nu]], {nu, n}]],
    {mu, n}, {a, n}, {c, n}];

cfLowerFirstFlat[om_, etaFlat_] := Module[{n = Length[etaFlat]},
  Table[Sum[etaFlat[[a, c]] om[[mu, c, b]], {c, n}, {mu, n}, {a, n}, {b, n}];

(* omega_mu^a_b *)
omegaCanonicalMixed =
  cfSpinConnection[frameCanonical, X, eta4488, GammaCanonical];
(* omega_{mu a b} *)
omegaCanonical = cfLowerFirstFlat[omegaCanonicalMixed, eta4488];
```

The same routine produces every connection of the notebook that comes from the vielbein postulate: the canonical one, those of Bridges 1 and 2, the Levi-Civita part `omegaTriLC` of Bridge 3 (to which Section 20 then adds the contortion), and the fable-5.1 connection. The Christoffel symbols are computed once and passed in. So the comparisons below compare results, not different pieces of code [**prose III §16, IV, V §21**].

What each certificate is worth. The notebook grades its certificates as follows [**prose III §16**]:

- [**proved III §16**] “CANONICAL [solver regression]: vielbein postulate residual is zero”. This is an algebraic identity of the solver. `cfSpinConnection` contracts the postulate with `Inverse[frame]` and `cfVielbeinResidual` contracts it back with `frame`, and the two cancel for any affine connection and any invertible frame.
- [**proved III §16**] “CANONICAL [structural]: torsion vanishes”. Once the residual is zero, this reduces to the symmetry of Γ in its lower indices, which is itself structural. The torsion 2-form used is $T^a{}_{\mu\nu} = \partial_\mu e_\nu^a - \partial_\nu e_\mu^a + \omega_\mu{}^a{}_b e_\nu^b - \omega_\nu{}^a{}_b e_\mu^b$ (`cfTorsion`).
- [**proved III §16**] “CANONICAL [has content]: `omega[mu,a,b] == -omega[mu,b,a]` (metric compatibility)”. This is not imposed anywhere. It holds only because the affine connection fed in is the Levi-Civita connection of the metric the frame builds, so a wrong Christoffel symbol would break it. The connection is therefore $\mathfrak{so}(4,4)$ -valued.
- [**proved III §16**] “INDEPENDENT CHECK: the frame-only derivation reproduces `omegaCanonical` exactly” and “INDEPENDENT CHECK: and it is antisymmetric in its two raised flat indices”. The connection is re-derived from the frame alone by the anholonomy formula, which never mentions Γ or g (`cfSpinConnectionFromFrame`, with $e^{a\nu} = (\eta^{-1} \cdot \text{coframe})^{a\nu}$):

$$\begin{aligned} \omega_\mu{}^{ab} = & \frac{1}{2}e^{a\nu}(\partial_\mu e_\nu^b - \partial_\nu e_\mu^b) - \frac{1}{2}e^{b\nu}(\partial_\mu e_\nu^a - \partial_\nu e_\mu^a) \\ & - \frac{1}{2}e^{a\rho}e^{b\sigma}(\partial_\rho e_\sigma^c - \partial_\sigma e_\rho^c)\eta_{cd}e_\mu^d. \end{aligned} \quad (55)$$

- The canonical frame is diagonal and so equals its own transpose. Section 16 therefore cannot detect a transposition of the frame’s two slots. Section 18 closes that gap on a frame that is not symmetric (the canonical frame times a constant boost of rapidity 1/3): [**proved IV §18**] “the slot-test frame is genuinely not symmetric”, “INDEPENDENT CHECK on a non-symmetric frame: the two solvers agree, so the curved/flat slots are placed correctly in both”, “and the result is the canonical connection conjugated by the constant boost, as the gauge law demands”.

6.6 The canonical spin connection: all its non-zero components

`omegaCanonical` has **24 non-zero components out of 512** [**displayed III §16**]. They are listed by `cfShowConnection[omegaCanonical, "omega"]`. In this subsection and the next only, write $c = \cos 6Hx_0$, $s = \sin 6Hx_0$ and $A = a_4(Hx_4)$. For $j = 1, 2, 3$ and $k = 5, 6, 7$, with all indices down, $\omega_{\mu ab} = \text{omega}[\text{mu}; \text{a}, \text{b}] = \text{omegaCanonical}[[\text{mu}+1, \text{a}+1, \text{b}+1]]$:

component (both flat indices down)	value
$\omega_{j0j} = -\omega_{jj0}$	$Hc^2/(e^A s^{13/6})$
$\omega_{j4j} = -\omega_{jj4}$	$Ha'_4/(e^A s^{1/6})$
$\omega_{k0k} = -\omega_{kk0}$	$-e^A Hc^2/s^{13/6}$
$\omega_{k4k} = -\omega_{kk4}$	$e^A Ha'_4/s^{1/6}$

That gives $6 \text{ directions} \times 2 \text{ planes} \times 2 \text{ orderings} = 24$ components. Every other component is zero. Part VII re-asserts the count and the pattern: **[proved VII §27] SPIN CONNECTION** [has content]: metric compatible ($\omega_{\mu}^{\{ab\}}$ antisymmetric) and every non-zero component is $\omega_{\mu}^{\{\mu 0\}}$ or $\omega_{\mu}^{\{\mu 4\}}$ (up to antisymmetry): $\omega_0 = \omega_4 = 0$. That label writes the connection with both flat indices up, ω_{μ}^{ab} . Neither the antisymmetry nor the pattern of non-zero components depends on that position: raising both flat indices multiplies each component by $\eta_{aa}\eta_{bb} = \pm 1$ **[derived here]**. The **sign** of a listed component does depend on it: raising both indices flips the components in the planes with exactly one timelike index — $(4, j)$ and $(0, k)$ here — and leaves the $(0, j)$ and $(4, k)$ components unchanged. That is why the table above is written with both flat indices down, as **omegaCanonical** is, and why Part VIII, Section 32, lists the components of the connections of its dynamical frames the same way (subsection 6.15).

Structure [derived here], from the table.

- $\omega_0 = \omega_4 = 0$: the connection vanishes along the hidden direction x_0 and along the observer's time x_4 .
- In each remaining direction μ ($1, 2, 3, 5, 6, 7$) only the two flat planes $(0, \mu)$ and $(4, \mu)$ are excited.
- Whether the generator of a plane is a rotation or a boost depends on the signs of **eta4488** in that plane. For $j = 1, 2, 3$, $(0, j)$ is a rotation (both directions spacelike) and $(4, j)$ is a boost. For $k = 5, 6, 7$, $(0, k)$ is a boost and $(4, k)$ is a rotation (both directions timelike).
- The x_0 -dependent components (planes $(0, \mu)$) do not involve a'_4 . The a_4 -dependent components (planes $(4, \mu)$) are proportional to a'_4 , so they all vanish when a_4 is constant.
- No component depends on $x_1, x_2, x_3, x_5, x_6, x_7$. The metric, the frame and the connection are invariant under translations along the observed and hidden sheets.
- The *mixed* components $\omega_{\mu}^{ab} = \text{omegaCanonicalMixed}$ follow by raising the first flat index with **eta4488**. For example, $\omega_j^{4j} = -\omega_{j4j}$ because $\eta_{44} = -1$.

Curvature. The curvature 2-form is $R_{\mu\nu}^{ab} = \partial_{\mu}\omega_{\nu}^{ab} - \partial_{\nu}\omega_{\mu}^{ab} + [\omega_{\mu}, \omega_{\nu}]^{ab}$ (**cfCurvature**). It has **156 non-zero components** **[displayed III §16]**. **[proved III §16]** “CANONICAL: curvature is antisymmetric in the two spacetime indices”. The Ricci scalar is **[displayed III §16]**

$$R = -3H^2 \left((55 + 7 \cos 12Hx_0) \cot^2(6Hx_0) \csc^2(6Hx_0) - 2a_4'^2 \right) \quad (56)$$

In the notebook:

```
RicciScalarCanonical = -3*H^2*((55 + 7*Cos[12*H*x0])*Cot[6*H*x0]^2*Csc[6*H*x0]^2
- 2*a4'[H*x4]^2)
```

This is equivalent to $-6H^2(24 \cot^2 6Hx_0 + 31 \cot^4 6Hx_0 - a_4'^2)$ **[derived here]**, using $\cos 12Hx_0 = 1 - 2 \sin^2 6Hx_0$ and $\csc^2 = 1 + \cot^2$. It depends on a_4 only through $a_4'^2$. It is the frame-independent invariant that every bridge below is checked against.

Summary of the certified properties of the canonical connection. It is metric compatible (antisymmetric in its flat indices, a statement with content), it is torsion-free (structural), it is reproduced exactly by an independent frame-only derivation, and it is not flat.

6.7 The spinor connection Γ_μ and its exact normalization

The 16×16 spin-connection matrix is built by `cfSpinMatrix` in Section 17:

```
cfSpinMatrix[om_, mu_Integer] :=
  (1/8) Sum[om[[mu, a, b]] (T16[a - 1] . T16[b - 1] - T16[b - 1] . T16[a - 1]),
    {a, 8}, {b, 8}];
GammaSpinCanonical =
  Table[cfSimpArray[cfSpinMatrix[omegaCanonical, mu]], {mu, 8}];
```

It is fed `omegaCanonical`, the connection with *both flat indices down*. Since $T_{16}[a] = \gamma^a$ has its flat index *up*, the exact convention is

$$\begin{aligned}\Gamma_\mu &= \frac{1}{8} \omega_{\mu ab} [\gamma^a, \gamma^b] = \frac{1}{4} \sum_{a,b=0}^7 \omega_{\mu ab} T_{16}[a] T_{16}[b] \\ &= \frac{1}{2} \sum_{a < b} \omega_{\mu ab} T_{16}[a] T_{16}[b] = \frac{1}{2} \omega_{\mu ab} \text{SAB}[[a+1, b+1]].\end{aligned}\tag{57}$$

The second line uses the antisymmetry of $\omega_{\mu ab}$ and $T_{16}[a]T_{16}[b] = -T_{16}[b]T_{16}[a]$ for $a \neq b$ [**derived here**]. The equivalent form with raised connection indices is $\frac{1}{4} \omega_\mu^{ab} \gamma_a \gamma_b$, with $\gamma_a = \eta_{ab} T_{16}[b]$.

Index warning. The expression $\frac{1}{4} \omega_\mu^{ab} T_{16}[a] T_{16}[b]$, with the raised connection contracted against the unlowered `T16`, is *not* the notebook's Γ_μ . It would multiply each plane by $\eta_{aa}\eta_{bb}$, and so flip the sign of the $(4, j)$ and $(0, k)$ terms [**derived here**].

The coefficient and the sign are not conventions left open. They are fixed against the vielbein postulate:

- [**proved III §17**] “[definition] $(1/8) \text{omega} [\text{gamma}^a, \text{gamma}^b] == (1/2) \text{omega SAB}$ (SAB is defined as $(1/4) [\text{gamma}, \text{gamma}]$, so this cannot fail)”.
- [**proved III §17**] “[has content] the spinor connection is compatible with omega: $[\text{Gamma}^{\text{spin_mu}}, \text{gamma}^a] + \text{omega_mu}^a{}_b \text{gamma}^b == 0$ ”. The spinor connection rotates the Dirac matrices exactly as ω rotates a vector. With $1/4$ in place of $1/8$, or with the opposite sign, this check fails.
- [**proved III §17**] “[control] with the opposite sign the compatibility check FAILS (witnessed): the sign is fixed by the vielbein postulate, not free”.
- [**proved VII §27**] the normalization used throughout Part VII, SPIN CONNECTION [definition]: $\text{Gamma}^{\text{spin_mu}} == \text{Sum}_{\{a < b\}} \text{omega_mu,ab} (1/2) T_{16}[a] T_{16}[b]$.

The explicit matrices [**derived here**], from the table of subsection 6.6 and the formula above. Only the planes $(0, \mu)$ and $(4, \mu)$ contribute. For $j = 1, 2, 3$ the ordered pair $(j, 4)$ gives $\omega_{jj4} T_{16}[j] T_{16}[4] = \omega_{j4j} T_{16}[4] T_{16}[j]$, so

$$\begin{aligned}\Gamma_0 &= \Gamma_4 = 0, \\ \Gamma_j &= \frac{1}{2} \left[\frac{Hc^2}{e^A s^{13/6}} T_{16}[0] T_{16}[j] + \frac{H a'_4}{e^A s^{1/6}} T_{16}[4] T_{16}[j] \right] \quad (j = 1, 2, 3), \\ \Gamma_k &= \frac{1}{2} \left[-\frac{e^A Hc^2}{s^{13/6}} T_{16}[0] T_{16}[k] + \frac{e^A H a'_4}{s^{1/6}} T_{16}[4] T_{16}[k] \right] \quad (k = 5, 6, 7).\end{aligned}\tag{58}$$

The provenance script `claude-fable/prov03_covariant_derivative.wls` (log `claude-fable/prov03.log`) prints the non-zero directions of $\text{Gamma}^{\text{spin}}$ as $\{1, 2, 3, 5, 6, 7\}$, each with 32 non-zero entries. That matches the two terms above: each $T_{16}[a]T_{16}[b]$ with $a \neq b$ is a signed permutation matrix with 16 entries, and the two terms occupy different positions.

The direct-sum structure. Every $T_{16}[a]$ is off-diagonal in the type-1/type-2 split, so every product of two of them is block-diagonal: $T_{16}[A].T_{16}[B] = \text{ArrayFlatten}[\{\{\text{taubar}[A].\text{tau}[B], 0\}, \{0, \text{tau}[A].\text{taubar}[B]\}\}]$. The spin connection therefore never mixes the two split-octonion spinors.

- [proved III §17] “Gamma^{spin}[mu] is block diagonal: it never mixes type-1 with type-2”.
- [proved III §17] “the type-1 block is (1/8) omega (taubar^a tau^b - taubar^b tau^a)”.
- [proved III §17] “the type-2 block is (1/8) omega (tau^a taubar^b - tau^b taubar^a)”.

Below, the upper (type-1) block is written $\Gamma_\mu^{(1)}$ and the lower (type-2) block $\Gamma_\mu^{(2)}$. In the notebook these are `GammaSpinType1` and `GammaSpinType2`.

6.8 The gauge-covariant derivative of the 16-component spinor

$$\begin{aligned} D_\mu \Psi &= \partial_\mu \Psi + \Gamma_\mu \Psi, & D_\mu \psi_1 &= \partial_\mu \psi_1 + \Gamma_\mu^{(1)} \psi_1, \\ D_\mu \psi_2 &= \partial_\mu \psi_2 + \Gamma_\mu^{(2)} \psi_2, & \text{Dirac}[\Psi] &= \gamma^\mu D_\mu \Psi. \end{aligned} \quad (59)$$

In the notebook these are `cfDcov16[psi, mu] = D[psi, X[[mu]]] + GammaSpinCanonical[[mu]] . psi`, `cfDcovType1`, `cfDcovType2` and `cfDiracOperator[psi] = Sum[gammaCurvedCanonical[[mu]] . cfDcov16[psi, mu], {mu, 8}]`.

- [proved III §17] “Dcov16 on the direct sum is the direct sum of Dcov1 and Dcov2” (on a generic spinor of all eight coordinates).

Covariant constancy of γ^μ . The Christoffel symbols act on the curved index, the spinor connection acts by commutator, and the frame connects the two. If any of the three had the wrong sign or index order, the curved Dirac matrices would not be covariantly constant:

$$D_\mu \gamma^\nu = \partial_\mu \gamma^\nu + \Gamma^\nu_{\mu\rho} \gamma^\rho + [\Gamma_\mu, \gamma^\nu] = 0. \quad (60)$$

- [proved III §17] “[has content] and therefore the curved Dirac matrices are covariantly constant: `D_mu gammaCurved^nu == 0`”.

The commutator identity. Contract $\mu = \nu$ and use $\Gamma^\mu_{\mu\rho} = \partial_\rho \ln \sqrt{g}$ [derived here]:

$$\frac{1}{\sqrt{g}} \partial_\mu (\sqrt{g} \gamma^\mu) = [\gamma^\mu, \Gamma_\mu]. \quad (61)$$

- [proved VI §24] “FABLE [has content]: in general $(1/\text{Sqrt}[g]) \text{d_mu}(\text{Sqrt}[g] \gamma^\mu) == [\gamma^\mu, \text{Gamma}^\text{spin_mu}]$ (covariant constancy of γ^μ , Section 17)”.

The connection term of the Dirac operator, direction by direction [derived here]. From the explicit Γ_μ of subsection 6.7 and the curved gammas of subsection 6.3, using $T_{16}[j]^2 = +\text{ID16}$, $T_{16}[k]^2 = -\text{ID16}$ and the anticommutation of distinct `T16`:

$$\begin{aligned} \gamma^j \Gamma_j &= \frac{1}{2q} T_{16}[j] (w_0 T_{16}[0] T_{16}[j] + w_4 T_{16}[4] T_{16}[j]) \\ &= -\frac{1}{2q} (w_0 T_{16}[0] + w_4 T_{16}[4]), \end{aligned} \quad (62)$$

$$\begin{aligned} \gamma^k \Gamma_k &= \frac{1}{2p} T_{16}[k] (v_0 T_{16}[0] T_{16}[k] + v_4 T_{16}[4] T_{16}[k]) \\ &= +\frac{1}{2p} (v_0 T_{16}[0] + v_4 T_{16}[4]), \end{aligned} \quad (63)$$

where w_0, w_4 and v_0, v_4 are the coefficients in Γ_j and Γ_k . Since $w_0/q = Hc^2/s^2 = H \cot^2 6Hx_0$, $w_4/q = Ha'_4$, $v_0/p = -H \cot^2 6Hx_0$ and $v_4/p = Ha'_4$:

μ	$\gamma^\mu \Gamma_\mu$ (no sum)	$\{\gamma^\mu, \Gamma_\mu\}$ (no sum)
0	0	0
$j = 1, 2, 3$	$-\frac{1}{2}H \cot^2(6Hx_0) T_{16}[0] - \frac{1}{2}Ha'_4 T_{16}[4]$	0
4	0	0
$k = 5, 6, 7$	$-\frac{1}{2}H \cot^2(6Hx_0) T_{16}[0] + \frac{1}{2}Ha'_4 T_{16}[4]$	0

The anticommutator column follows because $\Gamma_j \gamma^j$ equals $+\frac{1}{2q}(w_0 T_{16}[0] + w_4 T_{16}[4])$, the negative of $\gamma^j \Gamma_j$, and similarly for k .

The sum. Summing the table [derived here] gives $3(-\frac{1}{2}) + 3(-\frac{1}{2}) = -3$ for the $T_{16}[0]$ terms and $3(-\frac{1}{2}) + 3(+\frac{1}{2}) = 0$ for the $T_{16}[4]$ terms. The observed and hidden a'_4 terms cancel:

$$\gamma^\mu \Gamma_\mu = -3H \cot^2(6Hx_0) T_{16}[0], \quad \sum_\mu \{\gamma^\mu, \Gamma_\mu\} = 0. \quad (64)$$

- [proved VI §24] “FABLE [has content]: the HYPOTHESIS that makes it a product - the anticommutator $\{\gamma^\mu, \Gamma_\mu\} == 0$ on the canonical frame (no totally antisymmetric part of the connection)” (the sum over μ).
- [proved VI §24] “FABLE [has content]: hence $(1/2)(1/\text{Sqrt}[g]) \text{d}_\mu(\text{Sqrt}[g] \gamma^\mu) == \gamma^\mu \Gamma_\mu$, and on this geometry $== -3 H \cot^2[6 H x_0] T_{16}[0]$ ”.
- [proved VII §27] the per-direction form: every $\gamma^\mu \Gamma_\mu$ is $\alpha_\mu T_{16}[0] + \beta_\mu T_{16}[4]$, with $\sum \alpha_\mu = -3H \cot^2$ and $\sum \beta_\mu = 0$, SPIN CONNECTION [THE RESULT]: direction by direction $\gamma^\mu \Gamma_\mu == \alpha_\mu T_{16}[0] + \beta_\mu T_{16}[4]$; Sum $\alpha_\mu == -3 H \cot^2[6 H x_0]$ and Sum $\beta_\mu == 0$ (the observed and hidden a_4 terms cancel); and the anticommutator vanishes for *each* μ separately, LAGRANGIAN (c) [THE RESULT]: $\text{Lsym} - \text{Lsym}_d == (1/(2H)) \sum_\mu \text{Psibar} \{\gamma^\mu, \Gamma_\mu\} \text{Psi}$ identically, and $\{\gamma^\mu, \Gamma_\mu\} == 0$ for EACH μ on the canonical frame: $\text{Lsym} == \text{Lsym}_d$.

Why the anticommutator vanishes [derived here]. For $a \neq b$, $\{\gamma^c, \gamma^a \gamma^b\} = 2\gamma^a \gamma^b \gamma^c + 2\eta^{ca} \gamma^b - 2\eta^{cb} \gamma^a$. This is $2\gamma^{cab}$ (the antisymmetrized product) when c differs from both a and b , and 0 when $c = a$ or $c = b$. Hence

$$\sum_\mu \{\gamma^\mu, \Gamma_\mu\} = \frac{1}{2} \omega_{[cab]} \gamma^{cab}, \quad \omega_{cab} = e_c^\mu \omega_{\mu ab}, \quad (65)$$

and only the totally antisymmetric part of ω_{cab} survives. For the diagonal canonical frame, c equals the direction μ , and every non-zero $\omega_{\mu ab}$ has μ equal to a or b (the planes $(0, \mu)$, $(4, \mu)$). So the totally antisymmetric part is zero, and it is zero direction by direction.

Contrast with an expanding frame. On the separate 4-dimensional FLRW reference frame of Part VI, the same computation leaves a Hubble term: [proved VI §24] “FLRW [THE RESULT]: $\gamma^\mu \Gamma_\mu == (3/2) (a'/a) \gamma^4$ - the Hubble term of the FLRW Dirac equation”. On the canonical frame the analogous $T_{16}[4]$ terms cancel between the observed sheet and the hidden sheet ($\sum \beta_\mu = 0$), because qp does not depend on x_4 [derived here]: the per-direction coefficient $-\frac{1}{2}Ha'_4$ of the table equals $\frac{1}{2}\partial_4 \ln q$, and $+\frac{1}{2}Ha'_4$ equals $\frac{1}{2}\partial_4 \ln p$, so the sum is $\frac{3}{2}\partial_4 \ln(qp) = 0$.

On the model's own spinor Psi16 of Section 11.

- [proved III §17] “[definition] $D_{\text{cov}}[\mu] \text{ Psi16} - D[\text{Psi16}, x_{\mu}] == \text{Gamma}^{\text{spin}}[\mu] . \text{Psi16}$ ”.
- [proved III §17] “in the six directions on which Psi16 does not depend, $D_{\text{cov}} \text{Psi16}$ is purely the connection term”.
- [proved III §17] “the connection term does not vanish on Psi16: the connection matrices themselves are non-zero (witnessed)”.
- [proved VI §24] “FABLE [has content]: the connection term does not vanish on an x_0 -independent spinor - witnessed on the constant spinor e_1 : a spinor homogeneous in everything but x_4 is NOT a solution”.

Section 17 constructs the operator and applies it. It does not re-derive the flat field equations of Part II from a covariant Lagrangian [prose III §17].

6.9 The three bridges of Part IV

The gauge law [prose IV], verified per bridge. Suppose a new frame is $\text{frameNew} = \text{frameOld} . L$ with $L(x)$ invertible on the flat indices. Substituting into the vielbein postulate, with $M = \text{Transpose}[L]$, gives

$$\omega_{\mu}^{\text{new}} = M \omega_{\mu}^{\text{old}} M^{-1} - (\partial_{\mu} M) M^{-1} \quad (\text{mixed components}). \quad (66)$$

The second term is the inhomogeneous gauge term, and it vanishes exactly when L is constant. The law is not assumed anywhere. Each new connection is derived independently by cfSpinConnection , and the law is then verified against the result. A frame $\text{frameCanonical} . L$ with L preserving the flat metric describes the *same geometry*, and its Levi-Civita connection is the canonical one in a different gauge, *not* a new connection [prose IV].

6.9.1 Bridge 1: a locally boosted frame, the same connection in a local gauge (Section 18)

A new object, not in the original notebook, is introduced here: the rapidity $\theta = \text{cfBoostRapidity}[x_0, x_4]$, an arbitrary function. It generates a boost in the flat (0,4) plane:

$$\Lambda[\theta] = \text{ID8} + (\cosh \theta - 1)(E_{00} + E_{44}) + \sinh \theta (E_{04} + E_{40}) = e^{\theta K}, \quad (67)$$

where $K = E_{04} + E_{40}$ is the (0,4) boost generator on flat indices, and $\text{frameBoost} = \text{frameCanonical} . \Lambda$.

- [proved IV §18] “ Λ is in $O(4,4)$: $\Lambda . \text{eta} . \text{Transpose}[\Lambda] == \text{eta}$ ”, “ Λ is NOT in $O(8)$: it is a genuine boost, not a rotation”, “ $\Lambda == \text{MatrixExp}[\theta K]$ with K the (0,4) boost generator”.
- [proved IV §18] “BRIDGE 1 reproduces the SAME curved metric g ”, “BRIDGE 1 frame really is different from the canonical frame”.
- [proved IV §18] “BRIDGE 1: vielbein postulate holds identically”, “BRIDGE 1: $\omega_{\mu,a,b} == -\omega_{\mu,b,a}$ ”, “BRIDGE 1: torsion still vanishes”.
- [proved IV §18] “COMPARISON 1: $\omega_{\text{Boost}} == M \omega_{\text{Canonical}} \text{Inverse}[M] - D[M, x_{\mu}] \text{Inverse}[M]$ ”, “COMPARISON 1: the WHOLE difference is the inhomogeneous gauge term”, “COMPARISON 1: the difference equals $-D[\theta, x_{\mu}] * (\text{boost generator})$ ”.

- [proved IV §18] “COMPARISON 1: curvature transforms covariantly, $R' == M R \text{ Inverse}[M]$ ”, “COMPARISON 1: BRIDGE 1 has the SAME Ricci scalar as the canonical frame”.
- [displayed IV §18] ω_{Boost} has 28 non-zero components. The difference δ_{Boost} has 4: $-\partial_\mu \theta K$ for $\mu = 0$ and $\mu = 4$, in the two orderings of the plane $(0, 4)$.

At the spinor level, the spin lift is $S = \text{MatrixExp}[\theta \text{ SAB}[[1,5]]]$, with no extra factor $1/2$, because SAB already carries $1/4$ [prose IV §18]:

- [proved IV §18] “BRIDGE 1 spin lift: $\text{Inverse}[S] \cdot S == \text{ID}_{16}$ ”, “BRIDGE 1 spin lift: S preserves the spinor metric, $\text{Transpose}[S] \cdot \sigma_{16} \cdot S == \sigma_{16}$ ”, “BRIDGE 1 spin lift [has content]: $\text{Inverse}[S] \cdot \gamma^a \cdot S == \Lambda^a_b \gamma^b$ (S is the spin lift of Λ)”.
- [proved IV §18] “COMPARISON 1 at the spinor level [THE GAUGE LAW]: $\Gamma'_\mu == S \Gamma_\mu \text{ Inverse}[S] - D[S, x_\mu] \text{ Inverse}[S]$ ”, “COMPARISON 1 at the spinor level: the inhomogeneous term is $-D[\theta, x_\mu] \text{ SAB}[[1,5]]$, the spin image of $-D[\theta, x_\mu] K$ ”, “COMPARISON 1 at the spinor level: and that is exactly cfSpinMatrix applied to the vector difference δ_{Boost} ”.
- [proved IV §18] “BRIDGE 1 [has content]: $D'_\mu (S \psi) == S D_\mu \psi$ for a generic spinor - one covariant derivative, two gauges”, “BRIDGE 1: the curved Dirac matrices of the boosted frame are $S \gamma^\mu \text{ Inverse}[S]$ ”, “BRIDGE 1: so the Dirac operator is one operator in two gauges, $\text{Dirac}'[S \psi] == S \text{Dirac}[\psi]$ ”.

A solution of the Dirac equation in one gauge is therefore a solution in the other, after $\psi \rightarrow S\psi$. Part V identifies the inhomogeneous term as the fable-5.1 connection of the boosted frame (subsection 6.10).

6.9.2 Bridge 2: the null (light-cone) frame, the same connection in a constant gauge (Section 19)

$$U = \frac{1}{\sqrt{2}} \begin{pmatrix} \text{ID}_4 & \text{ID}_4 \\ \text{ID}_4 & -\text{ID}_4 \end{pmatrix}, \quad U = U^T, \quad U \cdot U = \text{ID}_8, \quad (68)$$

with $\eta_{\text{Null}} = \text{Transpose}[U] \cdot \eta_{4488} \cdot U$ and $\text{frameNull} = \text{frameCanonical} \cdot U$. The eight flat directions are paired as $(0, 4)$, $(1, 5)$, $(2, 6)$ and $(3, 7)$, one spacelike and one timelike in each pair, and each pair is replaced by its two null combinations [prose IV §19].

- [proved IV §19] “ U is an involution: $U \cdot U == \text{ID}_8$ ”, “ U is symmetric”, “THE NULL TANGENT METRIC IS EXACTLY THE SPINOR METRIC σ ”, “ η_{Null} is symmetric and squares to ID_8 ”, “all eight frame directions are null: η_{Null} has zero diagonal”. The identity $\eta_{\text{Null}} == \sigma$ is a statement about the flat metric, not about the connection.
- [proved IV §19] “BRIDGE 2 reproduces the SAME curved metric g ”, “BRIDGE 2: vielbein postulate holds identically”, “BRIDGE 2: $\omega_{[\mu, a, b]} == -\omega_{[\mu, b, a]}$ with indices lowered by η_{Null} ”, “BRIDGE 2: torsion vanishes”.
- [proved IV §19] “COMPARISON 2: $\omega_{\text{Null}} == U \cdot \omega_{\text{Canonical}} \cdot U$ exactly”, “COMPARISON 2: the inhomogeneous gauge term is ZERO because U is constant”, “COMPARISON 2: $R_{\text{null}} == U \cdot R_{\text{canonical}} \cdot U$ ”, “COMPARISON 2: BRIDGE 2 has the SAME Ricci scalar as the canonical frame”.
- [displayed IV §19] ω_{Null} has 48 non-zero components. A constant change of basis spreads the canonical 24 over more entries, and the difference from $U\omega U$ is zero.

6.9.3 Bridge 3: triality frame with totally antisymmetric split-octonion torsion, a different connection (Section 20)

The flat index is a *type-1 split-octonion spinor index*, reached through the constant triality bridge `triVecToSpin` of Section 9. [proved I §9] “THE TRIALITY METRIC IS EXACTLY THE SPINOR METRIC: `etaTri == sigma`”. The frame is `frameTri = frameCanonical . triVecToSpin`, with flat tangent metric σ , and the connection is

$$\omega_{\mu ab}^{\text{Oct}} = \omega_{\mu ab}^{\text{TriLC}} + \lambda m_{abc}^{\text{Skew}} (e_{\text{Tri}})_{\mu}^c, \quad (e_{\text{Tri}})_{\mu}^c = \text{frameTri}[[\mu, c]]. \quad (69)$$

Here `omegaTriLC` is the Levi-Civita connection in the triality frame. $m^{\text{Skew}} = \text{mSkewSpin}$ is the totally antisymmetrized split-octonion structure constant `mLower`, transported to the spinor basis. $\lambda = \text{lambdaOct}$ is a free real torsion-strength parameter introduced in Section 20, not in the original notebook.

- [proved IV §20] “the triality bridge is Euclidean-orthogonal: `Transpose[P] == Inverse[P]`”, “but it is NOT in $O(4,4)$: `P . eta4488 . Transpose[P] != eta4488` (it carries `eta4488` to `sigma`)”, “BRIDGE 3 reproduces the SAME curved metric `g`”, “the triality tangent metric is `sigma`”.
- [proved IV §20] “the antisymmetrized structure constants are totally antisymmetric”, “they agree with `mLower` on the seven imaginary directions”, “`mSkewSpin` is totally antisymmetric” (48 non-zero components [displayed IV §20]).
- [proved IV §20] “BRIDGE 3 ($\lambda=0$): vielbein postulate holds identically”, “BRIDGE 3 ($\lambda=0$): `omega[mu,a,b] == -omega[mu,b,a]`”, “BRIDGE 3 ($\lambda=0$) is the constant triality conjugate of the canonical connection”.
- [proved IV §20] “BRIDGE 3: `omegaOct[mu,a,b] == -omegaOct[mu,b,a]` (still metric compatible)”, “BRIDGE 3: at $\lambda = 0$ it reduces to the Levi-Civita connection”.
- [proved IV §20] “BRIDGE 3: torsion vanishes at $\lambda = 0$ ”, “BRIDGE 3: torsion is NOT zero for $\lambda \neq 0$ ”, “BRIDGE 3: the flat torsion `T[a,b,c]` is TOTALLY ANTISYMMETRIC”, “BRIDGE 3: `T[a,b,c] == -2 lambda mSkewSpin[a,b,c]`”.
- [proved IV §20] “COMPARISON 3: `omegaOct - (triality conjugate of omegaCanonical) == contortion`”, “COMPARISON 3: the difference is linear in λ and vanishes at $\lambda = 0$ ”. The difference is a *tensor* (a contortion), so this is a different connection.

Its curvature is quadratic in λ . With K the contortion, $R(\omega^{\text{Oct}}) = R(\omega^{\text{TriLC}}) + \lambda(dK + [\omega^{\text{TriLC}}, K]) + \lambda^2[K, K]$:

- [proved IV §20] “BRIDGE 3: at $\lambda = 0$ the curvature is the triality conjugate of the canonical curvature”, “BRIDGE 3 [has content]: the curvature DOES depend on λ (witnessed)”, “BRIDGE 3: the λ -dependence is a polynomial of degree exactly two”.
- [proved IV §20] “BRIDGE 3: the λ^1 term is the Levi-Civita covariant exterior derivative of the contortion, `dK + [omegaTriLC, K]`”. Written out, it is $\partial_{\mu} K_{\nu} - \partial_{\nu} K_{\mu} + [\omega_{\mu}^{\text{TriLC}}, K_{\nu}] - [\omega_{\nu}^{\text{TriLC}}, K_{\mu}]$.
- [proved IV §20] “BRIDGE 3: the λ^2 term is `[K_mu, K_nu]`, the contortion commuted with itself”.

The Ricci shift. $-42\lambda^2 = -\frac{1}{4}T_{abc}T^{abc}$.

- [proved IV §20] “BRIDGE 3: at $\lambda = 0$ the Ricci scalar is the canonical one”.
- [proved IV §20] “BRIDGE 3 [THE RESULT]: $R(\text{omegaOct}) - R(\text{canonical}) == -(1/4)T_{\{abc\}}T^{\{abc\}}$, a constant shift set by the torsion alone”.
- [proved IV §20] “BRIDGE 3: equivalently $-\lambda^2 (\text{mSkew} \cdot \text{mSkew})$, with $\text{mSkew} \cdot \text{mSkew} == 42$, so the shift is exactly $-42\lambda^2$ ”. Check [derived here]: $T = -2\lambda m^{\text{Skew}}$ gives $T \cdot T = 4\lambda^2 \cdot 42 = 168\lambda^2$, and $\frac{1}{4} \cdot 168 = 42$.

This connection resembles the Cartan–Schouten connections of a Lie group, but it differs in two ways that must not be assumed away [prose IV §20]. It is not flat. And the octonion structure constants violate the Jacobi identity, so they cannot be the anholonomy of any frame.

Spinor level. `cfSpinMatrix` is correct only for `eta4488` vector indices. Bridge 3’s flat index is a triality-spinor index, so the matching Dirac matrices are `T16Tri[a] = Sum_A triSpinToVec[[a,A]] T16[A-1]`, and they are contracted by `cfSpinMatrixTri` [prose IV §20]:

- [proved IV §20] “THE TRIALITY DIRAC MATRICES OBEY THE sigma CLIFFORD RELATION: $\{T16Tri[a], T16Tri[b]\} == 2 \text{sigma}[[a,b]] \text{ID16}$ ”, “they are NOT the vector-frame Dirac matrices, so the distinction is real”.
- [proved IV §20] “at $\lambda = 0$ the Bridge 3 spinor connection equals the canonical one”, “the extra spinor coupling vanishes at $\lambda = 0$ ”, “the extra spinor coupling is non-zero for $\lambda \neq 0$ ”, “the extra spinor coupling is still block diagonal (type-1 + type-2 preserved)”.
- The extra coupling $(\lambda/8) \text{mSkew}[a,b,c] \text{frameTri}[[\mu,c]] [T16Tri[a], T16Tri[b]]$ is *quadratic* in the gammas. The third index is contracted with the frame, not with a third gamma. The cubic Einstein–Cartan axial term would appear only after contracting with γ^μ in the Dirac operator, and the notebook does *not* carry that contraction out [prose IV §20]. The extra coupling has 160 non-zero entries [displayed IV §20].

6.10 The fable-5.1 bridge of Part V: the Weitzenböck (teleparallel) connection

The fable-5.1 bridge keeps the metric, the frame, `eta4488` and the bridge [d] unchanged, and changes only the rule for parallel transport. It declares the canonical frame itself to be parallel [prose V §21]:

$$(\Gamma^W)^\rho_{\mu\nu} = e_a^\rho \partial_\mu e_\nu^a, \quad (70)$$

in the notebook `GammaWeitzenboeck[[rho, mu, nu]] = Sum_a coframe[[a, rho]] D[frame[[nu, a]], x[mu]]`.

- [proved V §21] “FABLE-5.1 [definition]: the frame is parallel - $\nabla^W e == 0$ for all eight frame vectors”, “FABLE-5.1 [has content]: Γ^W is metric compatible, $\nabla^W g == 0$ ”, “FABLE-5.1 [has content]: Γ^W is NOT symmetric in its lower indices - it has torsion”, “FABLE-5.1 [has content]: Γ^W is NOT the Levi-Civita connection”.
- Γ^W has 13 non-zero components [displayed V §22].

The spin connection is zero, and so is the curvature. Γ^W is fed to the same solver. Because the first two terms of the vielbein postulate then cancel by construction, the solver returns $\omega = 0$:

- [proved V §21] “FABLE-5.1 [solver regression]: vielbein postulate residual is zero”, “FABLE-5.1 [THE RESULT]: the spin connection in the canonical frame is IDENTICALLY ZERO”, “FABLE-5.1: trivially antisymmetric, so trivially metric compatible”.
- [proved V §21] “FABLE-5.1 [has content]: the torsion is NOT zero”, “FABLE-5.1: $T^{\rho}_{\mu\nu} == \Gamma^W[\rho, \mu, \nu] - \Gamma^W[\rho, \nu, \mu]$ (torsion is the antisymmetric part of Γ^W)”. With $\omega = 0$ the torsion is $T^a_{\mu\nu} = \partial_\mu e_\nu^a - \partial_\nu e_\mu^a$.
- [proved V §21] “FABLE-5.1 [THE RESULT]: the curvature is IDENTICALLY ZERO - a flat connection on a curved metric”.

The canonical connection is minus the contortion of this torsion. The contortion is $K = \Gamma^W - \Gamma^{LC}$, and in terms of the torsion $K_{\rho\mu\nu} = \frac{1}{2}(T_{\rho\mu\nu} + T_{\mu\rho\nu} + T_{\nu\rho\mu})$ (all indices down, first index then raised). Substituting $\Gamma^{LC} = \Gamma^W - K$ into the vielbein postulate leaves $\text{omegaCanonical}[\mu, a, b] = -K[\rho, \mu, \nu] e[\rho, a] \text{coframe}[[b, \nu]]$ [prose V §21].

- [proved V §21] “COMPARISON F [has content]: $\Gamma^W - \Gamma^{LC} == (1/2)(T_{\text{rmn}} + T_{\text{mrn}} + T_{\text{nrm}})$ with the index raised, the contortion formula”.
- [proved V §21] “COMPARISON F [THE RESULT]: $\text{omegaCanonical} == -\text{contortion}(\text{fable-5.1 torsion})$, pulled back to flat indices, exactly”.
- [proved V §21] “COMPARISON F: so $\text{omegaCanonical} - \text{omegaFable51}$ is a TENSOR, not a gauge term ($\text{omegaFable51} == 0$)”.
- contortionFable51 has 24 non-zero components, the same 24 as omegaCanonical . $\text{torsionFable51Flat}$ has 24, and RiemannFable51 has 0 [displayed V §22].

The kind of torsion. A torsion tensor splits into a vector part, an axial part and a tensor part:

- [proved V §21] “FABLE-5.1 torsion vector $T[\mu] = T[\nu, \nu, \mu]$ is a GRADIENT: $T[\mu] == D[\text{Log}[\text{Sin}[6 H x_0]], x[\mu]]$ ”.
- [proved V §21] “FABLE-5.1 torsion vector has exactly one non-zero component, $6 H \text{Cot}[6 H x_0]$ along x_0 ”.
- [proved V §21] “FABLE-5.1: the AXIAL (totally antisymmetric) part of the torsion vanishes - the complement of Bridge 3”, “FABLE-5.1: the torsion itself does not vanish, so the tensor part carries the rest”.

For any diagonal frame the torsion vector is, component by component (no sum) [derived here],

$$T_\mu = \sum_\nu e_a^\nu (\partial_\nu e_\mu^a - \partial_\mu e_\nu^a) = \partial_\mu \ln \frac{e_\mu^\mu}{\det e}. \quad (71)$$

On the canonical frame this is $\partial_0 \ln(\tan 6Hx_0 / \sec 6Hx_0) = \partial_0 \ln \sin 6Hx_0 = 6H \cot 6Hx_0$ for $\mu = 0$. It is zero for $\mu = 4$, because $e_4^4 = 1$ and $\det e = \sec 6Hx_0$ do not depend on x_4 . It is zero for $\mu = 1, 2, 3, 5, 6, 7$, because nothing depends on those coordinates. This agrees with the assertion.

$R = -T + B$ in closed form. With the torsion scalar $T = \frac{1}{4}T^{\rho\mu\nu}T_{\rho\mu\nu} + \frac{1}{2}T^{\rho\mu\nu}T_{\nu\rho\mu} - T^\mu T_\mu$ and the boundary term $B = \frac{2}{\sqrt{\det g}} \partial_\mu (\sqrt{\det g} T^\mu)$, where $\sqrt{\det g} = \sec 6Hx_0$:

$$R = -3H^2 \left((55 + 7 \cos 12Hx_0) \cot^2(6Hx_0) \csc^2(6Hx_0) - 2a_4'^2 \right), \quad (72)$$

$$T = -6H^2(5 \cot^4(6Hx_0) + a_4'^2), \quad (73)$$

$$B = -36H^2(5 + \cos 12Hx_0) \cot^2(6Hx_0) \csc^2(6Hx_0). \quad (74)$$

These closed forms are **[displayed V §21]** (RicciScalarCanonical, torsionScalarFable51, boundaryTermFable51). The following are **[proved V §21]**: “COMPARISON F [THE RESULT]: $R == -T + B$ - the teleparallel equivalent of GR, in closed form for this geometry”, “FABLE-5.1: T is not zero and B is not zero - neither side is trivial”. The Einstein–Hilbert and teleparallel actions therefore differ by a boundary term on this geometry.

The spinor: one vector term, removed by a rescaling. In the fable-5.1 gauge the spinor connection is zero, so the Dirac operator is $\gamma^\mu \partial_\mu$. The canonical Dirac operator carries in addition $\gamma^\mu \Gamma_\mu$, and this whole term is one vector term:

$$\begin{aligned} \gamma^\mu \Gamma_\mu &= -\frac{1}{2} T_\mu \gamma^\mu = -\frac{1}{2} \partial_\mu (\ln \sin 6Hx_0) \gamma^\mu \\ &= -3H \cot^2(6Hx_0) T_{16}[0], \end{aligned} \quad (75)$$

$$\text{Dirac}_{LC}[\sqrt{\sin 6Hx_0} \Psi'] = \sqrt{\sin 6Hx_0} \gamma^\mu \partial_\mu \Psi' \quad \text{for every } \Psi'. \quad (76)$$

The first line checks against subsection 6.8 **[derived here]**: $-\frac{1}{2}(6H \cot)(\cot T_{16}[0]) = -3H \cot^2 T_{16}[0]$, because $\gamma^0 = \cot T_{16}[0]$. The matrix $\gamma^\mu \Gamma_\mu$ is $-3H \cot^2$ times the permutation matrix $T_{16}[0]$, so it has 16 non-zero entries.

- **[proved V §21]** “FABLE-5.1 [THE RESULT]: $\gamma^\mu \Gamma_\mu == -(1/2) T[\mu] \gamma^\mu$, one vector term”, “FABLE-5.1: equivalently $-(1/2) D[\text{Log}[\sin 6Hx_0], x_\mu] \gamma^\mu$ ”.
- **[proved V §21]** “FABLE-5.1 [THE RESULT]: $\text{DiracCanonical}[\sqrt{\sin 6Hx_0} \Psi'] == \sqrt{\sin 6Hx_0} \text{DiracFable51}[\Psi']$, for a generic Ψ' ”, “FABLE-5.1: the rescaling is NOT a no-op - the two Dirac operators differ on the constant spinor”.
- **[proved VI §24]** with a potential: “FABLE [THE RESULT]: $(\gamma^\mu D_\mu - H V'(s))[\sqrt{\sin 6Hx_0} \Psi'] == \sqrt{\sin 6Hx_0} (\gamma^\mu d_\mu \Psi' - H V'(s) \Psi')$ for a GENERIC $\Psi'(x_0, x_4)$: the rescaling removes the connection term exactly”.

The rescaling works because the torsion vector is a gradient. A vector-torsion term that is a gradient is removed by multiplying the spinor by $\exp(\frac{1}{2} \ln \sin 6Hx_0) = \sqrt{\sin 6Hx_0}$ **[derived here]**.

For the real spinor of Parts II–VI the connection drops out of the Lagrangian. σ_{16} . $T_{16}[a]$ is antisymmetric, so $\Psi^T \sigma_{16} T_{16}[a] \Psi = 0$ for real commuting Ψ . Since $\gamma^\mu \Gamma_\mu$ is a multiple of γ^μ , the connection contributes nothing to that Lagrangian:

- **[proved V §21]** “FABLE-5.1 [THE RESULT]: $\Psi^T \sigma_{16} \gamma^\mu \Gamma_\mu \Psi == 0$ for EVERY Ψ - the spin connection drops out of the Lagrangian”, “FABLE-5.1: the reason is Section 11’s identity, $\Psi^T \sigma_{16} T_{16}[a] \Psi == 0$ ($\sigma_{16} T_{16}[a]$ antisymmetric)”, “FABLE-5.1: but the connection term does NOT drop out of the Dirac OPERATOR (it is $-(1/2) T[\mu] \gamma^\mu$, non-zero)”.
- **[proved V §21]** “FABLE-5.1: the covariant Lagrangian with frame $\rightarrow$ ID8 and volume factor $\rightarrow$ 1 IS the author’s La of Section 11, term for term”, “FABLE-5.1: and the covariant Lagrangian has NO connection term to omit - its kinetic term is already the fable-5.1 one”. The author’s flat La omits only the volume factor $\sec 6Hx_0$ and the coframe factors inside γ^μ **[prose V §21]**.

This mechanism is specific to a *real* field. subsection 6.12 shows that it does not carry over to the complex fermion fable.

Frame covariance: Bridge 1 explained. In Bridge 1’s boosted frame, the same Γ^W fed to the same solver gives a fable-5.1 spin connection that is pure gauge:

- [proved V §21] “FABLE-5.1 in the boosted frame: vielbein postulate residual is zero”, “FABLE-5.1 in the boosted frame [THE RESULT]: $\omega == -D[M, x_\mu] \text{Inverse}[M]$, pure gauge and nothing else”, “FABLE-5.1 in the boosted frame: the lemma’s prediction itself satisfies the vielbein postulate with Γ^W ”, “FABLE-5.1 in the boosted frame: the curvature is STILL identically zero”.
- [proved V §21] “and Bridge 1’s inhomogeneous term of Section 18 IS this object: $\text{deltaBoost} == \text{omegaFable51Boost}$ ”, “so $\text{omegaBoost} == (\text{canonical}, \text{conjugated}) + (\text{fable-5.1 in the boosted frame})$ ”.
- $\text{omegaFable51BoostMixed}$ has 4 non-zero components, the same 4 as deltaBoost [displayed V §21].

6.11 Master comparison of the five connections (Section 22)

Section 22 prints two tables: the summary `cfConnectionSummary`, and a grid of the comparisons side by side [displayed V §22]. The table below is `cfConnectionSummary`, with three entries of the side-by-side grid added to the column “difference from canonical”: Bridge 1’s spinor-level law, and Bridge 3’s curvature and Ricci shift. Part VII restates the bridge results it relies on, so that it can be read alone: [proved VII §27] SPIN CONNECTION [has content]: Parts IV–V restated – Bridge 1’s difference is the pure gauge term, Bridge 2 is a constant conjugate, Bridge 3 has totally antisymmetric torsion that is not zero and SPIN CONNECTION [has content]: Part V restated – $\omega^W == 0$ in the canonical frame, $\omega^{\text{LC}} == -\text{contortion}$, $T_\mu == d_\mu \text{Log}[\text{Sin}[6 H x_0]]$, $R == -T + B$, and $\gamma^\mu \Gamma_\mu == -(1/2) T_\mu \gamma^\mu$. For all five connections the curved metric g is the same (“yes (reference)” for the canonical one), and the connection is metric compatible; those two columns of the notebook’s table read “yes” throughout and are not repeated below.

connection; frame field; flat metric	torsion zero? curvature zero?	difference from canonical	verdict
<code>omegaCanonical;</code> <code>frameCanonical (diagonal);</code> <code>eta4488</code>	torsion: yes; curvature: no	— (reference)	Levi-Civita
<code>omegaBoost (Bridge 1);</code> <code>frameCanonical . Lambda(x);</code> <code>eta4488</code>	torsion: yes; curvature: no	gauge term $-\partial_\mu \theta K$, which is the fable-5.1 connection of the boosted frame; at spinor level $\Gamma' = S \Gamma S^{-1} - (\partial S) S^{-1}$ with the explicit spin lift S	SAME connection, local gauge
<code>omegaNull (Bridge 2);</code> <code>frameCanonical . U; etaNull</code> <code>== sigma</code>	torsion: yes; curvature: no	constant similarity $U \omega U$	SAME connection, constant gauge
<code>omegaOct (Bridge 3);</code> <code>frameCanonical .</code> <code>triVecToSpin; etaTri ==</code> <code>sigma</code>	torsion: no , axial torsion $-2\lambda m^{\text{Skew}}$; curvature: no	contortion $\lambda m_{abc}^{\text{Skew}} (e_{\text{Tri}})_\mu{}^c$ (with $e_{\text{Tri}} = \text{frameTri}$), a tensor; curvature quadratic in λ ; Ricci shift $-42\lambda^2$	DIFFERENT connection
<code>omegaFable51 (fable-5.1);</code> <code>frameCanonical (parallel);</code> <code>eta4488</code>	torsion: no , vector + tensor torsion, axial part zero; curvature: yes , flat	$\omega^{\text{canonical}}$ $= -\text{contortion}(T_{\text{fable51}})$, a tensor; $\omega^{\text{fable51}} = 0$	DIFFERENT connection

Non-zero component counts (Section 22 fingerprint table and the counts printed in Sections 16–21, **[displayed]**):

object	non-zero components
GammaCanonical (Christoffel)	37
GammaWeitzenboeck	13
omegaCanonical	24
omegaBoost / its difference deltaBoost	28 / 4
omegaNull / its difference	48 / 0
omegaTriLC (Bridge 3, $\lambda = 0$)	84
contortionOct (Bridge 3)	78
torsionOctFlat (Bridge 3)	48
omegaFable51 (canonical frame)	0
contortionFable51 / torsionFable51Flat	24 / 24
RiemannCanonical	156
RiemannFable51	0

6.12 What the canonical connection does for the complex fermion fable

Everything in this subsection concerns the *complex* 16-component fermion fable of Part VII. Two kinds of statement appear. Some are consequences of results already proved in Parts I–VI, and the derivation is shown here. The others are assertions of the new Part VII, marked **[proved VII §n]**, with the label quoted verbatim from `claude-fable/run_fermion_fable_part7.log`.

6.12.1 The field and the symmetrized Lagrangian

Fermion fable is a complex 16-component Grassmann spinor Ψ . It is the minimal field that uses the author’s σ_{16} conjugation, propagates, admits a potential $V(s)$, and reduces to Part VI’s real field as its real commuting shadow. It is not the unique field with dynamics. This is the conclusion of Section 26 as a whole (section 3.1); no single assertion states it. It rests on **[proved VII §26]** REFINEMENT (d) **[THE RESULT]**: `s = Psibar Psi == i theta1^T sigma16 theta2: NON-ZERO (16 monomials) and HERMITIAN; p = Psibar T16[8] Psi` is non-zero and Hermitian too, REFINEMENT (b) **[THE RESULT]**: Majorana with C+: the kinetic term is genuine - Euler-Lagrange expression 2 C+ T16[a] phi with C+ T16[a] invertible - but every invariant bilinear vanishes (massless, no $V(s)$), REFINEMENT (c) **[THE RESULT]**: Weyl and Majorana-Weyl: `PL C T16[a] PL == PR C T16[a] PR == 0` for $C = C-, C+$ and every a (C chirality-diagonal, $T16[a]$ chirality-odd) - no kinetic term and FIDELITY [1] **[fidelity]**: `Psi -> psi, Psibar -> psi^T sigma16` turns the symmetrized covariant $Lhat$ into Part VI’s `cfLhatFable` EXACTLY (canonical frame, generic V), among the others quoted in section 3.1. Its Dirac conjugate and scalar bilinear are

$$\bar{\Psi} = \Psi^\dagger \sigma_{16}, \quad s = \bar{\Psi} \Psi = -\psi_1^\dagger \sigma \psi_1 + \psi_2^\dagger \sigma \psi_2, \quad (77)$$

where † is the classical conjugate transpose. The second form follows from $\sigma_{16} = \text{diag}(-\sigma, \sigma)$. In the quantized theory the Hilbert adjoint is $\Psi^\dagger := \Psi^\dagger J$ (subsection 6.12.6).

The only Lagrangian used is the *symmetrized covariant* one:

$$L = \sqrt{g} \hat{L}, \quad \hat{L} = \frac{1}{2H} \left[\bar{\Psi} \gamma^\mu D_\mu \Psi - (D_\mu \bar{\Psi}) \gamma^\mu \Psi \right] - V(s), \quad (78)$$

$$D_\mu \Psi = \partial_\mu \Psi + \Gamma_\mu \Psi, \quad D_\mu \bar{\Psi} = \partial_\mu \bar{\Psi} - \bar{\Psi} \Gamma_\mu.$$

$D_\mu \bar{\Psi}$ is the Dirac conjugate of $D_\mu \Psi$, and $\hat{L}$ is real, because $\sigma_{16} \gamma^a$ and $\sigma_{16} \Gamma_\mu$ are *real antisymmetric* matrices. That rests on **[proved I §5]** “`sigma16 . T16[A]` is antisymmetric for $A =$

0..7” and “sigma16 . SAB is antisymmetric”, together with $\Gamma_\mu = \frac{1}{2}\omega_{\mu ab}$ SAB with real ω (subsection 6.7). [proved VII §27] LAGRANGIAN (a) [definition]: every Γ_μ is real and sigma16 Γ_μ is antisymmetric ($\Gamma_\mu^T \text{ sigma16} == -\text{sigma16} \Gamma_\mu$): $D_\mu \Psi$ is the Dirac conjugate of $D_\mu \Psi$.

6.12.2 It drops out of the symmetrized Lagrangian because the matrix $\{\gamma^\mu, \Gamma_\mu\}$ vanishes

Expanding D in (78) gives [derived here]

$$\hat{L}(D) = \hat{L}(\partial) + \frac{1}{2H} \bar{\Psi} \left(\sum_\mu \{\gamma^\mu, \Gamma_\mu\} \right) \Psi. \quad (79)$$

The connection terms are $\bar{\Psi} \gamma^\mu \Gamma_\mu \Psi$ from the first product and $+\bar{\Psi} \Gamma_\mu \gamma^\mu \Psi$ from $-(D_\mu \bar{\Psi}) \gamma^\mu \Psi$. On the canonical frame $\{\gamma^\mu, \Gamma_\mu\} = 0$ for each μ (subsection 6.8), so $\hat{L}(D) = \hat{L}(\partial)$ identically. That is, “covariant = partial” holds for the symmetrized complex Lagrangian on this frame. It holds on any frame where the totally antisymmetric part $\omega_{[cab]}$ vanishes (65), and fails in general.

- [proved VII §27] LAGRANGIAN (c) [THE RESULT]: $L_{\text{sym}} - L_{\text{sym}_d} == (1/(2H)) \sum_\mu \bar{\Psi} \{\gamma^\mu, \Gamma_\mu\} \Psi$ identically, and $\{\gamma^\mu, \Gamma_\mu\} == 0$ for EACH μ on the canonical frame: $L_{\text{sym}} == L_{\text{sym}_d}$.

This is *not* Part VI’s mechanism. For the real commuting field, $\Psi^T \sigma_{16} \gamma^\mu \Gamma_\mu \Psi$ vanishes because $\sigma_{16} T_{16}[0]$ is antisymmetric (subsection 6.10). For a complex field, $\bar{\Psi} T_{16}[0] \Psi = \Psi^\dagger (\sigma_{16} T_{16}[0]) \Psi$ with $\sigma_{16} T_{16}[0]$ real antisymmetric is i times the Hermitian form of $-i \sigma_{16} T_{16}[0]$, which is not zero in general. So the drop-out for the complex field is a property of the *matrix* $\{\gamma^\mu, \Gamma_\mu\}$, not of a bilinear identity [derived here]. The other Lagrangians behave differently:

- The *unsymmetrized covariant* Lagrangian $\frac{1}{H} \bar{\Psi} \gamma^\mu D_\mu \Psi - V$ equals the symmetrized one plus $\frac{1}{2H} \frac{1}{\sqrt{g}} \partial_\mu (\sqrt{g} \bar{\Psi} \gamma^\mu \Psi)$, a total divergence. Its connection term is $\frac{1}{H} \bar{\Psi} \gamma^\mu \Gamma_\mu \Psi = -3 \cot^2(6Hx_0) \bar{\Psi} T_{16}[0] \Psi$, which is not zero. [proved VII §27] LAGRANGIAN (d) [has content]: $L_{\text{un}} - L_{\text{sym}} == (1/(2H)) (1/\text{Sqrt}[g]) d_\mu (\text{Sqrt}[g] \bar{\Psi} \gamma^\mu \Psi)$, a total divergence, LAGRANGIAN (e) [has content]: $L_{\text{un}} - L_{\text{un}_d} == (1/H) \bar{\Psi} \gamma^\mu \Gamma_\mu \Psi == -3 \cot^2(6Hx_0) \bar{\Psi} T_{16}[0] \Psi$, LAGRANGIAN (e) [control]: that term is NOT zero for a complex field (witness $u = e1, v = e13$), and IS zero in the real commuting limit (the scope of Parts V-VI).
- The *unsymmetrized partial* Lagrangian $\frac{1}{H} \bar{\Psi} \gamma^\mu \partial_\mu \Psi - V$ is *inadmissible*. Its $\bar{\Psi}$ and Ψ equations are not Dirac conjugates of each other: they differ by $\bar{\Psi} \sum_\mu [\gamma^\mu, \Gamma_\mu] = 2 \bar{\Psi} \gamma^\mu \Gamma_\mu = -6H \cot^2(6Hx_0) \bar{\Psi} T_{16}[0]$. [proved VII §27] LAGRANGIAN (f) [THE RESULT]: L_{un_d} varied with respect to $\bar{\Psi}$ gives $(\text{Sqrt}[g]/H) (\bar{\Psi} \gamma^\mu d_\mu \Psi - H V' \bar{\Psi})$: the connection term is MISSING, LAGRANGIAN (f) [THE RESULT]: varied with respect to Ψ it gives $-(\text{Sqrt}[g]/H) ((d_\mu \bar{\Psi}) \gamma^\mu + \bar{\Psi} [\gamma^\mu, \Gamma_\mu] + H V' \bar{\Psi})$, with $\bar{\Psi} [\gamma^\mu, \Gamma_\mu] == 2 \bar{\Psi} \gamma^\mu \Gamma_\mu$: the connection term is PRESENT, LAGRANGIAN (f) [control]: so L_{un_d} is INADMISSIBLE - its two field equations are not Dirac conjugates: they differ by $2 \bar{\Psi} \gamma^\mu \Gamma_\mu = -6H \cot^2(6Hx_0) \bar{\Psi} T_{16}[0]$, non-zero (witness $u = e1, v = 0, V = s^2$).

6.12.3 It enters the field equation as $-3H \cot^2 T_{16}[0]$, and the rescaling removes it

Varying $\bar{\Psi}$ and Ψ independently in the symmetrized action gives the field equation and its conjugate:

$$\gamma^\mu D_\mu \Psi = H V'(s) \Psi, \quad (D_\mu \bar{\Psi}) \gamma^\mu = -H V'(s) \bar{\Psi}. \quad (80)$$

[proved VII §27] FIELD EQUATION [THE RESULT]: $\text{EL_Psibar} == (\text{Sqrt}[g]/H)(\gamma^\mu \text{D}_\mu \text{Psi} - H V'(s) \text{Psi})$ for generic Psi, Psibar of all eight coordinates: the field equation is $\gamma^\mu \text{D}_\mu \text{Psi} == H V'(s) \text{Psi}$, FIELD EQUATION [THE RESULT]: $\text{EL_Psi} == -(\text{Sqrt}[g]/H)((\text{D}_\mu \text{Psibar}) \gamma^\mu + H V'(s) \text{Psibar})$: the adjoint equation is $(\text{D}_\mu \text{Psibar}) \gamma^\mu == -H V'(s) \text{Psibar}$. For the real field of Part VI the same equation is [proved VI §24] “FABLE [THE RESULT]: $\text{EL} == \text{Sqrt}[g] (2/H) \text{sigma16}$. ($\gamma^\mu \text{D}_\mu \text{Psi} - H V'(s) \text{Psi}$) : the field equation of fable is EXACTLY the Levi-Civita covariant Dirac equation $\gamma^\mu \text{D}_\mu \text{Psi} == H V'(s) \text{Psi}$ ”.

On the canonical frame the connection enters as follows [derived here], from (64). In the equation it is $\gamma^\mu \Gamma_\mu \Psi = -3H \cot^2(6Hx_0) T_{16}[0] \Psi$. In the conjugate it is $-\bar{\Psi} \Gamma_\mu \gamma^\mu = +\bar{\Psi} \gamma^\mu \Gamma_\mu = -3H \cot^2(6Hx_0) \bar{\Psi} T_{16}[0]$, using the per- μ anticommutator. Because $T_{16}[0] = \begin{pmatrix} 0 & \text{ID8} \\ \text{ID8} & 0 \end{pmatrix}$, in the 8+8 split-octonion form $\Psi = (\psi_1, \psi_2)$:

$$e_a^\mu \bar{\tau}_a (\partial_\mu + \Gamma_\mu^{(2)}) \psi_2 = H V'(s) \psi_1 \quad (\text{upper 8 rows}), \quad (81)$$

$$e_a^\mu \tau_a (\partial_\mu + \Gamma_\mu^{(1)}) \psi_1 = H V'(s) \psi_2 \quad (\text{lower 8 rows}), \quad (82)$$

where the connection term is $-3H \cot^2(6Hx_0) \psi_2$ in the upper rows and $-3H \cot^2(6Hx_0) \psi_1$ in the lower rows, and $\tau_a = \text{tau}[a]$, $\bar{\tau}_a = \text{taubar}[a]$, $s = -\psi_1^\dagger \sigma \psi_1 + \psi_2^\dagger \sigma \psi_2$, $\bar{\psi}_1 = -\psi_1^\dagger \sigma$ and $\bar{\psi}_2 = \psi_2^\dagger \sigma$. The conjugate equations are $(\partial \bar{\psi}_2 - \bar{\psi}_2 \Gamma^{(2)}) \tau_a e_a^\mu = -H V' \bar{\psi}_1$ and $(\partial \bar{\psi}_1 - \bar{\psi}_1 \Gamma^{(1)}) \bar{\tau}_a e_a^\mu = -H V' \bar{\psi}_2$. [proved VII §27] WRITTEN OUT (ii) [THE RESULT]: the split-octonion 8+8 form - the type-1 rows are the taubar equations for psi2, the type-2 rows the tau equations for psi1 ($\text{tau}[0] == \text{taubar}[0] == \text{ID8}$), WRITTEN OUT (ii) [THE RESULT]: the covariant 8+8 form - rows 1..8 are $e_a^\mu \text{taubar}[a] (\text{d}_\mu + \Gamma_\mu^{(2)}) \text{psi2} - H V' \text{psi1}$, rows 9..16 are $e_a^\mu \text{tau}[a] (\text{d}_\mu + \Gamma_\mu^{(1)}) \text{psi1} - H V' \text{psi2}$; and the conjugate rows, WRITTEN OUT (ii) [has content]: with $\text{Psibar} = \text{Psi}^\dagger \text{sigma16}$ and $\text{sigma16} = \text{diag}(-\text{sigma}, \text{sigma})$: $s == -\text{psi1}^\dagger \text{sigma psi1} + \text{psi2}^\dagger \text{sigma psi2}$ ($\text{psibar1} = -\text{psi1}^\dagger \text{sigma}$, $\text{psibar2} = \text{psi2}^\dagger \text{sigma}$).

The rescaling. The rescaling of Part V applies unchanged, because it is a statement about the operator. $\Psi = \sqrt{\sin 6Hx_0} \Psi'$ turns the equation into $\sqrt{\sin 6Hx_0} (\gamma^\mu \partial_\mu \Psi' - H V'(s) \Psi') = 0$ with $s = \sin(6Hx_0) \bar{\Psi}' \Psi'$. The connection term is gone. [proved VII §27] WRITTEN OUT (iv) [THE RESULT]: $\text{Psi} = \text{Sqrt}[\text{Sin}[6Hx0]] \text{Psi}'$ turns the field equation into $\text{Sqrt}[\text{Sin}] (\text{Cot } T_{16}[0] \text{d}_0 \text{Psi}' + (1/q) T_{16}[i] \text{d}_i \text{Psi}' + T_{16}[4] \text{d}_4 \text{Psi}' + (1/p) T_{16}[h] \text{d}_h \text{Psi}' - H V'(\text{Sin } s') \text{Psi}')$: NO connection term, WRITTEN OUT (iv) [THE RESULT]: and the adjoint equation into $\text{Sqrt}[\text{Sin}] (\text{Cot } \text{d}_0 \text{Psi}' \text{bar} T_{16}[0] + \dots + H V'(\text{Sin } s') \text{Psi}' \text{bar})$, with no connection term either. The x_4 -only family $\Psi = \sqrt{\sin 6Hx_0} \Psi'(x_4)$ solves the equation if and only if $V'' s' = 0$. That holds for linear V (the author's mass term), or for null data $s' = 0$. For the complex field s' is conserved on the x_4 -only equation, so null data give x_4 -only solutions for any V . [proved VII §29] PRE-UNIVERSE (2) [THE RESULT]: $\text{Psi} = \text{Sqrt}[\text{Sin}] \text{Psi}'(x_4)$ satisfies the field equation iff $\text{d}_x \text{V}'(\text{Sin}[6Hx0] s') == 0$, and $\text{d}_x \text{V}'(\text{Sin } s') == 6 H \text{Cos}[6 H x0] s' V''(\text{Sin } s')$: V linear OR $s' == 0$, PRE-UNIVERSE (2) [has content]: $s' = \text{Psi}' \text{bar} \text{Psi}'$ is conserved by the x_4 -only equations ($\text{d}_4 \text{Psi}' = -H V' T_{16}[4] \text{Psi}'$, $\text{d}_4 \text{Psi}' \text{bar} = H V' \text{Psi}' \text{bar} T_{16}[4]$) for ANY V ; so null data $s' = 0$ are solutions for any V , with $\rho == V(0)$; and for $V = s'^2$ with $s' != 0$ the x_0 -dependence is real (witness). The real-field version is [proved VI §24] “FABLE [has content]: for the mass term $V = -(2M/H) s$, V' is constant and $\text{Psi}'(x_4)$ IS an exact solution”.

The frame-independent form. Covariant constancy of γ^μ (61) together with a vanishing $\sum_\mu \{\gamma^\mu, \Gamma_\mu\}$ gives, for any frame on which that anticommutator vanishes,

$$\gamma^\mu \Gamma_\mu = \frac{1}{2} \frac{1}{\sqrt{g}} \partial_\mu (\sqrt{g} \gamma^\mu) = -\frac{1}{2} T_\mu \gamma^\mu, \quad T_\mu = \partial_\mu \ln \frac{e_\mu^\mu}{\det e} \quad (\text{no sum}), \quad (83)$$

where T_μ is the Weitzenböck torsion vector of a diagonal frame (71). The second equality is a one-line computation for a diagonal frame, since $\sqrt{g} = \det e$ and $\gamma^\mu = T_{16}[\mu]/e_\mu^\mu$ [derived here]. This links the canonical connection directly to the fable-5.1 bridge: the connection term of the Dirac operator is the Weitzenböck torsion vector. On the canonical frame $T_\mu = \partial_\mu \ln \sin 6Hx_0$ (subsection 6.10), and the rescaling by $\exp(\frac{1}{2} \ln \sin) = \sqrt{\sin 6Hx_0}$ removes it. [proved VII §27] SPIN CONNECTION [has content]: Part V restated - $\omega^\wedge W == 0$ in the canonical frame, $\omega^\wedge LC == -\text{contortion}$, $T_\mu == d_\mu \text{Log}[\text{Sin}[6 H x_0]]$, $R == -T + B$, and $\gamma^\mu \Gamma_\mu == -(1/2) T_\mu \gamma^\mu$ (on the canonical frame); and on the time-dependent warped frames of the coupled system, Part VIII, Section 32: WEITZENBOECK [THE RESULT]: the torsion vector of the warped frame is a GRADIENT, $T_\mu == d_\mu \text{Log}[\text{Sin}[6 H x_0]/(A^3 B^3 C)]$; on the canonical member it is Part V's torsionVectorFable51, WEITZENBOECK [THE RESULT]: $\gamma^\mu \Gamma_\mu == -(1/2) T_\mu \gamma^\mu$ on the warped frame - one vector term, a gradient.

6.12.4 It enters neither the canonical anticommutator nor the symmetrized Hamiltonian density

The admissible theory is a dimensional reduction $\Psi = \Psi(x_0, x_1, x_2, x_3, x_4)$. It uses a finite hidden coordinate volume $V_{\text{hid}} = \int dx_5 dx_6 dx_7$, which is an assumption: compact timelike directions contain closed timelike curves. Time is x_4 . The canonical anticommutator on a slice $x_4 = y_4$ is

$$\{\Psi_a(x), \Psi_b^\dagger(y)\} = \frac{H}{V_{\text{hid}} \sqrt{|g|}} G_{ab} \delta^4(x_{0..3} - y_{0..3}), \quad G = -i \sigma_{16} \gamma^4. \quad (84)$$

[proved VII §28] DIRAC BRACKET [THE RESULT]: $\{\Theta_i, \Theta_j\} = (\Omega^{-1})_{ij}$ gives $\{\Psi_a, \Psi_b^\dagger\} == i (K^{-1})_{ab} == (G^{-1})_{ab}$ and $\{\Psi_a, \Psi_b\} == 0$, ANTICOMMUTATOR [THE RESULT]: on the canonical frame ($c = \sqrt{g}/H$) $\{\Psi, \Psi^\dagger\} == (H/\sqrt{g}) J == H \cos[6 H x_0] J$ ($J^{-1} == J$), which IS the design's $(H/\sqrt{g})(-i \sigma_{16} \gamma^4)$. No spin connection appears in it. It is fixed by the coefficient of $\partial_4 \Psi$ in the Lagrangian, and $\Gamma_4 = 0$ in any case. On the canonical frame $\gamma^4 = T_{16}[4]$, so $G = -i T_{16}[0] T_{16}[1] T_{16}[2] T_{16}[3] T_{16}[4] = J$ [derived here]. J is the fundamental symmetry of subsection 6.12.6.

The Hamiltonian density of the symmetrized Lagrangian is

$$\mathcal{H} = \sqrt{g} \left[-\frac{1}{2H} \sum_{k \neq 4} (\bar{\Psi} \gamma^k \partial_k \Psi - \partial_k \bar{\Psi} \gamma^k \Psi) + V(s) \right], \quad (85)$$

and it contains *no spin connection* [proved VII §28] HAMILTONIAN [THE RESULT]: the Hamiltonian density $\Pi_\Psi d_4 \Psi + d_4 \bar{\Psi} \Pi_\Psi - L$ of the symmetrized Lagrangian is $\sqrt{g} [-(1/(2H)) \sum_{k \neq 4} (\bar{\Psi} \gamma^k d_k \Psi - d_k \bar{\Psi} \gamma^k \Psi) + V]$: NO spin connection (fields of all eight coordinates). This follows from subsection 6.12.2: the symmetrized $\hat{L}$ does not depend on the connection. The connection re-emerges in the equation of motion. Integrating the symmetrized spatial derivative terms by parts produces $\gamma^\mu \Gamma_\mu = \frac{1}{2} \frac{1}{\sqrt{g}} \partial_\mu (\sqrt{g} \gamma^\mu)$, because the volume factor and the curved gammas depend on x_0 . [proved VII §28] HAMILTONIAN [THE RESULT]: $(\sqrt{g}/H) \sigma_{16} (\gamma^\mu d_\mu \Psi - H V' \Psi) == i G d_4 \Psi - h \Psi$ for admissible $\Psi(x_{0..4})$: the field equation is $i G d_4 \Psi = h \Psi$, HAMILTONIAN [THE RESULT]: $(1/2) \sum_{\mu \neq 4} d_\mu (\sqrt{g} \sigma_{16} \gamma^\mu) == \sqrt{g} \sigma_{16} \sum \gamma^\mu \Gamma_\mu$, and $d_4 (\sqrt{g} \sigma_{16} \gamma^4) == 0$: the connection term is the one that makes h Hermitian.

The same rescaling organizes the one-particle space. $\sqrt{g} dx_0 |\Psi|^2 = |\Psi'|^2 dz$ for $\Psi' = \Psi/\sqrt{\sin 6Hx_0}$, where $z = -\ln \cos(6Hx_0)/(6H)$ is proper distance along x_0 . So the one-particle space of Ψ' is $L^2(dz)$. The x_0 -operator is regular at the wall $z = 0$ and needs a boundary

condition there. The Lorentz-covariant, J -compatible ones are $T_{16}[0]\Psi' = \pm\Psi'$ at $z = 0$. It is limit-point at $z \rightarrow \infty$, where no condition is needed. [proved VII §28] BOUNDARY [has content]: the x_0 direction in proper distance $z = -\text{Log}[\text{Cos}[6 H x_0]]/(6 H)$: $dz/dx_0 == \text{Tan}[6Hx_0] == \text{Sec Sin (so Sqrt[g] dx_0 |Psi|^2 = |Psi'|^2 dz)}$, $z(0) == 0$ and $z \rightarrow \infty$ at $6 H x_0 \rightarrow \pi/2$, $\gamma_0 d_0 == T_{16}[0] d_z$, and $\text{Sin}[6Hx_0] == \text{Sqrt}[1 - \text{Exp}[-12 H z]]$ (bounded frame factors), BOUNDARY [THE RESULT]: the bag condition $T_{16}[0] \Psi' = \pm\Psi'$ at the wall - $T_{16}[0]^2 == 1$, $\{\beta, T_{16}[0]\} == 0$, the normal-flux form vanishes on both eigenspaces ($P \beta T_{16}[0] P == 0$), and $[J, T_{16}[0]] == 0$.

6.12.5 It does not drop out of the energy-momentum tensor

The energy-momentum tensor operator is

$$\hat{T}_{\mu\nu} = -\frac{1}{4H} \left[\bar{\Psi} \gamma_\mu D_\nu \Psi - (D_\nu \bar{\Psi}) \gamma_\mu \Psi + (\mu \leftrightarrow \nu) \right] + g_{\mu\nu} \hat{L}, \quad \gamma_\mu = g_{\mu\nu} \gamma^\nu. \quad (86)$$

This is the Hilbert (symmetric-tetrad) tensor of the symmetrized action, and it equals the covariant tensor T_{cov} off shell. It is *built with D , not ∂* . The difference from the tensor T_{par} built with partial derivatives is [derived here]

$$T_{\mu\nu}^{\text{cov}} - T_{\mu\nu}^{\text{par}} = -\frac{1}{4H} \bar{\Psi} (\{\gamma_\mu, \Gamma_\nu\} + \{\gamma_\nu, \Gamma_\mu\}) \Psi. \quad (87)$$

The $g_{\mu\nu} \hat{L}$ terms agree by subsection 6.12.2. For $\mu = \nu$ the difference is $-\frac{1}{2H} \bar{\Psi} \{\gamma_\mu, \Gamma_\mu\} \Psi = 0$ by the per- μ anticommutator, because $\gamma_\mu = g_{\mu\mu} \gamma^\mu$ on the diagonal frame. So the energy density and all the pressures are the same with D or ∂ . Not every off-diagonal component differs. The $(0, 4)$ component coincides, because $\Gamma_0 = \Gamma_4 = 0$. The $(5, 6)$ component coincides too, because there the two anticommutators cancel [derived here]: $\gamma_5 = -p T_{16}[5]$, $\gamma_6 = -p T_{16}[6]$, the coefficients of Γ_5 and Γ_6 are equal, and $T_{16}[5] T_{16}[a] T_{16}[6] = -T_{16}[6] T_{16}[a] T_{16}[5]$ for $a = 0, 4$. The pair at which the notebook shows the two tensors to differ is given in the third item below.

- [proved VII §29] Hilbert = T_{cov} , via a first-order symmetric perturbation of the frame, EMT [THE RESULT]: the Hilbert (symmetric-tetrad) tensor of the symmetrized action IS That, OFF shell - components (x_1, x_1) , (x_4, x_4) , (x_1, x_4) , (x_0, x_5) , (x_5, x_4) .
- [proved VII §29] the piece coming from the variation of the spin connection is zero, EMT [has content]: the spin-connection variation contributes NOTHING - dropping it leaves the Hilbert tensor unchanged at (x_1, x_4) , (x_0, x_5) , (x_5, x_4) . The Lagrangian depends on the connection only through $\sum_\mu \{\gamma^\mu, \Gamma_\mu\} = \frac{1}{2} \omega_{[cab]} \gamma^{cab}$ (65), and under a symmetric variation of the tetrad $\delta\omega_{[cab]} = 0$. So the spin-connection variation contributes *exactly nothing*. The covariant terms of $\hat{T}$ come from the dependence of $\{\gamma^\mu, \Gamma_\mu\}$ on the frame. A partial-derivative Lagrangian must never be varied with respect to the frame. (Part VI's Text cell described this contribution as "a term antisymmetric in μ, ν that drops from the symmetrised tensor". The statement above is the sharper one.)
- [proved VII §29] Hilbert $\neq T_{\text{par}}$, witnessed on the non-vacuous pair (x_1, x_4) (the pairs $(0, 4)$ and $(5, 6)$ coincide on the canonical frame, as shown above), EMT [control]: the Hilbert tensor is NOT the partial-derivative tensor: at (x_1, x_4) they differ - witnessed on constant spinors ($\text{Psibar} = e_1$, $\Psi = e_{11}$: $(T_{16}[4] T_{16}[1] T_{16}[0])_{\{1, 11\}} \neq 0$).
- [proved VII §29] T_{cov} is conserved on shell, EMT CONSERVATION [THE RESULT]: $\text{nabla}^\mu \text{That}_{\{\mu \nu\}} == 0$ ON SHELL in all eight components, for generic $\Psi(x_0, x_4)$, $\text{Psibar}(x_0, x_4)$ and generic V .

The real-field analogues are already proved. [proved VI §24] “FABLE [definition]: T_{cov} is symmetric by construction”. [proved VI §24] “FABLE [has content]: the diagonal of T_{cov} equals the diagonal of the partial-derivative tensor, for the generic $\Psi(x_0, x_4)$: ρ and every pressure are the same in both”. [proved VI §24] “FABLE [has content]: but the two tensors DIFFER off the diagonal - witnessed on the (x_3, x_4) component $T[[4,5]]$, a momentum density along the observed sheet, for the constant spinor $e_1 + e_{13}$ and $V = s^2$ ”. [proved VI §24] “FABLE [THE RESULT]: $\nabla^\mu T_{\mu\nu} == 0$ ON SHELL in all eight components, for a generic $\Psi(x_0, x_4)$ - the covariant tensor is the conserved one”. That only the covariant tensor is conserved, and T_{par} is not, is [prose VI §24]: the divergence of T_{par} is not computed symbolically.

6.12.6 The connection and the fundamental symmetry J

$$J = -i T_{16}[0] T_{16}[1] T_{16}[2] T_{16}[3] T_{16}[4], \quad J^2 = \text{ID16}. \quad (88)$$

J is the fundamental symmetry that gives the positive quantization of the admissible sector. With $\beta = -i T_{16}[4]$, $\sigma_{16} = J\beta$ holds, and the Hilbert adjoint is $\Psi^\dagger = \Psi^\dagger J$. [proved VII §28] J [THE RESULT]: $\text{sigma16} == J \text{ beta}$ with $\text{beta} = -i T_{16}[4]$ Hermitian, $\text{beta}^2 == 1$; $T_{16}[0..3]$ Hermitian with $\text{beta } T_{16}[j] \text{ beta} == -T_{16}[j]$: beta is the observer’s Dirac adjoint, J [THE RESULT]: with $\Psi^\dagger := \Psi^\dagger J$: $\text{Psibar} = \Psi^\dagger \text{beta}$ ($J \text{ sigma16} == \text{beta}$), $\Psi^\dagger G = \Psi^\dagger$ ($J G == \text{ID16}$), $\{\Psi, \Psi^\dagger\} == (H/\text{Sqrt}[g]) \text{ID16}$ ($G^{-1} J == \text{ID16}$): POSITIVE.

$J^2 = \text{ID16}$ follows directly [derived here]. The product $P = T_{16}[0] \cdots T_{16}[4]$ of five mutually anticommuting matrices satisfies $P^2 = (-1)^{5 \cdot 4/2} T_{16}[0]^2 \cdots T_{16}[4]^2 = (+1)(1)(1)(1)(-1) = -\text{ID16}$, so $(-iP)^2 = +\text{ID16}$.

The commutation rules also follow by counting sign changes [derived here]. A single $T_{16}[a]$ anticommutes with the four factors of P other than itself and commutes with itself, if $a \in \{0..4\}$. It anticommutes with all five if $a \in \{5, 6, 7\}$. Hence J commutes with $T_{16}[0..4]$ and anticommutes with $T_{16}[5], T_{16}[6], T_{16}[7]$. Applying this to the explicit Γ_μ of subsection 6.7:

object	relation to J	reason
$\Gamma_0 = \Gamma_4 = 0$	commute	zero
Γ_j ($j = 1, 2, 3$)	commute	made of $T_{16}[0]T_{16}[j]$, $T_{16}[4]T_{16}[j]$, all factors in $\{0..4\}$
Γ_k ($k = 5, 6, 7$)	anticommute	made of $T_{16}[0]T_{16}[k]$, $T_{16}[4]T_{16}[k]$: one factor in $\{0..4\}$, one in $\{5, 6, 7\}$
each $\gamma^\mu \Gamma_\mu$ (no sum), and their sum	commute	each is $\alpha_\mu T_{16}[0] + \beta_\mu T_{16}[4]$ (subsection 6.8)

The hidden components Γ_5, Γ_6 and Γ_7 are J -odd. They enter the Dirac operator only through $\gamma^k \Gamma_k$, a product of two J -odd matrices, which is J -even. So every term of the admissible-sector Hamiltonian, the connection term included, is J -even. The J -odd objects, namely the $(0, h)$, (i, h) and $(4, h)$ components of T_{cov} and the hidden momenta P_h ($h = 5, 6, 7$), are nonzero operators with zero expectation in J -eigenmode Fock states. [proved VII §28] J TABLE [THE RESULT]: J commutes with sigma16 , $T_{16}[0..4]$, $\text{sigma16 } \gamma^\mu$ ($\mu = 0..4$), every $\gamma^\mu \Gamma_\mu$ and $\Gamma_{1..3}$; it ANTICommutes with $T_{16}[5..7]$, $\text{sigma16 } \gamma^h$ ($h = 5..7$), $\Gamma_{5..7}$, $T_{16}[8]$ and $C+$. The same J table holds on the dynamical (warped and Bianchi-I) frames of the coupled system. Part VIII, Section 32, proves it: J [THE RESULT]: $[J, \gamma^\mu \Gamma_\mu] == 0$ for EACH μ separately ($h = 5, 6, 7$ included), on the warped and on the Bianchi-I frame, J [THE RESULT]: $[J, \Gamma_\mu] == 0$ for $\mu = 0..4$ and $\{J, \Gamma_h\} == 0$ for $h = 5, 6, 7$, on the warped and on the Bianchi-I frame and J [has content]: the anticommuting Γ_h are not zero

(witnessed on $A = 1 + x^4/2$, $B = 1 + x^4/3$, $C = 1 + x^4/2/3$), so $\{J, \text{Gamma}_h\} == 0$ is not vacuous.

6.13 Status ledger

statement	status	where (notebook Part and Section; assertion tag)
frame e , $g = e \eta e^T$, stated line element, signature (4,4)	proved	III §15
a_4 is free and never given a value	proved	VI §25, VII §29 and VIII (controls); notice printed in III §15
$\det g = \sec^2 6Hx_0$; $\sqrt{ \det g } = \sec 6Hx_0$ on $0 < 6Hx_0 < \pi/2$	displayed; proved	displayed III §15; proved V §21
37 Christoffel symbols; symmetric in the lower indices; metric compatible	displayed; proved	III §16; VI §23
the vielbein postulate is solved by <code>cfSpinConnection</code> (residual zero)	proved (solver regression)	III §16
the 24 non-zero components of <code>omegaCanonical</code>	displayed; independent frame-only derivation proved; count 24 proved	III §16; VII §27 (SPIN CONNECTION [has content])
antisymmetry in the flat indices (metric compatibility)	proved (has content)	III §16; VII §27
torsion-free	proved (structural)	III §16
curvature: 156 components; Ricci scalar	displayed	III §16
$\Gamma_\mu = \frac{1}{8}\omega_{\mu ab}[T_{16}[a], T_{16}[b]]$; sign and coefficient fixed by compatibility	proved (with control)	III §17; VII §27 (SPIN CONNECTION [definition])
block-diagonal; type-1 and type-2 blocks; <code>Dcov16</code> = direct sum	proved	III §17
covariant constancy of γ^μ	proved	III §17
$\frac{1}{\sqrt{g}}\partial_\mu(\sqrt{g}\gamma^\mu) = [\gamma^\mu, \Gamma_\mu]$	proved	VI §24; VII §27 (FIELD EQUATION [has content])
$\sum_\mu \{\gamma^\mu, \Gamma_\mu\} = 0$; $\gamma^\mu \Gamma_\mu = -3H \cot^2(6Hx_0) T_{16}[0]$	proved	VI §24; VII §27
per-direction $\gamma^\mu \Gamma_\mu = \alpha_\mu T_{16}[0] + \beta_\mu T_{16}[4]$,	derived here;	VII §27 (SPIN CONNECTION [THE RESULT], LAGRANGIAN (c) [THE RESULT])
$\sum \alpha_\mu = -3H \cot^2(6Hx_0)$, $\sum \beta_\mu = 0$;	proved	
anticommutator zero for each μ		
Bridges 1 and 2: the same connection in a local / constant gauge, at vector and spinor level	proved	IV §18, IV §19; restated VII §27
Bridge 3: a different connection; torsion $-2\lambda m^{\text{Skew}}$;	proved	IV §20; restated VII §27
curvature of degree 2 in λ ; Ricci shift $-42\lambda^2 = -\frac{1}{4}T \cdot T$		
fable-5.1: $\omega = 0$, zero curvature, $\omega^{LC} = -\text{contortion}(T)$, $T_\mu = \partial_\mu \ln \sin 6Hx_0$, axial part zero, $R = -T + B$	proved	V §21; restated VII §27
$\gamma^\mu \Gamma_\mu = -\frac{1}{2}T_\mu \gamma^\mu$; the rescaling $\Psi = \sqrt{\sin 6Hx_0} \Psi'$	proved	V §21; VI §24; VII §27 (SPIN CONNECTION [has content], WRITTEN OUT (iv)); VIII §32 on the warped frames (WEITZENBOECK, RESCALING)
Bridge 1's gauge term is the fable-5.1 connection of the boosted frame	proved	V §21
master table of the five connections	displayed	V §22
complex fable: the connection drops out of the symmetrized L because the matrix $\{\gamma^\mu, \Gamma_\mu\}$ vanishes	proved	VII §27 (LAGRANGIAN (c) [THE RESULT])
unsymmetrized covariant L = symmetrized + divergence; unsymmetrized partial L inadmissible	proved	VII §27 (LAGRANGIAN (d), (e), (f))
field equations; connection term	proved	VII §27 (FIELD EQUATION, WRITTEN OUT (i)-(iv), SPIN CONNECTION); VIII §32
$-3H \cot^2(6Hx_0) T_{16}[0]$; 8+8 form; rescaling; frame-independent form		

statement	status	where (notebook Part and Section; assertion tag)
no connection in the canonical anticommutator or in the symmetrized Hamiltonian density; the connection makes h Hermitian	proved	VII §28 (QUANTIZATION [has content]: <code>Gamma_4 == 0</code> , ANTICOMMUTATOR, HAMILTONIAN)
EMT: $\text{Hilbert} = T_{\text{cov}}$; spin-connection variation zero; $\text{Hilbert} \neq T_{\text{par}}$; T_{cov} conserved	proved	VII §29 (EMT, EMT CONSERVATION)
J commutes with each $\gamma^\mu \Gamma_\mu$ and with $\Gamma_0.. \Gamma_4$, and anticommutes with $\Gamma_5.. \Gamma_7$	derived here; proved	VII §28 (J TABLE [THE RESULT]); VIII §32 on the warped and Bianchi-I frames (J [THE RESULT])

6.14 The interacting system: the same solver on two new frames

The spin connection of the frames on which fable acts back (section 5) is computed exactly as every connection of Parts III–VI was: the vielbein postulate of subsection 6.5, solved by `cfSpinConnection` with the Christoffel symbols of each frame, lowered with `eta4488` by `cfLowerFirstFlat`, checked by the postulate residual (a regression test of the solver) and by antisymmetry (metric compatibility, which has content), and turned into the 16×16 matrices $\Gamma_\mu = \frac{1}{8} \omega_{\mu ab} [T_{16}[a], T_{16}[b]]$ by Section 17's `cfSpinMatrix` [**prose VIII §32**]. This is Section 32 of the notebook (Input cells 208–210, 26 assertions, 3.893 s). The two frames are the warped frame of section 5.1,

$$e_{\text{warped}} = \text{diag}(\tan(6Hx_0) C, A S^{-1/6}, A S^{-1/6}, A S^{-1/6}, 1, B S^{-1/6}, B S^{-1/6}, B S^{-1/6}),$$

$$S = \sin 6Hx_0,$$

and the 8-dimensional Bianchi-I frame $e_{\text{BI}} = \text{diag}(C, A, A, A, 1, B, B, B)$ of section 5.4 (the connections took 0.550 s and 0.355 s). [**proved VIII §32**]

- SPIN CONNECTION [solver regression]: the vielbein-postulate residual is zero on the warped and on the Bianchi-I frame
- SPIN CONNECTION [has content]: `omega_mu[a,b] == -omega_mu[b,a]` on both frames (metric compatibility)

6.15 Every non-zero component of the connection of the interacting system

On the **warped** frame, with $c = \cos 6Hx_0$, $S = \sin 6Hx_0$ and $' = d/dx_4$, the complete list is thirteen independent components and their antisymmetric partners:

component (both flat indices down, as the array the notebook compares)	value
$\omega[0; 0, 4] = -\omega[0; 4, 0]$	$\tan(6Hx_0) C'$
$\omega[i; 0, i] = -\omega[i; i, 0], i = 1, 2, 3$	$H c^2 A / (C S^{13/6})$
$\omega[i; i, 4] = -\omega[i; 4, i], i = 1, 2, 3$	$A' / S^{1/6}$
$\omega[h; 0, h] = -\omega[h; h, 0], h = 5, 6, 7$	$-H c^2 B / (C S^{13/6})$
$\omega[h; 4, h] = -\omega[h; h, 4], h = 5, 6, 7$	$B' / S^{1/6}$

$1 + 3 + 3 + 3 + 3 = 13$ independent components, 26 non-zero entries of the 512. The first is **new**: the canonical frame ($C = 1$) has no connection along x_0 at all. On the **Bianchi-I** frame only seven survive:

$$\omega[0; 0, 4] = C', \quad \omega[i; i, 4] = A' \quad (i = 1, 2, 3), \quad \omega[h; 4, h] = B' \quad (h = 5, 6, 7).$$

[**proved VIII §32**]

- SPIN CONNECTION [THE RESULT]: the complete list of non-zero $\omega_{\mu ab}$ (both flat indices down) of the WARPED frame is the thirteen stated components (and their antisymmetric partners) - nothing else
- SPIN CONNECTION [THE RESULT]: the complete list for the Bianchi-I frame is $\omega_{\mu\{0\}4} = C'$, $\omega_{\mu\{i\}4} = A'$, $\omega_{\mu\{h\}4} = B'$ (both flat indices down) - nothing else
- SPIN CONNECTION [fidelity]: on the canonical member (labelled substitution) the warped connection IS $\omega_{\mu ab}$ of Section 16, and the new component $\omega_{\mu\{0\}4}$ vanishes there

Index position. The table and the Bianchi-I list are the arrays the assertions compare, `cfOmegaWarp = cfLowerFirstFlat[...]` and `cfOmegaBI`, whose two flat indices are both **down**, $\omega_{\mu ab}$, lowered with `eta4488` [file: `claude-fable/cells_part8.wl`, the lists `cfOmegaWarpList`, `cfOmegaBIList` and the assertions that use them]; $\Gamma_\mu = \frac{1}{8} \omega_{\mu ab} [T_{16}[a], T_{16}[b]]$ uses them in that position, with $T_{16}[a] = \gamma^a$ [**prose VIII §32**]. Until commit 3a5020d the notebook's Text cell and the three labels above wrote these components with the flat indices **up**, as ω_μ^{ab} ($\omega_{\mu\{0\}4}$, $\omega_{\mu\{i\}4}$, $\omega_{\mu\{h\}4}$); that commit corrected the Text cell and the labels to $\omega_{\mu ab}$ ($\omega_{\mu\{0\}4}$, $\omega_{\mu\{i\}4}$, $\omega_{\mu\{h\}4}$, both flat indices down) and changed no computation [**commit 3a5020d**]. The labels quoted above are the corrected ones, from the final run log; Part VIII's acceptance run `claude-fable/run_fermion_fable_part8.log` still prints the old ones. The position matters for the sign: raising both flat indices with $\eta_{4488} = \text{diag}(1, 1, 1, 1, -1, -1, -1, -1)$ multiplies the (a, b) component by $\eta_{aa}\eta_{bb}$, which leaves the $(0, i)$ and $(4, h)$ planes unchanged and flips the sign of the $(0, 4)$, $(i, 4)$ and $(0, h)$ planes [derived here: exactly one index of each of those three planes is timelike]. (The corrected Text cell says that raising both indices "would flip the sign of every plane that contains x4 or a hidden timelike direction"; for the $(4, h)$ plane, which contains both, the two flips cancel, as just derived.) **Recovery** [derived here, and asserted by the fidelity label above]: at $A = e^{-a_4}$, $B = e^{+a_4}$, $C = 1$, $A' = -Ha'_4 e^{-a_4}$ and $B' = +Ha'_4 e^{a_4}$, so $\omega[i; 4, i] = Ha'_4 / (e^{\mathcal{A}} s^{1/6})$ and $\omega[h; 4, h] = e^{\mathcal{A}} Ha'_4 / s^{1/6}$ with $\mathcal{A} = a_4(Hx_4)$: the 24 non-zero components of subsection 6.6.

6.16 The spinor connection of the interacting system

With $\Gamma_\mu = \frac{1}{2} \sum_{a < b} \omega_{\mu ab} T_{16}[a] T_{16}[b]$ (subsection 6.7), the table gives [**derived here**]:

$$\begin{aligned} \Gamma_0 &= \frac{1}{2} \tan(6Hx_0) C' T_{16}[0] T_{16}[4], \\ \Gamma_i &= \frac{1}{2} \left[\frac{H c^2 A}{C S^{13/6}} T_{16}[0] T_{16}[i] + \frac{A'}{S^{1/6}} T_{16}[i] T_{16}[4] \right] & i = 1, 2, 3, \\ \Gamma_4 &= 0, \\ \Gamma_h &= \frac{1}{2} \left[-\frac{H c^2 B}{C S^{13/6}} T_{16}[0] T_{16}[h] + \frac{B'}{S^{1/6}} T_{16}[4] T_{16}[h] \right] & h = 5, 6, 7. \end{aligned}$$

The first and last lines are asserted: $\Gamma_{x_4} = 0$, and Γ_{x_0} is not zero as soon as the x_0 direction is dynamical. Γ_μ rotates the Dirac matrices as ω rotates a vector, and the curved γ^μ are covariantly constant on the warped frame. [**proved VIII §32**]

- DIRAC [THE RESULT]: `Gamma_x4 == 0`, and `Gamma_x0 == (1/2) Tan[6 H x0] C' (x4) T16[0].T16[4]`, which is NOT zero when C varies (witnessed on `C = 1 + x4^2/3`)
- SPIN CONNECTION [has content]: `Gamma_mu` rotates the Dirac matrices as `omega` rotates a vector, `[Gamma_mu, T16[a]] == -omega_mu^a_b T16[b]`, on the warped frame

- **SPIN CONNECTION** [has content]: the curved γ^μ are covariantly constant on the warped frame, $D_\mu \gamma^\nu = 0$

6.17 What the connection of the interacting system does to the Dirac operator

The whole connection enters the Dirac operator as **one** matrix:

$$\begin{aligned}\gamma^\mu \Gamma_\mu &= -\frac{3H \cot^2(6Hx_0)}{C} T_{16}[0] + \frac{1}{2}(3H_A + 3H_B + H_C) \gamma^4, & \gamma^4 &= T_{16}[4] \quad (\text{warped frame}), \\ \gamma^\mu \Gamma_\mu &= \frac{1}{2}(3H_A + 3H_B + H_C) T_{16}[4] & & (\text{Bianchi-I frame}).\end{aligned}$$

The first term is the canonical term of subsection 6.8 divided by C ; the second is the Hubble (dilution) term of an expanding 7-volume — with the **plus** sign; the opposite sign fails. And $\{\gamma_\mu, \Gamma_\mu\} = 0$ for **each** μ separately on both frames, so the symmetrized Lagrangian of section 3.2 is the same with D_μ as with ∂_μ there, for fields of all eight coordinates and every V . [**proved VIII §32**]

- **DIRAC** [THE RESULT]: $\gamma^\mu \Gamma_\mu = -(3H \cot^2/C) T_{16}[0] + (1/2)(3H_A + 3H_B + H_C) \gamma^4$ on the warped frame, and $\gamma^4 = T_{16}[4]$
- **DIRAC** [control]: the opposite sign of the Hubble term FAILS (witnessed on $A = 1 + x^4/2$, $B = 1 + x^4/3$, $C = 1 + x^4/2/3$)
- **DIRAC** [THE RESULT]: on the Bianchi-I frame $\gamma^\mu \Gamma_\mu = (1/2)(3H_A + 3H_B + H_C) T_{16}[4]$: the pure Hubble term of an expanding 7-volume
- **DIRAC** [THE RESULT]: $\{\gamma_\mu, \Gamma_\mu\} = 0$ for EACH μ separately, on the warped and on the Bianchi-I frame
- **DIRAC** [THE RESULT]: hence Section 26's symmetrized Lagrangian is the same with D_μ as with ∂_μ on the warped frame, for Ψ_i , $\bar{\Psi}_i$ of all eight coordinates (cfPsiC8 , cfPsiB8) and every V

Direction by direction [derived here, from subsection 6.16 and the coframe $\gamma^0 = \cot T_{16}[0]/C$, $\gamma^i = S^{1/6} T_{16}[i]/A$, $\gamma^4 = T_{16}[4]$, $\gamma^h = S^{1/6} T_{16}[h]/B$, using $T_{16}[i]^2 = +1$, $T_{16}[h]^2 = -1$ and the anticommutation of distinct T_{16}]:

μ	$\gamma^\mu \Gamma_\mu$ (no sum), warped frame	on the canonical member
0	$\frac{1}{2} H_C T_{16}[4]$	0
$i = 1, 2, 3$	$-\frac{1}{2}(H \cot^2/C) T_{16}[0] + \frac{1}{2} H_A T_{16}[4]$	$-\frac{1}{2} H \cot^2 T_{16}[0] - \frac{1}{2} H a'_4 T_{16}[4]$
4	0	0
$h = 5, 6, 7$	$-\frac{1}{2}(H \cot^2/C) T_{16}[0] + \frac{1}{2} H_B T_{16}[4]$	$-\frac{1}{2} H \cot^2 T_{16}[0] + \frac{1}{2} H a'_4 T_{16}[4]$

The sum over the seven directions is the asserted $-(3H \cot^2/C) T_{16}[0] + \frac{1}{2} \Theta T_{16}[4]$, and the last column is subsection 6.8's table.

6.18 The Weitzenböck form, and the rescaling that removes the connection

Computed as in Part V — the torsion of the connection in which the frame is parallel, $T^a_{\mu\nu} = \partial_\mu e_\nu^a - \partial_\nu e_\mu^a$, pulled to curved indices and traced, $T_\mu = T^\nu_{\nu\mu}$ — the torsion vector of the warped frame is a **gradient**, and it carries the whole connection term of the Dirac operator:

$$T_\mu = \partial_\mu \ln \frac{\sin(6Hx_0)}{A^3 B^3 C}, \quad \gamma^\mu \Gamma_\mu = -\frac{1}{2} T_\mu \gamma^\mu$$

(on the canonical member it is Part V's torsion vector $\partial_\mu \ln \sin(6Hx_0)$). Hence the rescaling

$$\Psi = \sqrt{\sin 6Hx_0} (A^3 B^3 C)^{-1/2} \chi, \quad \gamma^\mu D_\mu \Psi = \sqrt{\sin 6Hx_0} (A^3 B^3 C)^{-1/2} \gamma^\mu \partial_\mu \chi$$

removes the connection from the Dirac operator **exactly**, for a generic $\chi(x_0, \dots, x_7)$. [**proved VIII §32**]

- WEITZENBOECK [THE RESULT]: the torsion vector of the warped frame is a GRADIENT, $T_\mu == d_\mu \text{Log}[\text{Sin}[6 H x_0]/(A^3 B^3 C)]$; on the canonical member it is Part V's torsionVectorFable51
- WEITZENBOECK [THE RESULT]: $\gamma^\mu \Gamma_\mu == -(1/2) T_\mu \gamma^\mu$ on the warped frame - one vector term, a gradient
- RESCALING [THE RESULT]: $\gamma^\mu D_\mu [\text{Sqrt}[\text{Sin}[6 H x_0]] (A^3 B^3 C)^{-(1/2)} \chi] == \text{Sqrt}[\text{Sin}[6 H x_0]] (A^3 B^3 C)^{-(1/2)} \gamma^\mu d_\mu \chi$ for a GENERIC $\chi(x_0, \dots, x_7)$: the connection is removed exactly

(the torsion took 0.064s). The frame-independent form of subsection 6.12.3, $T_\mu = \partial_\mu \ln(e_\mu^\mu / \det e)$ for a diagonal frame, gives the same vector [derived here: $\det e = \tan(6Hx_0) C A^3 B^3 / S$; for $\mu = 0$, $e_0^0 / \det e = S / (A^3 B^3)$, whose x_0 -derivative of the logarithm is $6H \cot(6Hx_0)$; for $\mu = 4$, $e_4^4 = 1$ and $-\partial_4 \ln \det e = -\Theta$, so $-\frac{1}{2} T_4 \gamma^4 = \frac{1}{2} \Theta T_{16}[4]$]. The factor $(A^3 B^3 C)^{-1/2}$ is the square root of the comoving 7-volume: the dilution of the density of an expanding 7-volume.

6.19 The anticommutator of the rescaled field is independent of x_4

The anticommutator of section 3.4 carries over to the warped family because its derivation is local in time and the lapse is still 1 with $\gamma^4 = T_{16}[4]$: the time-derivative part is $\Psi^\dagger K \partial_4 \Psi$ with $K = (\sqrt{|\det g|}/H) \sigma_{16} T_{16}[4]$, and Section 28's Dirac-bracket result for $K = c \sigma_{16} T_{16}[4]$ with $c = \sqrt{|\det g|}/H$ gives

$$\begin{aligned} \{\Psi_a(x), \Psi_b^\dagger(y)\} &= \frac{H}{\sqrt{|\det g|}} G_{ab} \delta^7(x-y), \\ G &= -i \sigma_{16} T_{16}[4] = J \quad (\text{the same matrix on every member}), \\ \sqrt{|\det g|} &= \tan(6Hx_0) C A^3 B^3 / \sin(6Hx_0), \\ \{\chi_a(x), \chi_b^\dagger(y)\} &= H \cot(6Hx_0) G_{ab} \delta^7(x-y) \quad (\text{independent of } x_4 \text{ and of } A, B, C) \end{aligned}$$

— the $\sqrt{\det g}$ cancels against the rescaling; on the canonical member this is Section 28's $H \cot(6Hx_0) J$. So the field to quantize on the dynamical frames is χ , whose canonical structure does not change as the universe expands. [**proved VIII §32**]

- RESCALING [has content]: $\det g == (\det e)^2$ and $\det e == \text{Tan}[6 H x_0] C A^3 B^3 / \text{Sin}[6 H x_0] > 0$, so $\text{Sqrt}[|\det g|] == \text{Tan}[6 H x_0] C A^3 B^3 / \text{Sin}[6 H x_0]$ on the warped frame
- RESCALING [has content]: on the warped frame $G = -i \sigma_{16} \gamma^4$ is Section 28's cfGK (= J) - the same matrix on every member of the family ($\gamma^4 == T_{16}[4]$, lapse 1)
- RESCALING [THE RESULT]: $\{\Psi, \Psi^\dagger\} == (H/\text{Sqrt}[|\det g|]) G$ on the warped frame (Section 28's cfAntiPsiPsiDd at $c = \text{Sqrt}[|\det g|]/H$), and for $\chi = \Psi/f$, $f = \text{Sqrt}[\text{Sin}](A^3 B^3 C)^{-(1/2)}$: $\{\chi, \chi^\dagger\} == H \text{Cot}[6 H x_0] G$ - the $\text{Sqrt}[\det g]$ cancels, independent of x_4 and of A, B, C

(Two uses of the letter χ must be kept apart on this page: here χ is the rescaled field of the dynamical frame; in sections 3.6 and 7 $\chi = \Psi/\sqrt{H}$ is the canonically normalized field of the Kohn-Sham gas.)

6.20 The fundamental symmetry J and the connection of the interacting system

$J = -i T_{16}[0]T_{16}[1]T_{16}[2]T_{16}[3]T_{16}[4]$ (Hermitian, $J^2 = 1$) commutes with $T_{16}[0..4]$ and anticommutes with $T_{16}[5..7]$. The connection of the interacting system respects this split, direction by direction, on the warped **and** on the Bianchi-I frame:

object	relation to J	on both frames
Γ_0 (now $\frac{1}{2} \tan C' T_{16}[0]T_{16}[4]$, non-zero when C varies)	commutes	both factors in $\{0..4\}$
$\Gamma_1, \Gamma_2, \Gamma_3$	commute	made of $T_{16}[0]T_{16}[i]$, $T_{16}[i]T_{16}[4]$
Γ_4	commutes	zero
$\Gamma_5, \Gamma_6, \Gamma_7$	anticommute , and they are not zero	made of $T_{16}[0]T_{16}[h]$, $T_{16}[4]T_{16}[h]$
each $\gamma^\mu \Gamma_\mu$ (no sum), $h = 5, 6, 7$ included	commutes	each is a combination of $T_{16}[0]$ and $T_{16}[4]$

So every connection term of the admissible-sector Hamiltonian is J -even on the frames of the interacting system, as on the canonical one. **[proved VIII §32]**

- J [definition]: Section 28's $J = -i T_{16}[0] \dots T_{16}[4]$ is Hermitian with $J^2 = 1$; it commutes with $T_{16}[0..4]$ and anticommutes with $T_{16}[5..7]$
- J [THE RESULT]: $[J, \gamma^\mu \Gamma_\mu] = 0$ for EACH μ separately ($h = 5, 6, 7$ included), on the warped and on the Bianchi-I frame
- J [THE RESULT]: $[J, \Gamma_\mu] = 0$ for $\mu = 0..4$ and $\{J, \Gamma_h\} = 0$ for $h = 5, 6, 7$, on the warped and on the Bianchi-I frame
- J [has content]: the anticommuting Γ_h are not zero (witnessed on $A = 1 + x_4^2/2$, $B = 1 + x_4^2/3$, $C = 1 + x_4^2/3$), so $\{J, \Gamma_h\} = 0$ is not vacuous

6.21 Recovery of the canonical connection, and why Part VI had no dilution

On the canonical member (labelled substitution) Γ_μ is Section 17's `GammaSpinCanonical` in every direction, and $\gamma^\mu \Gamma_\mu$ is $-3H \cot^2 T_{16}[0]$. The Hubble coefficient vanishes there: $3H_A + 3H_B + H_C = -3H a'_4 + 3H a'_4 + 0 = 0$ — the 7-volume of the canonical frame is constant in x_4 . That is why Part VI found no dilution ($ds/dx_4 = 0$) on the pre-universe: the dilution term is present in the general frame and cancels exactly on the author's. **[proved VIII §32]**

- RECOVERY [fidelity]: on the canonical member (labelled substitution) Γ_μ IS Section 17's `GammaSpinCanonical` for every μ , and $\gamma^\mu \Gamma_\mu$ IS Section 24's $-3 H \cot^2 T_{16}[0]$
- RECOVERY [THE RESULT]: the Hubble coefficient $3 H_A + 3 H_B + H_C$ vanishes on the canonical member — the canonical 7-volume is constant in x_4 , which is why Part VI had no dilution
- PART VIII [control]: `a4` is STILL UNDEFINED after Section 32, and the scale factors `scA`, `scB`, `scC` have no values either

6.22 Status ledger of the connection of the interacting system

statement	status	where
warped and Bianchi-I connections from the vielbein postulate; residual zero; antisymmetric	proved	VIII §32 (SPIN CONNECTION [solver regression], [has content])
the complete lists: 13 independent components (warped), 7 (Bianchi-I); the new $\omega[0;0,4] = \tan C'$ the explicit Γ_μ of 6.16	proved	VIII §32 (SPIN CONNECTION [THE RESULT])
$\gamma^\mu \Gamma_\mu = -(3H \cot^2 / C) T_{16}[0] + \frac{1}{2} \Theta \gamma^4$; the sign of the Hubble term	derived here; Γ_0 , Γ_4 proved	VIII §32 (the DIRAC assertion on Γ_{x_4} and Γ_{x_0})
per- μ $\{\gamma_\mu, \Gamma_\mu\} = 0$; symmetrized Lagrangian $D = \partial$ the per-direction table of 6.17	proved (with control)	VIII §32 (DIRAC)
Weitzenböck gradient form; exact rescaling by $\sqrt{\sin(A^3 B^3 C)}^{-1/2}$	proved	VIII §32 (DIRAC)
x_4 -independent anticommutator $H \cot(6Hx_0) G$ of χ	proved	VIII §32 (WEITZENBOECK, RESCALING)
the J table on both frames	proved	VIII §32 (RESCALING [THE RESULT])
recovery of the canonical connection; the Hubble coefficient vanishes on it	proved	VIII §32 (J)
		VIII §32 (RECOVERY)

7 The fable source in the Kohn–Sham ground state

This is Section 33 of the notebook, first half (Input cells 211–214, 25 assertions), with the numerical checks of notebook `05_fermion_fable_quantum_eos.ipynb` and the solver’s own statements [file: `fable-cosmology/rust/fable_fermion/src/kohn_sham.rs` and `potentials.rs`, module documentation]. The expectation value $\langle \hat{T}^\mu{}_\nu \rangle$ of section 4.1 is taken in the Kohn–Sham ground state of section 3.6: a homogeneous Fermi sea at zero temperature, $g = 8$ states per spatial momentum, momenta along the observed sheet only (zero modes along x_0, x_5, x_6, x_7 — the admissible sector of section 3.4), in the mean field $m = W'(\sigma)$ of the potential $W(\sigma) := V(H\sigma)$, $\sigma = \langle \bar{\chi}\chi \rangle$, $\chi = \Psi/\sqrt{H}$ [**prose VIII §33**]. In the asymptotic region of section 5.3 the metric is 8-dimensional Bianchi-I and this source is diagonal.

7.1 Densities per 7-volume, and the gap equation

The closed forms of section 3.6 are densities per unit **observed** 3-volume. Spread over the hidden 4-volume they become densities per 7-volume. With the hidden volume factor

$$v = \frac{B^3 C}{(B^3 C)_{\text{today}}} \quad (v = 1 \text{ today in every run; the solver's normalization}),$$

the fluid that sources the Einstein equations is [**prose VIII §33**]

$$\begin{aligned}
\sigma_7 &= \sigma_{\text{KS}}/v && \text{(the solver calls it `sigma8`:} \\
&&& \text{the scalar density per 7-volume),} \\
\text{gap: } m &= W'(\sigma_7), \\
\rho &= \varepsilon_{\text{KS}}/v + W(\sigma_7) - \sigma_7 W'(\sigma_7) && (= 3P_{\text{KS}}/v + W(\sigma_7) \text{ on the gap),} \\
P_{\text{obs}} &= P_{\text{KS}}/v + \sigma_7 W' - W && \text{(along } x_1, x_2, x_3), \\
P_{\text{hid}} &= P_C = \sigma_7 W' - W && \text{(along } x_5, x_6, x_7 \text{ and } x_0: \\
&&& \text{the gas has no momentum there).}
\end{aligned}$$

At $v = 1$ this is Section 29’s mean field; the massless limit is $\sigma \rightarrow 0$, $\varepsilon = 3P = gk_{\text{F}}^4/(8\pi^2)$. The couplings of W run as 4-dimensional couplings, $\lambda_4 = \lambda_8/V_{\text{hid}}$, and $\kappa_{4,0} = \kappa_8/V_{h,0}$ [file: `models.rs`, module documentation]. [**proved VIII §33**]

- KS FLUID [fidelity]: at $v = 1$ the fluid IS Section 29's mean field: $\rho = \epsilon_{\text{KS}} + W - \sigma W'$, $P_{\text{obs}} = P_{\text{KS}} + \sigma W' - W$, $P_{\text{hid}} = \sigma W' - W$
- KS FLUID [fidelity]: Section 29's identities, re-checked where they are used - $\epsilon + P == w_F n$, $\epsilon - 3P == m \sigma$, $T_s = \epsilon - m \sigma == 3P$, $d\epsilon/dm == \sigma$, $d\epsilon/dk_F == w_F dn/dk_F$
- KS FLUID [THE RESULT]: ON THE GAP $m = W'(\sigma_7)$, $\rho == 3 P_{\text{KS}}/v + W(\sigma_7)$ (Section 29's trace identity, per 7-volume), and $P_{\text{obs}} - P_{\text{hid}} == P_{\text{KS}}/v$, for every W
- KS FLUID [has content]: the massless limit $m \rightarrow 0+$: $\sigma_{\text{KS}} \rightarrow 0$ and $\epsilon_{\text{KS}} = 3 P_{\text{KS}} \rightarrow g k_F^4/(8 \pi^2)$

Negative m . σ_{KS} is odd in m and $\epsilon_{\text{KS}}, P_{\text{KS}}$ are even. Under $m \rightarrow -m$, $\sigma \rightarrow -\sigma$ and $W(x) \rightarrow \tilde{W}(x) := W(-x)$ the fluid is unchanged and the gap $m = W'(\sigma_7)$ goes into $-m = \tilde{W}'(-\sigma_7)$; so every statement proved for $m > 0$ and an arbitrary W holds for $m < 0$ as well. **[proved VIII §33]** KS FLUID [THE RESULT]: NEGATIVE m - the fluid is invariant under $\sigma \rightarrow -\sigma$, $W(x) \rightarrow \tilde{W}(x) = W(-x)$, and the gap $m = W'(\sigma_7)$ becomes $-m = \tilde{W}'(-\sigma_7)$; with σ_{KS} odd and $\epsilon_{\text{KS}}, P_{\text{KS}}$ even in m (Section 29) the case $m < 0$ reduces to $m > 0$.

Why per 7-volume, and not per 3-volume. The solver's unit test `kohn_sham::tests::per_3_volume_mean_field_violates_8d_conservation` checks that the per-3-volume prescription violates the 8-dimensional conservation law of section 5.5 [file: `fable-cosmology/rust/fable_fermion/src/kohn_sham.rs`]; the per-7-volume one obeys it exactly (section 7.5).

7.2 The Fermi-sea integrals, their stable forms, and their accuracy

The closed forms of section 3.6 cancel catastrophically for $x = k_F/|m| \ll 1$ ($\epsilon \sim mn$, $\sigma \sim n$, $P \sim nk_F^2/(5m)$ are small differences of large terms). The solver uses three regimes, written with

$$S(x) = 2 \int_0^x \frac{t^2}{\sqrt{1+t^2}} dt, \quad E(x) = 8 \int_0^x t^2 \sqrt{1+t^2} dt, \quad Q(x) = 8 \int_0^x \frac{t^4}{\sqrt{1+t^2}} dt,$$

$\sigma = g m^3 S/(4\pi^2)$, $\epsilon = g m^4 E/(16\pi^2)$, $P = g m^4 Q/(48\pi^2)$ [file: `kohn_sham.rs`, module documentation]:

regime	form used
$x < X_{\text{LO}} = 0.6$	the convergent binomial series (radius 1) in x^2 ; e.g. $\epsilon = m n \left[1 + \frac{3}{10}x^2 - \frac{3}{56}x^4 + \dots \right]$, $\sigma = \text{sgn}(m) n \left[1 - \frac{3}{10}x^2 + \dots \right]$
$0.6 \leq x \leq X_{\text{HI}} = 100$	the closed forms ($\text{asinh } x = \mathcal{L}$), losing at most about 1.3 digits at $x = 0.6$
$x > 100$	the ultra-relativistic series in $y = m /k_F$, with the logarithm kept exact
$m = 0$ exactly	$\sigma = 0$, $\epsilon = 3P = g k_F^4/(8\pi^2)$, $\chi = d\sigma/dm = g k_F^2/(4\pi^2)$

Notebook 05 compared them with adaptive quadrature from $x = 10^{-8}$ to 10^6 and with the Rust solver [nb05 §5.1]:

```
max relative error of the stable forms against quadrature over 1e-8 <= x <= 1e6: 9.88e-14 (at x = 0.251)
x = 1e-08: stable rel. errors (sigma, eps, P) = 0.0e+00 0.0e+00 0.0e+00 closed forms: 1.5e+00 2.4e-01 4.1e+16
x = 1e-06: stable rel. errors (sigma, eps, P) = 1.1e-16 1.1e-16 4.4e-16 closed forms: 7.8e-05 1.8e-06 2.6e+08
x = 0.0001: stable rel. errors (sigma, eps, P) = 0.0e+00 2.2e-16 2.2e-16 closed forms: 7.8e-09 1.5e-09 1.0e+00
x = 0.01: stable rel. errors (sigma, eps, P) = 2.2e-16 2.2e-16 3.3e-16 closed forms: 4.4e-13 1.1e-13 5.4e-09
x = 0.251: stable rel. errors (sigma, eps, P) = 4.3e-15 1.8e-15 9.9e-14 closed forms: 4.3e-15 1.8e-15 9.9e-14
x = 1: stable rel. errors (sigma, eps, P) = 0.0e+00 0.0e+00 2.2e-16 closed forms: 0.0e+00 0.0e+00 2.2e-16
x = 100: stable rel. errors (sigma, eps, P) = 0.0e+00 1.1e-16 2.2e-16 closed forms: 0.0e+00 1.1e-16 2.2e-16
x = 1e+06: stable rel. errors (sigma, eps, P) = 1.1e-16 2.2e-16 2.2e-16 closed forms: 1.1e-16 2.2e-16 2.2e-16
```

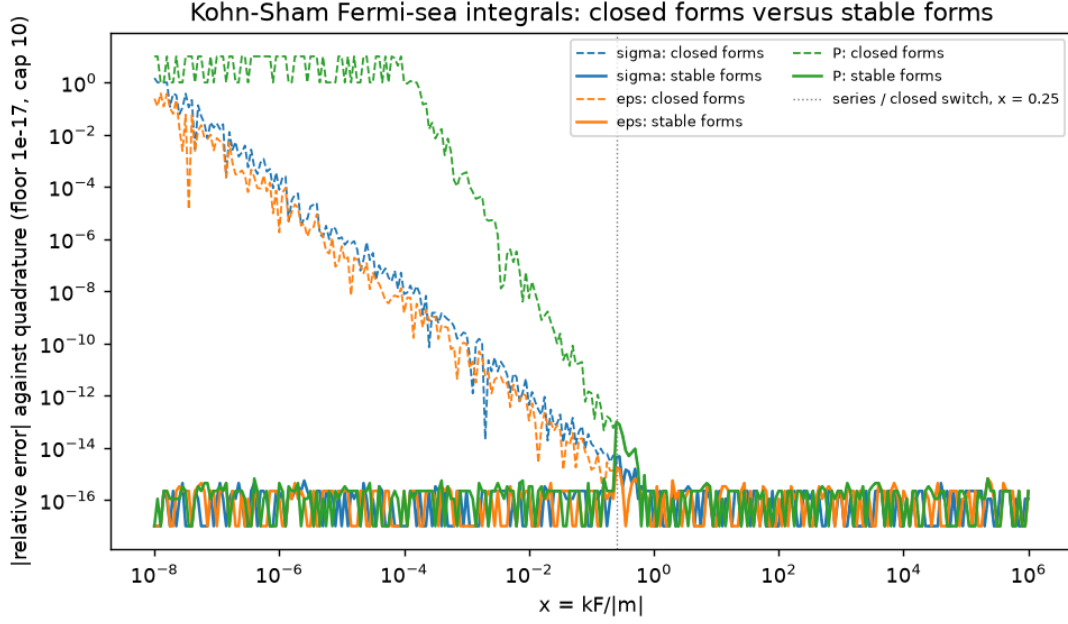


Figure 1: fable-cosmology/results/nb05_ks_accuracy.png (notebook 05, section 5.1): the accuracy of the stable forms of the Fermi-sea integrals against adaptive quadrature, and of the closed forms.

and against the Rust solver’s `gap` command at fifteen points from $x = 10^{-8}$ to 10^6 : largest relative difference Rust – Python over these 15 points: $3.11\text{e-}14$.

(the closed forms are wrong by a factor 4.1×10^{16} in P at $x = 10^{-8}$; the stable forms are exact there). The thermodynamic identities hold to rounding [nb05 §5.3, file: fable-cosmology/results/nb05_identities.csv]:

identity	largest residual
$\varepsilon - 3P - m\sigma$ (relative to ε)	$1.532\text{e-}15$
$\varepsilon + P - \omega_F n$ (relative)	$5.551\text{e-}16$
$d\varepsilon/dn _m - \omega_F$ (relative, central differences)	$7.439\text{e-}11$
$d\varepsilon/dm _{k_F} - \sigma$ (relative, central differences)	$3.610\text{e-}08$
$\chi - d\sigma/dm$ (relative)	$2.633\text{e-}07$
$\max \chi/\chi_0$	$1.000\text{e+}00$

and $w_{\text{KS}} = P_{\text{KS}}/\varepsilon_{\text{KS}}$ rises from $2.000000\text{e-}17$ at $x = 10^{-8}$ ($x^2/5 = 2.000000\text{e-}17$) through 0.121969 at $x = 1$ to 0.333333333332667 at $x = 10^6$ ($1/3 - 2/(3x^2)$) [nb05 §5.2]; figures fable-cosmology/results/nb05_ks_accuracy.png, nb05_w_ks.png (figures 1 and 2).

7.3 The gap equation: uniqueness, branch selection, parameter domains

$f(m) = m - W'(\sigma_{\text{KS}}(m, k_F)/v)$. Since σ_{KS} is odd and increasing in m with slope $\chi = d\sigma_{\text{KS}}/dm \leq \chi_0 = gk_F^2/(4\pi^2)$, $f' = 1 - W''\chi/v$: the root is **unique** when $W'' \leq 0$ (attractive, concave W) or when $W''\chi_0/v < 1$; otherwise there can be several roots [prose nb05 §4; file: kohn_sham.rs, module documentation]. Every potential states where its roots can lie and whether the root is provably unique there (Potential::gap_plan in potentials.rs). The adopted parameter domains (design review DFT-16): `expdamp` $m_0 > 0$, $s_1 > 0$ (the gap then pins $0 < \sigma < \min(n, s_1)$ strictly); `power` $\lambda > 0$, $0 < \nu < 1$, σ in $(0, n)$ (the bare mass m_0 may come out negative: section 9.8); `quadratic` $\lambda < 0$ unique, $\lambda > 0$ all roots enumerated.

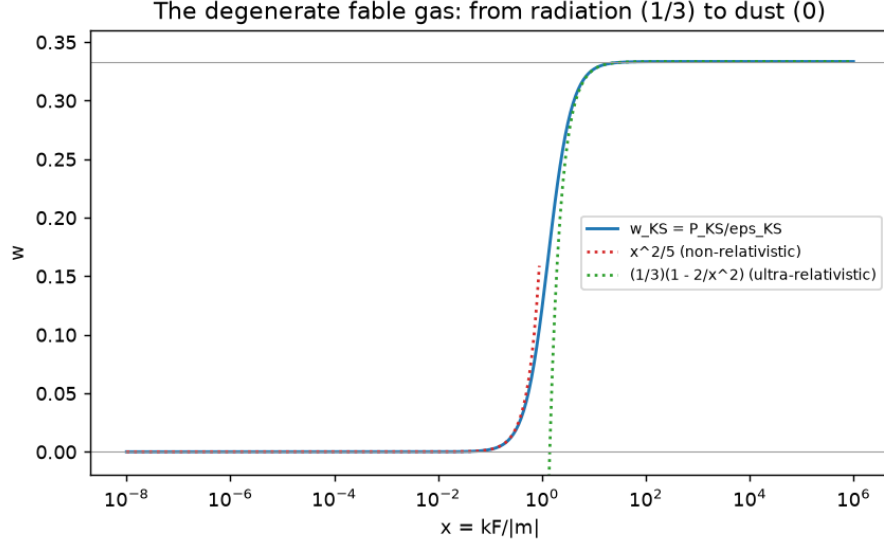


Figure 2: fable-cosmology/results/nb05_w_ks.png (notebook 05, section 5.2): $w_{\text{KS}}(x) = P_{\text{KS}}/\varepsilon_{\text{KS}}$ against $x = k_{\text{F}}/|m|$, from 0 (dust) to $1/3$ (radiation).

Branch selection. When several roots exist, the solver takes the one of **lowest** energy density ρ — not the minimum of the Kohn–Sham energy $E_n[\sigma]$ over σ in $(-n, n)$. The no-sea functional is a minimax: its kinetic part $T_s = \varepsilon_{\text{KS}}(m(\sigma)) - m(\sigma)\sigma$ is **concave** ($d^2T_s/d\sigma^2 = -1/\chi < 0$), so the physical root is a **maximum** of E_n over σ , and its infimum is the variational collapse $E_n \rightarrow -|m_0|n$ as $\sigma \rightarrow -n$ (every particle in the negative-mass band, i.e. in states of the Dirac sea). For attractive W the auxiliary-field **Walecka functional** $\Omega_n(m) = \varepsilon_{\text{KS}}(m) + W(\sigma_m) - m\sigma_m$, $W'(\sigma_m) = m$, is convex and minimized at the gap root, with $\Omega_n(m^*) = \rho$. The numbers are in section 8.1.

7.4 The energy–momentum tensor’s expectation: ρ , P_{obs} , P_{hid} , and w

On the gap:

$$\begin{aligned} \rho &= 3P_{\text{KS}}/v + W(\sigma_7) && \text{(Section 29’s trace identity } \varepsilon - 3P = m\sigma, \text{ per 7-volume),} \\ P_{\text{obs}} &= P_{\text{KS}}/v + \sigma_7 W' - W, \\ P_{\text{hid}} &= P_C = \sigma_7 W' - W, \\ P_{\text{obs}} - P_{\text{hid}} &= P_{\text{KS}}/v \geq 0 && \text{(for every } W: \text{ the Fermi gas can source} \\ &&& a_4'' < 0, \text{ section 4.6),} \\ w_f &= P_{\text{obs}}/\rho. \end{aligned}$$

The split $\rho = \rho_{\text{qp}} + \rho_U$ with $\rho_{\text{qp}} := \varepsilon_{\text{KS}}/v$ (quasiparticles, $w_{\text{qp}} = w_{\text{DM}} = P_{\text{KS}}/\varepsilon_{\text{KS}}$ in $[0, 1/3]$) and $\rho_U := W - \sigma_7 W'$ (the mean-field condensate, with $P = -\rho_U$ along **all seven** spatial directions: an 8-dimensional vacuum energy that varies with σ) is a **convention** [prose nb06 §3; file: kohn_sham.rs]. Section 7.10 defines the dark-matter and dark-energy equations of state built on it.

7.5 Theorem: the Kohn–Sham fluid is exactly conserved in 8-dimensional Bianchi-I

The fable number in a comoving 7-volume, $(n_{\text{KS}}/v) A^3 v = n_{\text{KS}} A^3$, is conserved, so $k_{\text{F}} = k_{\text{F}0}/A$. Take A , B , C and the mass $m(x_4) > 0$ as **arbitrary** functions and W arbitrary. Then, identically,

$$\rho' + 3H_A(\rho + P_{\text{obs}}) + (3H_B + H_C)(\rho + P_{\text{hid}}) = \sigma_7 \frac{d}{dx_4} [m - W'(\sigma_7)].$$

The right-hand side is the time derivative of the gap equation; on the self-consistent solution it vanishes at every x_4 , so the Kohn–Sham fluid obeys section 5.5’s conservation law (with $P_C = P_{\text{hid}}$) **exactly**, for every W — the Einstein equations with this source are consistent. The mechanism: $\rho + P_{\text{obs}} = \omega_F n_{\text{KS}}/v$ and $\rho + P_{\text{hid}} = \varepsilon_{\text{KS}}/v$ hold identically, $d\varepsilon = \omega_F dn + \sigma dm$, and the gap cancels the σdm term against the variation of W . Off the gap the fluid is **not** conserved. **[proved VIII §33]**

- CONSERVATION [definition]: the fable number per comoving 7-volume, $(n_{\text{KS}}/v) A^3 B^3 C$, is constant when $kF = kF_0/A$
- CONSERVATION [has content]: $\rho + P_{\text{obs}} == \omega_F n_{\text{KS}}/v$ and $\rho + P_{\text{hid}} == \varepsilon_{\text{KS}}/v$ identically (no gap needed)
- CONSERVATION [THE RESULT]: $\rho' + 3 H_A (\rho + P_{\text{obs}}) + (3 H_B + H_C)(\rho + P_{\text{hid}}) == \sigma_7 d/dx_4[m - W'(\sigma_7)]$ IDENTICALLY, for arbitrary A, B, C , $m(x_4) > 0$ and W : the KS fluid is exactly conserved on the gap
- CONSERVATION [control]: without the gap the fluid is NOT conserved – the left side is not identically zero (witnessed on $A = 1 + x_4/2$, $B = 1 + x_4/3$, $C = 1 + x_4^2/3$, $m = 2 + x_4$, $W = \sigma_7^2$, $kF_0 = 1$, $g = 8$)

The solver’s form of the same law: $d\rho_8 = \omega_F dn/v - (\varepsilon/v)d\ln v$, so at fixed v , $d\rho_8/dn_8 = \omega_F$ [file: `kohn_sham.rs`, module documentation]; its unit test `kohn_sham::tests::thermodynamic_identities_at_the_self_consistent_point` and the end-to-end test `covariant_conservation_along_real_fable8d_runs` check it [file: `kohn_sham.rs`, `tests/cosmology.rs`].

7.6 The hidden-sheet driver

With $P_C = P_{\text{hid}}$, section 5.4’s evolution equations for B and C both read

$$H'_B + H_B \Theta = H'_C + H_C \Theta = \kappa F/6, \quad F = \rho - 3P_{\text{obs}} + 2P_{\text{hid}}.$$

F is the driver of the hidden sheet:

source	F
radiation ($P_{\text{obs}} = \rho/3$, $P_{\text{hid}} = 0$)	0
dust	ρ
an 8-dimensional vacuum energy (all seven pressures $-\rho$; a bare V_0 , the condensate ρ_U)	2ρ
the Kohn–Sham fable, on the gap	$2W - \sigma_7 W'$ (off the gap: $\sigma_7(m - W') + 2W - \sigma_7 W')$
the author’s mass term $W = m_0 \sigma$	$m_0 \sigma_7$ (like dust)
$W = V_0 + m_0 \sigma$	$2V_0 + m_0 \sigma_7$

[proved VIII §33]

- DRIVER [THE RESULT]: with $P_C = P_{\text{hid}}$, $\kappa (P_{\text{hid}} - T/6) == \kappa (P_C - T/6) == \kappa F/6$, $F = \rho - 3 P_{\text{obs}} + 2 P_{\text{hid}}$: the right-hand side of the B and C equations
- DRIVER [has content]: $F == 0$ for radiation, $F == \rho$ for dust, $F == 2 \rho$ for an 8D vacuum
- DRIVER [THE RESULT]: for the KS fluid $F == \sigma_7 (m - W') + 2 W - \sigma_7 W'$ identically, i.e. $F == 2 W - \sigma_7 W'$ on the gap; the mass term gives $F == m_0 \sigma_7$ (dust-like), $W = V_0 + m_0 \sigma$ gives $2 V_0 + m_0 \sigma_7$

7.7 Theorem: a frozen hidden sheet makes the fable radiation

Given $H_B = H_C = 0$ at one time, the hidden sheet stays **frozen** for all time iff $F = 0$ along the history; for the Kohn–Sham fluid, for all σ_7 the history sweeps, iff $\sigma W' = 2W$, i.e. $W = c\sigma^2$. The gap is then $m = 2c\sigma_7$, and $m = 0$ is always a root. For $c < 0$ it is the only one ($m\sigma_{KS} > 0$ for $m \neq 0$, so a root $m \neq 0$ would give $m^2 = 2c(m\sigma_{KS})/v < 0$). For $c > 0$ a non-trivial root exists iff $cgk_F^2/(2\pi^2v) > 1$ (σ_{KS}/m decreases strictly, from $gk_F^2/(4\pi^2)$ at $m \rightarrow 0$ to 0 at $m \rightarrow \infty$), and it has **higher** energy: on it $\rho = (\varepsilon_{KS}(m) - m\sigma_{KS}(m)/2)/v$, and $\Delta(m) = \varepsilon_{KS}(m) - m\sigma_{KS}/2 - \varepsilon_{KS}(0)$ vanishes at $m = 0^+$ with

$$\frac{d\Delta}{dm} = \frac{gm^3}{4\pi^2} \left(\mathcal{L} - \frac{k_F}{\omega_F} \right) > 0$$

$(\mathcal{L} - k_F/\omega_F = \varphi(k_F/m)$, $\varphi(x) = \ln(x + \sqrt{1+x^2}) - x/\sqrt{1+x^2}$, which vanishes at 0 and has derivative $x^2/(1+x^2)^{3/2} > 0$). So the ground state of a frozen-compatible fable is the $m = 0$ branch, on which $\rho = \varepsilon_{KS}(k_F, 0)/v$, $P_{\text{obs}} = \rho/3$, $P_{\text{hid}} = 0$: **radiation**. A Kohn–Sham fable compatible with a frozen hidden sheet is neither dark matter nor dark energy. [**proved VIII §33**]

- FROZEN [THE RESULT]: $2W - \sigma W' = 0$ for all σ iff $W = c\sigma^2$ (the general solution of $\sigma W' = 2W$)
- FROZEN [has content]: $m\sigma_{KS} > 0$ for $m > 0$ (zero at $k_F = 0$, k_F -derivative $gk_F^2 m^2/(2\pi^2 vF) > 0$; even in m , so also for $m < 0$); for $c < 0$ a root $m \neq 0$ of $m = 2c\sigma_{KS}/v$ would give $m^2 = 2c(m\sigma_{KS})/v < 0$: only $m = 0$
- FROZEN [has content]: $\phi(x) = \text{Log}[x + \text{Sqrt}[1 + x^2]] - x/\text{Sqrt}[1 + x^2]$ is 0 at $x = 0$ and increasing (derivative $x^2/(1+x^2)^{(3/2)})$, hence positive for $x > 0$; and $L - k_F/w_F = \phi(k_F/m)$
- FROZEN [THE RESULT]: for $c > 0$, σ_{KS}/m decreases strictly ($d/dm = -(gm/(2\pi^2))(L - k_F/w_F)$) from $gk_F^2/(4\pi^2)$ at $m \rightarrow 0$ to 0 at $m \rightarrow \infty$: a non-trivial root of $1 = 2c(\sigma_{KS}/m)/v$ exists iff $cgk_F^2/(2\pi^2v) > 1$
- FROZEN [THE RESULT]: on the non-trivial root ($m = 2c\sigma_{KS}$) $\rho = (\varepsilon_{KS} - m\sigma_{KS}/2)/v$, and $\Delta = \varepsilon_{KS}(m) - m\sigma_{KS}/2 - \varepsilon_{KS}(0)$ has $\Delta(0^+) = 0$ and $d\Delta/dm = (gm^3/(4\pi^2))(L - k_F/w_F) > 0$: the $m = 0$ branch has LOWER ρ
- FROZEN [has content]: witness numbers - at $g = 8$, $v = 1$, $c = 15$, $k_F = 1$ the non-trivial root m^* exists (gap residual $< 10^{-20}$) and $\rho(m = 0) < \rho(m^*)$
- FROZEN [THE RESULT]: on the $m = 0$ branch the frozen-compatible fable is RADIATION: $\rho = \varepsilon_{KS}(k_F, 0)/v$, $P_{\text{obs}} = \rho/3$, $P_{\text{hid}} = 0$

The witness the run prints [**displayed VIII §33**], `claude-fable/run_fermion_fable_final.log`:

```
frozen-compatible witness (g = 8, v = 1, c = 15, kF = 1): non-trivial root m* =
↪ 3.97853200077825223420767254928521681893`12.; rho(m = 0) =
↪ 0.10132118364233777144387946320972763891`12. < rho(m*) =
↪ 0.28374208489508058306307225309709278221`12.
```

7.8 The classical limit, and the sign rule

The classical limit is the **non-relativistic** limit $k_F \ll |m|$ (not a coherent zero mode: the Pauli principle forbids that). Exactly, on the gap, with $s = H\sigma_7$ and $V(s) := W(s/H)$:

$$\rho = V(s) + 3P_{KS}/v, \quad P_{\text{obs}} = sV'(s) - V(s) + P_{KS}/v, \quad P_{\text{hid}} = sV'(s) - V(s),$$

and as $k_F/m \rightarrow 0$ ($m > 0$), $\sigma_{KS}/n_{KS} = 1 - \frac{3}{10} k_F^2/m^2 + \dots$, $(\varepsilon_{KS} - n_{KS}m)/(n_{KS}k_F^2/m) \rightarrow 3/10$, $P_{KS}/(n_{KS}k_F^2/m) \rightarrow 1/5$; $P_{KS}/\rho \rightarrow 0$ and the quantized fluid reduces to Part VI's $\rho = V(s)$, $P = sV'(s) - V(s)$ (its $K_h = 0$ branch); P_{hid} is exactly Part VI's on-shell pressure. With $s_0 = n_{KS}/v$, exactly on the gap,

$$\rho - W(s_0) = (\varepsilon - nm)/v - [W(s_0) - W(\sigma_7) - W'(\sigma_7)(s_0 - \sigma_7)] :$$

a second-order Taylor remainder in $s_0 - \sigma_7$. **The sign rule:** $m\sigma_{KS} \geq 0$, so on every self-consistent solution $\text{sign}(\sigma_7) = \text{sign}(m) = \text{sign}(W'(\sigma_7))$; for Part VI's mass term $W = -2M\sigma$ the gap is $m = -2M$ and $\rho \rightarrow 2|M|n/v > 0$ — the Part VI branch $Ms > 0$, of negative energy, is not a self-consistent state. **[proved VIII §33]**

- CLASSICAL LIMIT [THE RESULT]: exactly, on the gap and with $s = H \sigma_7$, $V(s) = W(s/H)$: $\rho == V(s) + 3 P_{KS}/v$, $P_{obs} == s V'(s) - V(s) + P_{KS}/v$, $P_{hid} == s V'(s) - V(s)$
- CLASSICAL LIMIT [fidelity]: Section 24's on-shell P_1 (cfP1Fable, re-checked here) is $s V'(s) - V(s)$; evaluated at $s = H \sigma_7$ with $V(s) = W(s/H)$ it IS the KS $P_{hid} = \sigma_7 W' - W$
- CLASSICAL LIMIT [THE RESULT]: NR series ($k_F/m \rightarrow 0$, $m > 0$): $\sigma_{KS}/n_{KS} = 1 - (3/10) k_F^2/m^2 + \dots$, $(\varepsilon_{KS} - n_{KS}m)/(n_{KS} k_F^2/m) \rightarrow 3/10$, $P_{KS}/(n_{KS} k_F^2/m) \rightarrow 1/5$
- CLASSICAL LIMIT [has content]: the design's form, exactly: on the gap ($m = W'(\sigma_7) > 0$) $\rho - W(s_0) == (\varepsilon - nm)/v - [W(s_0) - W(\sigma_7) - W'(\sigma_7)(s_0 - \sigma_7)]$, $s_0 = n/v$ — a second-order Taylor remainder in $s_0 - \sigma_7$
- SIGN RULE [THE RESULT]: $m \sigma_{KS} \geq 0$ gives $\text{sign}(\sigma_7) == \text{sign}(m) == \text{sign}(W'(\sigma_7))$; for Section 24's mass term $W = -2 M \sigma$ the gap is $m = -2 M$ and $W(\text{sign}(m) n/v) = -2 M \text{sign}(-2 M) n/v == 2 |M| n/v > 0$

Notebook 05 checks the series at $x = 0.001$ and the sign at $m = -2$ **[nb05 §5.4]**: $(\sigma/n - 1)/x^2 = -0.300000$ ($-3/10$), $(\varepsilon/(|m| n) - 1)/x^2 = +0.300000$ ($+3/10$), $P/(n k_F^2/|m|) = 0.200000$ ($1/5$); $m = -2$: $\sigma/n = -0.999999700$: $\text{sgn}(\sigma) = \text{sgn}(m)$.

7.9 $w_f \geq -1$: no phantom crossing

$$\rho + P_{obs} = \omega_F n_{KS}/v \geq 0 \quad \text{so} \quad w_f \geq -1 \quad \text{wherever } \rho > 0.$$

The quantized fermion fable **cannot cross the phantom divide**; the classical crossing of Part VI (section 2.12.7, at $V'(s^*) = 0$) does not survive quantization. **[proved VIII §33]** PHANTOM [THE RESULT]: $\rho + P_{obs} == w_F n_{KS}/v \geq 0$, so $w_f \geq -1$ wherever $\rho > 0$ — the quantized fable cannot cross the phantom divide. Notebook 05 compared the classical $w_{cl} = sV'/V - 1$ of five of Part VI's potential shapes with the quantum $w = P_{obs}/\rho$ of the lowest-energy gap root, over densities $10^{-3} \leq n \leq 30$ (in units where the classical s equals n), for $W = \mathcal{A}V$ with $\mathcal{A} = 10^4, 10^2, 1$ — a large $\mathcal{A}$ makes the quasiparticles heavy ($m = \mathcal{A}V'$), which is the classical limit (915 points) **[nb05 §5.4, file: fable-cosmology/results/nb05_classical_vs_quantum.csv, figure nb05_classical_vs_quantum.png]** (figure 3):

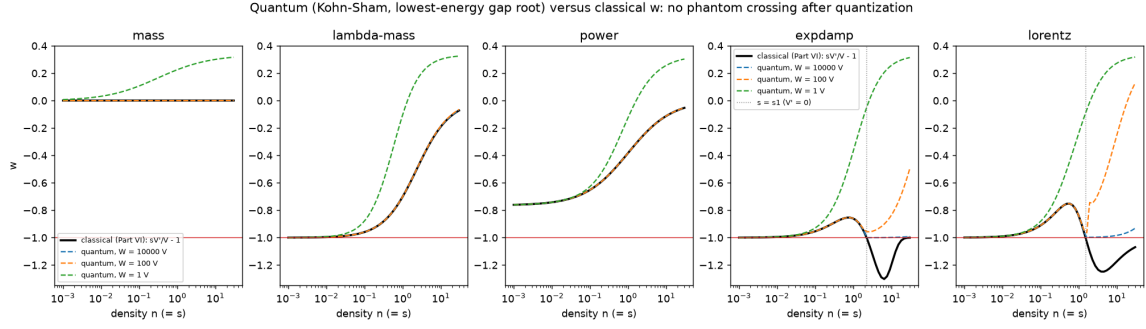


Figure 3: fable-cosmology/results/nb05_classical_vs_quantum.png (notebook 05, section 5.4): the classical $w_{\text{cl}} = s V'/V - 1$ against the quantum $w = P_{\text{obs}}/\rho$ of the lowest-energy gap root.

potential ($V(s)$)	classical min w_{cl}	quantum min w ($\mathcal{A} = 10^4, 100, 1$)
mass (s)	+0.0000, never crosses -1	+0.00000, +0.00000, +0.00741
lambda-mass ($0.7 + 0.3 s$)	-0.9996, never crosses -1	-0.99957, -0.99957, -0.99949
power ($s + s^{0.236}$)	-0.7601, never crosses -1	-0.76012, -0.76012, -0.76012
expdamp ($1.566 + 0.839 s e^{-s/2.21}$)	-1.3020, crosses -1 at $n = 2.2047$	-0.99952, -0.99947, -0.99945
lorentz ($1 + s/(1 + (s/1.5)^2)$)	-1.2481, crosses -1 at $n = 1.4999$	-0.99900, -0.99900, -0.99898

and it printed the quantum $w \geq -1$ at every density, every $\mathcal{A}$ and every potential; the classical expdamp and lorentz cross -1 . For expdamp the gap keeps $0 < \sigma < \min(n, s_1)$ strictly, with $\sigma \rightarrow s_1$ and $m \rightarrow 0$ only as $n \rightarrow \infty$ (solver test `kohn_sham::tests::expdamp_pins_sigma_below_s1_and_mass_goes_to_zero_at_high_density`).

7.10 The equations of state an observer can define

Four definitions are used, and they must not be confused [file: `models.rs`, documentation of the output columns; prose nb06 §5]:

1. $w_f = P_{\text{obs},f}/\rho_f$ — the intrinsic equation of state of the whole fable fluid;
2. $w_{\text{DM}} = w_{\text{qp}} = P_{\text{KS}}/\varepsilon_{\text{KS}}$ — the intrinsic equation of state of the quasiparticles, in $[0, 1/3]$;
3. $w_{\text{DM,eff}} = w_{\text{DM}} - \sigma_{\text{KS}}(dm/dN)/(3\varepsilon_{\text{KS}})$ — the effective equation of state of the quasiparticles when the mass varies, which exchange energy with the condensate at the rate $Q = \sigma_8 dm/dt$ (the exact exchange; $n dm/dt$ is its non-relativistic limit);
4. $w_{\text{DE,inf}}$ — the dark energy an observer **infers** by fitting cold dark matter plus dark energy: $\rho_{\text{DE,inf}} = H_A^2 - \Omega_{r0}A^{-4} - \Omega_{b0}A^{-3} - m_{\text{today}} n_{\text{today}} A^{-3}$ (radiation, baryons, and cold dark matter with **today's** mass $m_{\text{eff}}(A = 1)$ subtracted), $w_{\text{DE,inf}} = -1 - \frac{1}{3} d \ln \rho_{\text{DE,inf}} / d \ln A$.

The adiabatic sound speed of the fable is $c_s^2 = (dP_{\text{obs},f}/dN)/(d\rho_f/dN)$; $c_s^2 < 0$ is an adiabatic instability. The split into dark matter and dark energy is a convention (section 7.4): the condensate is an 8-dimensional vacuum energy.

7.11 The vacuum energy of mass-varying potentials

For a constant mass the Dirac-sea energy is a constant; it is absorbed into V and appears as a bare cosmological constant. **The cosmological-constant problem is not solved here.** For a mass that varies along the history the sea energy cannot be absorbed silently (design review DFT-4, adopted). The renormalized one-loop Dirac-sea energy that the no-sea functional drops (relativistic Hartree approximation, Chin 1977), per 3-volume, with reference mass M and counterterms through m^4 , is

$$\Delta E_{\text{vac}}(m) = -\frac{g}{16\pi^2} \left[m^4 \ln \frac{m}{M} + M^3(M-m) - \frac{7}{2} M^2(M-m)^2 + \frac{13}{3} M(M-m)^3 - \frac{25}{12} (M-m)^4 \right]$$

[file: kohn_sham.rs, function vacuum_energy; its test kohn_sham::tests::vacuum_energy_vanishes_to_fifth_order_at_the_reference_mass]. At $m = 0$ the bracket is $M^4(1 - \frac{7}{2} + \frac{13}{3} - \frac{25}{12}) = -M^4/4$, so $\Delta E_{\text{vac}}(0) - \Delta E_{\text{vac}}(M) = g M^4/(64\pi^2)$ [derived here]. With $g = 8$ and $E_c = \rho_{c0}^{1/4} = 2.46261 \times 10^{-3} \text{ eV}$ (asserted: fable4d RUNS [definition]: the units computed from CODATA 2018 - $E_c = \rho_{c0}^{1/4} == 2.46261\text{e-3 eV}$, $\Omega_{r0} == 9.2096\text{e-5}$, $\Omega_{b0} = 0.02237/h^2 == 0.049243$ (to the quoted digits)), a mass variation of order m changes the vacuum energy by

$$\frac{g m^4/(64\pi^2)}{\rho_{c0}} = 3.44, \quad 3.44 \times 10^4, \quad 3.44 \times 10^8, \quad 3.44 \times 10^{12}$$

at $m = 0.01, 0.1, 1, 10 \text{ eV}$ (it scales as m^4)

[derived here: $g/(64\pi^2) = 0.012665$, $(m/E_c)^4 = 271.9$ at $m = 0.01 \text{ eV}$]. So for $m \gtrsim 10 \text{ meV}$ it exceeds the critical density; notebook 05 prints the exact threshold [nb05 §5.8]: $\Delta E_{\text{vac}}(0; M) = g M^4/(64 \pi^2)$: check 1.0000000000000003; $g M^4/(64 \pi^2)$ exceeds ρ_{c0} for $M > (64 \pi^2/g)^{1/4}$ $E_c = 7.341 \text{ meV}$, and at $M = 1 \text{ eV}$ it gives $|\Delta E_{\text{vac}}|/\rho_{c0} = 2.760\text{e-02}, 2.802\text{e+03}, 9.454\text{e+06}, 3.444\text{e+08}$ at $m/M = 0.99, 0.9, 0.5, 0$ (the last is the 3.44×10^8 above); at $M = 1000 \text{ eV}$, 3.444e+20 at $m = 0$; the fifth-order zero at $m = M$ is confirmed by the ratio 31.9470 ($2^5 = 32$).

The design review's refinement (DFT-4, adopted): the sea energy **can** be absorbed formally by reading W as the fully renormalized effective potential, $\Omega(m) = \varepsilon_{\text{KS}} + \Delta E_{\text{vac}} + F = \text{no-sea}$ with $\tilde{F} = F + \Delta E_{\text{vac}}$; that reading is a **fine-tuning** of the bare potential against the sea energy. The vacuum also adds $\sigma_{\text{vac}} = d\Delta E_{\text{vac}}/dm$ to $\langle \bar{\Psi}\Psi \rangle$. The two options are (a) to include ΔE_{vac} in ρ and in the gap equation, or (b) to declare W the renormalized potential. **The runs use (b)**, and they print $\Delta E_{\text{vac}}(m(t)) - \Delta E_{\text{vac}}(m_{\text{today}})$ in units of ρ_{c0} (the solver's column `vac_over_rho_c`, $\Delta E_{\text{vac}}(m_{\text{eff}}; M = m_{\text{eff}}(A = 1))/v$). Along the power run $\nu = 0.236$, $m_{\text{today}} = 30 \text{ eV}$ notebook 05 found [nb05 §5.8] (figure 4):

```
power nu = 0.236, m_today = 30 eV: m_eff/m_today runs from 0.2007 (a = 1e-12) to 1; max |solver
  ↳ vac_over_rho_c - formula| / (g M^4/(64 pi^2)) = 5.4e-15
    |Delta E_vac(m(t)) - Delta E_vac(m_today)| / rho_total: max over a >= 0.3 = 1.314e+13, at a = 0.5:
  ↳ 1.149e+13
```

Section 9.6 gives the ratio for every run.

7.12 Exchange and correlation

The Kohn–Sham functional of section 8.1 has E_{xc} neglected (the Hartree mean field). For the author's mass term (W linear) the action is bilinear, the theory is free, and the mean field is **exact** ($E_{\text{xc}} = 0$). For a nonlinear W the design review (DFT-14) estimated that the Fock exchange of the contact interaction is suppressed relative to the Hartree term by $\sim 1/g = 1/8$ only in the non-relativistic regime $k_F \ll m$, and is of order one or larger for $k_F \gtrsim m$; in the relativistic era its neglect is controlled only by the smallness of the interaction energy relative to $3P_{\text{KS}}$. That ratio was not computed in the runs; the neglect of E_{xc} is a stated approximation of every mass-varying result on this page.

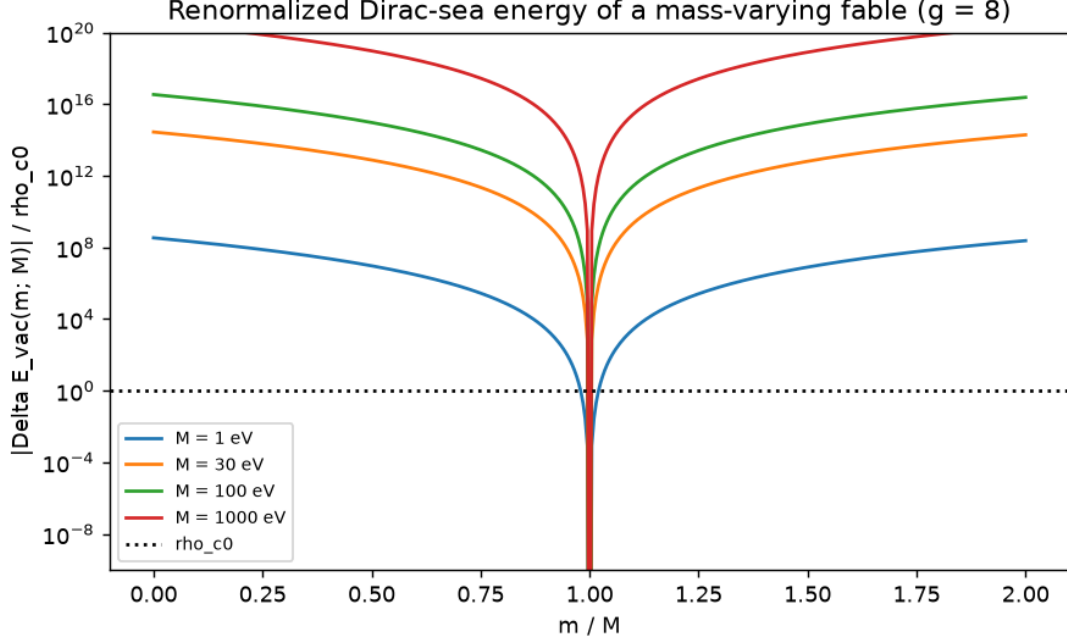


Figure 4: fable-cosmology/results/nb05_vacuum_energy.png (notebook 05, section 5.8): the renormalized Dirac-sea energy of a mass-varying fable.

8 Density-functional theory: the fable ground and first excited states

The author asked to “use some of the relevant ideas from Density Functional Theory to obtain the fable ground and first excited states”. Two systems are treated: the **homogeneous** fable gas of the present universe (sections 8.1–8.2), and the fable **along the hidden coordinate** x_0 of the primordial field, where the wall $x_0 = 0$ binds discrete states (sections 8.3–8.7). The homogeneous numbers are printed by notebook 05 [nb05 §5.5, §5.6]; the wall-state numbers are quoted from fable-cosmology/fermion/waveguide_REPORT.txt [WG n] and its logs, and notebook 07_fable_dft_states.ipynb, executed and committed in aca5102, reproduces them (a number it prints again carries the mark [nb07 §n], with the notebook’s section; the numbers it does not print again are listed at the end of section 8.7). The development logs that the report names are files under fable-cosmology/fermion/dev/, a folder the repository ignores; the report, which is tracked, quotes them.

8.1 The functional, and the homogeneous Kohn–Sham ground state

The functional. For the conserved charge n and the scalar density σ (the relativistic Hohenberg–Kohn–Sham scheme; the Serot–Walecka mean field is its Hartree limit), the energy of the gas at fixed n is

$$E_n[\sigma] = T_s(\sigma; n) + W(\sigma) + E_{xc}, \quad T_s = \varepsilon_{KS}(m(\sigma)) - m(\sigma) \sigma, \quad \sigma_{KS}(m(\sigma), k_F) = \sigma, \\ \frac{dE_n}{d\sigma} = W'(\sigma) - m(\sigma) \quad (\text{stationary exactly at the gap roots}), \quad \frac{d^2 T_s}{d\sigma^2} = -\frac{1}{\chi} < 0,$$

with E_{xc} neglected (section 7.12) and the Dirac sea excluded (the no-sea functional; its energy is section 7.11). The Kohn–Sham Slater determinant is the filled Dirac sea (renormalized away) plus the Fermi sea $|k| < k_F$ of the lowest positive band. **The ground state is the stationary point of this minimax functional** — a maximum in σ and a minimum in the occupations

— and **never** its minimum over σ [prose nb05 §5.6; solver: design departure D1 of the development report, and the unit tests `kohn_sham::tests::ks_functional_stationary_point_is_a_maximum_for_the_mass_term`, `walecka_functional_is_minimized_at_the_gap_root`, `multiple_gap_roots_are_found_and_the_lowest_energy_one_is_taken` in `kohn_sham.rs`]. For linear W the theory is free, $m = m_0$ exactly, and the Kohn–Sham state is the exact ground state.

The minimax, in numbers [nb05 §5.6, file: `fable-cosmology/results/nb05_minimax.csv`, figure `nb05_minimax.png`] (figure 5):

```
mass term W = 2 sigma, kF = 1.5: n = 0.4559453264, the gap root sigma* = 0.3961298703 (sigma*/n =
  ↪ 0.868810)
E_n(sigma*) = 1.052962422118 = rho = eps_KS(m0) = 1.052962422118; max of E_n over the grid =
  ↪ 1.052959281278
E_n/n at sigma = -(1 - 1e-7) n: -1.999480 (the collapse value -|m0| = -2.0)

attractive W = sigma - 10 sigma^2, kF = 1: unique gap root m* = 0.2112489356, rho = 0.1211932853;
  ↪ Omega_n(m*) = 0.121193285306
Omega_n on the grid: minimum 0.121193451922 at m = 0.2100; second differences all positive (convex):
  ↪ True
```

The root of the mass term is the **maximum** of E_n (the grid maximum lies just below it), and E_n/n falls to the collapse value $-|m_0|$ as $\sigma \rightarrow -n$. For the attractive W the convex Walecka functional $\Omega_n(m) = \varepsilon_{\text{KS}}(m) + W(\sigma_m) - m\sigma_m$ has its minimum at the gap root, equal to ρ .

Several roots, and a first-order transition. For the repulsive $W = m_0\sigma + (\lambda/2)\sigma^2$ with $(m_0, \lambda) = (0.05, 20)$ at $k_F = 1$, $W''\chi_0 = \lambda g k_F^2/(4\pi^2) = 4.0528 > 1$ and there are three roots; the Rust solver’s gap command and the Python solver agree [nb05 §5.5]:

```
# potential Quadratic { v0: 0.0, m0: 0.05, lam: 20.0 }, kF = 1e0, v = 1e0, n8 = 1.3509491152311703e-1,
  ↪ plan = Scan { lo: -2.651898230462341, hi: 2.7518982304623405 }
m,sigma8,rho8,P_obs,P_hid,gap_residual
-2.535353916022531e0,-1.292676958011266e-1,1.909782056219713e-1,1.772147779888931e-1,1.671013717773260e-1
  ↪ -1,8.506e-17
-1.640336879334347e-2,-3.320168439667173e-3,1.012381942988485e-1,3.387489103006236e-2,1.102351846776195
  ↪ e-4,-1.898e-17
2.644093986040535e0,1.297046993020268e-1,2.039272771420881e-1,1.779694075325236e-1,1.682330902102918e-1
  ↪ ,0.000e0
# selected (Lowest): m = -1.640336879334347e-2, rho8 = 1.012381942988485e-1, roots = 3
```

with $(\rho + P)/(\omega_F n) = 1.000000000000$ on every root (so $w \geq -1$ on every branch), and uniqueness criterion $\text{lam chi0} < 1$ holds for $k_F < 0.4967$; three roots first appear at $k_F = 0.625$ on this grid; the ground state jumps from $m = +0.4756$ ($k_F = 0.600$) to $m = -0.0998$ ($k_F = 0.625$) [nb05 §5.5, figure `nb05_gap_roots.png`] (figure 6): the ground state jumps between branches — a first-order transition. That is why the solver refuses a repulsive quadratic run (section 9.11): energy conservation across the jump would need a Maxwell construction, which it does not implement.

8.2 The first excited states of the homogeneous gas

The design review (DFT-15, adopted) states them; they are not assertions of the notebook, and the derivation is short [derived here]:

- **Particle–hole excitations** (a quasiparticle moved from just inside to just outside the Fermi surface) are **gapless**: $\Delta E \rightarrow 0$ as the momentum transfer goes to zero. The lowest excited state of the gas is degenerate with the ground state in the thermodynamic limit, and it has no effect on w .
- **Pair excitations** create a particle above the Fermi sea (energy $\geq \omega_F$, momentum $|k_1| \geq k_F$) and a hole in the Dirac sea, an antiparticle (energy $\sqrt{k_2^2 + m^2} \geq |m|$), with total momentum

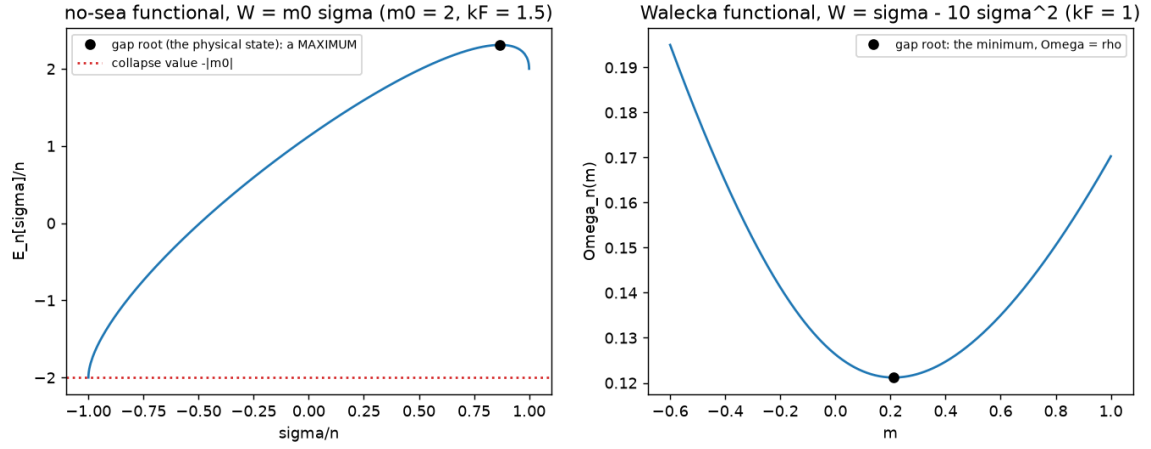


Figure 5: fable-cosmology/results/nb05_minimax.png (notebook 05, section 5.6): the no-sea Kohn–Sham functional $E_n[\sigma]$ of the mass term (its gap root is a maximum) and the Walecka functional $\Omega_n(m)$ of an attractive W (its gap root is a minimum).

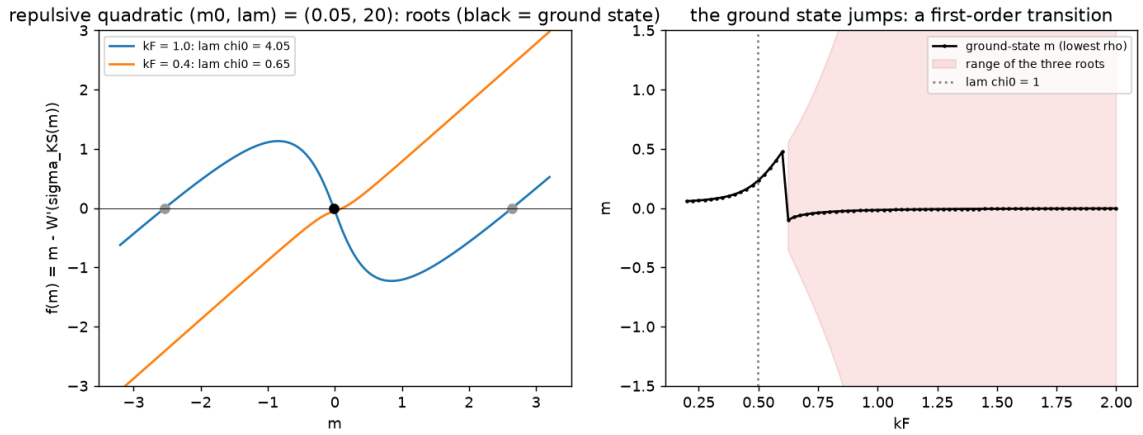


Figure 6: fable-cosmology/results/nb05_gap_roots.png (notebook 05, section 5.5): the roots of the gap equation of the repulsive quadratic W against k_F , and the jump of the ground state between branches.

$q = k_1 + k_2$. At $q = 0$, $k_2 = -k_1$, and the threshold is $2\omega_F$ (Pauli blocking pushes it from $2|m|$ to $2\omega_F$). The absolute threshold is $\omega_F + |m|$, at $|q| = k_F$ ($|k_1| = k_F$, $k_2 = 0$).

- **Delta-SCF** (moving one quantum and re-solving the gap): in the homogeneous gas one quantum changes σ by $O(1/\text{volume})$, so the self-consistent excitation energy of one quantum equals the single-particle one. A finite effect needs a finite fraction of the particles, which is how the wall problem below defines its first excited state (design review DFT-9).

8.3 The states along the hidden coordinate: the problem [WG 2]

The wall-state solver works at fixed observer's time on the canonical frame, with a_4 frozen to a constant (the adiabatic approximation). The spectrum depends on the transverse momentum k and the frozen a_4 only through $K_\infty = k e^{a_4}$, so every number below is a function of K_∞ ; the code works with K_∞ directly (internally it sets the frozen constant to zero, which only identifies k with K_∞ ; nothing here or in the notebook gives a_4 a value). Units: $H = 1$, $\hbar = c = 1$. Its derivation, `fable-cosmology/fermion/waveguide_derivation.wls`, loads the notebook's own Input cells 1–117 (with `DumpSave` blocked) and uses only the notebook's objects (`T16`, `sigma16`, `gammaCurvedCanonical`, `GammaSpinCanonical`, `cfDcov16`, `gCanonical`, `RicciScalarCanonical`, `eta4488`); its log records 94 checks, 94 PASS, 0 FAIL in 162.8s, and that the notebook's own 142 assertions of cells 1–117 passed while loading [file: `fable-cosmology/fermion/waveguide_derivation.log`, the run of 2026-09-25 00:06:38 committed in `aca5102`, which also wrote the committed `waveguide_T16.json`; the first committed run, in `18a2501`, had recorded the same 94 checks in 171.4s].

The reduction. With $\Psi = e^{ikx_1 - i\omega x_4} \sqrt{\sin 6Hx_0} \Psi'(x_0)$ the Dirac equation $\gamma^\mu D_\mu \Psi = m\Psi$ ($m = H V'(s)$; $m = -2M$ for the author's mass term) becomes **exactly**

$$(\gamma^\mu D_\mu - m) \Psi = \sqrt{\sin} e^{(\dots)} \left[\cot T_0 \partial_0 + i k e^{A_4} \sin^{1/6} T_1 - i \omega T_4 - m \right] \Psi',$$

$$T_0 \partial_z \Psi' + i K(z) T_1 \Psi' - i \omega T_4 \Psi' = m \Psi', \quad K(z) = K_\infty (1 - e^{-12Hz})^{1/12}, \quad K_\infty = k e^{a_4}$$

in the proper distance $z = -\ln \cos(6Hx_0)/(6H)$ ($\cot d/dx_0 = d/dz$; $z = 0$ at the wall, a finite proper distance away; $z \rightarrow \infty$ as $6Hx_0 \rightarrow \pi/2$). The measure is $\sqrt{|g|} |\Psi|^2 dx_0 = |\Psi'|^2 dz$, so the one-particle space is $L^2(dz)$. The transverse term rises from $K(0) = 0$ at the wall ($K = K_\infty (12Hz)^{1/12} (1 - Hz/2 + \dots)$, K' integrable) to K_∞ : a “pocket” of width $\sim 1/(12H)$ ($K^2/K_\infty^2 = 0.875776, 0.942018, 0.984275$ at $z = 0.05, 0.1, 0.2/H$). The wall is a curvature singularity at finite proper distance: $Rz^2 \rightarrow -31/24$ as $z \rightarrow 0$.

Hamiltonian form and the 2×2 blocks. $\omega \phi = h \phi$, $h = -i \alpha_z \partial_z + K(z) \alpha_x + m \beta$ with $\alpha_z = -T_4 T_0$, $\alpha_x = -T_4 T_1$, $\beta = -i T_4$ (Hermitian involutions, pairwise anticommuting); $J = -i T_0 T_1 T_2 T_3 T_4$ commutes with h . $w_3 = \alpha_z \alpha_x \beta$ ($w_3^2 = -1$) is central; the algebra $\langle \alpha_z, \alpha_x, \beta, J \rangle$ has dimension 16, and $\mathbb{C}^{16}$ splits into **8 invariant 2-dimensional blocks**: 4 with $w_3 = -i$ ($J = +1$ twice, $J = -1$ twice) and 4 with $w_3 = +i$. In an explicit block basis every block has $\alpha_z = \sigma_y$, $\beta = \sigma_z$, $T_0 = \sigma_x$, and with $\phi = (f, g)$:

$$\begin{aligned} \text{irrep } w_3 = -i : \quad & f' = -K(z) f + (m + \omega) g, \quad g' = K(z) g + (m - \omega) f, \\ \text{irrep } w_3 = +i : \quad & \text{the same with } K \rightarrow -K. \end{aligned}$$

The wall conditions. The self-adjoint conditions of a block are exactly the real lines $f(0) \cos \vartheta + g(0) \sin \vartheta = 0$, ϑ in $[0, \pi)$, and all of them have zero normal current. Only $\vartheta = \pm\pi/4$ give $\bar{\Psi} \Psi = 0$ at the wall, and only they are covariant under the wall's $\text{Spin}(1,3)$ of $x_1 \dots x_4$: the MIT (bag) conditions $T_{16}[0] \Psi = \pm \Psi$ (MIT+: $f = g$; MIT-: $f = -g$ in the basis above). The complete covariant, J -preserving family is $P \Psi(0) = \Psi(0)$ with $P = T_{16}[0] Q$, Q any Hermitian involution

in the 8-dimensional commutant of the Clifford algebra of $T_{16}[0..4]$ (which contains J); all of them have zero normal current and $\bar{\Psi}\Psi = 0$. Flavour symmetry (Q central: $\pm 1, \pm J$) together with charge conjugation (P real) select $Q = \pm 1$, the uniform MIT walls; a mixed Q gives one MIT+ sector and one MIT− sector with halved multiplicities. Without J -preservation there are more, e.g. $P = T_0 \exp(i v_t T_0 T_5)$. A **same-sign** wall is $T_{16}[0] \Psi = \text{sign}(m) \Psi$ (the limit of an outside mass of the same sign); an **opposite-sign** wall is $T_{16}[0] \Psi = -\text{sign}(m) \Psi$ (a domain wall); for the author’s mass term $m = -2M$, so both are physically possible.

The structure of the spectrum. With MIT walls the 16-component spectrum at transverse momentum K is $S(K) = A(K) \cup (-A(K))$, $A(K)$ the levels of the $w_3 = -i$ block equation, symmetric under $\omega \rightarrow -\omega$; a positive level is **4-fold** per transverse momentum vector (two blocks with $J = +1$, two with $J = -1$, of one irrep), and 8-fold only when ω and $-\omega$ both lie in $A(K)$. The 8 states per momentum of the homogeneous gas split at the wall into two generally non-degenerate 4-fold branches. The bulk continuum is $|\omega| \geq R = \sqrt{m^2 + K_\infty^2}$; $z = 0$ is a regular (limit-circle) endpoint and $z \rightarrow \infty$ limit point. Second-order form: $f'' = (K^2 - K' + m^2 - \omega^2)f$, a supersymmetric pair with superpotential K ; for $K > 0$ the f -channel has the attractive pocket $-K'$.

Exact solutions (used for validation): constant K , MIT−: $\omega = -K$, $\phi = e^{-mz}(1, -1)$; the true profile with $g(0) = 0$: $\omega = m$ for **every** $K_\infty > 0$ (a flat band); constant K , any ϑ : a level $\omega = -(m \cos 2\vartheta + K \sin 2\vartheta)$ exists iff $\kappa = m \sin 2\vartheta - K \cos 2\vartheta > 0$; constant $K > 0$, MIT+: **no** level.

8.4 Numerical method and its validation [WG 3, WG 5]

The block problem is solved by the Prüfer angle $(f, g) = r(\cos p, \sin p)$, $p' = m \cos 2p + K \sin 2p - \omega$, $p(0) = \vartheta + \pi/2$, matched to the exact decaying direction of the constant-coefficient problem beyond $z_a = 3/H$ (where $K(z)/K_\infty - 1 < 2 \times 10^{-17}$). The mismatch $\Delta(\omega)$ is strictly decreasing in ω , so the number of levels in the gap is exactly $\#\{j : \Delta(+R) < j\pi < \Delta(-R)\}$ — a rigorous count that catches shallow levels. Roots are found on a logarithmic grid of the decay rate κ and refined by Brent’s method; two integrators (adaptive DOP853 at 10^{-12} , and a 4th-order Magnus propagator) are compared. `python waveguide.py validate` runs 64 checks, all PASS (log dev/waveguide_run_validate.log, quoted in waveguide_REPORT.txt §3), among them: the exact solutions above (errors $\leq 6 \times 10^{-16}$); Hellmann–Feynman for the scalar density ($d\omega/dm = \int (f^2 - g^2) dz$, agreement 3×10^{-8}); the full 16×16 system with the notebook’s T16, in which every level has multiplicity exactly 4; and the small- K slope of the edge band against its closed form $\omega/K_\infty \rightarrow -\frac{m}{6H} B(\frac{m}{6H}, \frac{13}{12})$ (B the Euler beta function): -0.980822728 vs -0.980822733 at $m = H$.

The independent Mathematica check `fable-cosmology/fermion/waveguide_check.wls` (a different language, NDSolve at 30 digits, the normalized Wronskian with FindRoot, NullSpace for the asymptotic data) reports **11 checks, 11 PASS, 0 FAIL** and agreement with the Python levels to at most $4.68\text{e-}11$ [WG 5]. Its level table:

case	waveguide.py	Mathematica	difference
same-sign $K = 80$, $n = 0$ (irrep $-i$)	70.985307229908 [nb07 §5.5]	70.98530722995475448906	4.68e-11
same-sign $K = 80$, $n = 1$ (irrep $+i$)	78.1442493324274 [nb07 §5.5]	78.14424933243057439331	3.17e-12
same-sign $K = 80$, $n = 2$ (irrep $-i$)	79.3714957790936 [nb07 §5.5]	79.37149577909911904082	5.51e-12
same-sign $K = 80$, $n = 3$ (irrep $+i$)	79.9873085520475 [nb07 §5.5]	79.98730855204863978335	1.15e-12
same-sign $K = 8$, shallow (irrep $-i$)	8.05383822789729 [nb07 §5.7]	8.053838227898042799445	7.53e-13
same-sign $K = 40$, $n = 1$ (irrep $+i$)	39.9665602657784 [nb07 §5.7]	39.96656026577953705636	1.14e-12
opposite-sign $K = 3$, edge (irrep $-i$)	-2.93849067695725 [nb07 §5.3]	-2.938490676960405133909	3.16e-12
opposite-sign $K = 3$, edge (irrep $+i$)	2.9384906769572 [nb07 §5.8]	2.938490676960405133909	3.21e-12
$g(0) = 0$, $K = 3$, flat band (irrep $-i$)	1. [nb07 §5.3]	1.0000000000000000000000	0
$\vartheta = 1.0$, $K = 3$ (irrep $-i$)	-2.21043470304386	-2.210434703050103980776	6.24e-12
same-sign $m = 4$, $K = 20$ (irrep $-i$)	20.1427411133772	20.14274111338139405878	4.20e-12

It also confirms the 4-fold multiplicities in the full 16-component system (four singular values below 10^{-3} of the fifth, rising at $\omega \pm d\omega$) and the thresholds $K_c(1) = 6.70624065181$ and $K_c(2) = 33.9189585083$ (the Prüfer mismatch crosses an integer multiple of π between $(1 - 10^{-6})K_c$ and $(1 + 10^{-6})K_c$). At $K = 80$ NDSolve on the 16×8 matrix system did not finish within 30 minutes, so there the 16×16 check uses an independently written Magnus/MatrixExp propagator.

8.5 The wall levels: ground and first excited states, in tables [WG 4]

All at $m = H = 1$ unless stated; the levels are the **positive** levels of the full 16-component problem, each 4-fold per transverse momentum vector.

Thresholds of the same-sign wall (the transverse momentum $K_c(n)$ at which the n -th positive level appears; exact count, bisection to 10^{-9} ; dev/waveguide_run_threshold.log):

m/H	$K_c(1)/H$	irrep	$K_c(1)/m$	$K_c(2)/H$	irrep	$K_c(2)/m$
0.25	3.39259952 [nb07 §5.6]	$-i$	13.57040	33.87815035 [nb07 §5.6]	$+i$	135.51260
0.50	4.77916520 [nb07 §5.6]	$-i$	9.55833	33.89177761 [nb07 §5.6]	$+i$	67.78356
1.00	6.70624065 [nb07 §5.6]	$-i$	6.70624	33.91895851 [nb07 §5.6]	$+i$	33.91896
2.00	9.33759991 [nb07 §5.6]	$-i$	4.66880	33.97299731 [nb07 §5.6]	$+i$	16.98650
4.00	12.80358979 [nb07 §5.6]	$-i$	3.20090	34.07956537 [nb07 §5.6]	$+i$	8.51989
8.00	17.04604924 [nb07 §5.6]	$-i$	2.13076	34.28513351 [nb07 §5.6]	$+i$	4.28564

The first level always appears in irrep $-i$, with $K_c(1)$ growing roughly like $6.7(m/H)^{1/2}H$; the second always in the other irrep, at $K_c(2) = 33.9\text{--}34.3H$, almost independent of m . Within irrep $-i$ alone the second level appears only at $K = 49.540216755$ [nb07 §5.6] ($m = H$). For $K < K_c(1)$ **there is no bound level at all** — the design’s first guess (“for $k \neq 0$ the wall region is a potential pocket: bound states below $\sqrt{m^2 + k^2 e^{2a_4}}$ ”) was refuted by the review (DFT-7), and the refutation is confirmed.

Dispersion $\omega_n(K)$ of the same-sign wall ($[R - \omega] = \text{binding below the continuum } R = \sqrt{m^2 + K^2}$; dev/waveguide_dispersion_same_m1.csv):

K	R	$n = 0$ (irrep $-i$)	$n = 1$ (irrep $+i$)	$n = 2$ (irrep $-i$)	$n = 3$ (irrep $+i$)
7	7.07106781	7.07061457 [nb07 §5.7] [4.53e-4]			
8	8.06225775	8.05383823 [nb07 §5.7] [8.42e-3]			
10	10.04987562	9.99920505 [nb07 §5.7] [0.0507]			
20	20.02498439	19.39490065 [nb07 §5.7] [0.630]			
34	34.01470270	31.96226139 [nb07 §5.7] [2.05]	34.01469386 [nb07 §5.7] [8.8e-6]		
40	40.01249805	37.21339774 [nb07 §5.7] [2.80]	39.96656027 [nb07 §5.7] [0.0459]		
50	50.00999900	45.83408535 [nb07 §5.7] [4.18]	49.72238768 [nb07 §5.7] [0.288]	50.00982259 [nb07 §5.7] [1.8e-4]	
60	60.00833275	54.32136437 [nb07 §5.7] [5.69]	59.31765921 [nb07 §5.7] [0.691]	59.92357565 [nb07 §5.7] [0.0848]	
80	80.00624976	70.98530723 [nb07 §5.7] [9.02]	78.14424933 [nb07 §5.7] [1.86]	79.37149578 [nb07 §5.7] [0.635]	79.98730855 [nb07 §5.7] [0.0189]
100	100.00499988	87.32733251 [nb07 §5.7] [12.7]	96.59932036 [nb07 §5.7] [3.41]	98.43212058 [nb07 §5.7] [1.57]	99.66483153 [nb07 §5.7] [0.340]

(at $K = 100$ a fifth level 99.96879653 [nb07 §5.7] [0.0362], irrep $-i$). The band gap at fixed K ($\omega_1 - \omega_0$) of the bare wall: 2.05243246 [nb07 §5.14] ($K = 34$), 2.75316253 [nb07 §5.14] (40), 4.99629484 [nb07 §5.14] (60), 7.15894210 [nb07 §5.14] (80), 9.27198785 (100).

The ground and first excited level at $K = 80$ (the case the review examined, DFT-7): in the full 16-component problem the positive levels are 70.98531 [nb07 §5.5] ($n = 0$, irrep $-i$), 78.14425 [nb07 §5.5] ($n = 1$, from the **other** irrep, $+i$), 79.37150 [nb07 §5.5] ($n = 2$, irrep $-i$) and 79.98731 [nb07 §5.5] ($n = 3$, irrep $+i$), each 4-fold. The review’s “ $n = 1$ at 79.37150” labelled the second level of one irrep; it is $n = 2$ of the full problem, and the first excited level is 78.144249332 [nb07 §5.5]. The levels command printed, for both irreps, irrep $-i$: -79.987308552047, -78.144249332428, +70.985307229908, +79.371495779094; irrep $+i$: -79.371495779091, -70.985307229895, +78.144249332427, +79.987308552047 [nb07 §5.5] — the $\omega \rightarrow -\omega$ symmetry between the irreps to 10^{-11} [WG 1].

Wavefunctions (normalized in $L^2(dz)$): at $K = 80$, $n = 0$ has $\int(f^2 - g^2) = 0.24150473$ [nb07 §5.10] with f and g nodeless; $n = 1$ ($+i$) 0.01572127 [nb07 §5.10], g with one node; $n = 2$ ($-i$) 0.02345024, f and g one node each; they live within $z < 0.4/H$ (decay rates $\kappa = 36.9, 17.2, 10.1$) [WG 4.5].

The opposite-sign wall: a gapless edge band. For every $K > 0$ there is one edge level per block, deep inside the gap: $|\omega|/K \rightarrow \frac{m}{6H} B(\frac{m}{6H}, \frac{13}{12}) = 0.9808227327$ [nb07 §5.8] as $K \rightarrow 0$ (exact), 2.93849068 [nb07 §5.8] at $K = 3$, and the ratio falls for larger K [WG 4.3]:

K	R	edge $ \omega $	$ \omega /K$
0.10	1.004988	0.09808212 [nb07 §5.8]	0.980821
1.00	1.414214	0.98067175 [nb07 §5.8]	0.980672
3.00	3.162278	2.93849068 [nb07 §5.8]	0.979497
10.00	10.049876	9.69024420 [nb07 §5.8]	0.969024
20.00	20.024984	18.99601285 [nb07 §5.8]	0.949801
40.00	40.012498	36.76546650 [nb07 §5.8]	0.919137
80.00	80.006250	70.51329445 [nb07 §5.8]	0.881416
100.00	100.005000	86.85053394 [nb07 §5.8]	0.868505

The edge band lies inside the bulk gap $|\omega| < m$ for $K < \sim 1.02 m$: massless fermions bound to the wall, 4 positive states per transverse momentum, with a small **negative** scalar density ($\int(f^2 - g^2) = -0.0498$ at $K = 3$). Further positive levels of the opposite-sign wall appear at $K = 33.80968834$ [nb07 §5.6] (irrep $-i$) and 48.79382020 [nb07 §5.6] ($+i$).

The dependence on the wall condition (the uniform family $f(0) \cos \vartheta + g(0) \sin \vartheta = 0$, symmetry breaking except $\vartheta = \pm\pi/4$): at $K = 3$ the levels sweep the **whole** gap $(-R, R)$ as ϑ varies; at $K = 80$ the lowest positive level moves from $\omega = m = 1$ [nb07 §5.9] ($\vartheta = -\pi/2$, $g(0) = 0$) to 70.985 [nb07 §5.9] (MIT+) and 70.513 [nb07 §5.9] (MIT−). The wall condition, not the pocket, sets the low end of the spectrum [WG 4.4].

Convergence: the reference levels are accurate to about 10^{-10} (tolerance-limited; matching point, `rtol` from 10^{-9} to 10^{-13} , z_a from 2 to 6, the κ grid and the integrator each change ω at $K = 80$, $n = 0$ by at most 1.0×10^{-8}) [WG 4.6].

8.6 The Kohn–Sham (local-density) treatment of the wall [WG 6]

Setting. The functional per 8-volume is $W(\sigma) = m_0\sigma + (\lambda/2)\sigma^2$ with the local mean field $m(z) = W'(\sigma(z)) = m_0 + \lambda\sigma(z)$, $m_0 = 1 H$; $\lambda < 0$ is attractive. The admissible modes are zero modes along $x_5 \dots x_7$, normalized over the hidden coordinate volume, chosen as $V_{\text{hid}} = 1/H^3$ (so λ also stands for the effective 5-dimensional coupling λ/V_{hid}); N_s is the number per unit coordinate observed 3-volume. Per transverse momentum each positive level of either irrep contributes 4 degenerate states; a “band” is one level family $\omega_b(k)$. The scalar density of a block orbital is the Krein-weighted $|f|^2 - |g|^2$, the number density $|f|^2 + |g|^2$, and since the transverse proper volume is $1/\text{Sin}(z)$:

$$\sigma(z) = \text{Sin}(z) \sum_b 4 \int \frac{d^3k}{(2\pi)^3} \theta(\text{occupied}) (f^2 - g^2)_{b,k}(z), \quad n(z) \text{ likewise with } (f^2 + g^2),$$

$$E/A = \sum_{\text{occ}} \omega + \int (W - \sigma W') \frac{dz}{\text{Sin}} = \sum_{\text{occ}} \omega + \frac{|\lambda|}{2} \int \frac{\sigma^2}{\text{Sin}} dz,$$

no-sea, E_{xc} neglected, a common Fermi level μ , Anderson mixing; the particle number is reproduced to 10^{-8} in every iteration.

(a) The author’s mass term: exact. For $W = m_0\sigma$ ($\lambda = 0$) the action is bilinear, the theory is **free**, the single-particle levels of section 8.5 are the exact spectrum, and every many-body eigenstate is a Slater determinant of exact modes; no self-consistency is needed. Consequences: on the **same-sign** wall there is **no** bound positive level for $K < K_c(1) = 6.70624$ [nb07 §5.11]; every wall level lies above m_0 by more than $\sqrt{m_0^2 + K_c^2} - m_0 = 5.78 m_0$ [nb07 §5.11: 5.780388], while the bulk continuum starts at m_0 . The N_s -fermion ground state of the free theory therefore has **no wall-localized fermions**; the wall levels are excited states. On the **opposite-sign** wall the gapless edge band lies below m_0 for $K < \sim 1.02 m_0$; the free ground state of $N_s \leq \frac{4}{6\pi^2} (1.02)^3 = 0.072 H^3$ [nb07 §5.8, §5.11: 0.071622, with $K^* = 1.01971543$] fermions fills it — a genuine wall-bound massless $(3+1)$ -dimensional Fermi sea — and beyond that the particles spill into the bulk.

(b) Same-sign wall, attractive W : the wall-band sector ($\lambda = -0.01 H^{-3}$, $N_s = 100 H^3$, $m_0 = 1$). The self-consistency **converges** (9 iterations, $\max |m_{\text{new}} - m| = 1.5 \times 10^{-10}$). At the box size $Z = 30/H$:

```
E/A = 937.2277360325 H^4   (sum of occupied omega = 936.0540086871, interaction Int(W - sigma W') =
→ 1.1737273454)
```

$E/N = 9.37227736 \text{ H}$ $\mu = 11.6958993131 \text{ H}$
band $n = 0$ (irrep -i) holds all N_s : k in $[5.45898483, 11.80021359]$; band bottom $\omega = 5.54982120$ at
 $\hookrightarrow$ the self-consistent threshold $k_{lo} = 5.45898$ (bare 6.70624)

(every number in this block [nb07 §5.17], the bare threshold [nb07 §5.6]); the density profile has $\sigma = n = 0$ at the wall, a maximum $\sigma = 23.845616$ [nb07 §5.17] (mass minimum $m = 0.761543845$ [nb07 §5.17]) at $z = 0.167$, and an inverse-square tail. The truncation at Z converges slowly (a self-consistent tail $m - m_0 = -0.032/z^2$); extrapolated, $E/A \rightarrow 936.46\text{--}936.53$, $\mu \rightarrow 11.692$, $k_{lo} \rightarrow 5.439$; grid, quadrature and k -grid are converged to $\leq 4 \times 10^{-5}$ in E/A [WG 6.5]. **But this converged state is not the ground state.** (i) $\mu - m_0 = 10.696 > 0$: bulk continuum states lie below the Fermi level, so the global Kohn–Sham ground state has no wall layer at this coupling. (ii) Even within the wall-band sector the Walecka functional $\Omega[m] = \sum_{occ} \omega[m] + \int (m - m_0)^2 / (2|\lambda|) dz / \text{Sin}$ is **not stationary**: along $m = m_0 + s(m_{sc} - m_0)$, $\Omega = 938.1963767831, 937.2277359473, 936.2641446367$ [nb07 §5.17] at $s = 0.98, 1.00, 1.02$ and $d\Omega/ds = -48.305804$ [nb07 §5.17] — the threshold term $\frac{4}{2\pi^2} k_{lo}^2 (-dk_{lo}/ds) (R(k_{lo}) - \mu) = -48.306186$ ($dk_{lo}/ds = -1.301519$, $R(k_{lo}) = 5.549821$; agreement to 8×10^{-6} relative): a deeper well pulls new bound states out of the continuum at the band bottom (energy ~ 5.55) and they replace Fermi-surface states (energy $\mu = 11.70$). So the result is reported as the **self-consistent wall-band configuration**, not as a variational ground state.

The first excited state (Delta-SCF) of that configuration: a fraction x of N_s in band $n = 1$ (irrep $+i$), the rest in $n = 0$, each band filled from its bottom; all converged to 10^{-9} :

x	E/A	$\Delta E/A$	ΔE per promoted fermion
0.01	957.9790646375 [nb07 §5.17]	20.7513286050	20.75132861 [nb07 §5.17]
0.05	1041.8217858764 [nb07 §5.17]	104.5940498439	20.91880997 [nb07 §5.17]
0.10	1148.4782948320 [nb07 §5.17]	211.2505587995	21.12505588 [nb07 §5.17]
0.20	1367.8210883907 [nb07 §5.17]	430.5933523582	21.52966762 [nb07 §5.17]

The lowest Delta-SCF excitation is the $x \rightarrow 0^+$ limit, about $20.75 H$ per promoted fermion, close to the bottom of band $n = 1$ ($\omega = 31.99$ at $k_{lo} = 31.975$) minus the top of $n = 0$ (11.63), $= 20.36$, the remaining 0.4 being the mean-field rearrangement. Intra-band particle–hole excitations are gapless. The self-consistent band gaps at fixed k ($\omega_1 - \omega_0$): 2.08513398 [nb07 §5.17] ($k = 34$), 2.78024477 [nb07 §5.17] (40), 5.02368345 [nb07 §5.17] (60), 7.18485799 [nb07 §5.17] (80).

(c) **Opposite-sign wall: the edge band is a true ground state** ($m_0 = 1$, $N_s = 0.05 H^3$). The edge band starts at the Dirac point ($\omega = 0$, $k = 0$); $N_s = 0.05$ fills it to $k_F = 0.90459$ [nb07 §5.12] with $\mu < m_0$: every state below μ is a bound wall state, so this is a genuine ground state of N_s fermions.

	$\lambda = -0.01 H^{-3}$	$\lambda = -10 H^{-3}$
iterations	3 ($\max dm < 10^{-10}$)	6 ($\max dm < 10^{-10}$)
E/A	0.033268937058 H^4 [nb07 §5.12]	0.033267529405 H^4 [nb07 §5.12]
E/N	0.66537874 H [nb07 §5.12]	0.66535059 H [nb07 §5.12]
μ	0.8871343842 H [nb07 §5.12]	0.8870591549 H [nb07 §5.12]
$\sigma(z)$	in $[-7.964\text{e-}4, 0]$ (negative: $f \sim -g$)	in $[-7.986\text{e-}4, 0]$
$m(z)$	in $[1, 1.00000796]$	in $[1, 1.00798557]$ (the attraction raises m near the wall)
Walecka check	slope $-1.2\text{e-}14$, curvature $+5.6\text{e-}13$	slope $+2.0\text{e-}13$, curvature $+5.6\text{e-}10$

Stationary and a **minimum**: a variationally consistent Kohn–Sham ground state. $E/N = 0.6654$ is the massless value $\frac{3}{4} v k_F$ with $v = 0.9807$: the wall fermions are essentially massless.

It is fully converged ($Z = 45$, a finer grid and more quadrature points change E/A by at most 1.6×10^{-12}). **Its first excited state:** there is no bound band $n = 1$ at the occupied momenta (the next positive bands of the opposite-sign wall appear only at $K = 33.81$ and 48.79); the lowest excitations are gapless intra-band particle-hole pairs and promotion into the bulk continuum, which costs at least $m_0 - \mu = 0.113 H$ [nb07 §5.12: $m_0 - \mu$ from its μ] and unbinds the fermion. A Delta-SCF with a finite fraction in a bound band $n = 1$ would require $k \geq 33.81$, at $\omega - \mu \geq \sim 32.9 H$.

(d) **Strong attraction on the same-sign wall (exploratory; $\lambda = -100 H^{-3}$, $N_s = 0.03 H^3$): did not converge.** Homogeneous bulk matter is self-bound only for $|\lambda|$ between 60 and $80 H^{-3}$ or more (min $E/n - m_0 = +0.000033$ for $\lambda = -0.01 \dots -60$, -0.038261 for -80 , -0.072213 for -100). At $\lambda = -100$ the iteration **failed** (not converged after 60 iterations; max $|m_{\text{new}} - m|$ wandered between 6.6×10^{-3} and 0.71). What was obtained: the layer **is** bound ($\mu = 0.8796 H$, $\mu - m_0 = -0.1204 H$), the two irreps are occupied almost equally (the two 4-fold branches become degenerate away from the wall: 8 states per momentum, as in the homogeneous gas), and the layer drifts **outward** from $z_c = 2.07$ to 3.4 – 3.8 . A rigid-shift diagnostic shows why: the Walecka energy of the layer decreases monotonically with its distance from the wall (0.0318696 at $z_c = 0.6$ to 0.0286926 at 6.0): the same-sign wall **repels** the layer, which has no stationary position at finite distance. There is no wall-bound Kohn–Sham ground state on the same-sign wall at this coupling either.

8.7 What the density-functional treatment establishes, and what it does not

- **Homogeneous gas.** The Kohn–Sham ground state is the gap root of lowest ρ (a maximum of the no-sea $E_n[\sigma]$, a minimum of the Walecka functional for attractive W); for the author’s mass term it is the exact free Fermi sea. The first excitations are gapless particle-hole pairs; pair creation starts at $\omega_F + |m|$ ($q = k_F$) and at $2\omega_F$ for $q = 0$.
- **Along x_0 .** The geometry binds states only above a transverse-momentum threshold on the same-sign wall ($K_c(1) = 6.70624 H$ at $m = H$ [nb07 §5.6]), so the **ground state of the free fable has no wall-bound fermions there**; the wall levels (ground 70.98531 [nb07 §5.5], first excited 78.14425 [nb07 §5.5] at $K = 80$) are excited states of the many-body system. On the opposite-sign wall a gapless edge band of massless wall fermions exists at every K , and it carries a variationally consistent Kohn–Sham ground state; its first excitations are gapless.
- **What did not work** [WG 8]: on the same-sign wall the self-consistent wall-band configuration converges but is neither the global ground state nor a variational minimum within its sector; the strong-coupling layer does not converge (the wall repels it). A grand-canonical treatment with a bulk fable density and a position-constrained iteration were not done. Two runs of the derivation, before `DumpSave` was blocked, rewrote the notebook’s two Part II `.mx` files; their content was verified identical to the committed version (the final script blocks `DumpSave`).
- **Caveats** [WG 8]: a_4 frozen (adiabatic; with a_4 varying the momentum redshifts as $K_\infty = k e^{a_4}$); zero modes along $x_5 \dots x_7$ and along the transverse directions other than the momentum; the wall condition is a choice of self-adjoint extension and the results depend strongly on it; no-sea, E_{xc} neglected, local functional $m(z) = W'(\sigma(z))$; the Delta-SCF states are stationary configurations with constrained occupations, not eigenstates of an exact many-body Hamiltonian.

Notebook 07. `fable-cosmology/notebooks/07_fable_dft_states.ipynb` re-runs these computations with `waveguide.py` (its same-sign Kohn–Sham runs at a reduced resolution,

{'Z': 30.0, 'h_near': 0.006, 'n_gl': 12}, compared in its §5.17 with the solver's full-resolution run, which it starts in the background at its beginning), writes their tables and figures under `fable-cosmology/results/nb07_*`, and asserts every number it prints. Executed and committed in `aca5102`, it prints again every wall-state number above that carries the mark `[nb07 §n]` (to the digits it prints; a few only in a result file of that section). Its comparison with the full-resolution run ends `largest relative difference: opposite-sign (same resolution) 2.8e-09; same-sign ground state (reduced) 1.4e-04; Delta-SCF (reduced) 3.0e-06 [nb07 §5.17]`. The numbers of this section that it does **not** print again, and which are therefore quoted from the report alone `[WG n]`, are: the levels `-2.21043470304386` ($\vartheta = 1.0$, $K = 3$) and `20.1427411133772` ($m = 4$, $K = 20$) of the Mathematica comparison of section 8.4; the bare band gap `9.27198785` at $K = 100$ (it prints the two levels at $K = 100$ whose difference that is, `96.59932036 - 87.32733251 = 9.27198785 [derived here]`); $\int(f^2 - g^2) = 0.02345024$ of the level $n = 2$ at $K = 80$ and `-0.0498` of the opposite-sign edge level at $K = 3$; and the extrapolation of the same-sign Kohn–Sham state to $Z \rightarrow \infty$ ($E/A \rightarrow 936.46\text{--}936.53$, $\mu \rightarrow 11.692$, $k_{\text{lo}} \rightarrow 5.439$). For that last item notebook 07 prints instead the larger box $Z = 45$: `E/A = 937.06107832` ($Z = 30$: `937.22904016`; difference `-0.167962`, `-1.79e-04` relative); `mu = 11.69499658` (`-9.07e-04`); `k_lo = 5.454492` and the tail `sigma(20)/sigma(10) = 0.2529` (an inverse-square tail gives 0.25) `[nb07 §5.13]`, consistent with the slow convergence in Z described in section 8.6.

9 Solving the coupled equations from the beginning of the present universe to today

The coupled system of sections 4–7 is solved three times independently: by the Rust solver `fable_fermion` on the pure-Rust SUNDIALS 7.8.0 CVODE of the `rustSolveIt` repositories (`fable_fermion 0.1.0`, `sundials_rs 7.8.0` (pure Rust), CVODE BDF, its `-version` line), by the independent Python implementation `fable-cosmology/fermion/crosscheck.py` (adaptive quadrature, `brentq`, `DOP853`), and, for two reference cases, by Mathematica (Part VIII, Section 33, and `fable-cosmology/reference/make_reference_fermion.wls`). The runs, their tables and their figures are those of notebook 06 `[nb06]`, which writes every file named below under `fable-cosmology/results/`.

9.1 The two models, and the system handed to CVODE

fable8d — the unstabilized 8-dimensional system. The asymptotic region of the primordial field is the Bianchi-I metric of section 5.4; the sources are radiation (on the observed sheet, $P_{\text{obs}} = \rho_r/3$, $P_{\text{hid}} = P_{x_0} = 0$, $\rho_r = \Omega_{r0}A^{-4}/v$), baryons (dust in every direction, $\rho_b = \Omega_{b0}A^{-3}/v$) and the Kohn–Sham fable of section 7 ($k_F = k_{F0}/A$, per 7-volume, the gap solved at every step). The equations are the constraint (monitored, not imposed) and the evolution equations of section 5.4 with $\kappa_8\rho \rightarrow 3\hat{\rho}$ (densities per 7-volume in ρ_{c0}), so that $\kappa_8(P_i - T/6)$ becomes $F/2 + 3(P_i - P_{\text{hid}})$:

$$\frac{dH_A}{dt} = -H_A\Theta + \frac{F}{2} + 3(P_{\text{obs}} - P_{\text{hid}}), \quad \frac{dH_B}{dt} = -H_B\Theta + \frac{F}{2}, \quad \frac{dH_C}{dt} = -H_C\Theta + \frac{F}{2},$$

with the hidden driver F and $P_{\text{obs}} - P_{\text{hid}}$ formed per source analytically — radiation 0 and $\rho_r/3$, baryons ρ_b and 0, the fable $2W - \sigma_8 W'$ and P_{KS}/v — never by subtracting large numbers. The independent variable is $N = \ln A$; since the physical rates fall like A^{-2} by many orders of magnitude while H_B and H_C start at exactly zero, CVODE integrates the scaled variables $h_i = H_i A^2$ and $\tau = t/A^2$, state $y = (\ln B, \ln C, h_A, h_B, h_C, \tau)$:

$$\frac{d \ln B}{dN} = \frac{H_B}{H_A}, \quad \frac{d \ln C}{dN} = \frac{H_C}{H_A}, \quad \frac{dh_i}{dN} = 2h_i + \frac{A^2 (dH_i/dt)}{H_A}, \quad \frac{d\tau}{dN} = -2\tau + \frac{A^{-2}}{H_A}.$$

Initial state: $H_B = H_C = 0$, H_A from the constraint, $t(a_i) = 1/(2H_A(a_i))$. The forward run is a **boundary-value problem** ($H_B = H_C = 0$ at a_i ; $H_A = B = C = 1$ today), solved by an outer shooting (bracket and Brent) on the fable's amplitude and an inner secant iteration on $\ln v(a_i)$. It runs forward only (section 9.11). [prose nb06 §4, §4.1; file: `fable-cosmology/rust/fable_fermion/src/models.rs` and `run.rs`, module documentation.] (Since commit 3a5020d the module summary of `models.rs` names both: the physical state $(\ln B, \ln C, H_A, H_B, H_C, t)$, and the scaled state $(\ln B, \ln C, h_A, h_B, h_C, \tau)$ with $h_i = H_i A^2$ and $\tau = t/A^2$ that CVODE integrates; before, it listed only the physical one. The change is a comment only [commit 3a5020d].)

fable4d — the stabilized model, the physical one. `fable8d` plus a zero-energy stabilizing stress $P_{\text{stab}} = -F_{\text{total}}/2$ added to the hidden pressures of B and C . Then $F_{\text{total}} + 2P_{\text{stab}} = 0$, $H_B = H_C = 0$ is preserved exactly, $v = 1$, and the observer's equations are **exactly** the Friedmann equations $3H_A^2 = \kappa\rho$, $H'_A = -\frac{\kappa}{2}(\rho + P_{\text{obs}})$, with the hidden equations satisfied identically. The stabilizer is a **Lagrange multiplier** that enforces the frozen hidden sheet, not a stress derived from any field; it has zero energy density, drops out of the conservation law at $H_B = H_C = 0$, and it **violates the null energy condition along** x_0 whenever dust is present: for the null vector $k = e_0/C + e_4$ in the (x_0, x_4) plane $T_{\text{stab}}(k, k) = P_{\text{stab}} = -F/2$, which is $-\rho_{\text{dust}}/2 < 0$ for dust alone. [proved VIII §33]

- `fable4d` [THE RESULT]: with $P_{\text{stab}} = -F/2$ on B and C , the hidden drivers vanish, $F + 2 P_{\text{stab}} == 0$, so $H_B = H_C = 0$ is preserved exactly
- `fable4d` [THE RESULT]: with $B = C = 1$ (labelled substitution) the Einstein equations are EXACTLY $3 H_A^2 == \kappa \rho$ and $H_A' == -(\kappa/2)(\rho + P_{\text{obs}})$; the x_0 and hidden equations then hold with $P_{\text{hid}} + P_{\text{stab}}$
- `fable4d` [has content]: the Friedmann pair is self-consistent - $d/dx_4 (3 H_A^2 - \kappa \rho) == 0$ when $H_A' = -(\kappa/2)(\rho + P_{\text{obs}})$ and $\rho' = -3 H_A (\rho + P_{\text{obs}})$ (the conservation law at $H_B = H_C = 0$)
- `fable4d` [THE RESULT]: the stabilizer VIOLATES the null energy condition along x_0 : $T_{\text{stab}}(k, k) == P_{\text{stab}} == -F/2$ for the null k in the (x_0, x_4) plane, $== -\rho_{\text{dust}}/2 < 0$ for dust alone
- `fable4d` [has content]: it is a Lagrange multiplier with zero energy - $\rho_{\text{stab}} == 0$, and at $H_B = H_C = 0$ it drops out of the conservation law (its term is $(3 H_B + H_C) P_{\text{stab}}$)

(Without the stabilizer, dust would decelerate as $dH/dt = -\frac{5}{2}H^2$ instead of $-\frac{3}{2}H^2$ [file: `models.rs`, module documentation].) `fable4d` is **algebraic** in a : $k_F = k_{F0}/a$, the gap is local, $H_A(a)^2 = \Omega_{r0}a^{-4} + \Omega_{b0}a^{-3} + \rho_f(a)$, and only the age $t(a)$ is integrated ($d\tau/dN = -2\tau + A^{-2}/H_A(N)$). `fable8d -freeze-hidden` integrates the `fable8d` system with the stabilizer ($dH_A/dt = -3H_A^2 + \frac{3}{2}(\rho - P_{\text{obs}})$, H_A evolved, not algebraic) and must reproduce `fable4d`; it does (section 9.10).

CVODE settings: BDF, Newton iteration, a dense Jacobian, $\text{rtol} = 1\text{e-}10$, $\text{atol} = 1\text{e-}12$ [prose nb06 §4.1]. **Normalization:** `fable4d` by a closure today, $\rho_r + \rho_b + \rho_f = 1$ at $A = 1$, with one shooting parameter per potential (section 9.3). **The branch:** the gap root of lowest ρ (section 7.3).

Units (`fable_fermion -constants`, printed by notebooks 05 and 06; $\hbar = c = k_B = 1$):

```
h           = 0.674
H0          = 2.1842852411e-18 1/s = 6.8930799924e-11 1/yr = 1.4377226622e-33 eV
1/H0       = 1.4507302992e10 yr
M_pl (reduced, from G) = 2.4353234593e27 eV = 2.4353234593e18 GeV
rho_c0 = 3 H0^2 M_pl^2 = 3.6777719498e-11 eV^4
E_c = rho_c0^(1/4) = 2.4626131773e-3 eV (unit of m, kF; sigma, n in E_c^3; W, rho in E_c^4 =
↪ rho_c0)
```

```

1 eV                = 4.0607270732e2 E_c
T_CMB               = 2.7255 K = 2.3486541806e-4 eV;  T_nu0 = 1.6763891605e-4 eV
Omega_gamma0        = 5.4437728013e-5  (Omega_gamma0 h^2 = 2.4729753331e-5)
Omega_nu(1 species)0 = 1.2363206389e-5
Omega_r0 (N_eff=3.046) = 9.2096054673e-5  (Omega_r0 h^2 = 4.1837027333e-5)
Omega_b0 = 0.02237/h^2 = 4.9243191364e-2
(g_*(T) is not followed: radiation is Omega_r0 A^-4 at all times, which misstates rho_r A^4 by up to a
↪ factor ~0.39 before e+e- annihilation and the QCD transition)
g (fable states per momentum) = 8
a_BBN (T = 1 MeV)    = 1.6763891605e-10  (= T_nu0 / 1 MeV)
a_rec (z = 1090)     = 9.1659028414e-4

```

and the notebooks recompute it in Python from CODATA and agree to 10^{-9} . Part VIII computes the same constants in Mathematica and asserts them [displayed and proved VIII §33]: units: $1/H_0 = 1.4507303e10$ yr; $M_{\text{pl}} = 2.4353235e18$ GeV; $\rho_{\text{c}0} = 3.6777719e-11$ eV⁴; $E_{\text{c}} = \rho_{\text{c}0}^{(1/4)} = 0.0024626132$ eV, $\Omega_{\text{gamma}0} = 0.000054437728$, $\Omega_{\text{r}0} = 0.000092096055$, $\Omega_{\text{b}0} = 0.049243191$; fable4d RUNS [definition]: the units computed from CODATA 2018 - $E_{\text{c}} = \rho_{\text{c}0}^{(1/4)} == 2.46261e-3$ eV, $\Omega_{\text{r}0} == 9.2096e-5$, $\Omega_{\text{b}0} = 0.02237/h^2 == 0.049243$ (to the quoted digits). The coupling is $\kappa_{4,0} = \kappa_8/V_{h,0}$ with $v = 1$ today, and $\Omega_i := \kappa_8 \rho_{7,i}/(3H_A^2)$.

Assumptions stated with every run [prose nb06 §4]: x_0 is compactified in the asymptotic region with a Kaluza–Klein gap far above the temperatures considered; the hidden timelike sheet is a formal comoving volume (compactifying it gives closed timelike curves); every field is truncated to its zero modes along the hidden directions; gravity is semiclassical ($G = \kappa\langle\hat{T}\rangle$); the exchange–correlation energy is neglected; the Dirac-sea energy is absorbed into a renormalized W (section 7.11); the terms of order $\cot$ and $\cot^2$ of the asymptotic reading are neglected (section 5.3).

9.2 The interval, and why it is the right one

“From the beginning of the present universe.” The present universe is the asymptotic region of the primordial field (section 5.3), in which the primordial field is the 8-dimensional Bianchi-I metric. Its beginning, for this system, is the earliest time at which the model applies with ordinary matter: deep in the radiation era, before nucleosynthesis, with the fable ultra-relativistic. The runs start at

$$\begin{aligned}
a_i &\leq \min(10^{-10}, 0.01 a_{\text{nr}}(m)) \\
(\text{default } a_i &= 10^{-12}: T = T_{\text{CMB}}/a_i = 234.9 \text{ MeV;} \\
a_{\text{BBN}} &= 1.6763891605 \times 10^{-10} \text{ at } T = 1 \text{ MeV})
\end{aligned}$$

and end at $A = 1$, defined by $T_{\text{CMB}} = 2.7255$ K. The criterion is checked for every run [prose nb06 §4, §5.1]. Three facts justify it: (i) before a_{nr} the fable is radiation-like ($w = 1/3$ to 10^{-8} , asserted for the reference case in Part VIII) and its history is fixed by its conserved number; (ii) the pre-universe itself does **not** fix a_i — nothing in the notebook ties the author’s constant H or the position along x_0 to a time in the present universe (section 5.3); (iii) the results do not depend on a_i [nb06 §5.1, file: fable-cosmology/results/nb06_ai_insensitivity.csv]:

model	a_i	$T(a_i)$ [MeV]	$H_B(1)$	$\rho_U(1)$	q_0	t_0 [$1/H_0$]	$w_f(0.5)$	$G(a_{\text{BBN}})$ / $G(1)$
fable4d	1e-12	234.9	0.0000e+00	0.685664712581	-0.528451020843	0.951246512174	-0.244385837789	1.000000e+00
fable4d	1e-11	23.5	0.0000e+00	0.685664712581	-0.528451020843	0.951246510304	-0.244385846968	1.000000e+00
fable4d	1e-10	2.3	0.0000e+00	0.685664712581	-0.528451020843	0.951246512298	-0.244385836589	1.000000e+00
fable8d	1e-12	234.9	9.9990e-01	6.684880971343	-0.842486541055	0.652899266076	-0.164884834974	1.046057e+12
fable8d	1e-11	23.5	9.9990e-01	6.684880972924	-0.842486540972	0.652899265712	-0.164884977537	1.046057e+12
fable8d	1e-10	2.3	9.9990e-01	6.684880970384	-0.842486544662	0.652899266556	-0.164884818972	1.046057e+12

(**lambda-mass**, $m_0 = 30 \text{ eV}$.) The largest relative change between $a_i = 10^{-12}$ and 10^{-10} is 2.0×10^{-9} (t_0) and 3.8×10^{-8} ($w_f(0.5)$) in **fable4d**, 8.6×10^{-7} ($w_f(0.5)$) in **fable8d**. The one approximation the early start exposes is $g_*(T)$, which is not followed (the solver's constants line above).

9.3 The potentials, and what each run fixes

For the potentials that supply dark energy the split is imposed today: the quasiparticles carry $\Omega_{\text{qp}0} = 0.265$ (the dark matter) at the mass $m_{\text{today}} = W'(\sigma_t)$, and the amplitude closes the budget [prose nb06 §3; file: **run.rs**, module documentation]:

name	$W(\sigma)$	given	shot (to close the budget today)
mass (the author's)	$m_0\sigma$	m_0	$\ln k_{F0}$; the fable closes the budget alone, $\Omega_{f0} = 1 - \Omega_{r0} - \Omega_{b0}$ (Einstein–de Sitter-like)
lambda-mass	$V_0 + m_0\sigma$	$m_0, \Omega_{\text{qp}0} = 0.265$	V_0 (a bare Lambda, reported as such); k_{F0} from $\varepsilon_{\text{KS}}(m_0, k_{F0}) = \Omega_{\text{qp}0}$
power	$m_0\sigma + \lambda\sigma^\nu$	$m_{\text{today}}, \nu, \Omega_{\text{qp}0}$	the condensate energy today $U_t = (1 - \nu) \lambda \sigma_t^\nu$; then $\lambda = U_t / ((1 - \nu) \sigma_t^\nu)$, $m_0 = m_{\text{today}} - \nu \lambda \sigma_t^{\nu-1}$ (may be negative)
expdamp	$V_0 + m_0 \sigma e^{-\sigma/s_1}$	$m_{\text{today}}, x_t = \sigma_t/s_1$ (Part VI's 1/2.21), $\Omega_{\text{qp}0}$	V_0 ; $s_1 = \sigma_t/x_t$, $m_0 = m_{\text{today}} e^{x_t}/(1 - x_t)$
quadratic	$V_0 + m_0\sigma + (\lambda/2) \sigma^2$	$m_{\text{today}},$ $g_q = \lambda \sigma_t / m_{\text{today}}, \Omega_{\text{qp}0}$	V_0 ; $m_0 = m_{\text{today}} (1 - g_q)$
lorentz	$V_0 + m_0\sigma / (1 + (\sigma/s_1)^2)$	m_{today}, x_t (Part VI's 1/1.5), $\Omega_{\text{qp}0}$	V_0 ; $m_0 = m_{\text{today}} (1 + x_t^2)^2 / (1 - x_t^2)$

(The solver's parameter names: x_t is **xt**, g_q is **gq**, λ is **lam**, ν is **nu**.)

9.4 Runs (a): the author's mass term alone — an Einstein–de Sitter universe with a fable that cools

$W = m_0\sigma$, **fable4d**, $m_0 = 1, 30, 100, 1000, 2000 \text{ eV}$. There is no condensate ($U = 0$): the fable is a free degenerate gas and, with no Lambda, it closes the budget. Its $w_f = w_{\text{DM}} = P_{\text{KS}}/\varepsilon_{\text{KS}}$ runs from 1/3 to 0 through a_{nr} , the scale factor at which $k_F = m$ [**nb06 §5.2**, file: **results/nb06_runs.csv**, figure **results/nb06_a_mass_wdm.png**] (figure 7):

run	$k_{F0} [\text{eV}]$	a_{nr}	a_i ok	ΔN_{eff} BBN	ΔN_{eff} rec	$3P/\rho_\nu$ rec	q_0	$t_0 [\text{Gyr}]$	Ω_{f0}
mass 1 eV (EdS)	6.3727e-04	6.3727e-04	True	36.75	79.96	17.67	0.50005	9.6706	0.95066
mass 30 eV (EdS)	2.0509e-04	6.8364e-06	True	0.3943	70.49	0.002353	0.50005	9.6707	0.95066
mass 100 eV (EdS)	1.3730e-04	1.3730e-06	True	0.07918	70.48	9.489e-05	0.50005	9.6707	0.95066
mass 1000 eV (EdS)	6.3727e-05	6.3727e-08	True	0.003675	70.48	2.044e-07	0.50005	9.6707	0.95066
mass 2000 eV (EdS)	5.0580e-05	2.5290e-08	True	0.001459	70.48	3.22e-08	0.50005	9.6707	0.95066

(ΔN_{eff} is the whole fable energy in units of one massless neutrino species; at recombination it is large because the fable is the dark matter there, and $3P/\rho_\nu$ is its radiation-like part.) The solver's own log of the 30 eV run (**fable_fermion fable4d -potential mass -param m0_ev=30**, printed in notebook 05): # NOTE: W = m0 sigma alone (no V0): the closure forces $\Omega_{f0} = 1 - \Omega_{b0} - \Omega_{r0} = 0.950665$: an Einstein–de Sitter-like universe ($q_0 = 0.5000$, +1/2 for pure dust), and a_{nr} ($k_F = |m_{\text{eff}}|$) = 6.837070e-6

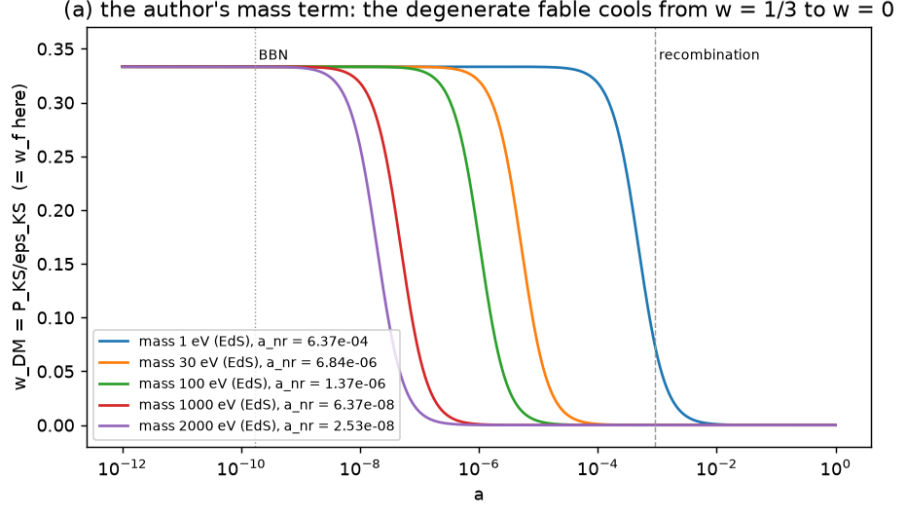


Figure 7: fable-cosmology/results/nb06_a_mass_wdm.png (notebook 06, section 5.2), runs (a): the author’s mass term, $w_{\text{DM}} = P_{\text{KS}}/\varepsilon_{\text{KS}}$ ($= w_f$ there) against a : the degenerate fable cools from $1/3$ to 0 .

($z_{\text{nr}} = 1.4626\text{e}5$), $a_{\text{eq}} = 9.2105\text{e}-5$: WARM / not hot (a_{nr} between $1\text{e}-6$ and a_{eq}) [nb05 §5.7]. This model has no dark energy; it is the author’s mass term as the only content of the dark sector.

9.5 Runs (b): a mass term plus a bare cosmological constant — Λ CDM with fable dark matter

$W = V_0 + m_0\sigma$, fable4d, $\Omega_{\text{qp}0} = 0.265$, V_0 (a bare Lambda) closing the budget [nb06 §5.3, figure results/nb06_b_lambda_mass.png] (figure 8):

run	a_{nr}	ΔN_{eff} BBN	ΔN_{eff} rec	q_0	t_0 [Gyr]	$V_0 = \rho_U$
lambda-mass 30 eV	4.4658e-06	0.07179	19.65	-0.52845	13.8000	0.685665
lambda-mass 100 eV	8.9687e-07	0.01442	19.65	-0.52845	13.8000	0.685665
lambda-mass 1000 eV	4.1629e-08	0.0006692	19.65	-0.52845	13.8000	0.685665
lambda-mass 2000 eV	1.6521e-08	0.0002656	19.65	-0.52845	13.8000	0.685665
lambda-mass 30 eV (reference)	4.4658e-06	0.07179	19.65	-0.52845	13.8000	0.685665

and the observer’s fits:

run	CPL $w_f(w_0, w_a)$	CPL $w_{\text{DE,inf}}(w_0, w_a)$	$\max w_{\text{DE,inf}} + 1 , a \geq 0.3$
lambda-mass 30 eV	(-0.7573, +1.0100)	(-1.0000000003, +1.33e-09)	1.47e-09
lambda-mass 100 eV	(-0.7573, +1.0100)	(-1.0000000000, +5.34e-11)	5.91e-11
lambda-mass 1000 eV	(-0.7573, +1.0100)	(-1.0000000000, +1.16e-13)	1.31e-13
lambda-mass 2000 eV	(-0.7573, +1.0100)	(-1.0000000000, +1.72e-14)	2.40e-14
lambda-mass 30 eV (reference)	(-0.7569, +1.0092)	(-1.0000000003, +1.31e-09)	1.44e-09

The mass does not vary ($w_{\text{DM,eff}} = w_{\text{DM}}$, $Q = 0$) and the condensate is the constant V_0 , so the dark energy an observer infers is a **cosmological constant**, to the precision of the quasiparticles’ kinetic energy that the observer counts as dark energy. The intrinsic w_f today is -0.72124767 [file: nb06_runs.csv] — it mixes the gas and V_0 and is not a dark-energy equation of state. The fifth run ($a_i = 10^{-10}$, 701 rows, results/nb06_fable4d_mass30eV.csv) is Part VIII’s reference case (section 9.12).

9.6 Runs (c): potentials with which the fable supplies the dark energy itself

power ($\nu = 0.236, 0.5$), expdamp (Part VI’s $x_t = 1/2.21$), attractive quadratic ($g_q = -0.5$), each at $m_{\text{today}} = 30$ and 100 eV, fable4d, and Part VIII’s mass-varying reference case (power, $\nu = 0.5$,

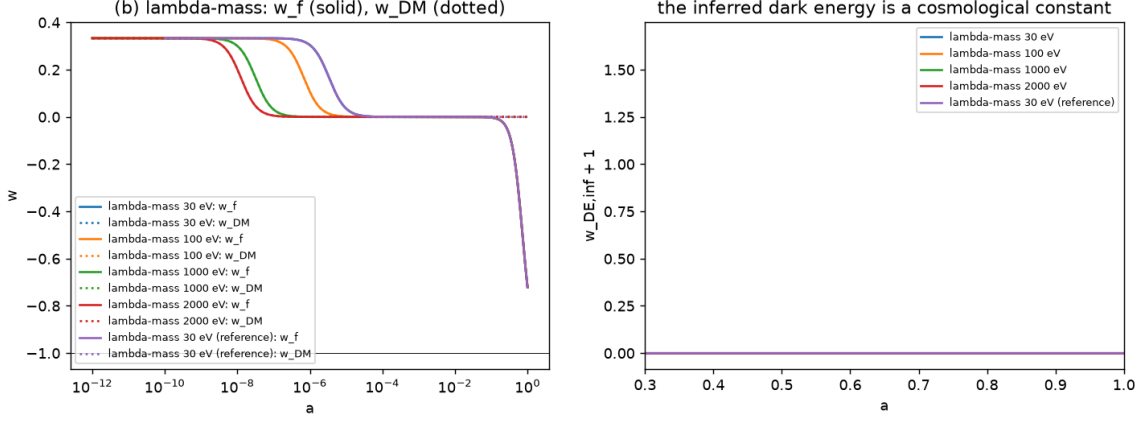


Figure 8: fable-cosmology/results/nb06_b_lambda_mass.png (notebook 06, section 5.3), runs (b): w_f (solid) and w_{DM} (dotted); and $w_{DE,inf} + 1$ on $0.3 \leq a \leq 1$: the inferred dark energy is a cosmological constant.

$m_{\text{today}} = 100 \text{ eV}$, $\Omega_{\text{qp0}} = 0.55$, 701 rows, results/nb06_fable4d_power.csv) [nb06 §5.4, figures results/nb06_c_de_potentials.png, results/nb06_c_exchange_vacuum_stabilizer.png] (figures 9 and 10):

run	a_{nr}	$\min m/m_{\text{today}}$ ($10^{-3} \leq a \leq 0.3$)	$w_f(1)$	$\min w_f$	CPL w_f	$\min c_s^2$
power nu=0.236, 30 eV	2.22e-05	2.01e-01	-0.7212	-0.7212	(-0.783, +0.445)	-0.611
power nu=0.236, 100 eV	4.47e-06	2.01e-01	-0.7212	-0.7212	(-0.783, +0.445)	-0.611
power nu=0.5, 30 eV	0.669	7.91e-12	-0.7212	-1.0000	(-0.804, -0.358)	-27
power nu=0.5, 100 eV	0.669	1.59e-12	-0.7212	-1.0000	(-0.804, -0.359)	-27
expdamp xt=1/2.21, 30 eV	0.712	6.58e-12	-0.7212	-1.0000	(-0.829, -0.317)	-21.8
expdamp xt=1/2.21, 100 eV	0.712	1.32e-12	-0.7212	-1.0000	(-0.829, -0.317)	-21.8
quadratic gq=-0.5, 30 eV	0.643	8.93e-12	-0.7212	-1.0000	(-0.730, -0.483)	-47.1
quadratic gq=-0.5, 100 eV	0.643	1.79e-12	-0.7212	-1.0000	(-0.730, -0.483)	-47.1
power nu=0.5, 100 eV, $\Omega_{\text{qp0}} = 0.55$	4.21e-06	2.72e-01	-0.4215	-0.4215	(-0.448, +0.264)	-0.364

and the observer's inferred dark energy:

run	poles of $w_{DE,inf}$	$w = -1$ crossings	phantom in $0.3 \leq a \leq 1$?	CPL $w_{DE,inf}$	fit from a
power nu=0.236, 30 eV	3.43e-06, 0.533	4.55966e-06, 1	True	(-0.082, -7.986)	0.559
power nu=0.236, 100 eV	6.89e-07, 0.533	9.15884e-07, 1	True	(-0.082, -7.986)	0.559
power nu=0.5, 30 eV	3.35e-06, 0.619	4.46748e-06, 1	True	(-0.249, -10.191)	0.650
power nu=0.5, 100 eV	6.73e-07, 0.619	8.97015e-07, 1	True	(-0.249, -10.191)	0.650
expdamp xt=1/2.21, 30 eV	3.35e-06, 0.625	4.46748e-06, 1	True	(-0.128, -11.778)	0.656
expdamp xt=1/2.21, 100 eV	6.73e-07, 0.625	8.97015e-07, 1	True	(-0.128, -11.779)	0.656
quadratic gq=-0.5, 30 eV	3.35e-06, 0.602	4.46748e-06, 1	True	(-0.180, -9.321)	0.632
quadratic gq=-0.5, 100 eV	6.73e-07, 0.602	8.97015e-07, 1	True	(-0.180, -9.321)	0.632
power nu=0.5, 100 eV, $\Omega_{\text{qp0}} = 0.55$	8.96e-07, 0.63	1.18957e-06, 1	True	(-0.327, -9.682)	0.662

and, for the same runs [nb06 §5.4]:

run	$c_s^2 < 0$ on ($a \geq 10^{-3}$)	$w_{\text{DM,eff}}$ range ($a \geq 10^{-3}$)	$ \Delta E_{\text{vac}} /\rho_{\text{tot}}, P_{\text{stab}}(1) [\rho_{c0}]$ $a \geq 0.3$	ΔN_{eff} BBN	
power nu=0.236, 30 eV	[0.00409, 1]	(-0.611, +9.85e-05)	1.314e+13	-0.8428	0.0718
power nu=0.236, 100 eV	[0.00194, 1]	(-0.611, +3.59e-06)	1.622e+15	-0.8428	0.0144
power nu=0.5, 30 eV	[0.608, 1]	(-27.6, +0.333)	2.229e+14	-0.8428	0.0718
power nu=0.5, 100 eV	[0.608, 1]	(-27.7, +0.333)	2.752e+16	-0.8428	0.0144
expdamp xt=1/2.21, 30 eV	[0.643, 1]	(-21.8, +0.333)	2.327e+14	-0.8428	0.0718
expdamp xt=1/2.21, 100 eV	[0.643, 1]	(-21.8, +0.333)	2.873e+16	-0.8428	0.0144
quadratic gq=-0.5, 30 eV	[0.592, 1]	(-47.1, +0.333)	2.031e+14	-0.8428	0.0718
quadratic gq=-0.5, 100 eV	[0.592, 1]	(-47.1, +0.333)	2.508e+16	-0.8428	0.0144
power nu=0.5, 100 eV, $\Omega_{\text{qp0}} = 0.55$	[0.001, 1]	(-0.364, -3.89e-05)	2.350e+14	-0.7003	0.0382

and smallest w_f over all 9 runs: -0.9999987921 (never below -1); every run has $c_s^2 < 0$ somewhere after $a = 1e-3$: True. Reading the tables:

- Only the two $\nu = 0.236$ runs and the reference run stay **massive** through the matter era. The $\nu = 0.5$, **expdamp** and **quadratic** runs with $\Omega_{\text{qp0}} = 0.265$ have $\min m/m_{\text{today}} \sim 10^{-12}$: the gap pins their mass near zero and the fable is a **massless gas** (radiation, $w = 1/3$) until $a_{\text{nr}} = 0.64\text{--}0.71$ — it supplies no dark matter at recombination (ΔN_{eff} at recombination equals its BBN value, e.g. 0.071792425 for **power nu=0.5, 30 eV** [file: nb06_runs.csv]). Their CPL fits of w_f look like the supernova fits (-0.861, -0.60), but they are not viable cosmologies.
- The observer's inferred dark energy $w_{\text{DE,inf}}$ has **poles** (its $\rho_{\text{DE,inf}}$ changes sign) inside $0.3 \leq a \leq 1$ in every run, and it is phantom and crosses -1 in every run; a CPL line is not a faithful description (the fits above the pole are in **results/nb06_cpl_fits.csv**).
- $c_s^2 < 0$ in every run: an **adiabatic instability**.
- The Dirac-sea energy that the no-sea functional drops changes along every run by 1.31×10^{13} to 2.87×10^{16} times the total density at $a \geq 0.3$.

9.7 The inferred dark energy of a massive power law, in closed form

Once the gas is non-relativistic, $\sigma_8 \approx n \propto a^{-3}$, the condensate is $U = U_t a^{-3\nu}$ and the mass $m(a) = m_0 + \nu U_t a^{3(1-\nu)} / ((1-\nu) \sigma_t)$. Subtracting cold dark matter with today's mass leaves $\rho_{\text{DE,inf}} = U + (m(a) - m_{\text{today}}) n = U_t [a^{-3\nu} - \nu a^{-3}] / (1-\nu)$ and $P_{\text{DE}} = -U$, hence

$$w_{\text{DE,inf}}(a) = -\frac{1-\nu}{1-\nu a^{-3(1-\nu)}},$$

a pole at $a_{\text{pole}} = \nu^{1/(3(1-\nu))}$, $w > 0$ before it, phantom ($w < -1$) between it and today, and $w = -1$ at $a = 1$ up to the quasiparticles' kinetic energy, which makes it cross -1 just before today. It depends on ν alone [prose nb06 §5.4]. The runs agree [**nb06 §5.4**]:

```
power nu=0.236, 30 eV      w_DE,inf against -(1-nu)/(1 - nu a^(-3(1-nu))) on a >= 0.3: max
→ deviation 1.2e-09; pole: formula 0.532600, run 0.532723; crossing of -1: 1.00000000
power nu=0.236, 100 eV    w_DE,inf against -(1-nu)/(1 - nu a^(-3(1-nu))) on a >= 0.3: max
→ deviation 4.9e-11; pole: formula 0.532600, run 0.532723; crossing of -1: 1.00000000
power nu=0.5, 100 eV, Omega_qp0=0.55 w_DE,inf against -(1-nu)/(1 - nu a^(-3(1-nu))) on a >= 0.3: max
→ deviation 4.2e-11; pole: formula 0.629961, run 0.630028; crossing of -1: 1.00000000
```

The mechanism is that of Das, Corasaniti and Khoury, Phys. Rev. D **73**, 083509 (2006): subtracting cold dark matter with today's mass when the mass was different leaves $\rho_{\text{DE}} + P_{\text{DE}} \approx (m_{\text{eff}}(a) - m_{\text{today}}) n$, which is negative whenever m_{eff} grows; an apparent phantom with no phantom field.

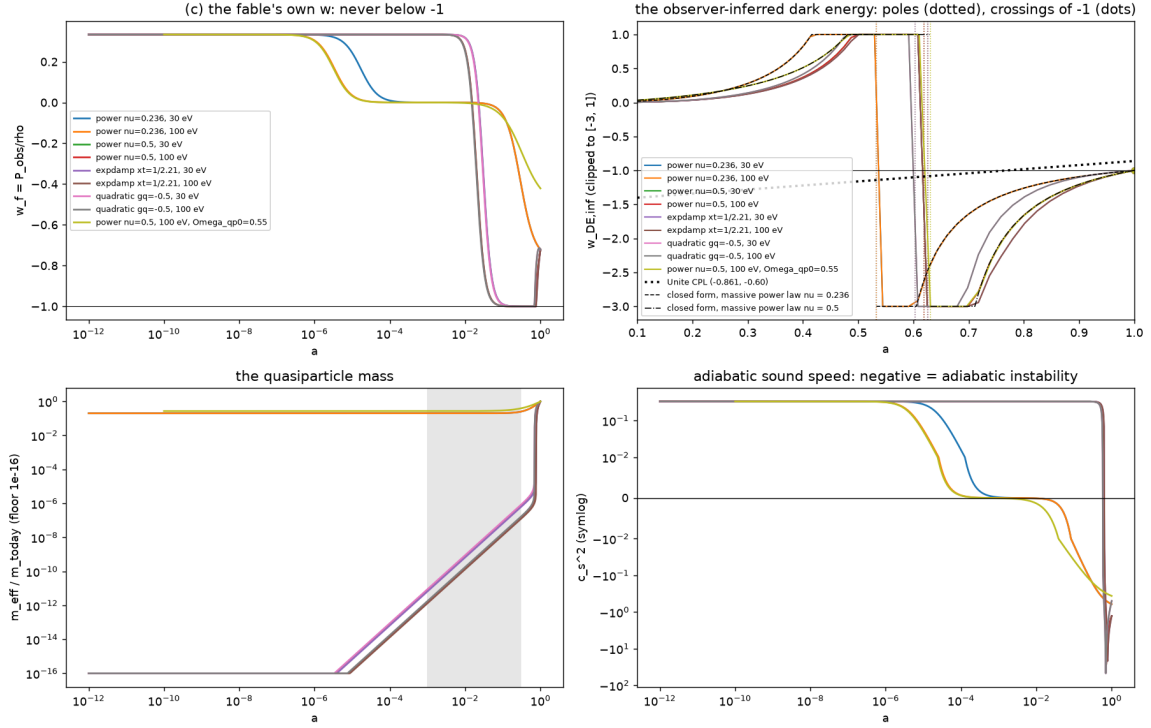


Figure 9: fable-cosmology/results/nb06_c_de_potentials.png (notebook 06, section 5.4), runs (c): w_f (never below -1), the observer's $w_{DE,inf}$ (clipped to $[-3, 1]$, poles and crossings marked), m_{eff}/m_{today} , and c_s^2 (negative below the line).

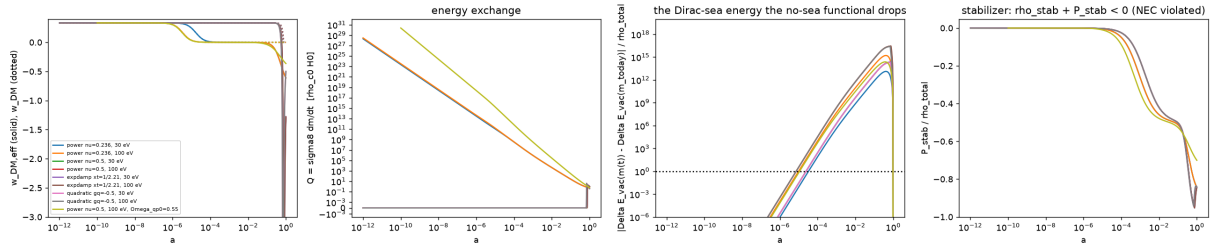


Figure 10: fable-cosmology/results/nb06_c_exchange_vacuum_stabilizer.png (notebook 06, section 5.4), runs (c): w_{DM} and $w_{DM,eff}$, the exchange Q , the Dirac-sea energy relative to the total density, and P_{stab}/ρ .

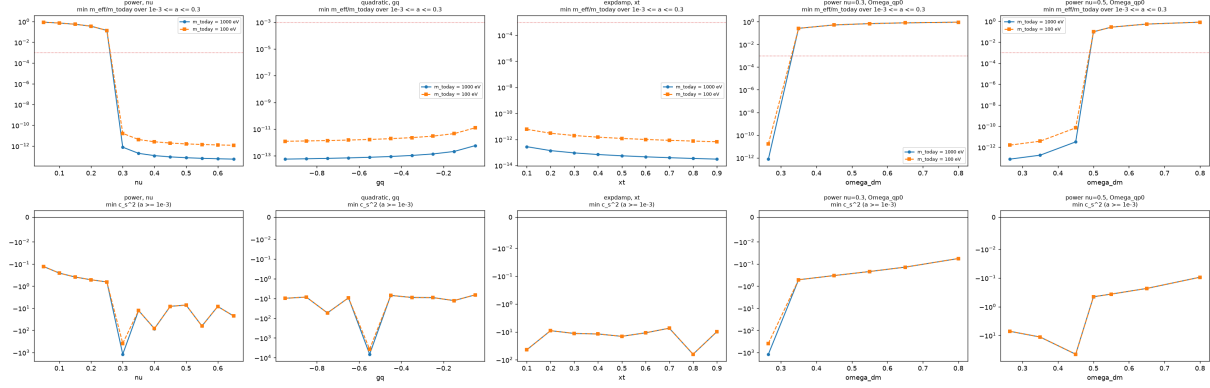


Figure 11: fable-cosmology/results/nb06_scan.png (notebook 06, section 5.6), the scans: per family, the smallest $m_{\text{eff}}/m_{\text{today}}$ over $10^{-3} \leq a \leq 0.3$ and the smallest c_s^2 ($a \geq 10^{-3}$) against the scanned parameter.

9.8 The scans: where does a dark-energy fable stay massive through the matter era?

Notebook 06 scanned the shape parameters — power $\nu = 0.05 \dots 0.65$, quadratic $g_q = -0.95 \dots -0.05$, expdamp $x_t = 0.1 \dots 0.9$, and for power $\nu = 0.3$ and 0.5 the quasiparticle share $\Omega_{\text{qp}0} = 0.265 \dots 0.8$ — at $m_{\text{today}} = 1000 \text{ eV}$ and 100 eV (fable4d, 401 output rows, 45 + 45 runs), and re-computed every scan run independently as it ran [nb06 §5.6, files results/nb06_scan.csv, figures results/nb06_scan.png, results/nb06_scan_cpl.png] (figures 11 and 12). Its conclusions, verbatim:

```
scan points whose fable stays massive through 1e-3 <= a <= 0.3 (min m/m_today > 1e-3): 28 of 90: power,
-> nu=0.05 (1000 eV), power, nu=0.1 (1000 eV), power, nu=0.15 (1000 eV), power, nu=0.2 (1000 eV),
-> power, nu=0.25 (1000 eV), power nu=0.3, Omega_qp0=0.35 (1000 eV), power nu=0.3, Omega_qp0=0.45 (1000
-> eV), power nu=0.3, Omega_qp0=0.55 (1000 eV), power nu=0.3, Omega_qp0=0.65 (1000 eV), power nu=0.3,
-> Omega_qp0=0.8 (1000 eV), power nu=0.5, Omega_qp0=0.5 (1000 eV), power nu=0.5, Omega_qp0=0.55 (1000
-> eV), power nu=0.5, Omega_qp0=0.65 (1000 eV), power nu=0.5, Omega_qp0=0.8 (1000 eV), power, nu=0.05
-> (100 eV), power, nu=0.1 (100 eV), power, nu=0.15 (100 eV), power, nu=0.2 (100 eV), power, nu=0.25
-> (100 eV), power nu=0.3, Omega_qp0=0.35 (100 eV), power nu=0.3, Omega_qp0=0.45 (100 eV), power
-> nu=0.3, Omega_qp0=0.55 (100 eV), power nu=0.3, Omega_qp0=0.65 (100 eV), power nu=0.3, Omega_qp0=0.8
-> (100 eV), power nu=0.5, Omega_qp0=0.5 (100 eV), power nu=0.5, Omega_qp0=0.55 (100 eV), power nu=0.5,
-> Omega_qp0=0.65 (100 eV), power nu=0.5, Omega_qp0=0.8 (100 eV)
of these, with c_s^2 >= 0 after a = 1e-3: 0; smallest c_s^2 among them: -0.6469; largest: -0.0565
power law: 'massive through the matter era' <=> 'bare m0 > 0' at all 52 power scan points: True
the estimate Omega_qp0/U_t > nu/(1 - nu) agrees with the sign of m0 at 52 of 52 points; with Omega_qp0
-> = 0.265 it allows nu < Omega_qp0/(1 - Omega_b0 - Omega_r0) = 0.2788
```

So: every quadratic and expdamp point is massless in the matter era; a power law stays massive exactly when its bare mass $m_0 = m_{\text{today}} - \nu \lambda \sigma_t^{\nu-1}$ is positive, i.e. when $\Omega_{\text{qp}0}/U_t > \nu/(1 - \nu)$; with the observed split $\Omega_{\text{qp}0} = 0.265$ that requires $\nu < 0.2788$ (massive up to $\nu = 0.25$, massless from $\nu = 0.3$ in the scan); a larger ν needs a larger quasiparticle share ($\Omega_{\text{qp}0} \geq 0.35$ for $\nu = 0.3$, ≥ 0.5 for $\nu = 0.5$). And every massive point has $c_s^2 < 0$ after $a = 10^{-3}$. (At high density $\nu \lambda \sigma_t^{\nu-1} \rightarrow 0$, so if $m_0 < 0$ the gap can only be met with $m \rightarrow 0^+$; with $\sigma_t \approx n_t$ and $\varepsilon_{\text{KS}} \approx m_{\text{today}} n_t$ today this happens exactly when $\Omega_{\text{qp}0}/U_t < \nu/(1 - \nu)$ [prose nb06 §5.6].) In the massless cases the smallest c_s^2 sits at the sharp release of the pinned gap, and its magnitude depends on how finely the output grid samples that release; its sign does not [prose nb06 §5.6]. The largest cross-check difference over the 90 scan runs is $4.43\text{e-}09$ [nb06 §5.10].

9.9 How heavy must a fable that is all of the dark matter be?

[nb06 §5.7, file results/nb06_dm_mass_bounds.csv, figure results/nb06_dm_mass_bounds.png] (figure 13)

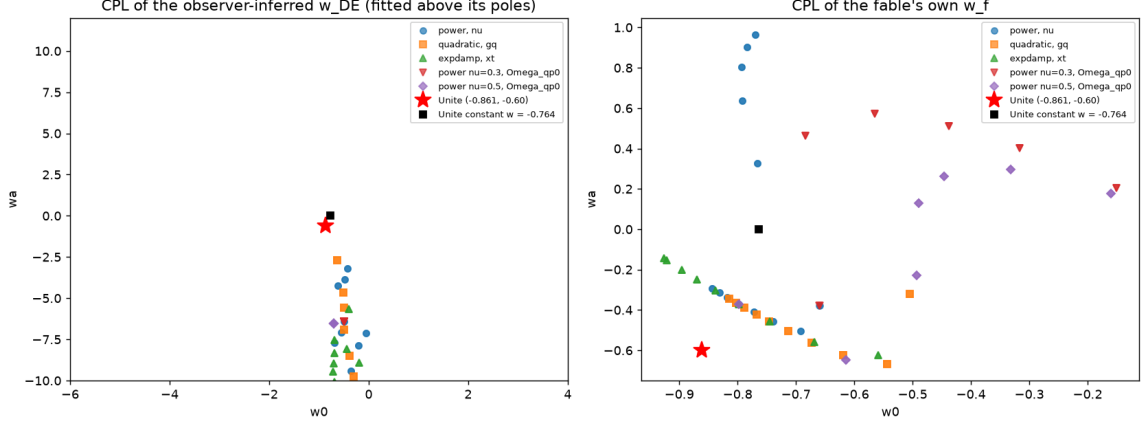


Figure 12: fable-cosmology/results/nb06_scan_cpl.png (notebook 06, section 5.6), the scans: the CPL points (w_0, w_a) of $w_{\text{DE,inf}}$ (fitted above its poles) and of w_f .

bound	fable mass	detail
$\Delta N_{\text{eff}}(\text{BBN}) < 0.3$	$m > 10.2645 \text{ eV}$	fit over 17 runs: $\Delta N_{\text{eff}} = 6.6922 (m/\text{eV})^{-1.33333}$, max $ \ln \text{residual} = 7.1\text{e-}05$
$\Delta N_{\text{eff}}(\text{BBN}) < 0.2$	$m > 13.9126 \text{ eV}$	same fit
free streaming: $v_{\text{rms}0}$ matched to thermal WDM of 3.3 keV (Viel et al. 2013)	$m \approx 1124.92 \text{ eV}$	$v_{\text{rms}0} = 8.2629\text{e-}03 \text{ km/s}$
free streaming: $v_{\text{rms}0}$ matched to thermal WDM of 5.3 keV (Iršič et al. 2017)	$m \approx 1806.69 \text{ eV}$	$v_{\text{rms}0} = 4.3932\text{e-}03 \text{ km/s}$
Tremaine–Gunn, $g = 8$, $\sigma = 10 \text{ km/s}$, $r_c = 0.3 \text{ kpc}$	$m > 231.315 \text{ eV}$	$m^4 > 9 h^3 / (2 (2\pi)^{5/2} g G \sigma r_c^2)$
Tremaine–Gunn, $g = 8$, $\sigma = 7 \text{ km/s}$, $r_c = 0.15 \text{ kpc}$	$m > 357.639 \text{ eV}$	same

$\Delta N_{\text{eff}} \propto m^{-4/3}$ because $\Delta N_{\text{eff}}(\text{BBN})$ depends only on k_{F0} , which $\varepsilon_{\text{KS}}(m, k_{F0}) = \Omega_{\text{qp}0}$ fixes, and $k_{F0} \propto m^{-1/3}$ for a non-relativistic gas today; at 1, 30, 100, 1000, 2000 eV it is 6.69227, 0.0717924, 0.0144181, 0.000669238, 0.00026561. A degenerate gas has the velocity dispersion of a filled Fermi sphere, $v_{\text{rms}} = \sqrt{3/5} k_F/m$: 4.4870e+00, 2.6132e+00, 2.0827e-01, 9.6670e-03, 3.8364e-03 km/s at 10, 15, 100, 1000, 2000 eV. The Lyman-alpha-allowed thermal-relic warm dark matter ($m_{\text{WDM}} \gtrsim 3.3 \text{ keV}$ and 5.3 keV) has $v_{\text{rms}0} = 8.2629\text{e-}03$ and 4.3932e-03 km/s; a degenerate fable with the same $v_{\text{rms}0}$ needs $m \approx 1.1\text{--}1.8 \text{ keV}$ — a velocity-matching estimate, not a transfer-function computation. The Tremaine–Gunn bound uses illustrative dwarf-spheroidal values (inputs, not results). The free-streaming bound is far stronger than ΔN_{eff} . At such masses the change of w_{DM} from 1/3 to 0 happens around $a_{\text{nr}} = 4.16 \times 10^{-8}$ (1 keV) or 1.65×10^{-8} (2 keV).

9.10 Runs (d): without the stabilizer, the hidden sheet moves and the Newton constant varies

fable8d for the mass term at 1 and 100 eV and lambda-mass at 30 eV and 1 keV, and two controls: radiation alone (`-no-fable -omega-b 0`), which must keep the hidden sheet exactly frozen ($F = 0$), and radiation plus baryons (`-no-fable`), in which dust alone drives it. $G(a)/G(1) = 1/v$ (the observed Newton constant is $G_4 = G_8/V_{\text{hid}} \propto 1/v$) [nb06 §5.8, file results/nb06_nogo_

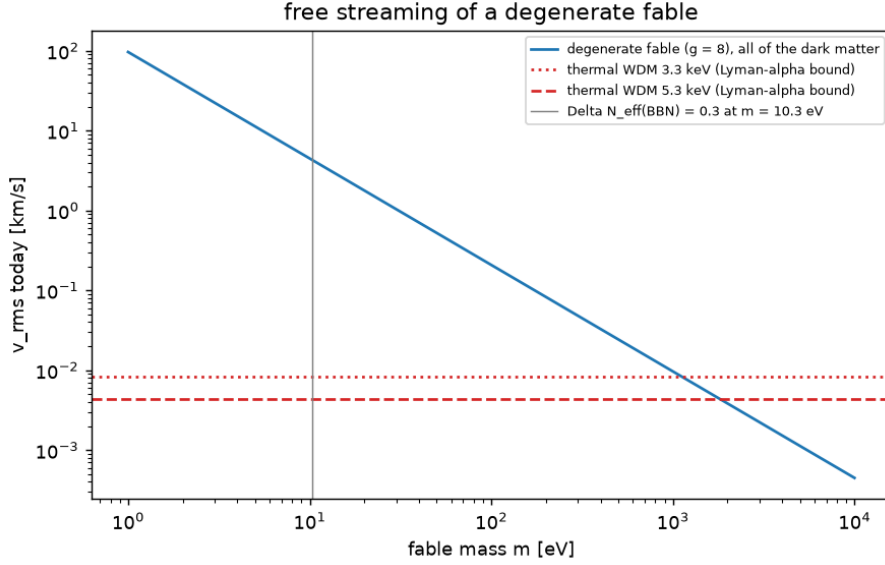


Figure 13: fable-cosmology/results/nb06_dm_mass_bounds.png (notebook 06, section 5.7): the free streaming of a degenerate fable: v_{rms} today against the fable mass, with the warm-dark-matter velocities.

8d.csv, figure results/nb06_d_nogo_8d.png] (figure 14):

run	$B(a_i) = C(a_i)$	$G(a_i)/G(1)$	$G(\text{BBN})/G(1)$	$\Delta G/G$ since BBN	$ d \ln G/dt $ today [1/yr]
mass 1 eV, fable8d	2.3224e-03	3.4376e+10	3.4376e+10	-1.0000e+00	2.7571e-10
mass 100 eV, fable8d	4.8207e-05	1.8516e+17	1.8516e+17	-1.0000e+00	2.7571e-10
lambda-mass 30 eV, fable8d	9.8881e-04	1.0461e+12	1.0461e+12	-1.0000e+00	2.7570e-10
lambda-mass 1 keV, fable8d	9.8389e-04	1.0671e+12	1.0671e+12	-1.0000e+00	2.7570e-10
radiation only, fable8d	1.0000e+00	1.0000e+00	1.0000e+00	0.0000e+00	0.0000e+00
radiation + baryons, fable8d	6.2532e-03	6.5402e+08	6.5402e+08	-1.0000e+00	2.3096e-11

and the state of the hidden sheet and of the observer today:

run	$H_B/H_A(1) = H_C/H_A(1)$	max constraint residual	$\Omega_{\text{sum}}(1)$	q_0	t_0 [Gyr]
mass 1 eV, fable8d	0.999959	1.16e-08	6.99967	2.5000	4.145
mass 100 eV, fable8d	0.999963	2.12e-08	6.99971	2.5000	4.145
lambda-mass 30 eV, fable8d	0.999902	2.47e-08	6.99922	-0.8425	9.472
lambda-mass 1 keV, fable8d	0.999902	2.07e-08	6.99922	-0.8425	9.472
radiation only, fable8d	0.000000	9.89e-16	1.00000	1.0000	755.851
radiation + baryons, fable8d	0.994782	1.04e-08	6.95831	2.4935	49.354

Radiation alone keeps $H_B = H_C = 0$ exactly ($\max |H_B/H_A| = 0.0e+00$: $F = 0$); dust or vacuum energy drives the hidden sheet to the 7-dimensional isotropic attractor ($H_B/H_A \rightarrow 1$), and the observed Newton constant changes by 6.5×10^8 to 1.9×10^{17} between the start and today, with $|d \ln G/dt|$ today 2.3×10^{-11} to 2.76×10^{-10} /yr against the lunar-laser-ranging bound 10^{-13} /yr, and $\Delta G/G$ since nucleosynthesis -1 against the bound of about 0.1. The closure today then needs $\Omega_{\text{sum}}(1) \approx 7$ ($\sum \Omega_i = 1 + (3H_B^2 + 9H_A H_B + 3H_A H_C + 3H_B H_C)/(3H_A^2)$, which is 7 on the attractor). The two controls have nothing to shoot, so their H_A today is not 1 and their ages are not those of our universe. **fable8d is a no-go for observers; that is why the physical model is the stabilized fable4d.** The last run, fable8d -freeze-hidden (lambda-mass 30 eV), agrees with the algebraic fable4d: $\max |dH_A/H_A| = 4.1e-09$, $\max |dt|/t = 9.8e-09$, $\max |dw_f| = 0.0e+00$, $\max |H_B| = 0.0e+00$.

9.11 The cases the solver refuses, and why

Each exits with code 1 and one line of physical reason [nb06 §5.9]:

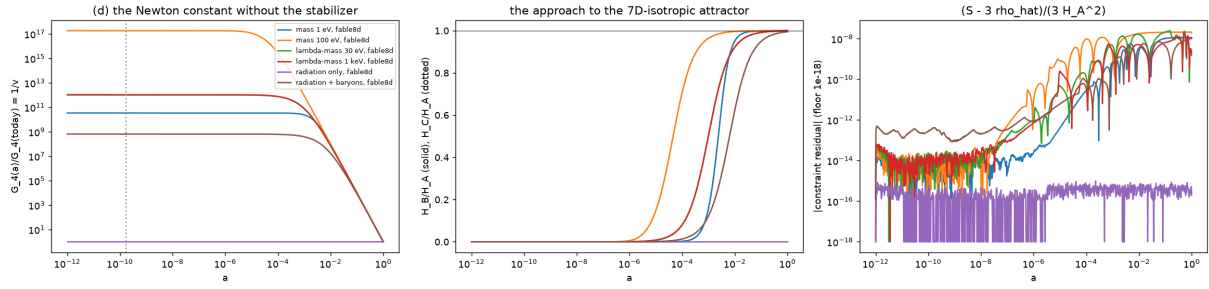


Figure 14: fable-cosmology/results/nb06_d_nogo_8d.png (notebook 06, section 5.8), runs (d): $G_4(a)/G_4(\text{today}) = 1/v$, H_B/H_A and H_C/H_A (the approach to the 7-dimensional isotropic attractor), and the constraint residual $(S - 3\hat{\rho})/(3H_A^2)$.

```
$ fable_fermion fable4d --potential lorentz --points 1001
exit code 1: fable_fermion: rho_hat = H_A^2 <= 0 from N = -0.7597808064 (a = 0.4677689478) on: the
  ↳ Kohn-Sham ground state has negative energy there (m_eff = -3.802078e-2, sigma8 = -9.789374e-4,
  ↳ rho_f = -4.830421e-1 (U = -4.847891e-1) against rho_r + rho_b = 4.830421e-1); H_A^2 = rho_hat is
  ↳ impossible (the onset located past the last accepted CVODE step and bisected)
$ fable_fermion fable4d --potential quadratic --param gq=0.5 --points 1001
exit code 1: fable_fermion: gap-branch jump at N = -0.0037262796 (a = 0.9962806544): a first-order
  ↳ transition of the Kohn-Sham ground state, the lowest-energy gap root jumps from the branch m_eff =
  ↳ -1.518260e0 (rho_f = 6.856759e-1) to the branch m_eff = 4.083552e2 (rho_f = 9.536521e-1); the
  ↳ right-hand side is discontinuous there, and energy conservation across it would need the Maxwell
  ↳ construction, which this solver does not implement (the transition detected inside the right-hand
  ↳ side and bisected in N to adjacent doubles)
$ fable_fermion fable8d --potential mass --direction backward
exit code 1: fable_fermion: fable8d runs forward only (design review E4): backward in time the shear
  ↳ modes grow as A^-3 v^-1 relative to H_A ~ A^-2, the solution runs to the 8D Kasner point H_B/H_A =
  ↳ -0.2929 and the constraint drifts to 0(1); use fable4d --direction backward
all three refused with exit code 1 and the expected reason
```

lorentz: the Kohn–Sham ground state moves to the negative- σ branch, whose energy is negative (a well of W on $\sigma < 0$ that the quantum gas reaches and the classical $s \geq 0$ field never did). **Repulsive quadratic:** a first-order transition (section 8.1). **fable8d backward:** the constraint grows backward (section 5.4). (These lines are the output of the committed notebook 06 (commit aca5102), executed on the final solver of commit 2bc936d. That commit detects both the negative-energy onset and the gap-branch jump inside the right-hand side, on every internal CVODE step, and bisects them to adjacent doubles, independently of the output grid: **lorentz** refuses at $a = 0.4677689478$ “whatever `-points` is”, the repulsive quadratic at $a = 0.9962806544$ “for any grid”, and **scipy** agrees to 8×10^{-12} ; before it, the negative-energy onset moved with the output grid, and `-points 11` died with a CVODE `mxstep` error [**commit 2bc936d**]. The repulsive-quadratic line above is that bisected refusal: at $N = -0.0037262796$ the lowest-energy gap root jumps from the branch $m_{\text{eff}} = -1.518260e0$ to the branch $m_{\text{eff}} = 4.083552e2$.)

9.12 Three implementations, one answer: CVODE, scipy and Mathematica

Part VIII’s own solution. Section 33 solves two **fable4d** cases in Mathematica, independently of the reference script: every tenth row (71 of 701) by exact algebra from Section 29’s closed forms (with **FindRoot** for the gap), the age of **mass30eV** by **NIntegrate** between the rows, and the age of **power** by **NDSolve** of the pair $(m(N), \ln t(N))$ with the **differentiated** gap equation

$$\frac{dm}{dN} = -\frac{W''(\sigma) (\partial\sigma/\partial k_F) k_F}{1 - W'''(\sigma) \partial\sigma/\partial m}$$

instead of solving it [**prose VIII §33**]. Besides the **PASS** lines quoted after it, the run printed [**displayed VIII §33**], **claude-fable/run_fermion_fable_final.log**:

```
[0.220 s] fable4d mass30eV: 71 rows by exact algebra, the age by NIntegrate between the rows
[0.538 s] fable4d power: NDSolve of (m(N), ln t(N)) with the differentiated gap equation
```

```
[0.045 s] fable4d power: 71 rows by exact algebra (FindRoot of the gap at every row)
mass30eV: m0 = 30 eV = 12182.18122 E_c; kF0 = 0.0544035545089 E_c = 0.00013397491 eV; V0 =
↳ 0.6856647126; a_nr = kF0/m0 = 4.46583e-6; t0 = 0.951246513195 / H0
power: m_today = 100 eV; kF0 = 0.0464558024014 E_c; lam = 217.736704888; m0 = 27.15187044 eV;
↳ U_today = 0.4006647126; w_f(1) = -0.421457436338; t0 = 0.9218031611 / H0
P_stab(a) [units of rho_c0] at a = 1e-10, 1e-6, 1e-3, 1: mass30eV {-2.4626e28, -6.53147e16,
↳ -1.57121e8, -0.842786}; power {-2.46243e28, -4.87299e16, -9.93078e7, -0.700286}
```

and asserted [proved VIII §33]

- fable4d RUNS [THE RESULT]: mass30eV - flat today ($H_A(1) == 1$), $\text{eps}_{KS}(1) == 0.265$, ultra-relativistic at the start ($w_f(a_i) = 1/3$ to $1e-8$), non-relativistic at $a_{nr} = kF0/m0 == 4.16e-4$ ($(m/eV)^{-4/3}$ (1%)), and $w_f \geq -1$ at every row
- fable4d RUNS [THE RESULT]: power - flat today, $m_{\text{eff}}(1) == 100$ eV, $m0 == 27.15$ eV > 0 , m_{eff} INCREASES monotonically from $m0$ (early) to m_{today} , and $w_f \geq -1$ at every row
- fable4d RUNS [has content]: the differentiated gap equation carries $m(N)$ along the history - NDSolve's m agrees with the per-row FindRoot to $1e-15$ at all 71 rows
- fable4d RUNS [THE RESULT]: the stabilizer $P_{\text{stab}} = -F_{\text{total}}/2$ is NEGATIVE at every row of both runs - the null-energy violation along $x0$ is present along the whole history
- fable4d RUNS [fidelity]: the reference CSVs have the expected header and 701 rows
- fable4d RUNS [fidelity]: at every tenth row ALL TEN columns agree with fable-cosmology/reference/mathematica_fable4d_mass30eV.csv and _power.csv to $1e-10$ (relative; P_{obs_f} relative to ρ_f ; N and w_f absolute)

The comparison with the reference CSVs printed max difference per column {0., 0., 0., 0., 0., 0., 0., 0., 0.} for both files (three-digit formatting). The final run was made after notebook 06 had written the Rust solver's CSVs results/nb06_fable4d_mass30eV.csv and nb06_fable4d_power.csv, so the conditional comparison of Section 33 with them ran, at every one of the 71 rows, and passed [displayed VIII §33], claude-fable/run_fermion_fable_final.log (<repo> stands for the absolute path of the clone):

```
Rust <repo>\claude-fable\..fable-cosmology\results\nb06_fable4d_mass30eV.csv: 71 rows compared; max difference per column {a, t, H_A,
↳ rho_f, P_obs_f, w_f, m_eff, sigma, kF_over_m}: {4.9e-14, 2.35e-9, 3.35e-11, 1.87e-13, 6.22e-15, 6.8e-15, 2.22e-15, 1.39e-13,
↳ 4.84e-14}
Rust <repo>\claude-fable\..fable-cosmology\results\nb06_fable4d_power.csv: 71 rows compared; max difference per column {a, t, H_A,
↳ rho_f, P_obs_f, w_f, m_eff, sigma, kF_over_m}: {4.9e-14, 1.39e-9, 3.36e-11, 1.92e-13, 1.34e-14, 1.39e-14, 5.04e-14, 1.41e-13,
↳ 8.82e-14}
```

[proved VIII §33]

- fable4d RUNS [fidelity]: the Rust solver's mass30eV run agrees with this cell to $1e-6$ in every column, at every one of the rows its grid covers
- fable4d RUNS [fidelity]: the Rust solver's power run agrees with this cell to $1e-6$ in every column, at every one of the rows its grid covers

In Part VIII's acceptance run, claude-fable/run_fermion_fable_part8.log, those CSVs did not yet exist, and the comparison was skipped, not failed (NOTE Rust CSV not found; that comparison was skipped, not failed): these two assertions are the difference between its Assertions run: 642 and the final run's Assertions run: 644. The three-way comparison below includes the same CSVs.

The reference script `fable-cosmology/reference/make_reference_fermion.wls` writes `mathematica_fable4d_mass30eV.csv` and `mathematica_fable4d_power.csv`: 701 values of N from $\ln 10^{-10}$ to 0, columns N , a , t , H_A , ρ_f , $P_{\text{obs},f}$, w_f , m_{eff} , σ , k_F/m ; every row is exact algebra at 40 significant digits (the power-law gap by Brent's method at 45 digits), the age by `NDSolve` at working precision 30 with the stable series of the Fermi-sea integrals for $k_F/|m| < 1/4$ [file: its header].

Three-way [nb06 §5.5, file `results/nb06_threeway.csv`] — `CVODE` (Rust), `scipy` (the independent implementation), and `Mathematica` (the reference CSVs), column by column, on both reference cases (the table of the notebook, transposed: one row per compared column):

column	nb06_fable4d_mass30eV	nb06_fable4d_power
<i>Rust – Mathematica:</i>		
t	2.352e-09	1.500e-09
H_A	3.338e-11	3.353e-11
ρ_f	1.271e-14	5.540e-14
$P_{\text{obs},f}$	1.603e-13	6.990e-13
m_{eff}	1.792e-15	5.032e-14
σ	9.938e-15	5.242e-14
k_F/m	6.506e-15	5.128e-14
w_f (abs)	3.608e-15	8.160e-15
<i>scipy – Mathematica:</i>		
H_A	3.347e-11	3.362e-11
ρ_f	1.419e-12	1.362e-12
$P_{\text{obs},f}$	1.172e-12	1.170e-12
m_{eff}	2.986e-16	4.586e-14
$t(A=1)$	2.665e-14	4.241e-14

written: `results/nb06_threeway.csv`; `CVODE`, `scipy` and `Mathematica` agree to better than $1e-6$ on both reference cases; in the power case m_{eff} runs from 27.1519 eV ($a = 1e-10$) to 100.0000 eV (today).

Every run re-computed independently [nb06 §5.10, file `results/nb06_crosscheck.csv`]: the independent implementation takes the normalized parameters from the `# params:` line of each CSV and re-computes the run — every Fermi-sea integral by adaptive quadrature, every gap root by `brentq`, `fable8d` by `DOP853` at `rto1 = 1e-11`, `fable4d` algebraically with $t(A=1)$ by adaptive quadrature — and checks the Friedmann consistency of the solver's columns and the covariant conservation of the independently computed fluid (Richardson-extrapolated central differences). The notebook fails if any difference exceeds 10^{-6} . It printed **largest difference** over the 29 kept runs: $1.08e-08$; over the 90 scan runs: $4.43e-09$; **largest conservation residual** of the independent fluid: $1.24e-07$.

9.13 The consolidated table of every kept run

[nb06 §5.11, files `results/nb06_runs.csv` (29 rows, 34 columns), `results/nb06_cpl_fits.csv`], verbatim (on the next page, which is turned to landscape so that its lines fit):

The consolidated table of every kept run, as notebook 06 printed it (its section 5.11)

run	m [eV]	a_nr	dNeff	BBN	CPL w_f	CPL w_DE,inf	min cs2	vac/rho	a>=.3 G(a_i)/G(1)	Gdot/G /yr
mass 1 eV (EdS)	1	0.000637	36.75	(-0.000, +0.000)	(+nan, +nan)	1.35e-07	0.000e+00	1.000e+00	0.000e+00	
mass 30 eV (EdS)	30	6.84e-06	0.3943	(-0.000, +0.000)	(+nan, +nan)	1.56e-11	0.000e+00	1.000e+00	0.000e+00	
mass 100 eV (EdS)	100	1.37e-06	0.07918	(-0.000, +0.000)	(+nan, +nan)	6.28e-13	0.000e+00	1.000e+00	0.000e+00	
mass 1000 eV (EdS)	1000	6.37e-08	0.003675	(-0.000, +0.000)	(+nan, +nan)	1.35e-15	0.000e+00	1.000e+00	0.000e+00	
mass 2000 eV (EdS)	2000	2.53e-08	0.001459	(-0.000, +0.000)	(+nan, +nan)	2.13e-16	0.000e+00	1.000e+00	0.000e+00	
lambda-mass 30 eV	30	4.47e-06	0.07179	(-0.757, +1.010)	(-1.000, +0.000)	6.65e-12	0.000e+00	1.000e+00	0.000e+00	
lambda-mass 100 eV	100	8.97e-07	0.01442	(-0.757, +1.010)	(-1.000, +0.000)	2.68e-13	0.000e+00	1.000e+00	0.000e+00	
lambda-mass 1000 eV	1000	4.16e-08	0.0006692	(-0.757, +1.010)	(-1.000, +0.000)	5.78e-16	0.000e+00	1.000e+00	0.000e+00	
lambda-mass 2000 eV	2000	1.65e-08	0.0002656	(-0.757, +1.010)	(-1.000, +0.000)	9.1e-17	0.000e+00	1.000e+00	0.000e+00	
lambda-mass 30 eV (reference)	30	4.47e-06	0.07179	(-0.757, +1.009)	(-1.000, +0.000)	6.65e-12	0.000e+00	1.000e+00	0.000e+00	
power nu=0.236, 30 eV	30	2.22e-05	0.07179	(-0.783, +0.445)	(-0.082, -7.986)	-0.611	1.314e+13	1.000e+00	0.000e+00	
power nu=0.236, 100 eV	100	4.47e-06	0.01442	(-0.783, +0.445)	(-0.082, -7.986)	-0.611	1.622e+15	1.000e+00	0.000e+00	
power nu=0.5, 30 eV	30	0.669	0.07179	(-0.804, -0.358)	(-0.249, -10.191)	-27	2.229e+14	1.000e+00	0.000e+00	
power nu=0.5, 100 eV	100	0.669	0.01442	(-0.804, -0.359)	(-0.249, -10.191)	-27	2.752e+16	1.000e+00	0.000e+00	
expdamp xt=1/2.21, 30 eV	30	0.712	0.07179	(-0.829, -0.317)	(-0.128, -11.778)	-21.8	2.327e+14	1.000e+00	0.000e+00	
expdamp xt=1/2.21, 100 eV	100	0.712	0.01442	(-0.829, -0.317)	(-0.128, -11.779)	-21.8	2.873e+16	1.000e+00	0.000e+00	
quadratic gq=-0.5, 30 eV	30	0.643	0.07179	(-0.730, -0.483)	(-0.180, -9.321)	-47.1	2.031e+14	1.000e+00	0.000e+00	
quadratic gq=-0.5, 100 eV	100	0.643	0.01442	(-0.730, -0.483)	(-0.180, -9.321)	-47.1	2.508e+16	1.000e+00	0.000e+00	
power nu=0.5, 100 eV, Omega_qp0=0.55	100	4.21e-06	0.03817	(-0.448, +0.264)	(-0.327, -9.682)	-0.364	2.350e+14	1.000e+00	0.000e+00	
power nu=0.236, 1 eV (Delta N_eff)	1	0.00207	6.692	(-0.783, +0.445)	(-0.082, -7.985)	-0.611	1.622e+07	1.000e+00	0.000e+00	
power nu=0.5, 1 eV (Delta N_eff)	1	0.67	6.692	(-0.807, -0.347)	(-0.251, -10.172)	-21.7	2.720e+08	1.000e+00	0.000e+00	
expdamp xt=1/2.21, 1 eV (Delta N_eff)	1	0.712	6.692	(-0.832, -0.305)	(-0.130, -11.753)	-21.7	2.850e+08	1.000e+00	0.000e+00	
quadratic gq=-0.5, 1 eV (Delta N_eff)	1	0.643	6.692	(-0.732, -0.473)	(-0.182, -9.306)	-46.4	2.490e+08	1.000e+00	0.000e+00	
mass 1 eV, fable8d	1	0.00124	521.5	(-0.000, +0.000)	(+0.394, -8.104)	3.93e-07	0.000e+00	3.438e+10	2.757e-10	
mass 100 eV, fable8d	100	2.66e-06	1.124	(-0.000, +0.000)	(+0.394, -8.104)	1.83e-12	0.000e+00	1.852e+17	2.757e-10	
lambda-mass 30 eV, fable8d	30	4.47e-06	0.07179	(-1.103, +1.715)	(-1.716, +5.459)	5.13e-12	0.000e+00	1.046e+12	2.757e-10	
lambda-mass 1 keV, fable8d	1000	4.16e-08	0.0006692	(-1.103, +1.715)	(-1.716, +5.459)	4.46e-16	0.000e+00	1.067e+12	2.757e-10	
radiation only, fable8d	nan	nan	0	(+nan, +nan)	(+nan, +nan)	nan	0.000e+00	1.000e+00	0.000e+00	
radiation + baryons, fable8d	nan	nan	0	(+nan, +nan)	(+0.386, -8.020)	nan	0.000e+00	6.540e+08	2.310e-11	

(The `vac/rho` column is 0 for constant-mass runs, where the sea energy is a constant absorbed into V_0 .)

9.14 The figures

All written by notebook 06 under `fable-cosmology/results/` (and those of notebook 05, which show the fluid itself):

figure	what it shows
<code>nb06_a_mass_wdm.png</code> (figure 7)	runs (a): the author's mass term, $w_{\text{DM}} = P_{\text{KS}}/\varepsilon_{\text{KS}}$ ($= w_f$ there) against a : the degenerate fable cools from $1/3$ to 0
<code>nb06_b_lambda_mass.png</code> (figure 8)	runs (b): w_f (solid) and w_{DM} (dotted); and $w_{\text{DE,inf}} + 1$ on $0.3 \leq a \leq 1$: the inferred dark energy is a cosmological constant
<code>nb06_c_de_potentials.png</code> (figure 9)	runs (c): w_f (never below -1), the observer's $w_{\text{DE,inf}}$ (clipped to $[-3, 1]$, poles and crossings marked), $m_{\text{eff}}/m_{\text{today}}$, and c_s^2 (negative below the line)
<code>nb06_c_exchange_vacuum_stabilizer.png</code> (figure 10)	runs (c): w_{DM} and $w_{\text{DM,eff}}$, the exchange Q , the Dirac-sea energy relative to the total density, and P_{stab}/ρ
<code>nb06_scan.png</code> , <code>nb06_scan_cpl.png</code> (figures 11, 12)	the scans: per family, the smallest $m_{\text{eff}}/m_{\text{today}}$ over $10^{-3} \leq a \leq 0.3$ and the smallest c_s^2 ($a \geq 10^{-3}$) against the scanned parameter; the CPL points (w_0, w_a) of $w_{\text{DE,inf}}$ (fitted above its poles) and of w_f
<code>nb06_dm_mass_bounds.png</code> (figure 13)	the free streaming of a degenerate fable: v_{rms} today against the fable mass, with the warm-dark-matter velocities
<code>nb06_d_nogo_8d.png</code> (figure 14)	runs (d): $G_4(a)/G_4(\text{today}) = 1/v$, H_B/H_A and H_C/H_A (the approach to the 7-dimensional isotropic attractor), and the constraint residual $(S - 3\hat{\rho})/(3H_A^2)$
<code>nb05_w_ks.png</code> , <code>nb05_ks_accuracy.png</code> , <code>nb05_classical_vs_quantum.png</code> , <code>nb05_gap_roots.png</code> , <code>nb05_minimax.png</code> , <code>nb05_preuniverse_cooling.png</code> , <code>nb05_vacuum_energy.png</code> (figures 2, 1, 3, 6, 5, 15, 4)	the Kohn–Sham fluid: $w_{\text{KS}}(x)$, the accuracy of the stable forms, classical vs quantum w , the gap roots, the minimax and Walecka functionals, the cooling on the pre-universe, the Dirac-sea energy

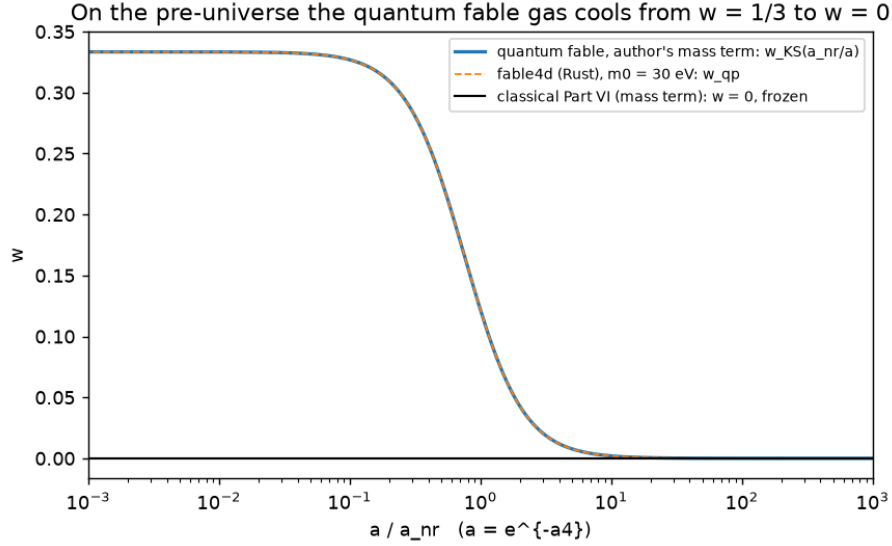


Figure 15: fable-cosmology/results/nb05_preuniverse_cooling.png (notebook 05): the cooling on the pre-universe — the quantum fable gas of the author’s mass term goes from $w = 1/3$ to $w = 0$ as the observed sheet expands, while the classical Part VI fable’s $w = 0$ is frozen (section 3.6).

10 The answers to [1] and [2]

The author asked: [1] “Does this system provide a physical mechanism for a time-varying dark energy equation of state (w)?” [2] “[...] time-varying dark matter equation of state (w)?” The answers below rest on the runs of section 9 and the theorems of sections 4–7. Notebook 06 composes its own statement of the answers from its tables and asserts every qualitative claim in it; its headlines read [nb06 §7]: “**Formally yes, in the stabilized model, but in none of the runs is it a viable one.**” for [1], and “**Yes, intrinsically, but for an allowed mass the variation is over long before recombination.**” for [2].

10.1 Answer [1]: a time-varying dark-energy equation of state

Yes, as a mechanism — in the stabilized model, and only with a mass-varying potential. None of the runs is a viable cosmology, and the mechanism carries every one of the following caveats.

What the mechanism is. In the Kohn–Sham ground state the fable’s energy is $\rho = \rho_{\text{qp}} + \rho_U$, and the mean-field condensate $\rho_U = W(\sigma_7) - \sigma_7 W'(\sigma_7)$ has $P = -\rho_U$ along all seven spatial directions: a vacuum energy. When W is not linear, ρ_U changes as the density falls, the quasiparticle mass $m_{\text{eff}} = W'(\sigma_7)$ changes with it, and the two parts exchange energy at the rate $Q = \sigma_8 dm/dt$ (section 7.10). The dark energy that an observer infers therefore varies in time. For a massive power law $W = m_0\sigma + \lambda\sigma^\nu$ it is, in closed form and confirmed by the runs to 1.2×10^{-9} ,

$$w_{\text{DE,inf}}(a) = -\frac{1 - \nu}{1 - \nu a^{-3(1-\nu)}},$$

with a pole at $a = \nu^{1/(3(1-\nu))}$ (0.5326 for $\nu = 0.236$) (section 9.7). The observer’s CPL fits of the eight dark-energy runs of section 9.6 lie at distances from the supernova fit $(w_0, w_a) = (-0.861, -0.60)$ of 7.43 (power $\nu = 0.236$: $(-0.082, -7.986)$) to 11.2 (expdamp: $(-0.128, -11.778)$) [file: fable-cosmology/results/nb06_cpl_fits.csv], and the fits are not faithful, because $w_{\text{DE,inf}}$ has a pole inside $0.3 \leq a \leq 1$ in all of them.

What does not give it. With the author’s mass term alone there is no dark energy at all: the fable is a free gas that closes the budget, an Einstein–de Sitter-like universe with $\Omega_{f0} = 0.95066$ and $q_0 = 0.50005$ (section 9.4). With a mass term plus a bare V_0 (**lambda-mass**), the observer infers **exactly a cosmological constant**: CPL $(w_0, w_a) = (-1.000000, +1.3 \times 10^{-9})$, $(-1.000000, +5.3 \times 10^{-11})$, $(-1.000000, +1.2 \times 10^{-13})$, $(-1.000000, +1.7 \times 10^{-14})$ at 30, 100, 1000, 2000 eV — no time variation (section 9.5). V_0 is a bare cosmological constant and is reported as such.

The caveats, each quantified:

1. **No true phantom — but an apparent phantom and phantom crossing in the observer’s fit.** The fable’s own equation of state never goes below -1 : $\rho + P_{\text{obs}} = \omega_F n/v \geq 0$ is an identity (section 7.9), and the smallest w_f over every run is -0.9999987921 . The classical Part VI crossing of -1 (at $V'(s^*) = 0$, section 2.12.7) does **not** survive quantization. The **observer-inferred** dark energy, however, is phantom or crosses -1 inside $0.3 \leq a \leq 1$ in 9 of 9 runs of section 9.6 and in 90 of 90 scan points: subtracting cold dark matter with today’s mass when the mass was different leaves $\rho_{\text{DE}} + P_{\text{DE}} \approx (m_{\text{eff}}(a) - m_{\text{today}}) n$, negative whenever m_{eff} grows (the mechanism of Das, Corasaniti and Khoury, Phys. Rev. D **73**, 083509 (2006)), with no phantom field. For the massive power law the crossing lies in the last output interval before today ($a > 0.9727$), where the quasiparticles’ kinetic energy, of relative size $\frac{3}{10}(k_F/m)^2$, finally exceeds $(m_{\text{today}} - m_{\text{eff}}(a)) n$ [**nb06 §7**].
2. **Mass-varying dark energy with massive dark matter needs a power law with $\nu < 0.279$.** In the scans 28 of 90 points keep the fable massive over $10^{-3} \leq a \leq 0.3$, all of them power laws; every **quadratic** and **expdamp** point is a massless gas (radiation) until late times and supplies no dark matter at recombination. A power law stays massive exactly when its bare mass m_0 is positive, i.e. when $\Omega_{\text{qp0}}/U_t > \nu/(1-\nu)$; with the observed split $\Omega_{\text{qp0}} = 0.265$ that is $\nu < 0.2788$ (section 9.8). The massive mass-varying example with a larger quasiparticle share (Part VIII’s case: **power**, $\nu = 0.5$, $m_{\text{today}} = 100$ eV, $\Omega_{\text{qp0}} = 0.55$, no bare Lambda) has m_{eff} rising from 27.15 eV to 100.00 eV, w_f today -0.4215 , $q_0 = -0.1010$, and matter today $\Omega_{\text{qp0}} + \Omega_{b0} = 0.599$, about twice the measured $\Omega_m \approx 0.315$ (Planck 2018): it is massive because its dark matter outweighs its dark energy, which the observed universe does not allow [**nb06 §7**].
3. **It is adiabatically unstable.** Every scan point that stays massive has $c_s^2 < 0$ after $a = 10^{-3}$ (smallest c_s^2 between -0.647 and -0.056), and so do all 9 runs of section 9.6 (-0.611 for $\nu = 0.236$, -0.364 for Part VIII’s case, down to -47.1 for **quadratic**). This is the adiabatic instability of mass-varying fermions (as found by Afshordi, Zaldarriaga and Kohri for mass-varying neutrinos).
4. **The Dirac-sea vacuum energy demands a fine-tuning.** The renormalized sea energy that the no-sea functional drops changes, for a mass variation of order m , by $g m^4/(64\pi^2)$: 3.44 , 3.44×10^4 , 3.44×10^8 , 3.44×10^{12} times ρ_{c0} at $m = 0.01, 0.1, 1, 10$ eV, more than the critical density for $m > 7.341$ meV (section 7.11). Along every mass-varying run it changes by **1.31×10^{13} to 2.87×10^{16} times the total density at $a \geq 0.3$** (over the scans at least 3.79×10^9) [**nb06 §7**]. The runs read W as the fully renormalized effective potential, which absorbs the sea energy only by fine-tuning the bare potential against it to that precision. The cosmological-constant problem is not solved.
5. **It needs the stabilizer, which violates the null energy condition.** All of this holds with the hidden sheet held by $P_{\text{stab}} = -F/2$ (today $-0.8428 \rho_{c0}$ in the Λ CDM-like runs, -0.7003 in Part VIII’s power case), a Lagrange multiplier, not a stress derived from any field, with zero energy density; it violates the null energy condition along x_0 on 100.0 % of the output rows of every stabilized run (section 9.1).

6. **Without the stabilizer it is excluded (the fable8d no-go).** Left free, the hidden sheet is driven by dust and vacuum energy to the 7-dimensional isotropic attractor, and the observed Newton constant $G_4 = G_8/V_{\text{hid}}$ varies: $G(a_i)/G(1) = 3.44 \times 10^{10}$, 1.85×10^{17} , 1.05×10^{12} , 1.07×10^{12} in the four fable runs and 6.54×10^8 for radiation plus baryons alone; $\Delta G/G$ since nucleosynthesis is -1 (against about 0.1), and $|d \ln G/dt|$ today $\geq 2.31 \times 10^{-11}$ /yr (against the lunar-laser-ranging bound 10^{-13} /yr) (section 9.10). The only frozen-compatible Kohn–Sham fable is radiation (section 7.7).
7. **The split into dark matter and dark energy is a convention.** The condensate ρ_U is an 8-dimensional vacuum energy, with $P = -\rho_U$ along the hidden directions as well; calling it dark energy and the quasiparticles dark matter is a choice of bookkeeping (section 7.4).
8. **The approximations under which all of this holds:** semiclassical gravity ($G = \kappa \langle \hat{T} \rangle$); the Kohn–Sham mean field with exchange–correlation neglected (section 7.12); zero modes along x_0 and the hidden timelike sheet, with x_0 compactified (a Kaluza–Klein gap above the temperatures considered) and the hidden timelike sheet a formal comoving volume (compactifying it gives closed timelike curves); the asymptotic Bianchi-I reading of the primordial field, which neglects terms of order $\cot(6Hx_0)$ and $\cot^2(6Hx_0)$ (section 5.3); $g_*(T)$ not followed.

10.2 Answer [2]: a time-varying dark-matter equation of state

Yes, intrinsically: the quantized fable’s dark-matter equation of state runs from 1/3 to 0. For an allowed mass the variation is over long before recombination.

1. **The mechanism.** The fable’s quasiparticles are a degenerate Fermi gas ($g = 8$ states per momentum), and their equation of state is $w_{\text{DM}} = P_{\text{KS}}/\varepsilon_{\text{KS}}$, a function of $x = k_{\text{F}}/|m|$ alone that rises monotonically from 0 (dust) to 1/3 (radiation) (section 3.6). As the universe expands, $k_{\text{F}} = k_{F0}/a$ falls, and w_{DM} falls from 1/3 to 0: **a time-varying dark-matter equation of state**, which is a property of the quantum gas and has no counterpart in the classical Part VI fable (whose w was frozen on the pre-universe and 0 for the mass term). Every run starts at $w_{\text{DM}} = 0.333333$ and ends today at most 8.1×10^{-8} [nb06 §7]; the transition is at $a_{\text{nr}} = k_{F0}/m$, 4.466×10^{-6} for 30 eV and 4.163×10^{-8} for 1 keV, scaling as $m^{-4/3}$ (sections 9.4–9.5). The same cooling happens on the pre-universe itself, on the canonical frame with a_4 free, whenever the observed sheet expands ($a'_4 < 0$, section 3.6).
2. **How heavy it must be, and hence when the variation happens.** A fable that is all of the dark matter is radiation before a_{nr} : $\Delta N_{\text{eff}}(\text{BBN}) = 6.692 (m/\text{eV})^{-4/3} < 0.3$ requires $m > 10.3 \text{ eV}$ (< 0.2 : $m > 13.9 \text{ eV}$). Free streaming is far more restrictive: matching the rms velocity of the Lyman-alpha-allowed thermal warm dark matter requires $m \approx 1125\text{--}1807 \text{ eV}$ ($\sim 1.1\text{--}1.8 \text{ keV}$, a velocity-matching estimate, not a transfer-function computation), and the Tremaine–Gunn phase-space bound for illustrative dwarf spheroidals gives $m > 231\text{--}358 \text{ eV}$ (section 9.9). At such masses the change of w_{DM} from 1/3 to 0 happens around $a_{\text{nr}} = 4.16 \times 10^{-8}$ (1 keV) or 1.65×10^{-8} (2 keV): **indistinguishable from cold dark matter in the late universe.**
3. **With a mass-varying potential the effective equation of state varies late.** The quasiparticles then exchange energy with the condensate ($Q = \sigma_8 dm/dt$), and their effective equation of state $w_{\text{DM,eff}} = w_{\text{DM}} - \sigma_{\text{KS}} (dm/dN)/(3\varepsilon_{\text{KS}})$ ranges, after $a = 10^{-3}$, over $[-0.611, +9.85 \times 10^{-5}]$ (power $\nu = 0.236$, 30 eV), $[-0.611, +3.59 \times 10^{-6}]$ (100 eV), $[-27.6, +0.333]$ and $[-27.7, +0.333]$ (power $\nu = 0.5$), $[-21.8, +0.333]$ (expdamp), $[-47.1, +0.333]$ (quadratic), and $[-0.364, -3.89 \times 10^{-5}]$ (Part VIII’s case) (section 9.6). This late variation is the same exchange that makes the inferred dark energy vary in [1], and it carries all of [1]’s caveats (adiabatic instability, the sea energy, the viability window $\nu < 0.2788$).

4. **The author's mass term alone is dark matter without dark energy.** $W = m_0\sigma$ gives an Einstein–de Sitter-like universe ($q_0 = +0.50005$, age 9.67 Gyr with $H_0 = 67.4$ km/s/Mpc), not ours; with a bare cosmological constant (`lambda-mass`) it is Λ CDM with fable dark matter ($q_0 = -0.52845$, age 13.8000 Gyr) (sections 9.4–9.5).
5. **The same caveats as [1] apply:** the stabilized model with its NEC-violating stabilizer (without it the Newton constant varies by 6.5×10^8 to 1.9×10^{17}); the split into dark matter and dark energy is a convention; semiclassical gravity, the Kohn–Sham mean field with E_{xc} neglected (exact for the author's mass term, which is free), zero modes along the hidden directions, the asymptotic reading of the primordial field.

11 Every command, in full

The commands run in Git Bash on Windows 11 (plain POSIX shell; they run the same way on Linux and macOS, with `.venv/bin/python` in place of `.venv/Scripts/python.exe` and without the `.exe` suffix). The toolchain recorded in the logs: Wolfram Language 15.0.1 for Microsoft Windows (64-bit) (July 2, 2026), called as `wolframscript` (the first lines of the run log and of `waveguide_derivation.log`); Rust with `cargo`; the solver `fable_fermion 0.1.0`, `sundials_rs 7.8.0` (pure Rust), `CVODE BDF`; Python 3 in the virtual environment `fable-cosmology/.venv` (`numpy`, `scipy`, `matplotlib`, `jupyter`); MiKTeX with `latexmk` for the LaTeX twin.

11.1 Set up, and build the solver

```
cd "$(git rev-parse --show-toplevel)"
bash fable-cosmology/setup.sh
```

`setup.sh` (idempotent) creates `fable-cosmology/.venv` and installs `requirements.txt` into it, printing python ok: `numpy <v> scipy <v> matplotlib <v>`; makes a sparse clone of the `rustSolveIt` repository for this platform (`rustSolveIt_Win11_SUNDIALS_7_8_0` on Windows, `rustSolveIt_macos-silicon_SUNDIALS_7_8_0` on Apple-silicon macOS, `rustSolveIt_linux_SUNDIALS_7_8_0` on Linux) into `rust/vendor/rustSolveIt`, restricted to its vendored pure-Rust SUNDIALS 7.8.0 (`sundials_rs`), on which the solver crates depend by path; then builds both solver crates in release mode and runs each with `-version`; it ends with `== setup complete` [file: `fable-cosmology/setup.sh`]. To rebuild the fermion solver alone:

```
cd "$(git rev-parse --show-toplevel)/fable-cosmology/rust/fable_fermion"
cargo build --release # -> target/release/fable_fermion(.exe)
./target/release/fable_fermion --version
```

which prints `fable_fermion 0.1.0`, `sundials_rs 7.8.0` (pure Rust), `CVODE BDF` (the first line of notebooks 05 and 06). The crate's `.cargo/config.toml` pins the FMA instruction on x86-64 (`rustflags = ["-C", "target-feature=+fma"]`), which its comment states is REQUIRED by the vendored `sundials_rs` engine, whose deterministic math library uses `f64::mul_add`; `cargo` reads that file from the directory the build is invoked in.

11.2 Test the solver

```
cd "$(git rev-parse --show-toplevel)/fable-cosmology/rust/fable_fermion"
cargo test --release
```

It runs the unit tests of `src/` and the end-to-end tests of `tests/cosmology.rs` (each end-to-end test runs the real binary, writes a CSV and reads it back). At commit `c5892bd` (the finished solver) the commit records `cargo test --release: 21 + 8 tests pass (re-run independently)`; at commit `2bc936d` (six defects fixed, 23:12 PDT) it records `34/34 (23 unit`

+ 11 integration, previously 21 + 8). That is the final solver: the only later change to the crate is a comment in the module summary of `src/models.rs` (commit 3a5020d, section 9.1). The 34 tests [listed from the source files]:

file	tests
<code>src/constants.rs</code>	<code>unit_system_matches_the_standard_values</code>
<code>src/cvode_driver.rs</code>	<code>exponential_decay_forward_and_backward</code>
<code>src/numerics.rs</code>	<code>brent_finds_simple_roots,</code> <code>gauss_kronrod_is_exact_on_smooth_functions</code>
<code>src/potentials.rs</code>	<code>derivatives_match_finite_differences,</code> <code>signs_that_the_bracket_proofs_use</code>
<code>src/kohn_sham.rs</code>	<code>fermi_integrals_match_quadrature,</code> <code>series_and_closed_forms_are_continuous_at_the_switch_points,</code> <code>nonrelativistic_and_ultrarelativistic_limits,</code> <code>per_3_volume_mean_field_violates_8d_conservation,</code> <code>thermodynamic_identities_at_the_self_consistent_point,</code> <code>derivative_along_the_trajectory_matches_finite_differences,</code> <code>expdamp_pins_sigma_below_s1_and_mass_goes_to_zero_at_high_density,</code> <code>multiple_gap_roots_are_found_and_the_lowest_energy_one_is_taken,</code> <code>quadrature_at_the_review_points_and_switch_continuity,</code> <code>nonrelativistic_classical_limit_of_the_mean_field,</code> <code>no_phantom_crossing_rho_plus_p_obs_is_nonnegative,</code> <code>root_counts_three_for_strong_repulsion_one_for_concave_w,</code> <code>walecka_functional_is_minimized_at_the_gap_root,</code> <code>vacuum_energy_vanishes_to_fifth_order_at_the_reference_mass,</code> <code>vacuum_energy_matches_high_precision_references_near_and_far_from_the_reference_mass,</code> <code>gap_residual_is_at_round_off_in_massless_phases,</code> <code>ks_functional_stationary_point_is_a_maximum_for_the_mass_term</code>
<code>tests/cosmology.rs</code>	<code>version_string,</code> <code>covariant_conservation_along_real_fable8d_runs,</code> <code>constraint_residual_stays_small_on_forward_fable8d_runs,</code> <code>frozen_hidden_theorem,</code> <code>fable8d_with_hidden_frozen_by_hand_reproduces_fable4d,</code> <code>backward_runs,</code> <code>physics_checks_along_runs,</code> <code>negative_energy_refusal_is_found_inside_the_rhs_independently_of_the_output_grid,</code> <code>first_order_transition_is_found_inside_the_rhs_independently_of_the_output_grid,</code> <code>gap_residual_and_vacuum_energy_at_round_off_in_1_kev_runs,</code> <code>insensitivity_to_a_start</code>

It prints, for each test binary, running `N` tests, one test `<name> ... ok` line per test, and test result: `ok. N passed; 0 failed; ...` Its output on the final pushed state is recorded with the fresh-clone verification of section 12.

11.3 The unit system, and one gap solve

```
cd "$(git rev-parse --show-toplevel)/fable-cosmology"
F=rust/fable_fermion/target/release/fable_fermion      # fable_fermion.exe on Windows
$F --constants
$F gap --form quadratic --param m0=0.05 --param lam=20 --kf 1
```

`--constants` prints the unit block quoted in section 9.1. `gap` solves the gap equation once and prints every root with `m`, `sigma8`, `rho8`, `P_obs`, `P_hid`, `gap_residual` (16 significant digits) and the selected root; its output for this case is quoted in section 8.1. Notebook 05 also solves

the mass term ($m_0 = 1$, so $k_F = x$) at the fifteen $x = k_F/m$ of its accuracy table (section 7.2), with the arguments Python’s `repr` gives:

```
for x in 1e-08 1e-06 0.0001 0.01 0.2 0.25 0.3 0.6 1.0 3.0 10.0 99.0 101.0 10000.0 1000000.0;
do
  $F gap --form mass --param m0=1 --kf $x
done
```

(the rows `sigma8`, `rho8`, `P_obs` of the mass term are σ_{KS} , ε_{KS} , P_{KS} ; they agree with the independent Python forms to 3.11×10^{-14}).

11.4 Every solver run behind the results of this page

The solver’s command line [file: `fable-cosmology/rust/fable_fermion/src/main.rs`, module documentation; prose nb05 §5, nb06 §5]:

```
fable_fermion fable4d|fable8d --potential NAME [--param KEY=VALUE ...] [--direction forward|backward]
  [--a-start A] [--points K] [--out FILE.csv] [--rtol R] [--atol A] [--method bdf|adams]
  [--branch lowest|positive|negative] [--freeze-hidden] [--no-shoot] [--no-fable]
  [--omega-b X] [--omega-r X]
fable_fermion gap --form NAME --param KEY=VALUE ... --kf KF [--v V]
fable_fermion --constants | --version
```

-param keys: mass: `m0_ev`; lambda-mass: `m0_ev`, `omega_dm`; power: `m_today_ev`, `nu`, `omega_dm`; expdamp: `m_today_ev`, `xt`, `omega_dm`; quadratic: `m_today_ev`, `gq`, `omega_dm`; lorentz: `m_today_ev`, `xt`, `omega_dm`. Defaults: -a-start 1e-12, -points 2001, rtol 1e-10, atol 1e-12, the lowest-energy gap branch, `omega_dm` = 0.265. A run writes a CSV that starts with # comment lines (the version; a machine-readable # params: line with the normalized parameters `kf0`, `m0`, `v0`, `lam`, `nu`, `s1`, `lnB_i`, `lnC_i`, `a_start`, `omega_r0`, `omega_b0`, ...; the definition of every column), then a header and one row per output point, 47 columns: `N`, `a`, `z`, `t`, `B`, `C`, `H_A`, `H_B`, `H_C`, `constraint_residual`, `rho_r`, `rho_b`, `rho_f`, `P_obs_f`, `P_hid_f`, `n_f`, `sigma`, `m_eff`, `kF_over_m`, `w_f`, `rho_qp`, `w_qp`, `rho_U`, `w_DE_eff`, `Omega_r`, `Omega_b`, `Omega_f`, `q_dec`, `G_ratio`, `cs2_adiabatic`, `N_eff_extra`, `P_stab`, `Omega_sum`, `rho_DE_inf`, `w_DE_inf`, `w_DM_intrinsic`, `w_DM_eff`, `Q`, `rho_DE_eff`, `F_hidden`, `dm_dN`, `m_eff_eV`, `kF_eV`, `gap_roots`, `branch`, `gap_residual`, `vac_over_rho_c`. Its log goes to stderr: the normalization, the result today, `a_nr`, `DeltaN_eff` at nucleosynthesis and recombination, the stabilizer’s range, the $t(A = 1)$ quadrature check, and a # stats: line with the CVODE step counts. Exit code 0 is success; 1 a solver or physics error with a one-line reason; 2 a usage error [prose nb06 §5].

One complete log — the author’s mass term at 30 eV, as notebook 05 ran and printed it [nb05 §5.7]:

```
$ fable_fermion fable4d --potential mass --param m0_ev=30 --points 1001 --out
↳ results/nb05_fable4d_mass30.csv
# normalization: closure today (A = B = C = 1): rho_r + rho_b + rho_f = 1; shooting variable [ln kf0
↳ (the fable number density)] = -2.485515249891274e0; rho_f(A=1; kf0) strictly increasing on the
↳ Lowest branch over ln kf0 in [-40, 40] (1601 grid points, 0 without a gap solution); unique
↳ solution ln kf0 = -2.485515249891274e0
# fable today (A = 1, v = 1): potential Mass { m0: 12182.181219597018 }, kf0 = 8.3282632088e-2 E_c =
↳ 2.050929e-4 eV, m_eff = 1.2182181220e4 E_c = 3.000000e1 eV, rho_qp = 9.5066471258e-1, U =
↳ 0.0000000000e0, rho_f = 9.5066471258e-1, w_f = 9.3473556871e-12, gap roots = 1
# result: H_A(A=1) = 1.000000000000000e0, B(1) = 1.000000000000000e0, C(1) = 1.000000000000000e0, q0 =
↳ 5.000460e-1, age t0 = 6.6660643059e-1/H0, max |constraint residual| = 4.303e-16, max |gap
↳ residual| = 0.0e0, rows with several gap roots = 0
# a_nr (kf = |m_eff|) = 6.837070e-6 (z_nr = 1.4626e5), a_eq = 9.2105e-5: WARM / not hot (a_nr between
↳ 1e-6 and a_eq)
# the fable is ultra-relativistic at a_start: kf/|m_eff| = 6.836e6
# Delta N_eff (fable energy density in units of one massless neutrino species): at BBN (a =
↳ 1.6764e-10, T = 1 MeV) = 3.942638e-1, at recombination (a = 9.1659e-4) = 7.048553e1
# stabilizer: P_stab = -F_hidden/2 ranges over [-2.4622e34, -4.9995e-1] (today -4.999540e-1); it is a
↳ Lagrange multiplier with zero energy density and violates the null energy condition along x0
↳ wherever dust is present (rho + P_C = -rho_dust/2)
```

```

# t(A=1) cross-check: CVODE 6.666064305910978e-1 vs Gauss-Kronrod quadrature 6.666064296244345e-1
↪ (rel. diff 1.45e-9, quadrature error estimate 2.4e-14)
# NOTE: W = m0 sigma alone (no V0): the closure forces Omega_f0 = 1 - Omega_b0 - Omega_r0 = 0.950665:
↪ an Einstein-de Sitter-like universe (q0 = 0.5000, +1/2 for pure dust)
# the DM/DE split (rho_qp = eps/v vs rho_U = W - sigma W') is a convention: the condensate has P =
↪ -rho_U in all seven spatial directions, i.e. it is an 8D vacuum energy
# stats: model=fable4d direction=Forward potential=mass steps=373 rhs_evals=459 nonlin_iters=456
↪ err_test_fails=18 jac_evals=7 wall=0.010s (summed over every CVODE integration of the run,
↪ shooting included)
# fable_fermion 0.1.0, sundials_rs 7.8.0 (pure Rust), CVODE BDF

```

Every run, run from fable-cosmology/ with F as in section 11.3, in the order the notebooks ran them, each followed by the # stats: line it printed (the CVODE steps, right-hand-side evaluations, Newton iterations, error-test failures, Jacobian evaluations and wall time, summed over every integration of the run, shooting included), as the notebooks recorded them:

```

# notebook 05, section 5.7: the author's mass term on the pre-universe (cooling cross-check)
$F fable4d --potential mass --param m0_ev=30 --points 1001 --out results/nb05_fable4d_mass30.csv
# prints: # stats: model=fable4d direction=Forward potential=mass steps=373 rhs_evals=459
↪ nonlin_iters=456 err_test_fails=18 jac_evals=7 wall=0.010s (summed over every CVODE integration of
↪ the run, shooting included)

# notebook 05, section 5.8: the vacuum-energy column of a mass-varying run
$F fable4d --potential power --param nu=0.236 --param m_today_ev=30 --points 1001 --out
↪ results/nb05_fable4d_power0.236_m30.csv
# prints: # stats: model=fable4d direction=Forward potential=power steps=390 rhs_evals=482
↪ nonlin_iters=479 err_test_fails=20 jac_evals=7 wall=0.010s (summed over every CVODE integration of
↪ the run, shooting included)

# notebook 06, section 5.1: the interval (a_i = 1e-12, 1e-11, 1e-10; CSV to stdout)
$F fable4d --potential lambda-mass --param m0_ev=30 --a-start 1e-12 --points 1001
# prints: # stats: model=fable4d direction=Forward potential=lambda-mass steps=430 rhs_evals=542
↪ nonlin_iters=539 err_test_fails=25 jac_evals=8 wall=0.010s (summed over every CVODE integration of
↪ the run, shooting included)
$F fable4d --potential lambda-mass --param m0_ev=30 --a-start 1e-11 --points 1001
# prints: # stats: model=fable4d direction=Forward potential=lambda-mass steps=433 rhs_evals=549
↪ nonlin_iters=546 err_test_fails=27 jac_evals=8 wall=0.010s (summed over every CVODE integration of
↪ the run, shooting included)
$F fable4d --potential lambda-mass --param m0_ev=30 --a-start 1e-10 --points 1001
# prints: # stats: model=fable4d direction=Forward potential=lambda-mass steps=438 rhs_evals=525
↪ nonlin_iters=522 err_test_fails=20 jac_evals=8 wall=0.010s (summed over every CVODE integration of
↪ the run, shooting included)
$F fable8d --potential lambda-mass --param m0_ev=30 --a-start 1e-12 --points 1001
# prints: # stats: model=fable8d direction=Forward potential=lambda-mass steps=37544 rhs_evals=42447
↪ nonlin_iters=42339 err_test_fails=654 jac_evals=647 wall=0.040s (summed over every CVODE integration
↪ of the run, shooting included)
$F fable8d --potential lambda-mass --param m0_ev=30 --a-start 1e-11 --points 1001
# prints: # stats: model=fable8d direction=Forward potential=lambda-mass steps=36759 rhs_evals=41619
↪ nonlin_iters=41511 err_test_fails=618 jac_evals=634 wall=0.039s (summed over every CVODE integration
↪ of the run, shooting included)
$F fable8d --potential lambda-mass --param m0_ev=30 --a-start 1e-10 --points 1001
# prints: # stats: model=fable8d direction=Forward potential=lambda-mass steps=34762 rhs_evals=39367
↪ nonlin_iters=39262 err_test_fails=576 jac_evals=593 wall=0.037s (summed over every CVODE integration
↪ of the run, shooting included)

# notebook 06, section 5.2: runs (a), the author's mass term
$F fable4d --potential mass --param m0_ev=1 --points 1001 --out results/nb06_mass_m1_4d.csv
# prints: # stats: model=fable4d direction=Forward potential=mass steps=392 rhs_evals=480
↪ nonlin_iters=477 err_test_fails=20 jac_evals=7 wall=0.010s (summed over every CVODE integration of
↪ the run, shooting included)
$F fable4d --potential mass --param m0_ev=30 --points 1001 --out results/nb06_mass_m30_4d.csv
# prints: # stats: model=fable4d direction=Forward potential=mass steps=373 rhs_evals=459
↪ nonlin_iters=456 err_test_fails=18 jac_evals=7 wall=0.010s (summed over every CVODE integration of
↪ the run, shooting included)
$F fable4d --potential mass --param m0_ev=100 --points 1001 --out results/nb06_mass_m100_4d.csv
# prints: # stats: model=fable4d direction=Forward potential=mass steps=387 rhs_evals=467
↪ nonlin_iters=464 err_test_fails=16 jac_evals=7 wall=0.010s (summed over every CVODE integration of
↪ the run, shooting included)
$F fable4d --potential mass --param m0_ev=1000 --points 1001 --out results/nb06_mass_m1000_4d.csv
# prints: # stats: model=fable4d direction=Forward potential=mass steps=377 rhs_evals=463
↪ nonlin_iters=460 err_test_fails=19 jac_evals=7 wall=0.010s (summed over every CVODE integration of
↪ the run, shooting included)

```

```

$F fable4d --potential mass --param m0_ev=2000 --points 1001 --out results/nb06_mass_m2000_4d.csv
# prints: # stats: model=fable4d direction=Forward potential=mass steps=377 rhs_evals=467
→ nonlin_iters=464 err_test_fails=18 jac_evals=7 wall=0.010s (summed over every CVODE integration of
→ the run, shooting included)

# notebook 06, section 5.3: runs (b), lambda-mass, and Part VIII's reference case
$F fable4d --potential lambda-mass --param m0_ev=30 --points 1001 --out results/nb06_lm_m30_4d.csv
# prints: # stats: model=fable4d direction=Forward potential=lambda-mass steps=430 rhs_evals=542
→ nonlin_iters=539 err_test_fails=25 jac_evals=8 wall=0.010s (summed over every CVODE integration of
→ the run, shooting included)
$F fable4d --potential lambda-mass --param m0_ev=100 --points 1001 --out results/nb06_lm_m100_4d.csv
# prints: # stats: model=fable4d direction=Forward potential=lambda-mass steps=418 rhs_evals=515
→ nonlin_iters=512 err_test_fails=21 jac_evals=8 wall=0.010s (summed over every CVODE integration of
→ the run, shooting included)
$F fable4d --potential lambda-mass --param m0_ev=1000 --points 1001 --out results/nb06_lm_m1000_4d.csv
# prints: # stats: model=fable4d direction=Forward potential=lambda-mass steps=407 rhs_evals=509
→ nonlin_iters=506 err_test_fails=21 jac_evals=8 wall=0.009s (summed over every CVODE integration of
→ the run, shooting included)
$F fable4d --potential lambda-mass --param m0_ev=2000 --points 1001 --out results/nb06_lm_m2000_4d.csv
# prints: # stats: model=fable4d direction=Forward potential=lambda-mass steps=425 rhs_evals=527
→ nonlin_iters=524 err_test_fails=21 jac_evals=8 wall=0.010s (summed over every CVODE integration of
→ the run, shooting included)
$F fable4d --potential lambda-mass --param m0_ev=30 --param omega_dm=0.265 --a-start 1e-10 --points 701
→ --out results/nb06_fable4d_mass30eV.csv
# prints: # stats: model=fable4d direction=Forward potential=lambda-mass steps=429 rhs_evals=548
→ nonlin_iters=545 err_test_fails=29 jac_evals=8 wall=0.008s (summed over every CVODE integration of
→ the run, shooting included)

# notebook 06, section 5.4: runs (c), the dark-energy potentials, and Part VIII's mass-varying reference
→ case
$F fable4d --potential power --param nu=0.236 --param m_today_ev=30 --points 1001 --out
→ results/nb06_power_0.236_m30_4d.csv
# prints: # stats: model=fable4d direction=Forward potential=power steps=390 rhs_evals=482
→ nonlin_iters=479 err_test_fails=20 jac_evals=7 wall=0.011s (summed over every CVODE integration of
→ the run, shooting included)
$F fable4d --potential power --param nu=0.236 --param m_today_ev=100 --points 1001 --out
→ results/nb06_power_0.236_m100_4d.csv
# prints: # stats: model=fable4d direction=Forward potential=power steps=401 rhs_evals=505
→ nonlin_iters=502 err_test_fails=23 jac_evals=7 wall=0.010s (summed over every CVODE integration of
→ the run, shooting included)
$F fable4d --potential power --param nu=0.5 --param m_today_ev=30 --points 1001 --out
→ results/nb06_power_0.5_m30_4d.csv
# prints: # stats: model=fable4d direction=Forward potential=power steps=430 rhs_evals=551
→ nonlin_iters=548 err_test_fails=26 jac_evals=8 wall=0.021s (summed over every CVODE integration of
→ the run, shooting included)
$F fable4d --potential power --param nu=0.5 --param m_today_ev=100 --points 1001 --out
→ results/nb06_power_0.5_m100_4d.csv
# prints: # stats: model=fable4d direction=Forward potential=power steps=433 rhs_evals=545
→ nonlin_iters=542 err_test_fails=24 jac_evals=8 wall=0.027s (summed over every CVODE integration of
→ the run, shooting included)
$F fable4d --potential expdamp --param xt=0.4524886877828054 --param m_today_ev=30 --points 1001 --out
→ results/nb06_expdamp_m30_4d.csv
# prints: # stats: model=fable4d direction=Forward potential=expdamp steps=423 rhs_evals=527
→ nonlin_iters=524 err_test_fails=21 jac_evals=8 wall=0.019s (summed over every CVODE integration of
→ the run, shooting included)
$F fable4d --potential expdamp --param xt=0.4524886877828054 --param m_today_ev=100 --points 1001 --out
→ results/nb06_expdamp_m100_4d.csv
# prints: # stats: model=fable4d direction=Forward potential=expdamp steps=434 rhs_evals=542
→ nonlin_iters=539 err_test_fails=23 jac_evals=8 wall=0.019s (summed over every CVODE integration of
→ the run, shooting included)
$F fable4d --potential quadratic --param gq=-0.5 --param m_today_ev=30 --points 1001 --out
→ results/nb06_quadratic_m30_4d.csv
# prints: # stats: model=fable4d direction=Forward potential=quadratic steps=451 rhs_evals=562
→ nonlin_iters=559 err_test_fails=23 jac_evals=8 wall=0.012s (summed over every CVODE integration of
→ the run, shooting included)
$F fable4d --potential quadratic --param gq=-0.5 --param m_today_ev=100 --points 1001 --out
→ results/nb06_quadratic_m100_4d.csv
# prints: # stats: model=fable4d direction=Forward potential=quadratic steps=451 rhs_evals=558
→ nonlin_iters=555 err_test_fails=22 jac_evals=8 wall=0.011s (summed over every CVODE integration of
→ the run, shooting included)
$F fable4d --potential power --param m_today_ev=100 --param nu=0.5 --param omega_dm=0.55 --a-start 1e-10
→ --points 701 --out results/nb06_fable4d_power.csv
# prints: # stats: model=fable4d direction=Forward potential=power steps=379 rhs_evals=483
→ nonlin_iters=480 err_test_fails=24 jac_evals=7 wall=0.008s (summed over every CVODE integration of
→ the run, shooting included)

```

```

# notebook 06, section 5.7: the Delta N_eff runs at 1 eV
$F fable4d --potential power --param nu=0.236 --param m_today_ev=1 --points 1001 --out
→ results/nb06_power_0.236_m1_4d.csv
# prints: # stats: model=fable4d direction=Forward potential=power steps=397 rhs_evals=487
→ nonlin_iters=484 err_test_fails=18 jac_evals=7 wall=0.011s (summed over every CVODE integration of
→ the run, shooting included)
$F fable4d --potential power --param nu=0.5 --param m_today_ev=1 --points 1001 --out
→ results/nb06_power_0.5_m1_4d.csv
# prints: # stats: model=fable4d direction=Forward potential=power steps=429 rhs_evals=553
→ nonlin_iters=550 err_test_fails=28 jac_evals=8 wall=0.019s (summed over every CVODE integration of
→ the run, shooting included)
$F fable4d --potential expdamp --param xt=0.4524886877828054 --param m_today_ev=1 --points 1001 --out
→ results/nb06_expdamp_m1_4d.csv
# prints: # stats: model=fable4d direction=Forward potential=expdamp steps=428 rhs_evals=560
→ nonlin_iters=557 err_test_fails=32 jac_evals=8 wall=0.018s (summed over every CVODE integration of
→ the run, shooting included)
$F fable4d --potential quadratic --param gq=-0.5 --param m_today_ev=1 --points 1001 --out
→ results/nb06_quadratic_m1_4d.csv
# prints: # stats: model=fable4d direction=Forward potential=quadratic steps=444 rhs_evals=576
→ nonlin_iters=573 err_test_fails=30 jac_evals=8 wall=0.011s (summed over every CVODE integration of
→ the run, shooting included)

# notebook 06, section 5.8: runs (d), the unstabilized no-go and its controls
$F fable8d --potential mass --param m0_ev=1 --points 1001 --out results/nb06_mass_m1_8d.csv
# prints: # stats: model=fable8d direction=Forward potential=mass steps=37019 rhs_evals=41186
→ nonlin_iters=41081 err_test_fails=499 jac_evals=634 wall=0.039s (summed over every CVODE integration
→ of the run, shooting included)
$F fable8d --potential mass --param m0_ev=100 --points 1001 --out results/nb06_mass_m100_8d.csv
# prints: # stats: model=fable8d direction=Forward potential=mass steps=40808 rhs_evals=45860
→ nonlin_iters=45758 err_test_fails=474 jac_evals=702 wall=0.042s (summed over every CVODE integration
→ of the run, shooting included)
$F fable8d --potential lambda-mass --param m0_ev=30 --points 1001 --out results/nb06_lm_m30_8d.csv
# prints: # stats: model=fable8d direction=Forward potential=lambda-mass steps=37544 rhs_evals=42447
→ nonlin_iters=42339 err_test_fails=654 jac_evals=647 wall=0.039s (summed over every CVODE integration
→ of the run, shooting included)
$F fable8d --potential lambda-mass --param m0_ev=1000 --points 1001 --out results/nb06_lm_m1000_8d.csv
# prints: # stats: model=fable8d direction=Forward potential=lambda-mass steps=38558 rhs_evals=42762
→ nonlin_iters=42657 err_test_fails=595 jac_evals=669 wall=0.040s (summed over every CVODE integration
→ of the run, shooting included)
$F fable8d --no-fable --omega-b 0 --points 1001 --out results/nb06_radiation_only_8d.csv
# prints: # stats: model=fable8d direction=Forward potential=none steps=7 rhs_evals=13 nonlin_iters=7
→ err_test_fails=0 jac_evals=2 wall=0.008s (summed over every CVODE integration of the run, shooting
→ included)
$F fable8d --no-fable --points 1001 --out results/nb06_radiation_baryons_8d.csv
# prints: # stats: model=fable8d direction=Forward potential=none steps=2109 rhs_evals=2379
→ nonlin_iters=2370 err_test_fails=37 jac_evals=38 wall=0.010s (summed over every CVODE integration of
→ the run, shooting included)
$F fable8d --potential lambda-mass --param m0_ev=30 --freeze-hidden --points 1001
# prints: # stats: model=fable8d direction=Forward potential=lambda-mass steps=395 rhs_evals=475
→ nonlin_iters=472 err_test_fails=15 jac_evals=7 wall=0.011s (summed over every CVODE integration of
→ the run, shooting included)

# notebook 06, section 5.9: the three refused runs
$F fable4d --potential lorentz --points 1001
# prints: exit code 1: fable_fermion: rho_hat = H_A^2 <= 0 from N = -0.7597808064 (a = 0.4677689478)
→ on: the Kohn-Sham ground state has negative energy there (m_eff = -3.802078e-2, sigma8 =
→ -9.789374e-4, rho_f = -4.830421e-1 (U = -4.847891e-1) against rho_r + rho_b = 4.830421e-1); H_A^2 =
→ rho_hat is impossible (the onset located past the last accepted CVODE step and bisected)
$F fable4d --potential quadratic --param gq=0.5 --points 1001
# prints: exit code 1: fable_fermion: gap-branch jump at N = -0.0037262796 (a = 0.9962806544): a
→ first-order transition of the Kohn-Sham ground state, the lowest-energy gap root jumps from the
→ branch m_eff = -1.518260e0 (rho_f = 6.856759e-1) to the branch m_eff = 4.083552e2 (rho_f =
→ 9.536521e-1); the right-hand side is discontinuous there, and energy conservation across it would
→ need the Maxwell construction, which this solver does not implement (the transition detected inside
→ the right-hand side and bisected in N to adjacent doubles)
$F fable8d --potential mass --direction backward
# prints: exit code 1: fable_fermion: fable8d runs forward only (design review E4): backward in time
→ the shear modes grow as A^-3 v^-1 relative to H_A ~ A^-2, the solution runs to the 8D Kasner point
→ H_B/H_A = -0.2929 and the constraint drifts to 0(1); use fable4d --direction backward

```

The 90 scan runs of section 9.8 (notebook 06, §5.6; CSV to stdout, 401 rows, each re-computed with the independent implementation as it ran; their # stats: lines are not printed):

[illegible]

```

$F fable4d --potential expdamp --param xt=0.6 --param m_today_ev=100 --points 401
$F fable4d --potential expdamp --param xt=0.7 --param m_today_ev=100 --points 401
$F fable4d --potential expdamp --param xt=0.8 --param m_today_ev=100 --points 401
$F fable4d --potential expdamp --param xt=0.9 --param m_today_ev=100 --points 401
$F fable4d --potential power --param omega_dm=0.265 --param nu=0.3 --param m_today_ev=100 --points 401
$F fable4d --potential power --param omega_dm=0.35 --param nu=0.3 --param m_today_ev=100 --points 401
$F fable4d --potential power --param omega_dm=0.45 --param nu=0.3 --param m_today_ev=100 --points 401
$F fable4d --potential power --param omega_dm=0.55 --param nu=0.3 --param m_today_ev=100 --points 401
$F fable4d --potential power --param omega_dm=0.65 --param nu=0.3 --param m_today_ev=100 --points 401
$F fable4d --potential power --param omega_dm=0.8 --param nu=0.3 --param m_today_ev=100 --points 401
$F fable4d --potential power --param omega_dm=0.265 --param nu=0.5 --param m_today_ev=100 --points 401
$F fable4d --potential power --param omega_dm=0.35 --param nu=0.5 --param m_today_ev=100 --points 401
$F fable4d --potential power --param omega_dm=0.45 --param nu=0.5 --param m_today_ev=100 --points 401
$F fable4d --potential power --param omega_dm=0.5 --param nu=0.5 --param m_today_ev=100 --points 401
$F fable4d --potential power --param omega_dm=0.55 --param nu=0.5 --param m_today_ev=100 --points 401
$F fable4d --potential power --param omega_dm=0.65 --param nu=0.5 --param m_today_ev=100 --points 401
$F fable4d --potential power --param omega_dm=0.8 --param nu=0.5 --param m_today_ev=100 --points 401

```

11.5 The independent Python implementation

```

cd "$(git rev-parse --show-toplevel)"
fable-cosmology/.venv/Scripts/python.exe fable-cosmology/fermion/crosscheck.py --jobs 7

```

It recomputes the unit system from CODATA, evaluates every Fermi-sea integral by adaptive quadrature (`scipy.integrate.quad`, `epsrel = 1e-12`, split at $k = |m|$), solves every gap by `brentq` on the documented brackets, integrates `fable8d` with DOP853 at `rto1 = 1e-11` (and `fable4d` algebraically), regenerates the Rust CSVs it compares against into `fable-cosmology/fermion/dev/`, and compares $\max|\Delta w_f|$, $\max|\Delta H_A|/H_A$, $\max|\Delta B|/B$, $\max|\Delta G_{\text{ratio}}|/G_{\text{ratio}}$; every number must be below 10^{-6} , otherwise it fails with exit code 1 [file: its docstring]. It prints a `units:` line, one row per case (`case`, $\max|dw_f|$, $\max|dH_A|/H$, $\max|dB|/B$, $\max|dG|/G$, `H_A(1)py`, `B(1)py`, `nfev`, `secs`) and `CROSSCHECK PASSED: every difference < 1e-06`; the commit of the finished solver (`c5892bd`) records that every run agreed with the Rust CSVs to below 10^{-8} . Notebook 06 imports the same module (`independent implementation imported from fermion/crosscheck.py`, `crosscheck.units()` agrees with the solver's unit system to $1e-9$) and runs it on every run of this page; its results are section 9.12.

11.6 The Mathematica reference runs

```

wolframscript -file "$(git rev-parse
↪ --show-toplevel)/fable-cosmology/reference/make_reference_fermion.wls"

```

Self-locating; it writes `mathematica_fable4d_mass30eV.csv` and `mathematica_fable4d_power.csv` (701 rows each) next to itself (section 9.12). It prints the units (`E_c = rho_c0^(1/4) = ... eV`; `1/H0 = ... yr`, the three `Omegas`), the agreement of the series with the closed forms at $k_F/m = 1/5$, $1/100$, the parameters of each case (`mass30eV: m0 = ... E_c = 30 eV`; `kF0 = ... E_c = ... eV`; `V0 = Omega_Lambda = ...`; `power: nu = 1/2, m_today = 100 eV, eps_today = 0.55`; `kF0 = ... E_c`; `lam = ...`), the timings and checks of each case (`|H_A(1) - 1|`, `w_f(a_i)`, the largest gap residual, `m_eff(a_i)`), and wrote `mathematica_fable4d_mass30eV.csv` and `mathematica_fable4d_power.csv` (701 rows each) [file: its Print statements]. Its console output was not captured in a repository log; the check of its files is Part VIII's assertion `fable4d RUNS [fidelity]: at every tenth row ALL TEN columns agree with fable-cosmology/reference/mathematica_fable4d_mass30eV.csv and _power.csv to 1e-10 (relative; P_obs_f relative to rho_f; N and w_f absolute)`, and the three-way table of section 9.12.

11.7 The wall-state solver, its derivation and its Mathematica check

From `fable-cosmology/fermion/` (`PY=../.venv/Scripts/python.exe`), with the durations the report records [**WG 1**]:

```

cd "$(git rev-parse --show-toplevel)/fable-cosmology/fermion"
PY=../.venv/Scripts/python.exe
wolframscript -file waveguide_derivation.wls > waveguide_derivation.log          # ~3.5 min; 94/94
→ PASS
$PY waveguide.py validate > dev/waveguide_run_validate.log                      # 45 s; 64/64
→ PASS
$PY waveguide.py threshold > dev/waveguide_run_threshold.log                  # 13 s
$PY waveguide.py figures > dev/waveguide_run_figures.log                      # dispersion,
→ edge, theta, wavefunctions; ~25 s
$PY waveguide.py convergence > dev/waveguide_run_convergence.log              # 18 s
$PY waveguide.py reference > dev/waveguide_run_reference.log                  # 7 s
wolframscript -file waveguide_check.wls > dev/waveguide_run_check.log          # 40 s; 11/11
→ PASS
$PY waveguide.py dft > dev/waveguide_run_dft.log                              # 9.3 min
$PY waveguide.py dft --scenario strong > dev/waveguide_run_dft_strong.log      # 10.9 min;
→ exploratory
$PY waveguide.py dft --scenario bulk > dev/waveguide_run_dft_bulk.log          # 1 s;
→ homogeneous self-binding
$PY waveguide.py dft --scenario convergence > dev/waveguide_run_dft_convergence.log # 12.1 min
$PY waveguide.py levels --K 80 --m 1 --wall same > dev/waveguide_run_levels_K80.log # 1.4 s
$PY waveguide.py all # everything except the exploratory dft scenarios

```

The subcommands [file: the module docstring of waveguide.py]: `validate` (analytic cases, Hellmann–Feynman, Magnus vs DOP853, 16×16 multiplicities; 64 checks), `levels` (all levels of both irreps at `-K -m -wall same|opposite [-theta th]`), `threshold` ($K_c(1)$, $K_c(2)$ against m/H), `dispersion` ($\omega_n(K)$, $K = 7..100$, and the opposite-sign spectrum), `edge` (the opposite-sign edge band), `theta` (the uniform self-adjoint family at $K = 3$ and 80), `convergence`, `wavefunctions`, `figures` (dispersion + edge + theta + wavefunctions), `reference` (the level table read by `waveguide_check.wls`), `dft -scenario all|free|same|opposite|strong|bulk|convergence`, and `all`. Outputs go to `fermion/dev/` with the prefix `waveguide_` (a folder the repository ignores; notebook 07 regenerates the final tables and figures).

What they print, quoted: `waveguide_derivation.wls` ends its log with

```

=====
waveguide_derivation.wls: 94 checks, 94 PASS, 0 FAIL
total time 162.8 s
=====

```

after [load] notebook Input cells: 217; evaluated cells 1..117 in 125. s, [load] cells that raised messages while loading (incidental to this script): {} and [load] the notebook’s own assertions in cells 1..117: 142 run, 142 passed; not passed: {} [file: `fable-cosmology/fermion/waveguide_derivation.log`, the run of 2026-09-25 00:06:38 committed in `aca5102`; the first committed run, in `18a2501`, took 171.4s with evaluated cells 1..117 in 132.3 s]. `validate` reports 64/64 PASS; `waveguide_check.wls` reports Summary: 11 checks, 11 PASS, 0 FAIL, no messages, with its level table as in section 8.4; `levels -K 80 -m 1 -wall same` prints the eight levels quoted in section 8.5 [WG 1, WG 3, WG 5].

The derivation’s section 8 writes the notebook’s `T16[0..8]`, `sigma16` and the block basis U to `fable-cosmology/fermion/waveguide_T16.json` (its log: 8. Export `T16[0..8]`, `sigma16` and the block basis U to `fermion/waveguide_T16.json` and `PASS export: T16` and `sigma16` read back identical from `waveguide_T16.json`), which `waveguide.py`, `waveguide_check.wls` and notebook 07 read. That file is committed (commit `aca5102`; it used to be in the ignored `dev/`, which broke a fresh clone without Mathematica, and the regenerated file is byte-identical [commit `aca5102`]). `waveguide.py validate` exits with code 1 whenever any of its 64 checks fails; if `waveguide_T16.json` is missing, the checks that need it (V7, the 16-component multiplicities, and V8, the block reduction by U) are recorded as failures, with the command that rewrites the file, and every other check still runs [file: `fable-cosmology/fermion/waveguide.py`, its module docstring, `validate` and `main`]. (Before commit `aca5102` the module docstring gave an earlier count of 81 checks for the derivation; it now quotes the last line of the committed log, 94 checks.)

11.8 The notebook: Parts VII and VIII

The notebook is never edited by hand. `build_tools.py` reads every `claude-fable/cells_part*.wl` in order and writes the notebook and `run_all.wls` (the same Input cells as a script):

```
cd "$(git rev-parse --show-toplevel)/claude-fable"
python build_tools.py          # use python3 where python is absent
```

With the eight manifests it prints (recorded on 2026-09-24 in a scratch copy of the directory, in 0.115s; the last line gives the absolute path of the directory it ran in):

```
manifest files : ['cells_part1.wl', 'cells_part2.wl', 'cells_part3.wl', 'cells_part4.wl',
↳ 'cells_part5.wl', 'cells_part6.wl', 'cells_part7.wl', 'cells_part8.wl']
cells parsed   : 384 {'Title': 8, 'Subtitle': 1, 'Subsubtitle': 1, 'Text': 123, 'Section': 34, 'Input':
↳ 217}
run_all.wls    : 217 Input cells
notebook       : 384 cells -> <repo>\claude-fable\claude-fable_Einstein-Rosen-2-Planes.nb
```

and the rebuilt notebook and `run_all.wls` were byte-identical to the committed ones. Re-run for this page on 2026-09-25 on the manifests of commit 3a5020d, it printed the same four lines, and the rebuilt notebook and `run_all.wls` were again byte-identical to the committed ones.

Run it end to end, straight out of the .nb:

```
cd "$(git rev-parse --show-toplevel)/claude-fable"
wolframscript -file run_from_nb.wls > run_fermion_fable_final.log 2>&1
git checkout -- claude-fable_Einstein-Rosen-2-Planes-eLa.mx
↳ claude-fable_Einstein-Rosen-2-Planes-eLazt.mx
```

`run_from_nb.wls` imports the `.nb`, takes its Input cells in order, and evaluates each as one unit through the harness `runner_header.wl`, which prints `CELL n t=... s` after each cell (with `MESSAGES: ...` if the cell raised any), every `PASS/FAIL` label, and the run summary. It must run inside `claude-fable/` (the author's helper packages `ConvertMapleToMathematicaV2.wl` and `EtoExp.wl` are found in the working directory). Section 12 of the notebook rewrites the two `.mx` files with `DumpSave`, whose bytes are not reproducible, so they are restored after every run. It prints first evaluating 217 Input cells straight out of the `.nb`, and it ends [file: `claude-fable/run_fermion_fable_final.log`, the final run, commit 3a5020d]:

```
identities accepted on numerical evidence alone : 0    (stage 3 returned True)
non-vanishing witnesses                        : 46    (cfNonZeroWitnessQ found a probe point at which
↳ every entry is a                               number and one of them is non-zero: a complete
                                                    ↳ proof of non-vanishing)

PASS no identity in this notebook was accepted on numerical evidence alone
Assertions run: 644   passed: 644   failed: 0
CELL 217 t=0.000 s
===== RUN SUMMARY =====
cells evaluated : 217
total seconds   : 198.352
cells w/ msgs   : 0
---- slowest cells ----
cell 80 68.159 s
cell 88 22.525 s
cell 204 9.360 s
cell 86 8.072 s
cell 191 7.887 s
cell 87 7.193 s
cell 137 6.985 s
cell 121 4.579 s
cell 160 3.822 s
cell 107 3.505 s
cell 196 3.334 s
cell 193 2.970 s
===== ASSERTIONS =====
assertions run : 644
passed         : 644
FAILED        : 0
=====
```

(The earlier run logs end with one more line, RUN-DONE, which the wrapper that ran them appended; the final run was captured without it.) The notebook's final assertion is **no identity in this notebook was accepted on numerical evidence alone**: every identity of Parts I–VIII is closed symbolically, and the 46 non-vanishing claims are complete proofs by witness. Part VIII (cells 199–217) took 20.994 s of the 198.352 s and added 116 assertions to Part VII's 528 [counted from the log]:

notebook Section	subject	Input cells	assertions	seconds	this page
30	the Einstein tensor of the canonical metric, and what it demands of its source	199–203	28	3.534	4
31	the generalized warped frame, its asymptotic region, and the 8-dimensional Bianchi-I limit	204–207	21	11.310	5
32	the canonical spin connection of the interacting system	208–210	26	3.893	6.14–6.22
33	the coupled equations of fable and the primordial field, and their solution from the radiation era to today	211–217	41	2.257	7, 9
Part VIII		19 cells	116	20.994	

The log's other non-assertion lines are the four EXPECTED MESSAGE(S) announcements of Parts II and V and the NOTE that a_4 is undefined (Part III); none is a message raised by a cell. Its lines for cells 1–198 are identical to those of Part VII's own acceptance run, `claude-fable/run_fermion_fable_part7.log`, apart from the first line (the cell count) and the timing lines [checked for this page with `diff`]; that log was produced by the same command with the notebook of Parts I–VII (198 cells, `Assertions run: 528 passed: 528 failed: 0, 232.146s`), and it remains the record of Part VII. Part VIII's acceptance run, `claude-fable/run_fermion_fable_part8.log` (commit `bc04ed1`), was made by the same command before notebook 06 had written the Rust solver's CSVs: 217 cells, 0 cells with messages, 233.289 s, 46 witnesses and `Assertions run: 642 passed: 642 failed: 0`, with two NOTE Rust CSV not found; that comparison was skipped, not failed lines in Section 33 (section 9.12) and the three spin-connection labels of section 6.15 in their earlier form, with upper flat indices; every other PASS label is identical to the final run's. The full run of commit 1671155, the first with both comparisons, recorded 644/644 with 0 cells with messages in 208.7 s [**commit 1671155**]; commit 3a5020d replaced its log by the run made after the label correction.

Check a log:

```
cd "$(git rev-parse --show-toplevel)/claude-fable"
grep -c '^ PASS' run_fermion_fable_final.log      # 644
grep -n '^ FAIL' run_fermion_fable_final.log     # no output: no assertion failed
grep -c 'MESSAGES: ' run_fermion_fable_final.log # 0: no cell raised a message
sed -n '/RUN SUMMARY/, $p' run_fermion_fable_final.log # the summary above
diff <(grep '^ PASS' run_fermion_fable_part8.log) <(grep '^ PASS'
↪ run_fermion_fable_final.log)
# the three relabelled lines and the
↪ two new ones
```

Check the notebook without evaluating it (about 4 s):

```
wolframscript -file "$(git rev-parse --show-toplevel)/claude-fable/verify_nb.wls"
```

prints [file: `claude-fable/verify_nb_part8.log`, rewritten in commit 3a5020d for the corrected notebook]:

```

file bytes: 545713
Head: Notebook
cells: 384
style tally: {{Title, 8}, {Subtitle, 1}, {Subsubtitle, 1}, {Text, 123}, {Section, 34}, {Input, 217}}
input cells with plain-string BoxData: 217
total input characters: 347237
input cells that FAIL to parse: {}
first cell: InputForm[(* --- provenance banner, as in the original notebook
→ -----]
last cell : InputForm[cfAssert["PART VIII [control]: a4 is STILL UNDEFINED -- nothing in Part VIII gave
→ it a val]]

```

The section-to-cell map (reads the `.nb`, evaluates nothing):

```

wolframscript -file "$(git rev-parse --show-toplevel)/claude-fable/nb_section_map.wls" > "$(git
→ rev-parse --show-toplevel)/claude-fable/nb_section_map.log"

```

Its Part VII and Part VIII lines [file: `claude-fable/nb_section_map.log`]:

```

== PART VII -- fermion fable: the complex 16-spinor, its canonical quantization in 4+4 dimensions, its
→ field equations in the primordial gravitational field, and its energy-momentum tensor operator
26. The refinement: why fermion fable is a complex 16-spinor | Input cells 173-179
27. The Lagrangian, the field equations in the primordial gravitational field, and the canonical spin
→ connection | Input cells 180-184
28. Canonical quantization in 4+4 dimensions: the Dirac bracket, the Krein structure J, and the
→ admissible sector | Input cells 185-191
29. The energy-momentum tensor operator of fermion fable: definition, conservation, and rho, P and w |
→ Input cells 192-198
== PART VIII -- fable as a source of the Einstein equations of the primordial gravitational field
30. The Einstein tensor of the canonical metric, and what it demands of its source | Input cells
→ 199-203
31. The generalized warped frame, its asymptotic region, and the 8-dimensional Bianchi-I limit |
→ Input cells 204-207
32. The canonical spin connection of the interacting system | Input cells 208-210
33. The coupled equations of fable and the primordial field, and their solution from the radiation era
→ to today | Input cells 211-217
total Input cells: 217

```

The TeX/text export of the fermion field's equations (section 3.3 quotes one of its files):

```

wolframscript -file "$(git rev-parse --show-toplevel)/claude-fable/export_fermion_fable_tex.wls"

```

It evaluates every Input cell of the notebook with the output suppressed, stops with exit code 1 if the Part VII objects are missing, and writes fourteen files into `provenance-latex/generated/` (the 16 field equations, the 16 adjoint equations, the $8 + 8$ form, the rescaled equations, the anticommutator, J and β , the Kohn–Sham closed forms; each as `.tex` and `.txt`), printing evaluating N Input cells of ... (output suppressed), assertions: ... failed: 0 cells with messages: 0, one wrote ... `.tex` / `.txt` line per object and done: 14 files in Like every evaluation of the notebook it rewrites the two `.mx` files, which are then restored as above.

11.9 Execute the notebooks 05, 06 and 07, and check them

```

cd "$(git rev-parse --show-toplevel)/fable-cosmology"
PY=.venv/Scripts/python.exe # .venv/bin/python on Linux/macOS
for nb in notebooks/05_fermion_fable_quantum_eos.ipynb notebooks/06_fable_primordial_gravity_8d.ipynb
→ notebooks/07_fable_dft_states.ipynb; do
    "$PY" -m nbconvert --to notebook --execute --inplace --ExecutePreprocessor.timeout=1800 \
        --ExecutePreprocessor.kernel_name=python3 "$nb"
done
"$PY" notebooks/_build/nbcheck.py

```

Each notebook is executed headlessly and its outputs are written back into the `.ipynb`; the notebooks locate the solver binary, read the unit system from it, run every case, write their CSV tables and PNG figures under `results/`, and assert every claim they print (a failed assertion stops the execution). The interactive last two cells (name the notebook, the save dialog) detect

the headless run and print `Headless execution: no prompt`; the notebook keeps its name `05_fermion_fable_quantum_eos` and `Headless execution: the save dialog is skipped`; the notebook stays where it is. (and likewise for 06 and 07). The notebooks themselves are generated from `notebooks/_build/nbgen.py`; they are never edited by hand. Notebook 05 prints its summary as ten numbered statements (sections 7–8 quote them), notebook 06 its answers (section 10), notebook 07 its summary as nine numbered statements (section 8). Their execution times were not recorded in a repository log.

`nbcheck.py` checks every notebook against eight requirements (a terminal launch explanation and how to rebuild its solver; no reference to another notebook; markdown before every code cell; the naming cell; the save dialog with a fallback; the field, Lagrangian, energy–momentum tensor, equations of motion and the first-order system handed to SUNDIALS, with a glossary; valid executed `nbformat-4` paired with its solver model; its own results with a plain-Python read-back) and prints a `FAIL` block per failing notebook and, last, the count, `{len(paths) - failed}/{len(paths)}` notebooks pass all eight requirements; it exits with code 1 if any notebook fails [file: `fable-cosmology/notebooks/_build/nbcheck.py`]. With the seven executed notebooks committed in `aca5102` (01–04 of the classical fable, 05–07 of this work) it passes: that commit records “All seven notebooks pass all eight requirements (nbcheck 7/7)” [`commit aca5102`], that is, the last line

```
7/7 notebooks pass all eight requirements
```

and exit code 0.

11.10 Reproduce everything

```
bash "$(git rev-parse --show-toplevel)/fable-cosmology/run_all.sh"
```

After `setup.sh`, it runs, printing a header per step [file: `fable-cosmology/run_all.sh`]: == 1. `solver tests` (`cargo test -release` of `rust/fable_cosmo`, its last three lines, and of `rust/fable_fermion`, its `running/test result` lines and any `FAILED` or `panicked` line); == 2. `notebooks` (executing `notebooks/0*.ipynb` for every notebook, in order, with the `nbconvert` command of section 11.9); == 3. `notebook requirements` (`nbcheck.py`); == 4. `the paper` (`latexmk` of the classical fable-cosmology paper, then its `ls -la` line); and == `all done`. It runs under `set -euo pipefail`, so any failing step stops it with a non-zero exit code. At commit `aca5102` it passed: the commit records that `bash fable-cosmology/run_all.sh` “passes end to end (both crates’ tests, all notebooks, the checker, the paper)” [`commit aca5102`]. Its output on the final pushed state is recorded with the fresh-clone verification of section 12.

11.11 Build the LaTeX twin of this page

```
bash "$(git rev-parse --show-toplevel)/provenance-latex/build_all.sh"
```

or `powershell -ExecutionPolicy Bypass -File provenance-latex\build_all.ps1`. For each of the three LaTeX documents of 2026-09-24, this page’s twin `PROVENANCE-15-FERMION-FABLE-AND-THE-PRIMORDIAL-GRAVITATIONAL-FIELD.tex` among them, it runs `latexmk -pdf -interaction=nonstopmode -halt-on-error <doc>.tex` (output in `provenance-latex/<doc>.latexmk.log`), stops on the first failure (`FAILED`; see `provenance-latex/<doc>.latexmk.log` and `<doc>.log`), warns on undefined references, and prints == `<doc>` and an `ls -la` line per document and finally == `all three PDFs built`. Every document begins with `\input{preamble}` (`provenance-latex/preamble.tex`). MiKTeX’s `latexmk` needs Perl on the `PATH`; Git for Windows ships one.

12 Push and repository verification

The repository. origin is https://github.com/once-ere/Pre-Universe_with_Claude.git, and the work is pushed to its main branch. (The remote named upstream, the author’s other repository, is never pushed to.) The repository carries a `.gitattributes` with `* -text`, so git never rewrites line endings and every file checks out with the bytes that were committed.

The commits of this work (`git log -oneline`, times from `git log -date=iso, PDT`; on 2026-09-24 up to 2bc936d, on 2026-09-25 from aca5102 on):

commit	time	what it contains
e34d454	18:52:36	the 2026-09-16 work as it stood: Part VI of the notebook (172 Input cells, 358/358), the classical fable-cosmology solver, notebooks, results and references
028b379	19:24:32	the classical reference generator no longer prints harmless underflow messages (its values are unchanged)
2f6d322	19:57:29	the 2026-09-16 gaps closed (the student instructions, the paper, the provenance page of that work); a fresh local clone ran its setup and full reproduction end to end
8459c57	20:02:04	a work-in-progress snapshot of the solver crate <code>fable-cosmology/rust/fable_fermion</code>
e130337	20:14:25	a second work-in-progress snapshot of that crate
0a789e2	20:17:18	the LaTeX build tree of the three provenance pages of 2026-09-24: <code>provenance-latex/preamble.tex</code> , <code>build_all.sh</code> , <code>build_all.ps1</code>
d0f2a99	20:18:53	<code>build_all.sh</code> tracked without the executable bit, like the repository’s other scripts
c5892bd	20:38:49	the solver of this effort: the Kohn–Sham fermion gas (stable integrals, per-7-volume mean field, gap equation, lowest-energy branch), the stabilized <code>fable4d</code> and the unstabilized <code>fable8d</code> , <code>cargo test</code> 21 + 8 passing, the independent <code>crosscheck.py</code> (every run below 10^{-8}) and <code>report.py</code>
4fd4c40	20:47:28	Part VII of the notebook (the fermion fable; 198 Input cells, 528/528) and its TeX export
b7ea18d	20:48:42	a wording correction about the canonical connection’s planes: for $k = 5, 6, 7$ the $(0, k)$ plane is a boost and the $(4, k)$ plane a rotation (stated in subsection 6.6 of this page)
bc04ed1	21:24:04	Part VIII of the notebook (this effort’s symbolic side , Sections 30–33; 217 Input cells, 642/642, 0 cells with messages; an independent re-run confirmed it), its run log, and the Mathematica reference runs <code>make_reference_fermion.wls</code> with their two CSVs
18a2501	22:13:42	the wall-state solver <code>fable-cosmology/fermion/waveguide.py</code> , its derivation (94/94) with its log, its Mathematica check (11/11) and its report
fee5603	22:31:20	the page of Effort A (draft), with its LaTeX twin
2bc936d	23:12:51	six solver defects fixed: the negative-energy refusal and the first-order transition located inside the right-hand side and bisected, independently of the output grid; the <code>fable8d</code> shooting no longer accepts $ H_A(1) - 1 > 10^{-6}$; <code>vac_over_rho</code> evaluated stably (a series in $(m - M)/M$, 1.8×10^{-15} against 120-digit references); <code>gap_residual</code> measured against the scale of W' ; one CODATA $\hbar$ for all units (<code>E_c = 2.46261317732986325e-3 eV</code> , <code>Omega_r0 = 9.20960546728882807e-5</code>); <code>cargo test</code> 34/34 (23 unit + 11 integration); the scipy cross-check still passes (largest difference 1.1×10^{-8}). This is the final solver

commit	time	what it contains
aca5102	00:23:46	notebooks 05–07 of this work , executed on the final solver, with every <code>results/nb05_*</code> , <code>nb06_*</code> , <code>nb07_*</code> table and figure they write; fixes of the notebook generator for the classical notebooks 01–04 (their 59 result files byte-identical to before); the checker and conventions for both solvers and all seven notebooks (<code>nbcheck</code> 7/7); setup and reproduction scripts that build and test both crates (<code>run_all.sh</code> passes end to end); <code>fable-cosmology/fermion/waveguide_T16.json</code> committed and the derivation re-run (94/94, 162.8 s); <code>waveguide.py validate</code> exits 1 on any failure
1671155	00:28:54	the full run of the notebook after notebook 06 had written the Rust solver's CSVs, so that Part VIII's two comparisons with them ran and passed: 217 Input cells, 644/644, 0 cells with messages, 208.7 s, 46 witnesses
38701ef	05:40:19	this page and the page of the complete solution, as drafts, with their LaTeX twins and PDFs, the figures the PDFs use, and the repository's front page updated for the pages of this work
3a5020d	05:54:25	Part VIII's spin-connection labels corrected : the Text cell of Section 32 and three assertion labels write the listed components with both flat indices down, $\omega_{\mu ab}$ (section 6.15; no computation changed); the final run recorded again (<code>claude-fable/run_fermion_fable_final.log</code> : 217 Input cells, 644/644, 0 cells with messages, 198.352 s, 46 witnesses, 0 identities on numerical evidence alone); <code>verify_nb_part8.log</code> rewritten (384 cells, none failing to parse); a comment of <code>src/models.rs</code> (section 9.1)

Checking that the remote has what was pushed:

```
cd "$(git rev-parse --show-toplevel)"
git fetch origin
git rev-parse HEAD origin/main          # both must print the same hash
git ls-remote origin refs/heads/main    # the hash GitHub serves for main
git status --short                      # what is not yet committed
```

When this page was brought to its final state (2026-09-25, 06:04 PDT), `git ls-remote origin refs/heads/main` printed

```
3a5020d766d6e306e29ef469c9c0800a16b82e54      refs/heads/main
```

which is the local `main` (3a5020d). Everything of this effort that exists as a notebook part, a log, the reference runs, the solver, the wall-state solver or a notebook is in that pushed history: Part VIII in `bc04ed1` and `3a5020d`, its final run log in `3a5020d`, the wall-state solver in `18a2501` and `aca5102`, the solver crate in `c5892bd` and `2bc936d`, notebooks 05–07 and their results in `aca5102`.

What is committed, and what is not yet. Every code file, log, notebook and result that this page quotes is committed and pushed. The numbers of sections 7–10 are quoted from the committed notebooks 05 and 06 and their `results/nb05_*` and `results/nb06_*` files, and those of section 8 from the committed wall-state report, its derivation log and notebook 07 with its `results/nb07_*` files (commit `aca5102`); the notebooks were executed on the final solver of `2bc936d`, and the refusal of section 9.11 is the bisected one of that solver. The Dirac-sea column `vac_over_rho_c` is evaluated stably since that commit; the old closed form had left rounding noise of $1.6 \times 10^4 - 5 \times 10^4 \rho_{c0}$ at 1 keV [**commit 2bc936d**], far below every ratio of that column that this page quotes, the smallest being the scan minimum 3.79×10^9 . This final version of the page, its LaTeX twin and its PDF replace the drafts of `38701ef` with the final push below.

The final push and the fresh-clone verification. The final state is verified from a fresh clone of the pushed `main`, with these commands (<scratch> is any empty directory outside the repository):

```
cd <scratch>
git clone https://github.com/once-ere/Pre-Universe_with_Claude.git fresh
cd fresh
git log --oneline -1                                # the hash of the pushed main
cd claude-fable
cp claude-fable_Einstein-Rosen-2-Planes.nb committed.nb
python build_tools.py                                # the four lines of section 11.8
cmp claude-fable_Einstein-Rosen-2-Planes.nb committed.nb && echo "notebook rebuilds byte for byte"
wolframscript -file verify_nb.wls                    # section 11.8
wolframscript -file run_from_nb.wls > fresh_run.log 2>&1
sed -n '/RUN SUMMARY/, $p' fresh_run.log             # 217 cells, 644/644, cells w/ msgs 0
diff <(grep '^ PASS' fresh_run.log) <(grep '^ PASS' run_fermion_fable_final.log) && echo "every label
↪ identical"
cd ../fable-cosmology
bash setup.sh                                         # section 11.1
bash run_all.sh                                       # section 11.10: cargo test, notebooks 01-07,
↪ nbcheck, the paper
cd .. && bash provenance-latex/build_all.sh           # section 11.11
```

The final commit hashes, the `git ls-remote` line of the final push, and the results of this fresh-clone verification (the notebook's summary, the solver's test counts, the notebooks' check, the byte-for-byte comparison of the regenerated results) are recorded here: `{{FINAL-PUSH}}`